

Discontinuous spectral/*hp* element methods: development, analysis and applications to compressible flows

by

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*A papà,
tra ponti, calli
e campielli*

Declaration of Originality

This is to certify that the work presented in this thesis has been carried out at Imperial College London and has not been previously submitted to any other university or technical institution for a degree or award. The thesis comprises only my original work, except where due acknowledgement is made in the text.

Gianmarco Mengaldo

London, July 2015

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Abstract

This thesis is concerned with the development and analysis of discontinuous spectral/ hp element methods and their applications to compressible aerodynamics with special focus on boundary-layer flows. In this thesis, we provide a detailed analysis on the connections between the discontinuous Galerkin method and the flux reconstruction approach for multidimensional nonlinear systems of conservation laws on irregular meshes (i.e. meshes with deformed and/or curved elements). The results help a better understanding of the broader class of discontinuous spectral/ hp element methods and allow the direct applications to the flux reconstruction approach of the existing and more established techniques used in the discontinuous Galerkin community for tackling various issues of this class of schemes, including their aliasing problems.

From this perspective, we present two dealiasing strategies based on the concept of consistent integration of the nonlinear terms (also referred to as over-integration of the linear terms). The first is a localised approach and it targets in each element the nonlinearities arising in the problem, while the second is a more global approach which involves a higher quadrature of the overall right-hand side of the discretised equation(s). The two dealiasing strategies have been observed to be effective in enhancing the numerical stability of both schemes, the flux reconstruction and the discontinuous Galerkin approaches.

We finally present the direct numerical simulation of a high-speed subsonic flow past a roughness element, achieved by means of the discontinuous spectral/ hp element methods developed. These results were successively compared to some data obtained from the asymptotic triple-deck theory. This work, besides demonstrating that the class of schemes analysed and developed is attractive for such aerodynamic problems, also addresses the lack of comparisons between theoretical models and numerical simulations.

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Chapter 1

Introduction

In this chapter we provide an overview of the underlying foundations of the current work. We successively detail the motivations and the objectives of this thesis as well as its main scientific and computational contributions. We finally present the outline of the manuscript.

1.1 Overview

In recent years, numerical methods for the solution of partial differential problems which make use of a tessellation of the domain in separate elements (*h-type* refinement), with spectral-like resolution properties (*p-type* refinement) in each element, namely spectral/*hp* element methods, have gained significant attention, both in academia and industry. The reason for this increased interest is multifold - on one hand, spectral/*hp* element methods offer an arbitrary order of spatial accuracy, whilst maintaining a relatively lower computational cost (for a given error threshold one aims to achieve) than traditional low-order finite element schemes; on the other hand, they are well-suited for applications involving complex geometries, which can be a challenge for finite difference schemes. The combination of these two factors makes spectral/*hp* element methods competitive in various fields, including computational fluid dynamics (CFD), computational aero-acoustics (CAA) and magnetohydrodynamics (MHD) to cite just some of them (Cockburn *et al.*, 2000). In CFD for instance, the use of spectral/*hp* element methods is particularly tailored to bluff body problems, vortical flows and unsteady aerodynamics because of their arbitrary dissipation and dispersion properties due to the spectral-like resolution or *p-type* refinement of these schemes. However, the use of these methods can be also advantageous for problems where a high spatial resolution is required, such as the direct numerical simulation (DNS) or the large-eddy simulation (LES) of

boundary layer flows at high-Reynolds numbers.

A spectral/*hp* element method traditionally involves the variational form of the governing equations and, inside each element of the tessellated domain, the definition of an arbitrary-degree ‘P’ polynomial describing the solution onto a set of ‘Q’ quadrature points. The elements are then connected by means of some elemental coupling. From the latter perspective, we distinguish between two different families of spectral/*hp* element methods: ‘continuous’ and ‘discontinuous’ methods. The first family enforces C^0 -continuity of the solution between adjacent elements, while the second allows the solution to be discontinuous yet enforcing continuity on the fluxes by calculating a common interface flux commonly called *numerical flux*. This thesis is concerned with ‘discontinuous’ spectral/*hp* element methods and their applications to compressible boundary-layer flows.

1.1.1 Discontinuous spectral/*hp* element methods

In the past few decades many discontinuous spectral/*hp* element methods have been developed. In the following we will provide a brief account of history of the three main classes of schemes which have constituted the foundations of this work, namely the discontinuous Galerkin method, the spectral difference scheme and the flux reconstruction approach. Although we have mainly explored the first and the third method, it seems appropriate to introduce also the spectral difference scheme since it can be recast within the broader class of discontinuous spectral/*hp* element methods and it can be included in the flux reconstruction framework as we shall see in chapter 2.

Discontinuous Galerkin method

Among the most widely adopted discontinuous spectral/*hp* element methods, is the family deriving from the discontinuous Galerkin (DG) method originally proposed by Reed & Hill (1973) to solve the neutron transport equation. The DG scheme has undergone significant development in various fields during the years. An important milestone is the work by Cockburn *et al.* (1989*b*) who, for the first time, successfully applied the Runge-Kutta schemes to the DG method. The same authors have later provided a more general framework (Cockburn & Shu, 1989; Cockburn *et al.*, 1989*a*, 1990) and they applied the concept to more complex hyperbolic system of conservation laws including the compressible Euler equations (Cockburn & Shu, 1998*b*). The DG method has been also applied to elliptic problems; the work by Bassi & Rebay (1997) represents the first of a series of papers where a second order equation is split into

two first order equations maintaining the compact and discontinuous nature of the DG scheme also for second order problems. Based on this concept, other works include (Cockburn & Shu, 1998a), where the well-known local discontinuous Galerkin (LDG) approach is presented for the first time and (Arnold *et al.*, 2002), where the authors provide a unified analysis of the discontinuous Galerkin method applied to elliptic equations.

Classically, the DG method starts with an expression of the underlying problem in a variational form, whereby the governing system is multiplied by a test function and integrated over the domain, which is itself partitioned into non-overlapping elements. The DG discretisation then allows selecting both a basis of polynomials that represent the solution locally on each element and a set of quadrature points on which the inner products arising in the variational formulation can be calculated. In the DG method, the integration of the nonlinear terms can be numerically exact (this usually produces a computationally costly implementation) or can be inexact; this is the case of collocation-based DG methods, where the use of a collocation projection allows a reduction of the computational costs. Examples of collocation-based DG methods are the so-called nodal DG schemes (Hesthaven & Warburton, 2008), whereby Lagrange interpolants are combined with a set of nodal solution points on a given element, producing an efficient implementation of the DG method.

Spectral difference scheme

While the class of DG methods is probably the most well-known and widely adopted, more recently, another discontinuous spectral/*hp* element method is becoming increasingly popular. This relatively recent approach, namely the spectral difference (SD) scheme, was first introduced by Kopriva (1996) under the name of ‘staggered grid Chebyshev multi domain’ method, and has been further developed some years later by Liu *et al.* (2006), who applied the method to both triangular and quadrilateral elements and named it SD scheme. The SD scheme is conceptually similar to the DG method but is based on the differential form of the problem. This aspect has offered an alternative route that avoids the need for quadrature rules, making these schemes potentially more efficient to implement.

Flux reconstruction approach

The most recent discontinuous spectral/*hp* element method is however the flux reconstruction (FR) approach, first proposed by Huynh (2007). This approach is also based

on the differential form of the governing equations and shares the same benefits as the SD scheme, potentially more efficient than variational-based methods, where there is the necessity of calculating integrals. As shown by Huynh, the FR approach can recover a wide range of numerical schemes including the SD and the nodal DG methods (at least for linear problems) and has been significantly developed in the past few years. Vincent *et al.* (2011b), for instance, have identified a new class of energy-stable FR schemes, referred to as Vincent-Castonguay-Jameson-Huynh (VCJH) schemes on quadrilateral tensor-product meshes. Here, a single real-valued parameter ‘ c ’ dictates the scheme that is recovered and enables one to recover differential- and variational-based schemes, such as a particular SD method or a nodal DG scheme. The same authors, have extended the VCJH schemes to triangular elements (Castonguay *et al.*, 2012) and applied this new class of energy-stable schemes to the compressible Euler equations (Castonguay *et al.*, 2011). They have also gained some insights into the dispersion and dissipation properties of the VCJH schemes (Vincent *et al.*, 2011a) as well as into the nonlinear stability properties of these schemes (Jameson *et al.*, 2012). Some extensions to elliptic problems have also been presented in the literature, see for example (Williams *et al.*, 2011), where the FR approach has been applied to the compressible Navier-Stokes equations.

Given the vast variety of discontinuous spectral/ hp element methods and their development in different time frames, with the DG method being the oldest and probably the most established, it is crucial to understand whether there are some connections between these relatively similar schemes and, if this is the case, how we can use these connections to attain some advantages.

1.1.2 Connections between discontinuous spectral/ hp element methods and their practical implications

All the numerical schemes above are similar, although the discontinuous Galerkin is based on the variational form of the equations while the other two are based on the differential form, and they constitute the wider class of discontinuous spectral/ hp element methods. It seems therefore reasonable to ask what the connections between these schemes are and how we can exploit them. Some works in this sense have already been carried out; as already mentioned, Huynh (2007) showed that the FR approach can recover a particular SD or DG scheme for one-dimensional problems. Allaneau & Jameson (2011) have gone further, showing

that, the VCJH schemes can be seen as special DG schemes, where a linear filter is applied to the right-hand side of the DG method. However, these works are restricted to linear problems and the underlying connections between the FR approach and the DG method for nonlinear problems on generic tensor-product irregular meshes (i.e. meshes formed by deformed or curvilinear elements) have so far remained unexplored. This is of particular interest, especially for practical applications, such as the DNS or LES of high-Reynolds numbers compressible flows, where the discretisation of the nonlinear terms of the governing equations plays a key role in the numerical stability of a given discontinuous spectral/ hp element method. In addition, these nonlinear terms intrinsically related to the (set of) equation(s) being solved may couple with the deformed/curvilinear elements possibly present in the mesh casting further complexity into the problem.

In the previous section, we have mentioned that, for the DG method, it is possible to achieve different numerical properties depending on the choice of the quadrature and the solution points. For instance, one may choose to use a collocation projection, leading to the class of collocated DG schemes, or use a higher quadrature, leading to a better integration of the nonlinear terms (when present). This is a particularly relevant point in the context of discontinuous spectral/ hp element methods because the use of collocation-based methods, such as the nodal DG schemes as well as the FR approach and the SD scheme, as they have been originally presented, under-integrate the nonlinear terms of the discretised equations, and therefore introduce errors, yielding to the so-called polynomial aliasing (or simply aliasing). Polynomial aliasing have been investigated in a number of papers, especially in the spectral and spectral element communities. These works include (Orszag, 1971; Kirby & Karniadakis, 2003; Kirby & Sherwin, 2006a), where the authors argue that the non-exact integration of the nonlinear terms also referred to as ‘insufficient quadrature’ leads to a degradation of the accuracy. The accuracy degradation can be quantified by theoretical estimates (Canuto & Quarteroni, 1982) and might be negligible for well-resolved problems. However, when dealing with badly- or marginally-resolved simulations (which is often the case for high-Reynolds number flows in complex geometries), these errors besides degrading the accuracy, also affect the numerical stability and therefore the reliability of discontinuous spectral/ hp element methods, leading to the so-called aliasing-driven instabilities. These aliasing-driven instabilities, in turn, affect the numerical stability of this class of schemes.

A lack of numerical stability is one of the most important shortcoming of discontinuous spectral/ hp element methods (along with the mesh generation and the algorithmic complexity) and it is crucial to develop new or optimise existing strategies to address this issue.

Finding the underlying connections between the FR approach and the DG scheme and their practical implications is a promising strategy, given the existing machinery developed in the DG community for addressing such problem, which include the adoption of spectral vanishing viscosity (SVV) (Kirby & Karniadakis, 2002; Pasquetti, 2006; Kirby & Sherwin, 2006*b*), polynomial filtering (Fischer & Mullen, 2001; Fischer & Kruse, 2002), more stable formulations of the nonlinear terms (Blaisdell *et al.*, 1996) and over-integration (i.e. consistent integration of the nonlinear terms) (Kirby & Karniadakis, 2003).

To conclude this section, we want to provide an example on how the connections between two class of schemes, developed in two different communities, has been of practical application. This example is related to the recent work by Gassner (2013), who has shown how the collocated DG scheme on a set of Gauss-Lobatto-Legendre points can be seen as an SBP-SAT (summation-by-parts simultaneous approximation term) finite difference scheme. Based on this finding, the same author has constructed a conservative, skew-symmetric and energy stable collocated DG formulation for the nonlinear Burgers' equation and for the compressible Euler equations (Gassner, 2014).

This example demonstrates how, the numerical methods developed in two different scientific communities can be effectively linked (whether they are not the same), and some concepts can be directly applied to improve the properties of one, the other or both schemes. The example presented was specifically concerned with the stability properties of the nodal DG schemes. The derivation of a skew-symmetric form of the DG method allows better numerical stability properties than the standard conservative form adopted in discontinuous spectral/*hp* element methods.

1.1.3 Applications to compressible boundary-layer flows

As briefly mentioned above, in CFD, discontinuous spectral/*hp* element methods are particularly suited for applications where a high spatial resolution and favourable dispersion and dissipation properties are required. Some examples of recent applications of this class of methods in CFD are (Landmann *et al.*, 2008; Uranga *et al.*, 2011; Bolemann *et al.*, 2015).

Compressible boundary-layer flows at high-Reynolds numbers represent a potentially interesting area of applicability of discontinuous spectral/*hp* element methods. In fact, the correct analysis of this type of flows usually requires a very high spatial resolution and computationally efficient algorithms in order to provide an accurate solution in a reasonable amount of time.

Particularly interesting from this point of view is the analysis of the flow past roughness

elements, steps and gaps. These features are very common in aircraft wings and their implications in the onset mechanisms of laminar-turbulent transition is crucial for reducing the skin-friction drag - one of the big target of many aircraft manufacturers, including Airbus, Boeing and Embraer.

This class of problems have been widely studied in the past by means of various methodologies, such as the asymptotic triple-deck theory (Smith, 1973; Rizzetta *et al.*, 1978), experiments (Klebanoff & Tidstrom, 1972) and DNS (Rizzetta & Visbal, 2007; Rizzetta *et al.*, 2010; Rizzetta & Visbal, 2014) just to cite a few. Whilst various comparisons exist between experiments and numerical simulations as well as between theoretical models and experiments, little work has been carried out in comparing theoretical models, such as the asymptotic triple-deck theory, with DNS or LES. Given the importance that triple-deck theory is assuming in receptivity calculations, having such a comparison can provide additional insights into the capabilities and limitations of the theory and verify the applicability of discontinuous spectral/*hp* element methods to this class of challenging problems.

1.2 Motivations for and objectives of the thesis

In the SherwinLab (<http://sherwinlab.ae.ic.ac.uk>), of which the author is a member, a robust effort has been made in the past few years to develop a comprehensive numerical tool in order to solve a wide range of partial differential problems. These efforts have come together in the creation of the C++ object-oriented spectral/*hp* element library Nektar++ (<http://www.nektar.info>). The Nektar++ project began ten years ago with the main scope of developing a flexible and efficient spectral element framework to solve various differential problems, with a special focus on fluid dynamics.

At the beginning of this thesis work, the Nektar++ project was seen as potentially strategic for the so-called ‘Laminar Flow Control UK’ (LFC-UK) project (<http://www.imperial.ac.uk/lfc-uk>) and the library was being considered as the possible tool for DNS and LES simulations within the LFC-UK group at Imperial College London (ICL). Since, the primary aim of the ‘LFC-UK’ project is the development of underpinning technology for laminar flow control in a compressible regime, from a DNS/LES perspective it was crucial to develop the core building blocks to produce a reliable, accurate and efficient numerical tool for compressible aerodynamics, using cutting-edge and developing novel numerical technologies.

The library, at that time, included a well-developed implementation of the continuous Galerkin (CG) method applied to the incompressible Navier-Stokes equations, and a relatively

well-developed DG method applied to the compressible Euler equations. The relatively new discontinuous spectral/*hp* element methods based on the differential form of the governing equations, namely the SD and FR approaches, had been consistently promoted as potentially competitive compared to the well established DG schemes for compressible flow applications. The FR approach in particular, given its ability of recovering a broad range of existing schemes, including the SD and DG methods, was a very attractive option and its integration within the Nektar++ library provided a solid starting point from which we could compare the two schemes within the same numerical environment and possibly produce new strategies for further developing and improving these methods. From the last perspective, an interesting question to answer was what the connections between the DG method and the FR approach are. Although some literature on this topic already existed, this was confined to linear problems and the connections between the two numerical frameworks for nonlinear equations and irregular meshes constituted an important gap which prevented to fully understand which the implications of adopting one or the other approach were for more complex applications.

The numerical stability of discontinuous spectral/*hp* element methods and efficient yet effective strategies for improving it constituted an important driver of this thesis from an engineering perspective, because one of the aims was to construct a reliable numerical framework based on this class of methods for the DNS/LES of compressible flows. Therefore, understanding the connections between the two classes of schemes was a promising candidate to gain further insights into the underlying numerics and into the strategies for improving the numerical stability, yet maintaining the advantages in terms of efficiency of these schemes.

From a more fundamental point of view, the connections between these schemes allowed us to achieve a broader and more comprehensive picture on the overall class of discontinuous spectral/*hp* element methods, giving us the possibility of drawing some considerations on the advantages and disadvantages of one or the other numerical framework.

Given what discussed above, the overarching goal of the research project was to understand the state-of-the-art of the current numerical tools in the context of discontinuous spectral/*hp* element methods used in compressible aerodynamics, analyse their intrinsic connections, investigate strategies for solving the existing shortcomings, with the ultimate goal of deploying a robust, reliable and efficient discontinuous spectral/*hp* element numerical framework for compressible aerodynamics and provide a more comprehensive view on discontinuous spectral/*hp* element methods. The additional, yet no less important, goal regarded providing some DNS results (relevant to the LFC-UK group) in order to prove the effectiveness of the new numerical framework developed. This resulted in the DNS of a flow past a roughness

element in subsonic and transonic conditions and the comparison of these results with the triple-deck theory. This work aimed to address the lack in terms of comparisons between DNS and triple-deck data, with the latter method (i.e. triple-deck theory) being increasingly adopted in recent years for receptivity calculations which are considered a key for understanding the onset mechanisms of the laminar-turbulent transition.

1.3 Contributions and literature survey

The contribution of this research is the development of the Nektar++ project in the SherwinLab at ICL and, as a part of this work, considerable research effort was made in the investigation on the connections between existing and novel discontinuous spectral/ hp element methods as well as into robust strategies to improve the numerical stability of this class of schemes. In addition, some aerodynamic problems concerning separated flows were explored with a view to making a direct comparison between DNS and triple-deck data. These efforts have resulted in a series of journal publications and conference presentations.

In the following, we highlight the main contributions of the thesis to the existing literature and we briefly present the additional features which have been integrated into the Nektar++ library as part of this thesis.

1.3.1 Connections between DG and FR schemes

The connections between the DG and the FR schemes have been thoroughly investigated in this work. The original idea came from a gap in the literature regarding the connections between the DG method and the FR schemes for multi-dimensional nonlinear problems. This resulted in a first paper published in ‘International Journal for Numerical Methods in Fluids’, (De Grazia *et al.*, 2014) by the author and his co-workers, where we explored the connections between three nodal versions of tensor product discontinuous Galerkin spectral element approximations and two types of flux reconstruction schemes for solving systems of conservation laws on quadrilateral meshes. The different types of discontinuous Galerkin approximations arose from the choice of the solution points of the Lagrange basis representing the solution and from the quadrature approximation used to integrate the mass matrix and the other terms of the discretisation. We considered both a linear and a nonlinear advection equation on a regular mesh, examining the mathematical properties which connect these discretisations. These arguments were further confirmed by the results of an empirical numerical study. In this work we additionally made some considerations concerning the aliasing

sources due to the nonlinearity of the flux functions and their intrinsic implications on the connections between the two class of schemes and we presented a comparative analysis of the performance of the FR and DG schemes investigated.

In a second paper which is currently under review in ‘Journal of Scientific Computing’, the author and co-workers extended the work in (De Grazia *et al.*, 2014) by considering irregular/curvilinear meshes and therefore non-constant geometric terms arising in the discretised equations due to the mapping between the reference and the physical space. In this second paper, the author and co-workers also made some additional considerations in terms of aliasing issues due to the irregularity of the mesh and explored their implications.

This study is presented in chapter 4.

1.3.2 Dealiasing techniques for discontinuous spectral/*hp* element methods

After having explored the connections between the DG and the FR approaches for nonlinear multidimensional conservation laws on irregular meshes and identified the common aliasing sources, which may lead to numerical instabilities, we explored some dealiasing strategies to be applied to both classes of schemes.

A common choice in the DG community is, among others, the use of additional quadrature points to eliminate the errors due to the under-integration of the nonlinear terms. This technique, also referred to as over-integration (Kirby & Karniadakis, 2003), seemed particularly attractive and its deployment to the FR approach, given the connections found in the previous works by the author and co-workers, was a natural step.

These considerations resulted in a paper published in ‘Journal of Computational Physics’ (Mengaldo *et al.*, 2015a), where we detailed two dealiasing strategies based on the concept of over-integration (of the linear terms) or consistent integration (of the nonlinear terms), the first of which uses a localised approach that is useful when the nonlinearities only arise in parts of the problem and the second is based on the more traditional approach of using a higher quadrature. The main goal of both dealiasing techniques was to improve the numerical stability in the discontinuous spectral/*hp* element methods explored, thereby reducing aliasing-driven instabilities. We also addressed the role of the volumetric and boundary terms contribution naturally arising in the formulation of discontinuous spectral/*hp* element methods. This work is presented in chapter 5, where we show the main results of the paper.

1.3.3 Guide to the robust implementation of boundary conditions

The nature of boundary conditions, and how they are implemented, can have a significant impact on the stability and accuracy of a CFD solver. Given the objectives of this thesis, the implementation of boundary conditions has represented a crucial aspect. The author and co-workers noted that practical guidelines for the robust implementation of boundary conditions for this class of methods could have been useful for the scientific community and especially for young scientists facing the challenge of implementing a compressible flow solver by means of discontinuous spectral/*hp* element methods. This resulted in a conference paper accepted at the 2014 AIAA conference in Atlanta. The aim of this paper was to assess how different boundary condition implementations impact the performance of discontinuous spectral/*hp* element methods, when these schemes are used to solve the Euler and compressible Navier-Stokes equations on irregular grids. Specifically, the paper investigated inflow/outflow and wall boundary conditions enforced by modifying the boundary flux - therefore in a weak manner. In this paper we provided implementation tips as well as guidelines for the successful integration of weak boundary conditions into a discontinuous spectral/*hp* element framework.

Part of the contents of this paper are reported in chapter 3.

1.3.4 Applications to compressible flows

The numerical framework developed and analysed during this thesis has been thoroughly tested, since the ultimate goal was to produce a DNS/LES compressible solver for the LFC-UK centre at ICL. The tests have been carried out in several two-dimensional geometries and the numerical framework has proven successful being applied to a problem relevant to the LFC-UK group. This problem was specifically concerned with the boundary-layer separation in subsonic and transonic flows caused by a two-dimensional isolated wall roughness. The process of the separation was analysed by means of two approaches: the DNS of the compressible Navier-Stokes equations (through the discontinuous spectral/*hp* element methods developed throughout the thesis), and the numerical solution of the triple-deck equations (which has been performed using an existing solver developed by a co-worker at the LFC-UK centre). The calculations were conducted by using two different free-stream Mach numbers $Ma_\infty = 0.5$ and $Ma_\infty = 0.87$ and the Reynolds number adopted was $Re = 4 \times 10^5$ based on the stream wise position of the roughness element. We used different roughness element heights, some of which were large enough to cause a well-developed separation region behind the roughness. The requirement in terms of spatial resolution were quite severe and the DNS

performed by using the discontinuous spectral/*hp* element methods developed have proven particularly reliable, accurate and ultimately robust in terms of numerical stability. In particular, in order to achieve stable simulations we needed to adopt a localised dealiasing strategy for the first stages of each DNS, which allowed maintaining relatively efficient simulation performance.

This work has been published in ‘Journal of Fluid Mechanics’ (Mengaldo *et al.*, 2015b) and is presented in chapter 6.

1.3.5 Nektar++: the compressible flow solver

Nektar++ is an open-source software framework designed to support the development of high-performance scalable solvers for partial differential equations using the spectral/*hp* element method. The Nektar++ project has been designed to make this class of numerical approaches accessible to the broader scientific and industrial communities within an efficient object-oriented C++ framework. The software supports a variety of functionalities, including advanced pre- and post-processing tools as well as pre-written solvers relying on the underlying spectral/*hp* element method numerics.

The author developed the LDG approach for second order problems and both the advection and diffusion FR classes. The implementation of these operators is general and can be applied to the majority of the solvers within the library. The author also constructed the localised dealiasing strategy for improving the numerical stability of both the DG and FR approaches and he was responsible for developing the compressible flow solver. From the latter perspective, he developed the viscous term of the compressible Navier-Stokes equations and played an important role in developing the Riemann solvers, the boundary conditions and the overall layout of the compressible flow solver.

Several of the routines implemented have been used to obtain the results presented in the published paper ‘Nektar++: An open-source spectral/*hp* element framework’ where the author and co-developers have presented the Nektar++ functionalities (Cantwell *et al.*, 2015).

1.4 Outline of the thesis

This thesis is organised as follows. In chapter 2, after a brief account of the most common spatial discretisations one can adopt for a partial differential problem, we introduce the foundations of spectral/*hp* element methods, highlighting the main operations required, such as numerical integration and differentiation. We also present the nomenclature used throughout

this thesis and we finally detail the two discontinuous spatial discretisations analysed in this work, the DG method and the FR approach, for both first and second order problems. At the end of the chapter we show some verification results on an unsteady advection-diffusion equation. In chapter 3, we present the compressible Euler and Navier-Stokes equations and their discretisations by means of the DG and FR approaches. In this chapter we also show the implementation of the boundary conditions and some results to verify the implementation of the compressible flow solver. In chapter 4, we detail the connections between the DG method and the FR approach for nonlinear multidimensional systems of conservation laws on irregular meshes. In chapter 5, we describe the dealiasing strategies applied to both the DG and FR approaches and how they can help enhancing the numerical stability of these methods. In chapter 6, we present the applications to high-speed subsonic flows past a roughness element. Finally, in chapter 7 we draw the conclusions.

Also to be noted that, although part of the relevant literature has been presented in this introductory chapter, in the rest of the thesis, at the beginning of each chapter, we will provide a detailed literature review concerning the specific topic explored in the chapter.

Chapter 2

Numerical approximation of PDEs

In this chapter, after a brief review of the most common numerical discretisations employed for approximating partial differential equations (PDEs), we focus on the spectral/*hp* element method, describing the main operations required to construct the spatial discretisation and introducing the time discretisation techniques usually adopted. Successively, we introduce the two discontinuous spectral/*hp* element methods developed and analysed in this thesis, namely the discontinuous Galerkin (DG) method and the flux reconstruction (FR) approach. We finally present some results on the unsteady advection-diffusion equation, which demonstrate the correct implementation of the two methods and highlight their main features.

2.1 Preliminaries

When approximating numerically a partial differential problem, the continuous PDEs are replaced by their discrete representation. The discrete equations are satisfied only on a finite number of points through a set of prescribed conditions, which, in turn, define the numerical method employed (also referred to as projection operator). The so-called method of weighted residuals shows how the choice of different test (or weight) functions can be used to construct many of the most popular numerical methods for solving a set of PDEs (Karniadakis & Sherwin, 2005; Finlayson, 2013).

Let us consider a linear partial differential equation in a domain Ω , denoted by

$$\mathcal{L}(u) = 0 \tag{2.1}$$

and subject to appropriate initial and boundary conditions. We now assume that the solution

$u(\mathbf{x}, t)$, which, in general, depends on space and time, can be accurately approximated by the following finite expansion

$$u^\delta(\mathbf{x}, t) = u_0(\mathbf{x}, t) + \sum_{i=1}^{N_{dof}} u_i(t) \phi_i(\mathbf{x}), \quad (2.2)$$

where $\phi_i(\mathbf{x})$ are the so-called trial (or expansion) functions which depend on space only, $u_i(t)$ are the N_{dof} unknown coefficients which depend on time only and $u_0(\mathbf{x}, t)$ is chosen such that it satisfies the initial and boundary conditions of the problem. If we now substitute Eq. (2.2) into Eq. (2.1), the original continuous problem is not satisfied everywhere and contains an error denoted by:

$$\mathcal{L}(u^\delta) = R(u^\delta), \quad (2.3)$$

where R is called residual. In order to solve Eq. (2.3), we need to place a restriction on the residual R . This will reduce Eq. (2.3) into a system of ordinary differential equations in $u_i(t)$ and, if Eq. (2.1) is independent of time, the coefficient can be directly calculated from a system of algebraic equations. There exist various restrictions we can place on R , each of which will lead to different numerical schemes. These restrictions are usually applied through the Legendre inner product, defined as:

$$(h, g) = \int_{\Omega} h(\mathbf{x}) g(\mathbf{x}) d\mathbf{x}, \quad (2.4)$$

where h and g are two general functions which depend on space only. In particular, we require that the inner product of the residual with respect to a test function $v_j(\mathbf{x})$ is equal to zero, that is

$$(v_j(\mathbf{x}), R) = 0, \quad j = 1, \dots, N_{dof}. \quad (2.5)$$

We note that the residual $R(\mathbf{x})$ tends to zero as $N_{dof} \rightarrow \infty$ since the discrete solution $u^\delta(\mathbf{x}, t)$ asymptotically approaches the exact solution $u(\mathbf{x}, t)$. Depending on the choice of the trial function $\phi_j(\mathbf{x})$ and of the test function $v_j(\mathbf{x})$, we can obtain different numerical techniques. The most popular numerical approaches and the associated choice of the trial and test functions are reported in the following.

1. *Collocation methods:*

In collocation methods the test functions $v_j(\mathbf{x})$ are equal to the Dirac delta function

$$v_j(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}_j), \quad (2.6)$$

where $\mathbf{x}_j$ denotes a set of collocation points at which the differential equation is exactly satisfied, i.e. the residual is set to zero: $R(\mathbf{x}_j) = 0$. The collocation projection of the residual is usually employed in spectral methods (Gottlieb *et al.*, 1977; Canuto *et al.*, 1988a) and in finite-difference methods (LeVeque, 2007; Lele, 1992).

2. *Finite volume methods:*

In finite volume (FV) methods, the domain Ω is divided into N_{dof} non-overlapping subdomains Ω_j and the test function is chosen to be of the form

$$v_j(\mathbf{x}) = \begin{cases} 1, & \text{inside } \Omega_j \\ 0, & \text{outside } \Omega_j. \end{cases} \quad (2.7)$$

FV methods are one of the most popular discretisations in computational fluid dynamics and there exists various versions, both low and high-order. A broad overview of finite volume methods for hyperbolic problems can be found in (LeVeque, 2002) while some specific references for high-order FV schemes include (Barth & Frederickson, 1990) for k - *exact* methods, (Harten *et al.*, 1987; Abgrall, 1994) for FV type essentially non-oscillatory (ENO) methods and (Liu *et al.*, 1994; Hu & Shu, 1999) for FV type weighted essentially non-oscillatory (WENO) methods.

3. *Least squares methods:*

In least squares methods the residual is set to $v_j = \partial R / \partial u_j$ and one has to minimise the quadratic functional (R, R) . In this way it is possible to find the unknown coefficients u_j . This projection has been used in traditional finite element methods (Bochev & Gunzburger, 1998; Jiang, 1998) and has been recently extended to spectral/*hp* element methods (Proot & Gerritsma, 2002).

4. *Galerkin methods:*

In Galerkin methods, (also known as Bubnov-Galerkin methods), the test functions are chosen to be identical to the trial functions, $v_j(\mathbf{x}) = \phi_j$. This type of projection is a minimisation of the residual over the domain Ω . The majority of finite element and spectral/*hp* element methods use this type of projection (Karniadakis & Sherwin, 2005).

5. *Petrov-Galerkin methods:*

Petrov-Galerkin methods can be seen as an extension of the more traditional Galerkin (or Bubnov-Galerkin) methods where the test functions are different from the trial

functions $v_j(\mathbf{x}) \neq \phi_j$. Specifically, they are chosen such that they are perturbations upon the trial functions which improve the numerical stability properties of the resulting schemes (Brooks & Hughes, 1982; Hughes *et al.*, 1986).

By defining the type of projection operator, the method of weighted residuals shows how to construct different numerical techniques which can be applied to any partial differential problem. However, it does not define the trial functions and the approximation space. For instance, spectral methods make use of expansion functions which are non-zero throughout the solution domain (also referred to as global expansion functions). Finite element methods, instead, use expansion functions defined in a local finite region of the domain.

After having defined both the projection operator and the approximation space, the spatial discretisation of the problem is complete and the PDE(s) has become an algebraic system of equations (or a system of ordinary differential equations (ODE)s if the original problem depended on time).

In this work, the main focus is on spectral/*hp* element methods, which combine the local nature of the expansion functions typical of finite element methods and the arbitrary expansion functions common in spectral methods.

2.2 Spectral/*hp* element methods

The starting point from which spectral/*hp* element methods can be introduced is the meaning of their name. The word ‘*spectral*’ and the letter ‘*p*’ derive from spectral methods. In these methods, the order of accuracy increases exponentially with the polynomial order. This particular feature is also referred to as *p-type* refinement. On the other hand, the letter ‘*h*’ and the word ‘*element*’ come from finite element methods, where the spatial discretisation is achieved by dividing the computational domain into non-overlapping subdomains and by approximating the solution through a piecewise interpolation function on each subdomain (or element). The spatial accuracy in this case is achieved by increasing the number of elements of the spatial discretisation. This particular feature is also referred to as *h-type* refinement and allows the discretisation of complex geometries.

Spectral/*hp* element methods combine together the *h-type* refinement, thus the geometrical flexibility of finite element methods, and the superior spatial accuracy properties of spectral methods. In particular, a spectral/*hp* element method involves a decomposition of the domain into subdomains or elements, where each of these elements is mapped into a reference domain (also referred to as standard element) through a suitable parametric mapping

between the original and the standard space. In each standard element, the solution is represented through an arbitrary-degree polynomial function (also referred to as expansion basis function) and the operations required, namely multiplications, integrals and differentiations, are performed. Successively, the global problem is finalised by means of specific connectivity rules at the interfaces between adjacent elements. The original PDE, if it does not depend on time, has now been transformed into a system of algebraic equations. Equivalently, if the original problem depended on time, the PDE has been transformed into a system of ODEs which needs to be advanced in time via a suitable time-integration scheme.

In the following, we first introduce the spatial discretisation along with the principal concepts for implementing a spectral/ hp element method and successively we review the most common time integration schemes adopted for solving time-dependent problems.

2.2.1 Spatial discretisation

As just mentioned, the main building blocks involved in the construction of spectral/ hp element methods are the domain decomposition, through which the computational domain is divided into non-overlapping elements, the local to standard element mappings by which the local element is transformed into a reference element, the expansion bases describing the solution inside each reference element, the numerical integration and differentiation and the connectivity rules. An illustrative diagram of the main operations involved when constructing a spectral/ hp element method is shown in figure 2.1.

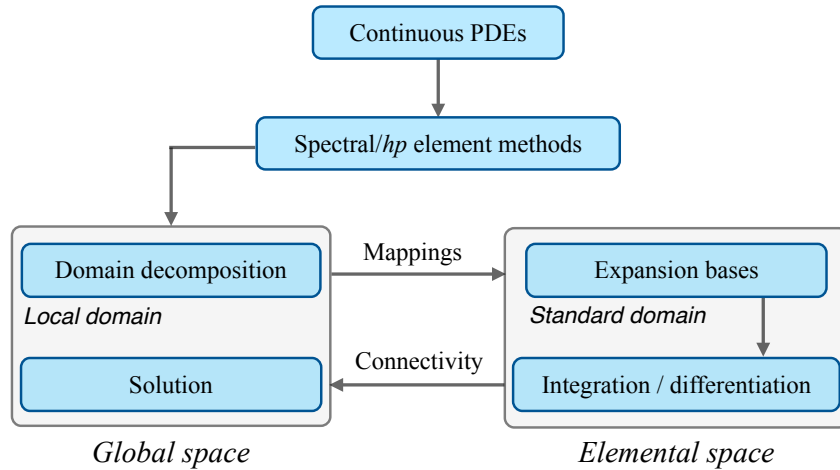


Figure 2.1: Illustrative diagram of the main building blocks underlying spectral/ hp element methods.

These operations are briefly introduced in the next few sections for one-dimensional and quadrilateral tensor-product elements. The extension to three-dimensional tensor-product elements follows easily and it is not presented in this chapter for the sake of compactness. We do not present these concepts for other element shapes such as triangles because the main focus of the thesis has been on quadrilateral meshes.

Domain decomposition

If we denote the computational domain by Ω , its decomposition (or tessellation) consists of defining N non-overlapping subdomains (or elements) Ω_n as follows

$$\Omega = \bigcup_{n=0}^{N-1} \Omega_n, \quad \bigcap_{n=0}^{N-1} \Omega_n = \emptyset. \quad (2.8)$$

There exist various element shapes which can be used in conjunction with spectral/*hp* element methods, such as segments in one-dimensional problems, quadrilaterals and triangles in two-dimensional problems and hexahedra, tetrahedra, prisms and pyramids for three-dimensional problems. To facilitate the implementation of the main operations needed, in spectral/*hp* element methods it is common to associate to each shape a reference element usually called standard domain/element and denoted by Ω_s .

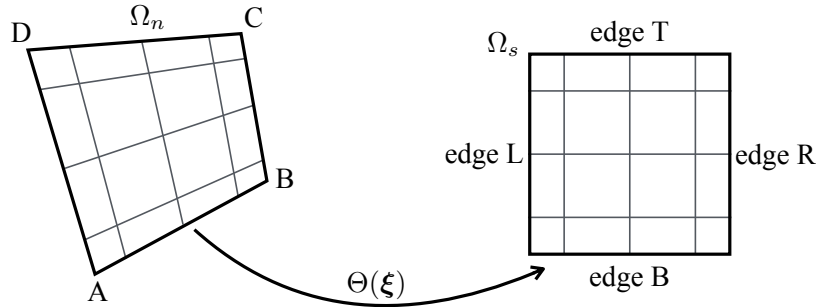


Figure 2.2: Representation of and conventions adopted for the quadrilateral standard element.

On each standard element Ω_s , we then specify our expansion bases and the main operations required, such as numerical integration and differentiation. For segments the standard element is defined as

$$\Omega_s = (\xi) \in [-1, 1], \quad (2.9)$$

with ξ being the one-dimensional coordinate associated to Ω_s . The standard quadrilateral

element is instead defined as

$$\Omega_s = (\xi_1, \xi_2) \in [-1, 1] \times [-1, 1], \quad (2.10)$$

with ξ_1 and ξ_2 being the orthogonal coordinates associated to Ω_s . Figure 2.2 illustrates the nomenclature and conventions adopted for a quadrilateral standard element. Note that, throughout this thesis, we will indicate with the letter ‘B’ and the letter ‘T’ the bottom and top edges, respectively, while, with the letter ‘L’ and the letter ‘R’, the left and right edges. Note also that $\Theta(\xi)$ denotes the mapping between the local and the standard element described in the next section.

Local to standard element mappings

In order to transform the local domain Ω_n into a standard domain Ω_s , we need to define the mappings of the transformation. For one-dimensional elements, the mapping simply reads

$$x = \Theta_n(\xi) = \phi_0(\xi) x_{n-1} + \phi_1(\xi) x_n \quad \xi \in \Omega_s, \quad (2.11)$$

where we used generic expansions ϕ_0 and ϕ_1 to map the coordinates. Equation (2.11) is known as *parametric mapping* and, if the order of the mapping expansions ϕ_0 and ϕ_1 is the same as the dependent variables, then the mapping is defined as iso-parametric, while if the order is lower or higher, the mapping is defined as sub- or super-parametric, respectively.

The concepts expressed for one-dimensional elements can be easily transferred to two-dimensional tensor-product elements. In particular, if we define the following representations of the Cartesian coordinates x_1 and x_2 :

$$x_1 = \Theta_{n,1}(\xi_1, \xi_2) \quad x_2 = \Theta_{n,2}(\xi_1, \xi_2), \quad (2.12)$$

the elemental mapping for a general straight-sided quadrilateral element is

$$\begin{aligned} x_i = \Theta_{n,i}(\xi_1, \xi_2) = & x_i^A \frac{1 - \xi_1}{2} \frac{1 - \xi_2}{2} + x_i^B \frac{1 + \xi_1}{2} \frac{1 - \xi_2}{2} + \\ & + x_i^D \frac{1 - \xi_1}{2} \frac{1 + \xi_2}{2} + x_i^C \frac{1 + \xi_1}{2} \frac{1 + \xi_2}{2}, \quad i = 1, 2, \end{aligned} \quad (2.13)$$

where the vertices A , B , C and D of the quadrilateral are depicted in figure 2.2. If the

quadrilateral element have curvilinear edges, then the mapping can be represented as follows:

$$x_i = \Theta_{n,i}(\xi_1, \xi_2) = \sum_{p=0}^{Q_1-1} \sum_{q=0}^{Q_2-1} \bar{x}_{pq}^i \phi_p(\xi_1) \phi_q(\xi_2), \quad (2.14)$$

where ϕ_p and ϕ_q are the same basis functions used for representing the solution. We note that, if x_i is a polynomial of order $P < Q_1, Q_2$, the mapping is sub-parametric while if $P > Q_1, Q_2$ the mapping is super-parametric.

From the mappings defined above, it is possible to obtain the geometric terms usually needed when transforming the equations from the physical to the standard space and vice versa. From this perspective, we define the following geometric terms for the two-dimensional case:

$$\mathbf{G}_n = \begin{bmatrix} \frac{\partial x_1}{\partial \xi_1} & \frac{\partial x_1}{\partial \xi_2} \\ \frac{\partial x_2}{\partial \xi_1} & \frac{\partial x_2}{\partial \xi_2} \end{bmatrix} = J_n \begin{bmatrix} \frac{\partial \xi_2}{\partial x_2} & -\frac{\partial \xi_1}{\partial x_2} \\ -\frac{\partial \xi_2}{\partial x_1} & \frac{\partial \xi_1}{\partial x_1} \end{bmatrix}, \quad J_n = |\mathbf{G}_n|. \quad (2.15)$$

where $\mathbf{G}_n$ is the deformation gradient and J_n is the Jacobian of the transformation and is defined as the determinant of the deformation gradient. Note that the one-dimensional case simply reduces to $\mathbf{G}_n = \partial x / \partial \xi = J_n$. These geometric terms will be extensively used in section 2.3 and section 2.4 as well as in chapter 4.

Expansion bases

After having introduced the mappings and the geometric terms needed for transforming the equations from the local to the standard space, it is necessary to define the expansion bases representing the quantities in the discretised equations.

In spectral/ hp element methods, the expansion bases employed in each standard element are usually polynomials and can be of *modal* or *nodal* type. To highlight the differences between a modal and a nodal expansion, we define the following two complete sets of polynomials up to order P :

$$\begin{aligned} \phi_p^M(\xi) &= \xi^p, & p &= 0, \dots, P, \\ \phi_p^N(\xi) &= \frac{\prod_{q=0, q \neq p}^P (\xi - \xi_q)}{\prod_{q=0, q \neq p}^P (\xi_p - \xi_q)}, & p &= 0, \dots, P. \end{aligned} \quad (2.16)$$

The first expansion $\phi_p^M(\xi)$ is referred to as a *modal* or a *hierarchical* expansion because the

order (P-1) expansion set is contained within the order P expansion set. A modal expansion, therefore, satisfies the following relationship

$$\chi_{P-1}^\delta \subset \chi_P^\delta. \quad (2.17)$$

In simple modal bases, usually all the modes (or polynomials) influence the boundary points of a given element. This can be particularly inefficient when we need to apply connectivity rules between the various element of the spatial discretisation. It is therefore common to redefine these expansions in a *boundary-interior decomposition* fashion, where only two modes are non-zero at the boundaries, while all the others are identically zero at the boundaries and assume non-zero values at the interior points only. An example of modal basis using a boundary-interior decomposition is the following:

$$\phi_p(\xi) = \begin{cases} \psi_0^a(\xi) = \frac{1-\xi}{2}, & p = 0, \\ \psi_p^a(\xi) = \left(\frac{1-\xi}{2}\right)\left(\frac{1+\xi}{2}\right)P_{p-1}^{1,1}, & 0 < p < P, \\ \psi_P^a(\xi) = \frac{1+\xi}{2}, & p = P, \end{cases} \quad (2.18)$$

which is also one of the most adopted modal basis in spectral/*hp* element methods (Karniadakis & Sherwin, 2005) and was developed by Peano (1976), Szabo & Babuška (1991) and Oden (1994) .

The second expansion in Eq. (2.16), $\phi_p^N(\xi)$, is a Lagrange polynomial defined on a set of (P+1) nodal points ξ_q . In contrast to the modal expansion set, the Lagrange polynomial is a non-hierarchical basis since it consists of (P+1) polynomials of order P, thus $\chi_{P-1}^\delta \not\subset \chi_P^\delta$. The Lagrange expansion has the important property that $\phi_p^N(\xi_q) = \delta_{pq}$, where δ_{pq} is the Kronecker delta function, which in turn implies

$$u^\delta(\xi_q) = \sum_{p=0}^P u_p \phi_p^N(\xi_q) = \sum_{p=0}^P u_p \delta_{pq} = u_q, \quad (2.19)$$

that is, the coefficients of the Lagrange expansion basis are the physical values of the discrete solution u^δ at the nodal points ξ_q . In the Lagrange basis, the boundary-interior decomposition is not necessary because already satisfies this requirement. It is only sufficient that the nodal points include the element boundaries. Each of the expansions defined above, regardless if modal or nodal, can be seen as one-dimensional tensors. Therefore the extension to

quadrilateral elements can be simply achieved through the product of two one-dimensional tensors as follows:

$$\phi_{pq}(\xi_1, \xi_2) = \phi_p(\xi_1)\phi_q(\xi_2), \quad 0 \leq p, q; p \leq P_1, q \leq P_2, \quad (2.20)$$

where, in general, we allow the polynomials to be different in each coordinate direction. Note that the modal expansion on a quadrilateral (equivalently on a hexahedron) maintains its hierarchy property while the nodal basis maintains the Kronecker delta property.

We now introduce the matrix form of the expansion bases, which will be extensively used throughout this thesis. Specifically, when using a tensor-based expansion, we usually require the evaluation of the quantities constituting the equations on a set of nodal points. For instance, the one-dimensional function $u(\xi_q)$ on the set of Q nodal points $\boldsymbol{\xi} = [\xi_0, \xi_1, \dots, \xi_{Q-1}]^T$ is denoted by

$$\mathbf{u} = [u(\xi_0), u(\xi_1), \dots, u(\xi_{Q-1})]^T. \quad (2.21)$$

The two-dimensional case is analogous:

$$\mathbf{u} = [u(\xi_{1,0}, \xi_{2,0}), \dots, u(\xi_{1,Q_1-1}, \xi_{2,0}), u(\xi_{1,0}, \xi_{2,1}), \dots, u(\xi_{1,Q_1-1}, \xi_{2,Q_2-1})]^T, \quad (2.22)$$

where Q_1 and Q_2 represent the points in the ξ_1 - and ξ_2 -directions, respectively.

Concerning the expansion bases, the matrix form for the one-dimensional case is written as follows:

$$\mathbf{B} = \begin{bmatrix} \phi_0(\xi_0,) & \dots & \phi_p(\xi_0) & \dots & \phi_P(\xi_0) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_0(\xi_{1,Q-1}) & \dots & \phi_p(\xi_{Q-1}) & \dots & \phi_P(\xi_{1,Q-1}) \end{bmatrix}, \quad (2.23)$$

where P is the order of the expansion and $\mathbf{B}$ is referred to as the basis matrix. The two-dimensional case is analogous and reads:

$$\mathbf{B} = \begin{bmatrix} \phi_{00}(\xi_{1,0}, \xi_{2,0}) & \dots & \phi_{pq}(\xi_{1,0}, \xi_{2,0}) & \dots & \phi_{P_1P_2}(\xi_{1,0}, \xi_{2,0}) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_{00}(\xi_{1,Q_1-1}, \xi_{2,0}) & \dots & \phi_{pq}(\xi_{1,Q_1-1}, \xi_{2,0}) & \dots & \phi_{P_1P_2}(\xi_{1,Q_1-1}, \xi_{2,0}) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi_{00}(\xi_{1,Q_1-1}, \xi_{2,Q_2-1}) & \dots & \phi_{pq}(\xi_{1,Q_1-1}, \xi_{2,Q_2-1}) & \dots & \phi_{P_1P_2}(\xi_{1,Q_1-1}, \xi_{2,Q_2-1}) \end{bmatrix}, \quad (2.24)$$

where P_1 and P_2 are the orders of the expansion in the ξ_1 - and ξ_2 -directions, respectively.

In this work, we mainly used nodal bases. However, because of the general design frame-

work behind the Nektar++ library, all the implementations carried out can also use modal bases. From this perspective, a particularly interesting topic (not explored in this thesis) is the extension of the mathematical background behind the flux reconstruction approach to modal bases.

After having introduced the one-dimensional expansion bases, showed how to apply them to a standard quadrilateral element through a tensor product approach and explained the conventions adopted, it is now important to define the main operations needed in spectral/ hp element methods, namely numerical integration and differentiation.

Numerical integration

The integrals arising in spectral/ hp element methods usually need to be evaluated in the physical space and therefore it is common to have integrals of the type

$$\int_{x_n}^{x_{n+1}} u(x) dx = \int_{-1}^1 J_n u(\Theta(\xi)) d\xi, \quad (2.25)$$

where $u(\xi)$ is the integrand and can be a combination of polynomial bases, like $u(\xi) = \phi_p(\xi)\phi_q(\xi)$, while J_n is the one-dimensional Jacobian of the transformation from the physical to the standard space, defined in Eq. (2.15). The associated numerical integration or quadrature of Eq. (2.25) is:

$$\int_{-1}^1 J_n u(\Theta(\xi)) d\xi \approx \sum_{i=0}^{Q-1} w_i J_{n_i} u(\xi_i), \quad (2.26)$$

where w_i are the so-called *weights*, J_{n_i} are the Jacobians, ξ_i is the abscissa representing the Q distinct quadrature points in the interval $-1 \leq \xi_i \leq 1$. There exists various types of numerical integration, however, in this work we concentrate on *Gaussian quadrature* rules which are the one used in Nektar++. For additional details on other forms of numerical integration, the interested reader can refer to Appendix B in (Karniadakis & Sherwin, 2005).

Gaussian quadrature is widely adopted in spectral/ hp element methods and it employs a Lagrange polynomial onto a set of Q quadrature points to represent the integrand $u(\xi)$

$$u^\delta(\xi) = \sum_{i=0}^{Q-1} u(\xi_i) \ell_i(\xi) + \epsilon(u), \quad (2.27)$$

where $\epsilon(u)$ is the approximation error (note that we did not assume the polynomial order P

for the integrand $u(\xi)$. By substituting Eq. (2.27) into Eq. (2.25) we obtain the following summation over the Q quadrature points

$$\begin{aligned} \int_{-1}^1 J_n u(\xi) d\xi &= \\ &= \sum_{i=0}^{Q-1} \int_{-1}^1 J_{n_i} [u(\xi_i) \ell_i(\xi) + \epsilon(u)] d\xi = \sum_{i=0}^{Q-1} w_i J_{n_i} u(\xi_i) + \int_{-1}^1 J_{n_i} \epsilon(u) d\xi, \end{aligned} \quad (2.28)$$

where the weights w_i are the integrals of the Lagrange polynomials $\ell_i(\xi)$. The integral associated to the Gaussian quadrature in Eq. (2.28) is known as Legendre integration and to characterise it we need to define a set of quadrature points or zeros ξ_i . From this perspective, we distinguish between three quadratures which differ for the choice of the quadrature points:

Gauss-Legendre, whose point distribution is defined only inside the element;

Gauss-Radau-Legendre, whose point distribution includes only one end of the element;

Gauss-Lobatto-Legendre, whose points distribution include both the ends of the element.

Each of these quadratures allows the exact integration of polynomials of different orders. The Gauss-Legendre quadrature permits the exact integration of a polynomial integrand of order $(2Q-1)$, that is $\int_{-1}^1 \epsilon(u) d\xi = 0$ if $u^\delta(\xi) \in \mathbb{P}_{2Q-1}(-1, 1)$ or less. The Gauss-Radau-Legendre quadrature allows the exact integration if $u^\delta(\xi) \in \mathbb{P}_{2Q-2}(-1, 1)$ or less, whereas Gauss-Lobatto-Legendre integrates exactly integrand polynomials $u^\delta(\xi) \in \mathbb{P}_{2Q-2}(-1, 1)$ or less. In this work, we considered Gauss and Gauss-Lobatto quadratures.

The extension of Eq. (2.28) to a quadrilateral tensor-product element, $\Omega_s = (\xi_1, \xi_2) \in [-1, 1] \times [-1, 1]$, is straightforward and reads as follows:

$$\iint_{-1}^1 J_n u(\xi_1, \xi_2) d\xi_1 d\xi_2 = \sum_{i=0}^{Q_1-1} w_i \left\{ \sum_{j=0}^{Q_2-1} w_j J_{n_{ij}} u(\xi_{1_i}, \xi_{2_j}) \right\}, \quad (2.29)$$

where Q_1 and Q_2 are the number of quadrature points in the ξ_1 - and ξ_2 -direction, respectively and $J_{n_{ij}}$ are the two-dimensional point-wise Jacobians of the transformation between the physical to the standard space. The considerations regarding the exact integration hold also for the two-dimensional case.

It is now convenient to introduce the matrix form of the weight matrix, as done for the

expansion bases before. Specifically, for the one-dimensional case the weight matrix $\mathbf{W}$ reads

$$\mathbf{W} = \begin{bmatrix} w_0 J_{n_0} & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 \\ 0 & 0 & w_i J_{n_i} & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & w_{Q-1} J_{n_{Q-1}} \end{bmatrix}. \quad (2.30)$$

Analogously, for the two-dimensional case we have:

$$\mathbf{W} = \begin{bmatrix} w_{00} J_{n_{00}} & 0 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 & 0 \\ 0 & 0 & w_{ij} J_{n_{ij}} & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & w_{(Q_1-1)(Q_2-1)} J_{n_{(Q_1-1)(Q_2-1)}} \end{bmatrix}. \quad (2.31)$$

By using either Eq. (2.30) or Eq. (2.32), we can rewrite the one- or two-dimensional integration in compact form as follows:

$$\mathcal{I} = \mathbf{W} \mathbf{u}^\delta, \quad (2.32)$$

where $\mathcal{I}$ is either Eq. (2.28) or Eq. (2.29) and we used the definitions in Eq. (2.21) and Eq. (2.22).

As a final remark, we point out that the error associated to the inexact integration (also referred to as under-integration) can lead to numerical instabilities because of the (possibly) strong *aliasing* of the under-resolved higher modes, which inject energy into the lower modes. This aspect has been well explained by Karniadakis & Sherwin (2005) and the investigation of under-integration errors or simply aliasing errors is carried out in this thesis (chapter 5), where we analyse the implications of these aliasing issues in discontinuous spectral/*hp* element methods and we propose possible strategies to alleviate them.

Numerical differentiation

Another important aspect when implementing spectral/*hp* element methods is the numerical differentiation. Let us consider a generic function $u(\xi)$ and represent it in terms of Lagrange

polynomials $\ell_i(\xi)$ on a set of Q nodal points ξ_i

$$u^\delta(\xi) = \sum_{i=0}^{Q-1} u(\xi_i) \ell_i(\xi), \quad (2.33)$$

where $Q \geq P + 1$ and where we assume $u^\delta \in \mathbb{P}_P([-1, 1])$. The representation of the function $u^\delta(\xi)$ in this case is exact and its derivative at the nodal points ξ_i can be defined as follows:

$$\frac{du^\delta(\xi_i)}{d\xi} = \sum_{j=0}^{Q-1} \left. \frac{d\ell_j(\xi)}{d\xi} \right|_{\xi=\xi_i} u(\xi_j). \quad (2.34)$$

To properly characterise the differentiation matrix, it is necessary to choose (as for the numerical integration) a set of quadrature points. Without entering into details (the interested reader can refer to Karniadakis & Sherwin (2005)), the differentiation matrices employed in this work use either *Gauss-Legendre* or *Gauss-Lobatto-Legendre* quadrature points.

The extension of Eq. (2.34) to two-dimensional tensor-product elements is straightforward and reads:

$$\begin{aligned} \frac{du^\delta(\xi_{1i}, \xi_{2j})}{d\xi_1} &= \sum_{p=0}^{Q_1-1} \sum_{q=0}^{Q_2-1} \left[\left. \frac{d\ell_p(\xi_1)}{d\xi_1} \right|_{\xi_{1i}} \ell(\xi_{2j}) u_{pq} \right] \\ \frac{du^\delta(\xi_{1i}, \xi_{2j})}{d\xi_2} &= \sum_{p=0}^{Q_1-1} \sum_{q=0}^{Q_2-1} \left[\left. \frac{d\ell_q(\xi_2)}{d\xi_2} \right|_{\xi_{2j}} \ell(\xi_{1i}) u_{pq} \right], \end{aligned} \quad (2.35)$$

where Q_1 and Q_2 are the quadrature points in the ξ_1 - and ξ_2 -direction, respectively. Also in this case, as for the numerical integration, we can introduce the matrix form of the differentiation matrices for the one-dimensional and two-dimensional cases. Specifically, in one dimension we have

$$\mathbf{D}_\xi = \left. \frac{d\ell_j(\xi)}{d\xi} \right|_{\xi=\xi_i}, \quad 0 \leq i, j \leq Q - 1. \quad (2.36)$$

The two-dimensional case reads:

$$\mathbf{D}_{\xi_1} = \left. \frac{d\ell_p(\xi_1)}{d\xi_1} \right|_{\xi_{1i}} \ell(\xi_{2j}) \quad \mathbf{D}_{\xi_2} = \left. \frac{d\ell_q(\xi_2)}{d\xi_2} \right|_{\xi_{2j}} \ell(\xi_{1i}) \quad (2.37)$$

where the matrices $\mathbf{D}_{\xi_1}$ and $\mathbf{D}_{\xi_2}$ are the one-dimensional derivative matrices in the ξ_1 - and ξ_2 -direction, respectively and the indices associated to the ξ_1 -direction are defined on Q_1 points: $0 \leq i, p \leq Q_1 - 1$, while those associated to the ξ_2 -direction are defined on Q_2 points: $0 \leq j, q \leq Q_2 - 1$.

Forward and backward transformations

In this section we introduce the transformation to evaluate the physical values from the expansion basis coefficients (backward transformation) and vice-versa (forward transformation).

Backward transformation

Reusing the matrix form defined above for the basis functions and the solution, we can write the polynomial expansion u^δ (such as the expansion in Eq. (2.19)) in vector form as follows:

$$\mathbf{u}^\delta(\boldsymbol{\xi}) = \mathbf{B}(\boldsymbol{\xi}) \mathbf{u}, \quad (2.38)$$

where $\mathbf{u}^\delta$ is either represented by a one- or a two-dimensional expansion. Equation (2.38) is the discrete backward transformation and involves the transformation from the coefficient space to the physical space. Note that, if we use a nodal expansion basis, such as Lagrange polynomials, through a set of nodal points and the basis is evaluated at the same nodal points, then $\mathbf{B} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix. In this case we recover the Kronecker delta property, which in matrix form reads

$$\mathbf{u}^\delta = \mathbf{B} \mathbf{u} = \mathbf{I} \mathbf{u} = \mathbf{u}. \quad (2.39)$$

However, if the basis is evaluated at a set of points which does not correspond to the nodal points, then the matrix $\mathbf{B}$ is full. This arises in the case of quadrilateral tensor-product basis when the quadrature order is not equal to the polynomial order plus one, that is $Q \neq P + 1$.

Forward transformation

The forward transformation can be formulated by reusing some concepts of the method of weighted residuals introduced in section 2.1. In particular, we first define the error between the continuous function and its discrete representation as follows

$$u^\delta(\xi_1, \xi_2) - u(\xi_1, \xi_2) = \sum_{pq} u_{pq} \phi_{pq} - u = R(u), \quad (2.40)$$

where $R(u)$ is the residual introduced in Eq. (2.3).

Following the method of weighted residuals, we multiply both sides of Eq. (2.40) by a test

function v and set the right-hand side to zero:

$$\left(v, \sum_{pq} u_{pq} \phi_{pq} \right) - (v, u) = \underbrace{(v, R(u))}_{=0} \longrightarrow \left(v, \sum_{pq} u_{pq} \phi_{pq} \right) = (v, u). \quad (2.41)$$

The choice of the test function determines the type of projection (i.e. of elemental forward transformation). In particular, we distinguish between a collocation and a Galerkin projection.

The first is simply obtained by setting $v = \delta(\xi_{1_i}, \xi_{2_j})$, where δ is the Kronecker delta function. In matrix form this becomes:

$$\mathbf{u} = \mathbf{B}_{\mathcal{N}}^{-1} \mathbf{u}^\delta \quad (2.42)$$

and it is known as collocation forward transformation. Note that the matrix $\mathbf{B}_{\mathcal{N}}$ is defined as: $\mathbf{B}_{\mathcal{N}} = \phi_{pq}(\xi_{1_i}, \xi_{2_j})$

The second is instead obtained by setting the test functions equal to the expansion functions, that is $v = \phi$. The Galerkin transformation can be written in matrix form as follows:

$$\mathbf{u} = (\mathbf{B}^T \mathbf{W} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{W} \mathbf{u}^\delta = \mathbf{M}^{-1} \mathbf{B}^T \mathbf{u}^\delta, \quad (2.43)$$

where $\mathbf{W}$ is the weight matrix introduced in Eqs. (2.30 and 2.32) and $\mathbf{M} = \mathbf{B}^T \mathbf{W} \mathbf{B}$ is the elemental mass matrix. The manipulations that produce the matrix form in Eq. (2.43) are not reported here for the sake of brevity. However, the interested reader can refer to chapter 4 in (Karniadakis & Sherwin, 2005). Note that also in this case, if we use a nodal expansion onto a set of nodal points and $Q = P + 1$ where Q is the number of points and P is the order of the expansion, we recover the Kronecker delta property, therefore Eq. (2.43) simplifies as follows:

$$\mathbf{u} = (\mathbf{B}^T \mathbf{W} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{W} \mathbf{u}^\delta = (\mathbf{I}^T \mathbf{W} \mathbf{I})^{-1} \mathbf{I}^T \mathbf{W} \mathbf{u}^\delta = \mathbf{W}^{-1} \mathbf{W} \mathbf{u}^\delta = \mathbf{u}^\delta, \quad (2.44)$$

where we used $\mathbf{B} = \mathbf{I}$, with $\mathbf{I}$ being the identity matrix.

Connectivity

After having decomposed the domain Ω into N non-overlapping subdomains Ω_n and on each of these subdomain performed the required operations, it is necessary to impose connectivity across the elements in order to construct the global problem. Connectivity can be imposed in different ways. For example, when dealing with continuous Galerkin methods, we require

the solution to be continuous between two adjacent elements, whereas when dealing with discontinuous-type approaches, such as the discontinuous Galerkin and the flux reconstruction approaches, we require the flux and not the solution to be continuous between two adjacent elements.

Therefore, we describe separately how to reinforce the connectivity for a continuous discretisation, e.g. continuous Galerkin method, and a discontinuous one - e.g. discontinuous Galerkin and flux reconstruction methods.

Continuous discretisation

For describing the connectivity on a continuous numerical framework, we focus on the classical continuous Galerkin method. In this numerical approach, C^0 -continuity is imposed across the elements by using the so-called *global assembly* strategy (also referred to as *direct stiff summation*). We can describe this methodology by defining a generic function $u(x_1, x_2)$ expanded as follows

$$u^\delta(x_1, x_2) = \sum_{n=0}^{N-1} \sum_{p=0}^{P_1-1} \sum_{q=0}^{P_2-1} u_{pq_n} \phi_{pq_n}(x_1, x_2), \quad (2.45)$$

where N is the number of elements, u_{pq_n} are the local expansion coefficients and $\phi_{pq_n}(x_1, x_2)$ are the local expansion modes for a given element n . In vectorial notation, Eq. (2.45) can be represented as

$$\mathbf{u}_l = \underline{\mathbf{u}}_n = \begin{bmatrix} \mathbf{u}_0 \\ \mathbf{u}_1 \\ \cdot \\ \cdot \\ \mathbf{u}_{N-1} \end{bmatrix}, \quad (2.46)$$

where $\underline{\mathbf{u}}_n$ represents the concatenation of the local expansion coefficients. The relation between the global expansion coefficient $\mathbf{u}_g$ and the local one, $\mathbf{u}_l$, can be represented through a matrix-vector multiplication:

$$\mathbf{u}_l = \mathbf{A} \mathbf{u}_g, \quad (2.47)$$

where $\mathbf{A}$ is called the assembly matrix and it applies the connectivity rules across the elements. The assembly matrix is generally sparse having entries equal to ± 1 and it concatenates all the elements into a global expansion.

Discontinuous discretisation

In the case of a discontinuous discretisation, such as the discontinuous Galerkin and flux reconstruction approaches, the solution is allowed to be discontinuous across two adjacent elements but the fluxes are required to be continuous. The connectivity across the elements in this case is obtained through a boundary term which is composed by a numerical flux $\mathbf{f}^{\delta I}(u_+^\delta, u_-^\delta)$. The numerical flux depends on the solution on the left, u_+^δ , and on the right, u_-^δ , element with respect to the given interface, as illustrated in figure 2.3. The form of the numerical flux depends on the problem being solved. For instance, if we are solving the linear advection equation, where the flux is $\mathbf{f}(u^\delta) = \mathbf{v} u^\delta$, we can use the following upwind numerical flux

$$\mathbf{f}^{\delta I}(u_+^\delta, u_-^\delta) = \begin{cases} \mathbf{v} u_+^\delta, & \mathbf{v} \cdot \mathbf{n}_+ \geq 0, \\ \mathbf{v} u_-^\delta, & \mathbf{v} \cdot \mathbf{n}_+ < 0, \end{cases} \quad (2.48)$$

where $\mathbf{n}_+$ is the outward pointing normal of the left element and $\mathbf{v}$ is the advection velocity, as illustrated in figure 2.3.

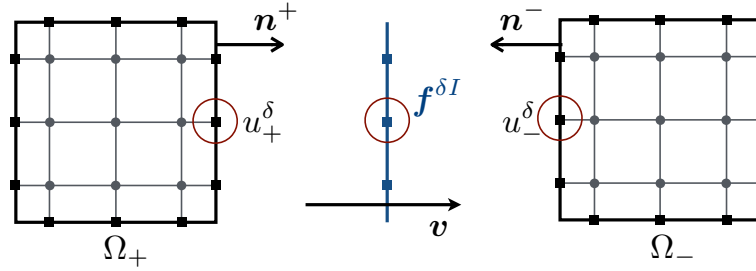


Figure 2.3: Conventions adopted for calculating the numerical fluxes in discontinuous spectral/ hp element methods.

For more complicated hyperbolic problems, such as compressible flows, we need to apply connectivity through more complex forms of the numerical fluxes which are based on the physics of the problem. Specifically, for discretising a first order flux (e.g. the advective flux of the compressible Navier-Stokes equations or the flux characterising the compressible Euler equations) we need to use either approximated or exact Riemann solvers. This implies solving a Riemann problem at each interface between two elements. In chapter 3, we thoroughly describe the form of the numerical fluxes when approximating the advection operator of the equations governing compressible flows.

When dealing with a second order flux (e.g. the diffusion equation or the viscous flux of the compressible Navier-Stokes equations), connectivity is again achieved through a numerical flux but with a slightly different form than the advection term. The numerical flux associated to a second order problem is reviewed later on, when discussing the discontinuous Galerkin

and flux reconstruction diffusion operators in section 2.4.

2.2.2 Time discretisation

Once the spatial derivatives have been discretised and if the problem is time-dependent, the partial differential problem reduces to a set ordinary differential equations (ODEs), where the only differentiation variable is the time t . There exist two main categories of time-integration methods for integrating an ODE or a system of ODEs,

- multi-step methods: They combine the information from previous time-levels (or time-steps) to calculate the solution at the new time-level;
- multi-stage methods: They combine the information at various stages between two time-steps discarding the information from previous time-steps.

The first strategy is more memory demanding because it is necessary to store the information of the previous time-steps but, on the other hand, it requires a relatively limited number of floating point operations. The second strategy is instead more CPU intensive (it requires more floating point operations per time-step) but needs less memory. Both strategies can be explicit - the current time-step is calculated using information from the previous time-step(s) only - or implicit - the current time-step is computed solving a nonlinear problem which involve the current time-step. Multi-step and multi-stage methods, can be grouped together by means of the unified General Linear (GL) method presented by Butcher (1987). The time-integration strategy adopted in this work makes use of explicit time-integration schemes, specifically explicit Runge-Kutta schemes. In the following we introduce the GL method and we successively focus on the explicit Runge-Kutta schemes which can be directly derived from the GL method.

If we consider the discretised linear differential equation $\mathcal{L}(u^\delta) = R(u^\delta)$ and we assume that we have chosen one of the spatial discretisations described in the previous section, the problem reduces to an ODE dependant only on time t (i.e. it effectively reduces to an initial-value problem), that is

$$\frac{du^\delta}{dt} = f(u^\delta), \quad u^\delta(t_0) = u_0^\delta \quad (2.49)$$

where f is the approximation of the spatial derivatives of the differential problem. The task is to find a suitable approximation to solve Eq. (2.49) in order to propagate the solution in

time. If we define the discrete time as $t \rightarrow t^k$ and we introduce a given and finite interval of time or time-step $\Delta t = (t^{k+1} - t^k)$, the GL method is stated as finding $u^\delta(t^{k+1})$ such that

$$\begin{cases} u_i^{\delta,s} = \Delta t \sum_{j=1}^s z_{ij}^{(1)} g_j + \sum_{j=1}^r z_{ij}^{(3)} f_j^k, & i \in [1, s], \\ u_i^\delta(t^{k+1}) = \Delta t \sum_{j=1}^s z_{ij}^{(2)} g_j + \sum_{j=1}^r z_{ij}^{(4)} u_j^\delta(t^k), & i \in [1, r] \end{cases} \quad (2.50)$$

where $z_{ij}^{(1)}, z_{ij}^{(2)}, z_{ij}^{(3)}$ and $z_{ij}^{(4)}$ are matrices which defines the time-integration scheme, s and r are the stages and the steps, respectively, $u_i^{\delta,s}$ are the stage values and g_j are the stage derivatives. The explicit Runge-Kutta schemes adopted in this work can be defined within the GL method framework by rewriting Eq. (2.52) in matrix form

$$\begin{bmatrix} \mathbf{u}^{\delta,s} \\ \mathbf{u}^\delta(t^{k+1}) \end{bmatrix} = \begin{bmatrix} \mathbf{Z}^{(1)} \otimes \mathbf{I} & \mathbf{Z}^{(3)} \otimes \mathbf{I} \\ \mathbf{Z}^{(2)} \otimes \mathbf{I} & \mathbf{Z}^{(4)} \otimes \mathbf{I} \end{bmatrix} \begin{bmatrix} \Delta t \mathbf{g} \\ \mathbf{u}^\delta(t^k) \end{bmatrix} \quad (2.51)$$

where $\mathbf{I}$ is the identity matrix. The matrices $\mathbf{Z}^{(1)}, \mathbf{Z}^{(2)}, \mathbf{Z}^{(3)}$ and $\mathbf{Z}^{(4)}$ for a 4th-order Runge-Kutta scheme are as follows

$$\left[\begin{array}{c|c} \mathbf{Z}^{(1)} & \mathbf{Z}^{(3)} \\ \hline \mathbf{Z}^{(2)} & \mathbf{Z}^{(4)} \end{array} \right] = \left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 1 \\ 1/2 & 0 & 0 & 0 & 1 \\ 0 & 1/2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ \hline 1/6 & 1/3 & 1/3 & 1/6 & 1 \end{array} \right]. \quad (2.52)$$

In Nektar++ the implementation of the time-integration schemes is performed using the GL method strategy and therefore by defining a matrix whose coefficients change depending on the scheme chosen. In theory, we could have also used other explicit methods for the time-integration, such as forward Euler, Adams-Bashforth and 2nd-order Runge-Kutta schemes. In practice, although more expensive than the others, we mainly used the 4th-order Runge-Kutta scheme whose stability region was satisfactory for the majority of the cases treated in this thesis.

2.2.3 Summary

The concepts introduced in the previous two sections will be used extensively in the rest of the thesis along with the associated nomenclature. Specifically, in the rest of this chapter we will detail two discontinuous spatial discretisations, the discontinuous Galerkin and the flux reconstruction approaches for first and second order problems by reusing some of the concepts and conventions presented in section 2.2.1. In addition, in chapter 5, we will thoroughly investigate the aliasing issues (briefly introduced in the numerical integration section above) arising in discontinuous spectral/*hp* element methods along with some strategies to address them.

2.3 DG and FR schemes for first order problems

In this section, we describe the discontinuous Galerkin (DG) method and the flux reconstruction (FR) approach applied to a general first order problem - i.e. we describe the DG and FR advection operators. The implementation of the FR approach in the spectral/*hp* element library Nektar++ has been carried out for one-dimensional and two-, three-dimensional tensor-product elements - specifically quadrilaterals and hexahedra. The DG approach was already existing. For describing the two approaches we consider the following two-dimensional first order conservation law

$$\frac{\partial u}{\partial t} + \nabla_{\mathbf{x}} \cdot \mathbf{f}(u) = 0, \quad (2.53)$$

within a domain $\Omega \in \mathbb{R}^2$, with $\mathbf{f} = [f_1, f_2]$, where $f_1 = f_1(u)$ and $f_2 = f_2(u)$ are the advection fluxes in the x_1 and x_2 Cartesian directions, respectively and where t is the time, $u = u(x_1, x_2, t)$ is the conserved variable and $\nabla_{\mathbf{x}} = [\partial/\partial x_1, \partial/\partial x_2]$.

For both, the discontinuous Galerkin method and the flux reconstruction approach, we partition the domain Ω into N non-overlapping subdomains Ω_n , as shown in Eq. (2.8), and we define the approaches on a local element Ω_n . The solution of the global problem is the summation of all the elemental contributions across the computational domain.

We remark that the steps described in the following two sections are very general and they can therefore be applied to any first order PDE, such as the linear advection equation, the Burgers' equation and the compressible Euler equations. They can also be applied to mixed-type problems where we need to discretise an advection term, (e.g. compressible Navier-Stokes equations). In the latter case, these steps need to be complemented by the discretisation of the diffusion operator, which we will be described in section 2.4.

2.3.1 Discontinuous Galerkin (DG) approach

To describe the DG method applied to Eq. (2.67), we first define the space of the test functions ℓ :

$$\mathcal{V}^\delta := \{\ell \in L^2(\Omega) : \ell|_{\Omega_n} \in \mathbb{P}_P(\Omega_n), \quad \forall \Omega_n\}, \quad (2.54)$$

where $\mathbb{P}_P$ is the space of two-dimensional piecewise continuous polynomials of order $P \geq 1$ on Ω and

$$L^2(\Omega) = \left\{ \ell : \Omega_n \rightarrow \mathbb{R} \mid \iint_{\Omega_s} |\ell(\mathbf{x})|^2 d\mathbf{x} \leq \infty \right\}, \quad (2.55)$$

with $\mathbb{R}$ being the space of the real numbers. Note that in general the approximation space can use both modal and nodal polynomial bases. If we use a nodal basis, such as Lagrange polynomials, we define the common class of nodal DG schemes. We now approximate the solution u by a piecewise two-dimensional polynomial, that is $u^\delta \in \mathcal{V}^\delta$, we multiply Eq. (2.67) by the test function ℓ and integrate over a local element Ω_n , such that:

$$\iint_{\Omega_n} \ell(\mathbf{x}) \frac{\partial u_n^{\delta D}(\mathbf{x})}{\partial t} d\mathbf{x} + \iint_{\Omega_n} \ell(\mathbf{x}) (\nabla_{\mathbf{x}} \cdot \mathbf{f}_n^{\delta D}) d\mathbf{x} = 0, \quad (2.56)$$

where the superscript ‘ D ’ indicates discontinuous quantities. We then integrate by parts the second term of Eq. (2.56) obtaining the weak form of the DG method

$$\iint_{\Omega_n} \ell(\mathbf{x}) \frac{\partial u_n^{\delta D}(\mathbf{x})}{\partial t} d\mathbf{x} - \underbrace{\iint_{\Omega_n} \nabla_{\mathbf{x}} \ell(\mathbf{x}) \cdot \mathbf{f}_n^{\delta D} d\mathbf{x}}_{\text{volumetric term}} + \underbrace{\int_{\partial\Omega_n} \ell(\mathbf{x}) (\mathbf{f}^{\delta I} \cdot \mathbf{n}) ds}_{\text{boundary term}} = 0, \quad (2.57)$$

where $\mathbf{f}_n^{\delta D}$ is the flux calculated from the solution and it can be discontinuous between the elements, $\mathbf{n}$ is the outward pointing normal with respect to a given edge of the local element Ω_n and the solution $u_n^{\delta D}(\mathbf{x})$ satisfies Eq. (2.57) for all $\ell(\mathbf{x}) \in \mathcal{V}^\delta$.

Equation (2.57) involves only elemental operations. In order to couple the various elements of the computational domain together, thus solve the global problem, it is necessary to propagate the solution across the domain by means of some elemental coupling. As seen in section 2.2.1, the elemental coupling for a discontinuous approach like the DG method involves defining the numerical flux $\mathbf{f}^{\delta I}(u_-^\delta, u_+^\delta)$. This numerical flux is represented by the third term of Eq. (2.57) and the values u_-^δ and u_+^δ represent the solution on the edges of two adjacent elements as shown in figure 2.3. The boundary conditions are also enforced through the numerical flux $\mathbf{f}^{\delta I}(u_-^\delta, u_+^\delta)$. In this case, u_-^δ represents the value of the boundary condi-

tion on a ghost point. To generate a coupling which produces a stable numerical scheme, it is necessary to make some considerations based on the natural propagation of the information. Specifically, if we are for instance considering the linear advection equation, then we could use a simple upwind approach as that presented in Eq. (2.48). For more complicated cases, such as the compressible Euler equations, we need to use one-dimensional Riemann solvers, either approximated or exact. In chapter 3, we will detail some one-dimensional Riemann solvers which are commonly used in practical applications in the context of compressible flows. For additional details, the interested reader can refer to Toro (2009).

Equation (2.57) can be rewritten in compact form as follows

$$\left(\ell(\mathbf{x}), \frac{\partial u_n^{\delta D}(\mathbf{x})}{\partial t} \right)_{\Omega_n} - \left(\nabla_{\mathbf{x}} \ell(\mathbf{x}), \mathbf{f}_n^{\delta D} \right)_{\Omega_n} + \left\langle \ell(\mathbf{x}), \mathbf{f}^{\delta I}(u_-^\delta, u_+^\delta) \right\rangle_{\partial\Omega_n} = 0, \quad (2.58)$$

where the elemental inner products are:

$$\begin{aligned} (\mathbf{g}_1, \mathbf{g}_2)_{\Omega_n} &= \iint_{\Omega_n} \mathbf{g}_1 \cdot \mathbf{g}_2 \, d\mathbf{x}, & (\mathbf{g}_1, \mathbf{g}_2)_{\Omega_n} &= \iint_{\Omega_n} \mathbf{g}_1 \cdot \mathbf{g}_2 \, d\mathbf{x}, \\ \langle \mathbf{g}_1, \mathbf{g}_2 \rangle_{\partial\Omega_n} &= \int_{\partial\Omega_n} \mathbf{g}_1 \cdot \mathbf{n} \, ds. \end{aligned} \quad (2.59)$$

For implementation convenience, it is common to transform Eq. (2.57) into the standard space. This means redefining each elemental contribution of Eq. (2.57) into the standard element Ω_s by using the mapping $\mathbf{x} = \boldsymbol{\Theta}_n(\boldsymbol{\xi})$ defined in section 2.2.1, such that:

$$\begin{aligned} \iint_{\Omega_s} \ell(\boldsymbol{\Theta}_n(\boldsymbol{\xi})) \frac{\partial u_n^{\delta D}(\boldsymbol{\Theta}_n(\boldsymbol{\xi}))}{\partial t} J_n \, d\boldsymbol{\xi} - \iint_{\Omega_s} \nabla_{\boldsymbol{\xi}} \ell(\boldsymbol{\Theta}_n(\boldsymbol{\xi})) \cdot \widehat{\mathbf{f}}^{\delta D} J_n \, d\boldsymbol{\xi} \\ + \int_{\partial\Omega_s} \ell(\boldsymbol{\Theta}_n(\boldsymbol{\xi})) \left(\widehat{\mathbf{f}^{\delta I} \cdot \mathbf{n}} \right) J_{1d} \, d\widehat{s} = 0, \end{aligned} \quad (2.60)$$

where $\widehat{\mathbf{f}}^{\delta D} = J_n \mathbf{G}^{-1} \mathbf{f}_n^{\delta D}$, J_{1d} are the one-dimensional Jacobians referred to the edge of the element and where we used the Nanson formula for the normals:

$$\left(\widehat{\mathbf{f}^{\delta I} \cdot \mathbf{n}} \right) = \frac{1}{J_{1d}} \left[\mathbf{f}^{\delta I} \cdot (\mathbf{G}_n^T \mathbf{n}) \right]. \quad (2.61)$$

Equation (2.60) can be rewritten in matrix form using the nomenclature introduced in section 2.2.1. In particular, the $(P+1) \times (P+1)$ equations defined at each solution point within

a given element are

$$\begin{aligned} \mathbf{M} \frac{d\mathbf{u}^D}{dt} + \left[\mathbf{S}_{\xi_2} \boldsymbol{\Lambda} \left(\frac{\partial x_2}{\partial \xi_1} \right) - \mathbf{S}_{\xi_1} \boldsymbol{\Lambda} \left(\frac{\partial x_2}{\partial \xi_2} \right) \right] \mathbf{f}_1^D + \\ + \left[\mathbf{S}_{\xi_1} \boldsymbol{\Lambda} \left(\frac{\partial x_1}{\partial \xi_2} \right) - \mathbf{S}_{\xi_2} \boldsymbol{\Lambda} \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \mathbf{f}_2^D + \mathbf{b}^{\text{DG}} = 0, \end{aligned} \quad (2.62)$$

where we dropped the subscript ‘ n ’ and the various terms in Eq. (2.62) are defined in the following. $\mathbf{S}_{\xi_1}$, $\mathbf{S}_{\xi_2}$ are the elemental stiffness matrices:

$$\mathbf{S}_{\xi_1} = (\mathbf{D}_{\xi_1} \mathbf{B})^T \mathbf{W} \mathbf{B}, \quad \mathbf{S}_{\xi_2} = (\mathbf{D}_{\xi_2} \mathbf{B})^T \mathbf{W} \mathbf{B}, \quad (2.63)$$

$\mathbf{M}$ is the elemental mass matrix:

$$\mathbf{M} = \mathbf{B}^T \mathbf{W} \mathbf{B}, \quad (2.64)$$

$\boldsymbol{\Lambda}(\cdot)$ are diagonal matrices containing the geometric terms and $\widehat{\mathbf{b}}^{\text{DG}}$ is the boundary term.

In the above, $\mathbf{D}_{\xi_1}$ and $\mathbf{D}_{\xi_2}$ are the differentiation matrices defined in Eq. (2.35), $\mathbf{W}$ is the matrix of the weights defined in Eq. (2.32) and we used the discrete backward transformation in Eq. (2.38) to represent the discrete solution polynomial (note that the Jacobian J_n in Eq. (2.60) has been incorporated into the weight matrix $\mathbf{W}$).

The matrix form of the DG method can be further reduced by defining the following two matrices:

$$\mathbf{S}_{x_1} = \mathbf{S}_{\xi_2} \boldsymbol{\Lambda} \left(\frac{\partial x_2}{\partial \xi_1} \right) - \mathbf{S}_{\xi_1} \boldsymbol{\Lambda} \left(\frac{\partial x_2}{\partial \xi_2} \right), \quad \mathbf{S}_{x_2} = \mathbf{S}_{\xi_1} \boldsymbol{\Lambda} \left(\frac{\partial x_1}{\partial \xi_2} \right) - \mathbf{S}_{\xi_2} \boldsymbol{\Lambda} \left(\frac{\partial x_1}{\partial \xi_1} \right) \quad (2.65)$$

and substituting them into Eq. (2.62), such that

$$\mathbf{M} \frac{d\mathbf{u}^D}{dt} + \mathbf{S}_{x_1} \mathbf{f}_1^D + \mathbf{S}_{x_2} \mathbf{f}_2^D + \mathbf{b}^{\text{DG}} = 0. \quad (2.66)$$

The final form of the DG method can then be advanced in time via any explicit time discretisation scheme, such as the explicit Runge-Kutta methods.

2.3.2 Flux Reconstruction (FR) approach

The approximation space for the FR approach is identical to the DG method and is defined in Eq. (2.54). Unlike the DG method, the FR approach is formulated starting from the differential form of Eq. (2.67). From this perspective, it is convenient to map each subdomain

Ω_n into a standard element Ω_s via the generic mapping in Eq. (2.13) or Eq. (2.14). The procedure to solve Eq. (2.53) can be explained within the standard element Ω_s since the only difference between one element and another is the mapping Θ_i , which may change from element to element. The conservation law, Eq. (2.53), in the transformed space becomes

$$\frac{\partial \hat{u}}{\partial t} + \nabla_{\boldsymbol{\xi}} \cdot \hat{\mathbf{f}}(\hat{u}) = 0, \quad (2.67)$$

where the transformed solution is represented as:

$$\hat{u}(\boldsymbol{\xi}, t) = J_n u_n(\Theta_n(\boldsymbol{\xi}), t), \quad (2.68)$$

the transformed fluxes are:

$$\hat{\mathbf{f}}(\boldsymbol{\xi}, t) = J_n \mathbf{G}_n^{-1} \mathbf{f}_n = \left(\frac{\partial x_2}{\partial \xi_2} f_{1,n} - \frac{\partial x_1}{\partial \xi_2} f_{2,n}, -\frac{\partial x_2}{\partial \xi_1} f_{1,n} + \frac{\partial x_1}{\partial \xi_1} f_{2,n} \right) \quad (2.69)$$

and $\nabla_{\boldsymbol{\xi}} = [\partial/\partial \xi_1, \partial/\partial \xi_2]$. The metric terms $\mathbf{G}_n$ and J_n in the previous equations are the elemental deformation gradient and its determinant and they have been defined in Eq. (2.15). Hereafter, we will drop the subscript ‘ n ’ for the sake of clarity.

The first step to construct the FR approach is to project the known solution onto a set of $(P+1)^2$ solution points

$$\hat{u}^{\delta D}(\boldsymbol{\xi}, t) = \sum_{i,j=0}^P \ell_{ij}(\boldsymbol{\xi}) \hat{u}_{ij}^D = \sum_{i,j=0}^P \ell_i(\xi_1) \ell_j(\xi_2) \hat{u}_{ij}^D, \quad (2.70)$$

within Ω_s , where $\hat{u}_{ij}^{\delta D}$ are given values of the known solution and $\ell_{ij} = \ell_{ij}(\boldsymbol{\xi})$ are two-dimensional continuous polynomials (such as Lagrange polynomials) of order P and we use the superscript ‘ D ’ to indicate that the solution can be discontinuous between adjacent elements. Note that, the solution expansion in Eq. (2.70) incorporates the Jacobian J_n . This expansion is identical to expanding the Jacobian and the solution separately in the case of a collocation projection (and if the number of points is equal to $Q=P+1$, where P is the polynomial order of the expansion) or if the Jacobian is constant. In general, when $Q > P+1$, expanding the Jacobian and the solution separately produces a different result than Eq. (2.70). In Nektar++, the implementation has been carried out using two different expansions for the Jacobian (including the metric terms) and the solution.

The second step involves calculating an order P polynomial, called *discontinuous flux* and

denoted by $\hat{\mathbf{f}}^{\delta D}(\boldsymbol{\xi}) = [\hat{f}_1^{\delta D}(\boldsymbol{\xi}), \hat{f}_2^{\delta D}(\boldsymbol{\xi})]$, using the known solution defined at the previous step:

$$\hat{f}_1^{\delta D}(\boldsymbol{\xi}, t) = \sum_{i,j=0}^P \ell_{ij}(\boldsymbol{\xi}) \hat{f}_{1ij}^D = \sum_{i,j=0}^P \ell_i(\xi_1) \ell_j(\xi_2) \hat{f}_{1ij}^D, \quad (2.71a)$$

$$\hat{f}_2^{\delta D}(\boldsymbol{\xi}, t) = \sum_{i,j=0}^P \ell_{ij}(\boldsymbol{\xi}) \hat{f}_{2ij}^D = \sum_{i,j=0}^P \ell_i(\xi_1) \ell_j(\xi_2) \hat{f}_{2ij}^D, \quad (2.71b)$$

for the ξ_1 - and ξ_2 -direction, respectively. Note that here, the same considerations made for the metric terms hold. In particular, the expansions in Eq. (2.71) include the geometric factors and they are identical to representing the fluxes and the metric terms separately if we use a collocation projection or if the element is not deformed.

The third step is the calculation of the numerical flux denoted by $\mathbf{f}^{\delta I} = \mathbf{f}^{\delta I}(u_-^\delta, u_+^\delta)$. The procedure is identical to that adopted for the DG method.

The fourth step of the FR approach involves adding a polynomial of degree $P+1$, called *correction flux* and denoted by $\hat{\mathbf{f}}^{\delta C} = [\hat{f}_1^{\delta C}(\boldsymbol{\xi}), \hat{f}_2^{\delta C}(\boldsymbol{\xi})]$, to the discontinuous flux $\hat{\mathbf{f}}^{\delta D}$ calculated at the second step. The sum of these two fluxes together guarantees that the final flux, called *total flux*, is continuous across the elements. In order to compute the correction flux, it is necessary to define four correction functions, one for each edge of the quadrilateral standard element Ω_s considered.

If we define the correction functions as Ψ_B and Ψ_T for the bottom and top edges of the standard element Ω_s , and as Ψ_R and Ψ_L for the right and left edges, we require that these functions satisfy the following constraints:

$$\begin{aligned} \Psi_B(-1) &= 1, & \Psi_B(1) &= 0; & \Psi_T(-1) &= 0, & \Psi_T(1) &= 1; \\ \Psi_R(-1) &= 0, & \Psi_R(1) &= 1; & \Psi_L(-1) &= 1, & \Psi_L(1) &= 0; \\ \Psi_B(\xi_2) &= \Psi_T(-\xi_2); & \Psi_R(\xi_1) &= \Psi_L(-\xi_1) \end{aligned} \quad (2.72)$$

As already mentioned, the FR approach allows the recovery of a wide-range of numerical schemes. From this point of view, Vincent *et al.* (2011b) have identified a class of linearly energy stable FR schemes called VCJH schemes. In particular, these VCJH schemes are

recovered using the following expressions for the correction functions:

$$\begin{aligned}\Psi_B = \Psi_L &= \frac{(-1)^P}{2} \left[L_P - \left(\frac{\eta_P L_{P-1} + L_{P+1}}{1 + \eta_P} \right) \right], \\ \Psi_T = \Psi_R &= \frac{1}{2} \left[L_P + \left(\frac{\eta_P L_{P-1} + L_{P+1}}{1 + \eta_P} \right) \right],\end{aligned}\tag{2.73}$$

where L_P is a Legendre polynomial of order P ,

$$\eta_P = \frac{c(2P+1)(a_P P!)^2}{2}, \quad a_P = \frac{(2P)!}{2^P (P!)^2}\tag{2.74}$$

and c is a free scalar parameter which defines the VCJH scheme recovered. The free parameter c is comprised in the following interval:

$$\frac{-2}{(2P+1)(a_P P!)^2} < c < \infty.\tag{2.75}$$

From an implementation perspective, particular attention has been devoted to three numerical schemes recovered by three different values of the parameter c .

Nodal discontinuous Galerkin scheme (FR_{DG})

If $c = c_{DG} = 0$, then $\eta_P = 0$ and the correction functions are the left and right Radau polynomials:

$$\begin{aligned}\Psi_B = \Psi_L &= \frac{(-1)^P}{2} (L_P - L_{P+1}), \\ \Psi_T = \Psi_R &= \frac{1}{2} (L_P + L_{P+1}),\end{aligned}\tag{2.76}$$

In this case a particular nodal discontinuous Galerkin scheme is recovered.

Spectral difference scheme (FR_{SD})

In order to recover a spectral difference scheme it is necessary that the correction functions have symmetrical zeros with respect to $\xi = 0$ of a standard element. To satisfy this requirement the parameter c has to be equal to:

$$c_{SD} = \frac{2P}{(2P+1)(P+1)(a_P P!)^2}, \quad \text{then} \quad \eta_P = \frac{P}{P+1}.$$

The correction functions then become:

$$\begin{aligned}\Psi_B = \Psi_L &= \frac{(-1)^P}{2} \left[L_P - \left(\frac{PL_{P-1}L_P + (P+1)L_{P+1}}{2P+1} \right) \right], \\ \Psi_T = \Psi_R &= \frac{1}{2} \left[L_P + \left(\frac{PL_{P-1}L_P + (P+1)L_{P+1}}{2P+1} \right) \right],\end{aligned}\tag{2.77}$$

Huynh or g2 scheme (FR_{g2})

A third important scheme recovered from the FR approach is the Huynh or g2 scheme presented for the first time by Huynh (2007). This particular scheme is recovered using c equal to:

$$c_{g2} = \frac{2(P+1)}{(2P+1)P(a_PP!)^2}, \quad \text{then} \quad \eta_P = \frac{P+1}{P}.$$

The expressions of the correction functions become:

$$\begin{aligned}\Psi_B = \Psi_L &= \frac{(-1)^P}{2} \left[L_P - \left(\frac{(P+1)L_{P-1}L_P + PL_{P+1}}{2P+1} \right) \right], \\ \Psi_T = \Psi_R &= \frac{1}{2} \left[L_P + \left(\frac{(P+1)L_{P-1}L_P + PL_{P+1}}{2P+1} \right) \right],\end{aligned}\tag{2.78}$$

Having defined the correction functions Ψ_B , Ψ_T , Ψ_L and Ψ_R , it is now possible to calculate the correction flux $\widehat{\mathbf{f}}^{\delta C}$. In particular, by evaluating the transformed flux jumps, defined as the difference between the numerical flux and the discontinuous flux evaluated at the interfaces between two adjacent elements, it is possible to calculate the correction fluxes as

$$\widehat{f}_1^{\delta C} = \underbrace{\left((\widehat{\mathbf{f}}^{\delta I} \cdot \mathbf{n})_L - \widehat{f}_{1,L}^{\delta DI} \right)}_{\widehat{\Delta f_{1,L}^\delta}} \Psi_L + \underbrace{\left((\widehat{\mathbf{f}}^{\delta I} \cdot \mathbf{n})_R - \widehat{f}_{1,R}^{\delta DI} \right)}_{\widehat{\Delta f_{1,R}^\delta}} \Psi_R,\tag{2.79a}$$

$$\widehat{f}_2^{\delta C} = \underbrace{\left((\widehat{\mathbf{f}}^{\delta I} \cdot \mathbf{n})_B - \widehat{f}_{2,B}^{\delta DI} \right)}_{\widehat{\Delta f_{2,B}^\delta}} \Psi_B + \underbrace{\left((\widehat{\mathbf{f}}^{\delta I} \cdot \mathbf{n})_T - \widehat{f}_{2,T}^{\delta DI} \right)}_{\widehat{\Delta f_{2,T}^\delta}} \Psi_T,\tag{2.79b}$$

where $\mathbf{n}$ is the outward normal with respect to the edges of the standard element and $\widehat{f}^{\delta DI}$ are the discontinuous fluxes evaluated at the edges of the standard element. We can appreciate how the flux jumps have been transformed into the standard space by using the Nanson formula previously introduced in (Eq. (2.61)) for the DG method.

The fifth step is calculating the divergence of the discontinuous flux:

$$\nabla_{\xi} \cdot \hat{\mathbf{f}}^{\delta D} = \sum_{i,j=0}^P \hat{f}_{1ij}^D \frac{\partial \ell_i(\xi_1)}{\partial \xi_1} \ell_j(\xi_2) + \sum_{i,j=0}^P \hat{f}_{2ij}^D \ell_i(\xi_1) \xi_1 \frac{\partial \ell_j(\xi_2)}{\partial \xi_2}, \quad (2.80)$$

and of the correction flux:

$$\nabla_{\xi} \cdot \hat{\mathbf{f}}^{\delta C} = (\widehat{\Delta f_{1,L}^{\delta}}) \frac{\partial \Psi_L}{\partial \xi_1} + (\widehat{\Delta f_{1,R}^{\delta}}) \frac{\partial \Psi_R}{\partial \xi_1} + (\widehat{\Delta f_{2,B}^{\delta}}) \frac{\partial \Psi_B}{\partial \xi_2} + (\widehat{\Delta f_{2,T}^{\delta}}) \frac{\partial \Psi_T}{\partial \xi_2}. \quad (2.81)$$

Finally, we can write the FR approach in matrix form on the $(P+1) \times (P+1)$ solution points as follows:

$$\mathbf{B} \frac{d\hat{\mathbf{u}}^D}{dt} + (\mathbf{D}_{\xi_1} \mathbf{B})^T \hat{\mathbf{f}}_1^D + (\mathbf{D}_{\xi_2} \mathbf{B})^T \hat{\mathbf{f}}_2^D + \mathbf{b}^{\text{FR}}, \quad (2.82)$$

where $\hat{\mathbf{b}}^{\text{FR}}$ is the boundary term vector which is also defined on the $(P+1) \times (P+1)$ and assumes the expression in Eq. (2.81). Note that, as done for the DG method, we can explicitly write the expressions of the transformed quantities between the physical to the standard space in order to highlight the metric terms as follows

$$\begin{aligned} \mathbf{B} \Lambda(J_n) \frac{d\mathbf{u}^D}{dt} + \left[(\mathbf{D}_{\xi_1} \mathbf{B})^T \Lambda \left(\frac{\partial x_2}{\partial \xi_1} \right) - (\mathbf{D}_{\xi_2} \mathbf{B})^T \Lambda \left(\frac{\partial x_2}{\partial \xi_1} \right) \right] \mathbf{f}_1^D + \\ + \left[(\mathbf{D}_{\xi_2} \mathbf{B})^T \Lambda \left(\frac{\partial x_1}{\partial \xi_1} \right) - (\mathbf{D}_{\xi_1} \mathbf{B})^T \Lambda \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \mathbf{f}_2^D + \mathbf{b}^{\text{FR}} = 0. \end{aligned} \quad (2.83)$$

The last form of the FR approach can be further reduced by defining the following expressions:

$$\begin{aligned} \mathbf{D}_{x_1} &= (\mathbf{D}_{\xi_1} \mathbf{B})^T \Lambda \left(\frac{\partial x_2}{\partial \xi_1} \right) - (\mathbf{D}_{\xi_2} \mathbf{B})^T \Lambda \left(\frac{\partial x_2}{\partial \xi_1} \right), \\ \mathbf{D}_{x_2} &= (\mathbf{D}_{\xi_2} \mathbf{B})^T \Lambda \left(\frac{\partial x_1}{\partial \xi_1} \right) - (\mathbf{D}_{\xi_1} \mathbf{B})^T \Lambda \left(\frac{\partial x_1}{\partial \xi_2} \right) \end{aligned} \quad (2.84)$$

and substituting them into Eq. (2.83), such that

$$\mathbf{B} \Lambda(J_n) \frac{d\mathbf{u}^D}{dt} + \mathbf{D}_{x_1} \mathbf{f}_1^D + \mathbf{D}_{x_2} \mathbf{f}_2^D + \mathbf{b}^{\text{FR}} = 0. \quad (2.85)$$

Equation (2.85) can be advanced in time via an explicit time integration scheme (e.g. explicit Runge-Kutta methods).

2.4 DG and FR schemes for second order problems

After having introduced the DG and FR approaches for discretising a first order conservation law, we now present the discretisation of second order problems - i.e. we describe the DG and FR diffusion operators. The implementation of the DG and FR diffusion operators in the spectral/*hp* element library Nektar++ has been carried out for one-dimensional and two-, three-dimensional tensor-product elements - specifically quadrilaterals and hexahedra. For describing the two approaches we consider the following two-dimensional second order problem

$$\frac{\partial u}{\partial t} - \nabla_{\mathbf{x}} \cdot \mathbf{f}(u, \nabla_{\mathbf{x}} u) = 0, \quad (2.86)$$

within a two-dimensional domain Ω , where t is the time, $u = u(x_1, x_2; t)$ is the conserved variable, $\mathbf{f} = [f_1(u, \nabla_{\mathbf{x}} u), f_2(u, \nabla_{\mathbf{x}} u)]$ is the diffusive flux of the variable u in the x_1 and x_2 Cartesian directions, respectively and $\nabla_{\mathbf{x}} = [\partial/\partial x_1, \partial/\partial x_2]$. We can appreciate how the fluxes in this case depend also on the first order derivatives of the independent variables.

To construct both the DG and FR diffusion operators, it is convenient to rewrite the second order problem in Eq. (2.86) into a system of first order equations

$$\frac{\partial u}{\partial t} - \mathbf{f}(u, \mathbf{q}) = 0 \quad (2.87a)$$

$$\mathbf{q} - \nabla_{\mathbf{x}} u = 0, \quad (2.87b)$$

within Ω , where $\mathbf{q}$ are the auxiliary variables. Hereafter, we will refer to Eq. (2.87a) as the principal equation and to Eq. (2.87b) as the auxiliary equation.

Similar to the advection operators, we partition the domain Ω into N non-overlapping subdomains Ω_n , on which, we will define the diffusion operators.

In the following, we first review the discontinuous Galerkin method applied to Eq. (2.87). We then focus on the flux reconstruction approach. We also remark that the steps described in the following two sections are very general and they can be applied to any elliptic or second order problem, such as the diffusive flux of the advection-diffusion equation or the viscous part of the compressible Navier-Stokes equation.

2.4.1 DG diffusion operator

To describe the DG diffusion operator we first define the spaces of the test functions ℓ and $\boldsymbol{\ell}$ associated to the system in Eq. (2.87),

$$\mathcal{V}^\delta := \{\ell \in L^2(\Omega) : \ell|_{\Omega_n} \in \mathbb{P}_P(\Omega_n), \}, \quad (2.88)$$

$$\mathcal{W}^\delta := \{\boldsymbol{\ell} \in [L^2(\Omega)]^2 : \boldsymbol{\ell}|_{\Omega_n} \in [\mathbb{P}_P(\Omega_n)]^2, \}, \quad (2.89)$$

where $\mathbb{P}_P$ is the space of piecewise continuous polynomials of degree $P \geq 1$ on Ω and $L^2(\Omega)$ has been defined in Eq. (2.55). Note that the space in Eq. (2.88) is associated to the principal equation, Eq. (2.87a), while the space in Eq. (2.89) is associated to the auxiliary equation, Eq. (2.87b).

We now approximate the solution u by the piecewise two-dimensional polynomial defined in Eq. (2.88), that is $u^\delta \in V^\delta$, multiply Eq. (2.87a), by the test function ℓ and the auxiliary equation, Eq. (2.87b), by the test function $\boldsymbol{\ell}$ and integrate over a local element Ω_n , such that:

$$\iint_{\Omega_n} \ell(\mathbf{x}) \frac{\partial u^{\delta D}(\mathbf{x})}{\partial t} d\mathbf{x} - \iint_{\Omega_n} \ell(\mathbf{x}) (\nabla_{\mathbf{x}} \cdot \mathbf{f}^{\delta D}) d\mathbf{x} = 0, \quad (2.90a)$$

$$\iint_{\Omega_n} \boldsymbol{\ell}(\mathbf{x}) \cdot \mathbf{q}^{\delta D}(\mathbf{x}) d\mathbf{x} - \iint_{\Omega_n} \boldsymbol{\ell}(\mathbf{x}) \cdot \nabla_{\mathbf{x}} u_{\mathbf{q}}^{\delta D}(\mathbf{x}) d\mathbf{x} = 0, \quad (2.90b)$$

where we used the superscript ‘ D ’ to denote discontinuous variables across adjacent element. We successively integrate by parts the second term of both the equations

$$\iint_{\Omega_n} \ell(\mathbf{x}) \frac{\partial u^{\delta D}}{\partial t} d\mathbf{x} + \overbrace{\iint_{\Omega_n} \nabla_{\mathbf{x}} \ell(\mathbf{x}) \cdot \mathbf{f}^{\delta D} d\mathbf{x}}^{\text{volumetric term}} - \overbrace{\int_{\partial\Omega_n} \ell(\mathbf{x}) (\mathbf{f}^{\delta I} \cdot \mathbf{n}) ds}^{\text{boundary term}} = 0, \quad (2.91a)$$

$$\iint_{\Omega_n} \boldsymbol{\ell}(\mathbf{x}) \cdot \mathbf{q}^{\delta D} d\mathbf{x} + \overbrace{\iint_{\Omega_n} (\nabla_{\mathbf{x}} \cdot \boldsymbol{\ell}(\mathbf{x})) u_{\mathbf{q}}^{\delta D} d\mathbf{x}}^{\text{volumetric term}} - \overbrace{\int_{\partial\Omega_n} (\boldsymbol{\ell}(\mathbf{x}) \cdot \mathbf{n}) u_{\mathbf{q}}^{\delta I} ds}^{\text{boundary term}} = 0, \quad (2.91b)$$

obtaining the weak form of the DG approach for the second order problem represented in Eq. (2.86). In Eq. (2.91) we have used the same nomenclature adopted for describing the DG method in the context of first order problems. In particular, $u^{\delta D}$ denotes the discretised discontinuous solution, $\mathbf{f}^{\delta D} = \mathbf{f}^{\delta D}(u^{\delta D}, \mathbf{q}^{\delta D})$ is the diffusive flux calculated from the discontinuous solution and its gradient, $\mathbf{q}^{\delta D}$ is the vector of the auxiliary variables and $\mathbf{f}^{\delta I}$, $u_{\mathbf{q}}^{\delta I}$

are the numerical fluxes associated to the principal and the auxiliary equation, respectively, with $\mathbf{n}$ being the outward pointing normal with respect to a given element edge.

The procedure to solve Eq. (2.91) starts by solving the auxiliary equation. Specifically, we need to find the numerical flux $u_q^{\delta I}$ via an appropriate methodology. Suitable techniques for calculating the numerical flux in the context of a second order problem include the *Bassi-Rebay* (BR) flux (Bassi & Rebay, 1997), the *Interior Penalty* (IP) flux (Arnold, 1982), the compact DG (CDG) method (Peraire & Persson, 2008) and the *Local Discontinuous Galerkin* (LDG) flux (Cockburn & Shu, 1998a). In Nektar++ we have implemented the latter, that reads

$$u_q^{\delta I} = \{\{u_q^\delta\}\} \pm \beta \cdot [[u_q^\delta]], \quad (2.92)$$

where $\{\{ \cdot \}\}$ and $[[\cdot]]$ are the average and the jump operator, respectively and they are defined as follows:

$$\{\{g\}\} = \frac{g_+ + g_-}{2}, \quad [[g]] = \frac{g_+ \mathbf{n}_+ + g_- \mathbf{n}_-}{2}, \quad (2.93)$$

while β was chosen such that it produces a compact stencil¹, that is $\beta = (1/2)\mathbf{n}_+$. The quantities g_- and g_+ denotes respectively the variable g on the left and on the right side of the interface between two adjacent elements and $\mathbf{n}_+$ and $\mathbf{n}_-$ are the associated outward pointing normals.

To complete the solution of the auxiliary equation we need to calculate the volumetric term in the identical way as done in the case of the advection operator.

After having solved the auxiliary equation and, therefore, the derivatives needed for computing the flux $\hat{\mathbf{f}}^{\delta D}$, the principal equation, Eq. (2.95a), reduces to a simple first order problem and can be solved using the same procedure used for a first order problem, except for what concern the numerical flux $\mathbf{f}^{\delta I}$. In particular, this needs to be calculated by means of the same approach employed for the numerical flux of the auxiliary equation. Therefore, in our case we have:

$$\mathbf{f}^{\delta I} = \{\{\mathbf{f}^\delta\}\} \mp \beta \cdot [[\mathbf{f}^\delta]] - \tau[[u^\delta]], \quad (2.94)$$

where we can add a penalty term, $\tau[[u^\delta]]$, to control the jump in the solution and we note that the sign prior to the jump operator has changed.

¹Note that we could have used different values of β which would have produced wider stencils.

2.4.2 FR diffusion operator

The procedure to solve Eq. (2.87) by means of the FR approach can be explained within the standard element Ω_s as done for the FR advection operator. Within the standard domain Ω_s , Eq. (2.87) becomes the following transformed system of equations

$$\frac{\partial \hat{u}}{\partial t} - \nabla_{\boldsymbol{\xi}} \cdot \hat{\mathbf{f}}(\hat{u}, \hat{\mathbf{q}}) = 0 \quad (2.95a)$$

$$\hat{\mathbf{q}} - \nabla_{\boldsymbol{\xi}} \hat{u} = 0, \quad (2.95b)$$

where $\hat{u}$ and $\hat{\mathbf{f}}$ have been defined in Eq. (2.68) and Eq. (2.69), respectively and are reported here for the sake of clarity:

$$\hat{u}(\boldsymbol{\xi}, t) = J_n u_n(\Theta_n(\boldsymbol{\xi}), t), \quad \hat{\mathbf{f}}(\boldsymbol{\xi}, t) = J_n \mathbf{G}_n^{-1} \mathbf{f}_n$$

while $\hat{\mathbf{q}}$ is defined as follows:

$$\hat{\mathbf{q}}(\boldsymbol{\xi}, t) = \nabla_{\boldsymbol{\xi}} \hat{u} = J_n \mathbf{G}_n^T \mathbf{q}_n. \quad (2.96)$$

The implementation of the FR diffusion operator involves projecting the solution onto $(P+1)^2$ points within Ω_s (as done for the advection operator), using Eq. (2.70), here reported for the sake of clarity

$$\hat{u}^{\delta D}(\boldsymbol{\xi}, t) = \sum_{i,j=0}^P \ell_{ij}(\boldsymbol{\xi}) \hat{u}_{ij}^D = \sum_{i,j=0}^P \ell_i(\xi_1) \ell_j(\xi_2) \hat{u}_{ij}^D,$$

where the superscript ‘ D ’ denotes that the solution can be discontinuous across the elements. We now need to solve the auxiliary equation first in order to calculate the auxiliary variable vector $\hat{\mathbf{q}}$ required for calculating the diffusive flux $\hat{\mathbf{f}}(\hat{u}, \hat{\mathbf{q}})$. The procedure is analogous to that adopted for the advection term. In particular we calculate the numerical flux for the auxiliary equation $u^{\delta I}$ in order to propagate the numerical solution from element to element by using Eq. (2.92). We then add a polynomial of order $P+1$ called auxiliary correction flux and denoted by $\hat{u}^{\delta C}$ to the discontinuous solution $\hat{u}^{\delta D}$, in order to obtain the following total auxiliary flux:

$$\hat{u}^{\delta} = \hat{u}^{\delta D} + \hat{u}^{\delta C}. \quad (2.97)$$

The auxiliary correction flux is defined as follows:

$$\hat{u}^{\delta C} = (\hat{u}_B^{\delta I} - \hat{u}_B^{\delta DI})\Psi_B + (\hat{u}_L^{\delta I} - \hat{u}_L^{\delta DI})\Psi_L + (\hat{u}_R^{\delta I} - \hat{u}_R^{\delta DI})\Psi_R + (\hat{u}_T^{\delta I} - \hat{u}_T^{\delta DI})\Psi_T \quad (2.98)$$

where $u_{B/R/T/L}^{\delta I}$ are the auxiliary numerical fluxes at the bottom, right, top and left edges of the standard element, $u_{B/R/T/L}^{\delta DI}$ are the associated values of the solution $u^{\delta D}$ at the same edges and where we used the correction functions defined in Eq. (2.73).

After having calculated the total auxiliary flux, we can calculate the auxiliary variables $\hat{\mathbf{q}}^{\delta D}$ by differentiating Eq. (2.97) within Ω_s as follows

$$\hat{\mathbf{q}}^{\delta D} = \begin{bmatrix} \sum_{i,j=0}^P \hat{u}_{1ij}^{\delta D} \frac{\partial \ell_i(\xi_1)}{\partial \xi_1} \ell_j(\xi_2) + \left(\hat{u}_L^{\delta qI} - \hat{u}_L^{\delta DI} \right) \frac{\partial \Phi_L}{\partial \xi_1} + \left(\hat{u}_R^{\delta I} - \hat{u}_R^{\delta DI} \right) \frac{\partial \Phi_R}{\partial \xi_1} + \\ \sum_{i,j=0}^P \hat{u}_{2ij}^{\delta D} \ell_i(\xi_1) \xi_1 \frac{\partial \ell_j(\xi_2)}{\partial \xi_2} + \left(\hat{u}_B^{\delta I} - \hat{u}_B^{\delta DI} \right) \frac{\partial \Phi_B}{\partial \xi_2} + \left(\hat{u}_T^{\delta I} - \hat{u}_T^{\delta DI} \right) \frac{\partial \Phi_T}{\partial \xi_2} \end{bmatrix}. \quad (2.99)$$

This concludes the work on the auxiliary equation and permits the construction of the flux $\hat{\mathbf{f}}^{\delta}$. Therefore, we can now solve the principal equation, Eq. (2.95a), which results in performing the same steps as for the FR advection operator. Specifically, we first define the discontinuous fluxes $\hat{f}_1^{\delta D}$ and $\hat{f}_2^{\delta D}$ as in Eq. (2.71a) and Eq. (2.71b), respectively. We then calculate the correction fluxes $\hat{f}_1^{\delta C}$ and $\hat{f}_2^{\delta C}$ as in Eq. (2.79a) and Eq. (2.79b). The only difference at this stage is the calculation of the numerical flux $\mathbf{f}^{\delta I}$ needed for calculating the correction fluxes. In order to have a stable numerical scheme, they must be calculated using the same approach used for the numerical flux of the auxiliary equation. In our case, this means using Eq. (2.94).

We finally calculate the divergence of the resulting total flux by using Eq. (2.80) and Eq. (2.81), and we assemble the overall right-hand side.

2.5 Verification of the implementation

The implementation of the FR approach carried out in Nektar++ and the assessment of the DG method already present in the library, have been thoroughly tested in this thesis work. Additional verification results can be found in (Mengaldo, 2012) and (De Grazia, 2013). In this section, we report the results obtained on a two-dimensional unsteady advection diffusion equation, in order to verify the implementation of both the advection and the diffusion terms.

The test case is defined as follows:

$$\begin{cases} \frac{\partial u}{\partial t} + a_x \frac{\partial u}{\partial x} + a_y \frac{\partial u}{\partial y} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \\ u(x, y, t = 0) = \sin(\pi x) \sin(\pi y) \\ u(x_b, y_b, t) = e^{-2\pi^2 t} \sin(\pi(x - a_x t)) \sin(\pi(y - a_y t)), \end{cases} \quad (2.100)$$

where u is the independent variable, a_x and a_y are the advection velocities along x and y and (x_b, y_b) denotes the boundaries of the numerical domain where we used the exact solution of the problem $u_{exact}(x, y, t) = e^{-2\pi^2 t} \sin(\pi(x - a_x t)) \sin(\pi(y - a_y t))$ as boundary condition.

For all the simulations, we employed $a_x = a_y = 20$ and a final time $T = 0.1s$ in order for the initial condition to travel for one convective length without being strongly dissipated. The time-integration scheme used was a fourth-order Runge-Kutta scheme and the time-step was chosen such that the time-integration errors were negligible.

To verify the implementation of the FR approach, we considered the three different schemes presented in section 2.3.2, namely the FR_{DG}, the FR_{SD} and the FR_{g2} schemes on a set of Gauss-Lobatto-Legendre points. All the FR schemes considered, employ a collocation projection at the solution points to reconstruct the flux. Concerning the DG method, we took into account its collocated version - i.e. for a polynomial of order $P=3$ we use four quadrature points. Note that this approach associated to the use of Gauss-Lobatto-Legendre points leads to a non-exact (or lumped) mass matrix.

Throughout the following, we will make extensive use of the L_2 error defined as follows

$$E_{L2} = \sqrt{\sum_{i=0}^{Q-1} (u_i - u_{i,exact})^2 w_i}, \quad (2.101)$$

where u_i is the numerical solution calculated at the i -th quadrature point, $u_{i,exact}$ is the associated exact solution, Q is the number of quadrature points and w_i are the associated weights. The four meshes employed are depicted in figure 2.4. They are increasingly finer, with a number of elements equal to $N_x \times N_x$ with $N_x = 2, 4, 8, 16$, respectively.

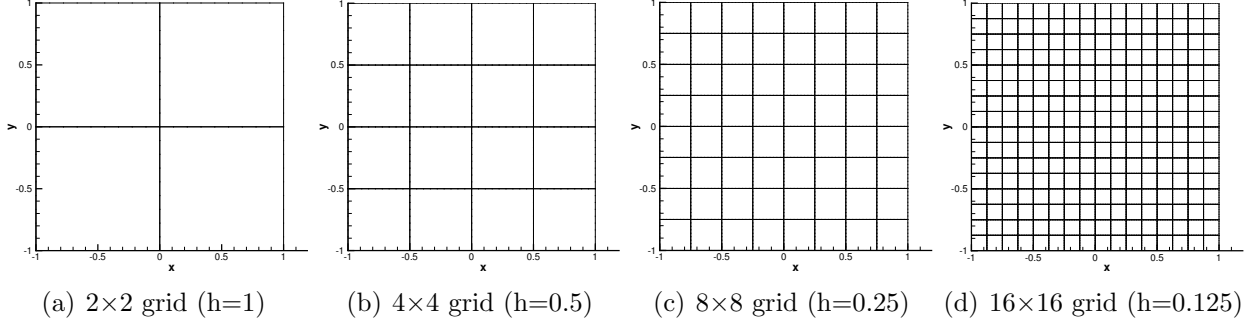


Figure 2.4: Unsteady advection diffusion test case: regular meshes.

In figure 2.5(a), we show the L_2 error vs. the element size ‘ h ’ for the four regular meshes depicted in figure 2.5 and for four different polynomial orders, $P=3,5,7,9$. The curves are obtained in a logarithmic scale and it is possible to observe an $\mathcal{O}(h^{P+1})$ convergence for all the polynomial orders considered depicted as well as for both the FR schemes and the DG method.

In figure 2.5(b), we show the L_2 error vs. the polynomial order. The curves are obtained in a semi-logarithmic scale and it is possible to see an exponential convergence of the error for all the meshes considered as well as for both the FR schemes and the DG method. It is also possible to note an odd-even behaviour of the error for the coarsest mesh, which has been widely reported in the literature for discontinuous spectral/ hp element methods (see for example (Babuška *et al.*, 1999; Baumann & Oden, 1999; Larson & Niklasson, 2004; Tu *et al.*, 2012)).

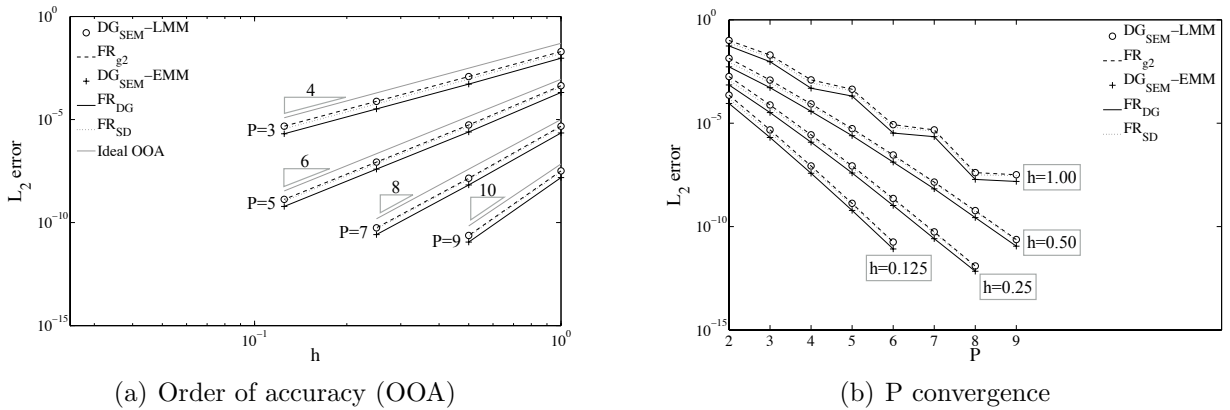


Figure 2.5: Order of accuracy (OOA) and P convergence for the regular meshes.

In table 2.1, we report the values of the L_2 errors, E_{L2} , and the maximum time-step

possible to achieve a stable simulation for the mesh in figure 2.4(b). From this table, it is possible to appreciate how the error associated to the FR_{DG} and the DG_{SEM} -EMM schemes are the lowest amongst the schemes considered for a given polynomial order, while the largest time-step limit is obtained for FR_{g2} and DG_{SEM} -LMM. The results are in agreement with the finding by Vincent *et al.* (2011a), where the authors showed how the FR_{DG} scheme is the most accurate but also the scheme with the most severe restriction in terms of time-step, while the FR_{g2} and FR_{SD} are slightly less accurate but have more favourable time-step restrictions. Interestingly, we can note that the DG_{SEM} -LMM scheme provides the same results of the FR_{g2} scheme in terms of both L_2 and maximum time-step as shown in (De Grazia *et al.*, 2014), where the author of this thesis and co-workers have shown how, for regular meshes, also the FR_{DG} scheme and the DG_{SEM} approach with an exact mass matrix are identical, for both linear and nonlinear problems.

Polynomial order			DG_{SEM} -EMM	FR_{DG}	DG_{SEM} -LMM	FR_{g2}	FR_{SD}
P=2	$\mathbf{E}_{L2}$	[-]	5.31×10^{-3}	5.31×10^{-3}	1.38×10^{-2}	1.38×10^{-2}	9.13×10^{-3}
	Δt_{max}	[s]	0.0015	0.0015	0.0036	0.0036	0.0027
P=3	$\mathbf{E}_{L2}$	[-]	5.23×10^{-4}	5.23×10^{-4}	1.21×10^{-3}	1.21×10^{-3}	9.14×10^{-4}
	Δt_{max}	[s]	0.00060	0.00060	0.0010	0.0010	0.00094
P=4	$\mathbf{E}_{L2}$	[-]	3.75×10^{-5}	3.75×10^{-5}	8.59×10^{-5}	8.59×10^{-5}	6.85×10^{-5}
	Δt_{max}	[s]	0.00027	0.00027	0.00041	0.00041	0.00038
P=5	$\mathbf{E}_{L2}$	[-]	2.49×10^{-6}	2.49×10^{-6}	5.44×10^{-6}	5.44×10^{-6}	4.55×10^{-6}
	Δt_{max}	[s]	0.00014	0.00014	0.00019	0.00019	0.00018
P=6	$\mathbf{E}_{L2}$	[-]	1.31×10^{-7}	1.31×10^{-7}	2.90×10^{-7}	2.90×10^{-7}	2.47×10^{-7}
	Δt_{max}	[s]	0.000079	0.000079	0.00010	0.00010	0.000098
P=7	$\mathbf{E}_{L2}$	[-]	6.69×10^{-9}	6.69×10^{-9}	1.42×10^{-8}	1.42×10^{-8}	1.25×10^{-8}
	Δt_{max}	[s]	0.000047	0.000047	0.000061	0.000061	0.000059
P=8	$\mathbf{E}_{L2}$	[-]	2.73×10^{-10}	2.73×10^{-10}	5.92×10^{-10}	5.92×10^{-10}	5.25×10^{-10}
	Δt_{max}	[s]	0.000030	0.000030	0.000038	0.000038	0.000037
P=9	$\mathbf{E}_{L2}$	[-]	1.13×10^{-11}	1.13×10^{-11}	2.37×10^{-11}	2.37×10^{-11}	2.14×10^{-11}
	Δt_{max}	[s]	0.000020	0.000020	0.000024	0.000024	0.000024

Table 2.1: L^2 error and maximum Δt vs. polynomial order for the different types of DG and FR schemes tested, on the mesh with element size $h=0.5$.

These two identities are particularly important because they indicate that, having already an existing implementation of the DG method allows recovering two schemes with different accuracy and time-restriction properties. To better explain this point, figure 2.6 shows the CFL limit and the super-accuracy² of the three FR schemes considered for a one-dimensional

²The super-accuracy for the current implementation of the FR approach and of the DG method has

linear advection equation with $P=3$. We can see how the FR_{g2} scheme allows the largest time-step limit. The FR_{SD} has a similar time-step limit while the FR_{DG} scheme is the most restrictive. On the other hand the FR_{DG} scheme is the most accurate while the FR_{g2} scheme is the least accurate among the FR schemes considered. In chapter 4, we will show how these connections holds also for deformed/curvilinear meshes.

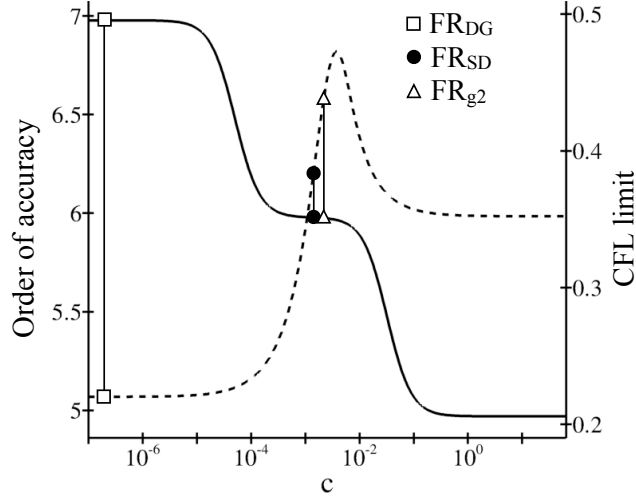


Figure 2.6: Order of accuracy (OOA) and CFL limit for the three FR schemes considered. Figure re-adapted from (Vincent *et al.*, 2011a) for a one-dimensional test case with $P=3$.

In terms of computational performance, from figure 2.7 we can note how the FR approach and the DG method have the same trend as the polynomial order used for solving the problem increases. The only difference between the two schemes is a shift in the CPU time. This difference is likely to be due to the different data structures used in the DG method and in the FR approach within Nektar++. Note that we do not report the results for the other schemes shown in table 2.1 because identical to the curved reported.

Concerning the scalability, we measured two indexes, the speed-up S_p and the efficiency E_p . The first is defined as $S_p = T_1/T_p$, where the T_1 is the CPU time required using one CPU while T_p is the cpu time required using ‘ p ’ CPUs. The second is the speed-up normalised with respect to the number of CPUs. In figure 2.8, we show both indexes as a function of the number of elements per CPU for the FR_{DG} scheme. We can note a relatively good scalability up to 16 elements per CPU.

also been tested for the one-dimensional linear advection equation (De Grazia, 2013) providing the expected results.

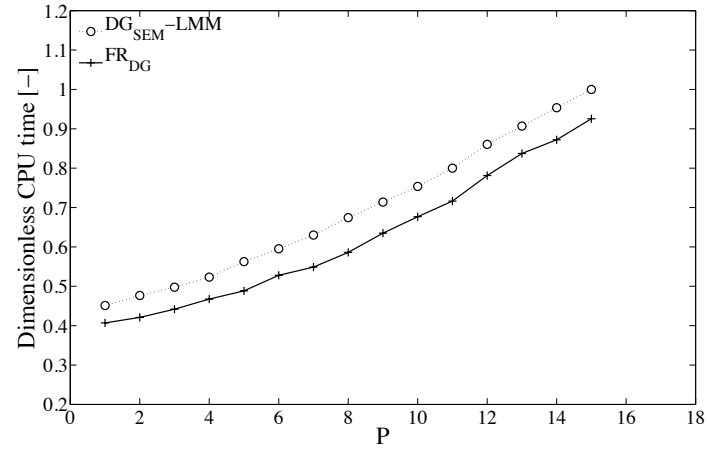
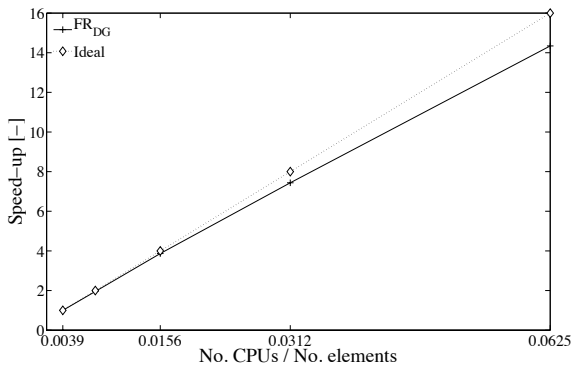
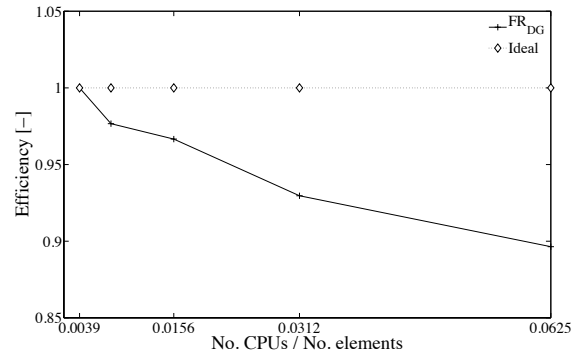


Figure 2.7: Normalised CPU time as a function of the polynomial order: comparison between the DG and the FR approaches.



(a) Speed-up



(b) Efficiency

Figure 2.8: Scalability performance of the methods: speed-up (figure 2.8(a)) and efficiency (figure 2.8(b)).

Chapter 3

Governing equations

In this chapter, we present the equations governing inviscid and viscous compressible flows, namely the compressible Euler and Navier-Stokes equations. We also introduce briefly their numerical discretisation by means of the DG and the FR approaches as well as the implementation strategy adopted for the boundary conditions. In the final part of the chapter, we show some results to verify the compressible flow solver, whose implementation has been an important part of this thesis work.

3.1 Euler equations

The Euler equations are a first-order hyperbolic system of equations and they describe an inviscid flow with no thermal conductivity. They are a subset of the compressible Navier-Stokes equations and can be used to describe flows where the viscosity effects can be neglected - for example the free-stream region of high-Reynolds number flows. Their two-dimensional conservative form on a physical domain Ω and in a Cartesian frame of reference (x_1, x_2) reads

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}_{i,1}}{\partial x_1} + \frac{\partial \mathbf{f}_{i,2}}{\partial x_2} = 0, \quad (3.1)$$

where $\mathbf{u}$ is the vector of the conserved variables and $\mathbf{f}_{i,1} = \mathbf{f}_{i,1}(\mathbf{u})$, $\mathbf{f}_{i,2} = \mathbf{f}_{i,2}(\mathbf{u})$ are the vectors of the inviscid fluxes

$$\mathbf{u} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ E \end{pmatrix}, \quad \mathbf{f}_{i,1} = \begin{pmatrix} \rho u \\ p + \rho u^2 \\ \rho uv \\ u(E + p) \end{pmatrix}, \quad \mathbf{f}_{i,2} = \begin{pmatrix} \rho v \\ \rho uv \\ p + \rho v^2 \\ v(E + p) \end{pmatrix}, \quad (3.2)$$

with ρ being the density, u and v being the velocity components in x_1 - and x_2 -directions, p being the pressure and E being the total energy. Throughout this work, we consider a perfect gas law for which the pressure is related to the total energy by the following expression

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho(u^2 + v^2), \quad (3.3)$$

where γ is the ratio of specific heats.

3.1.1 Numerical discretisation

Following the steps presented in chapter 2 for the discretisation of a first order problem (section 2.3), the numerical discretisation of the Euler equations (Eq. (3.1)) by means of the DG method and the FR approach involves dividing the computational domain Ω into N non-overlapping subdomains, $\Omega \rightarrow \Omega_n$, and, on each subdomain, representing the solution by a polynomial of degree P , $(\rho_n, \rho u_n, \rho v_n, E_n)^T \in \mathbb{P}_P$. As shown previously, the solution over the entire domain is allowed to be discontinuous and the coupling between the elements is obtained through the boundary term. The formulation of the DG method and of the FR approach is reported in the following by using a compact form, where we use the index m to indicate the m -th equation of the system in Eqs. (3.1, 3.2).

DG scheme:

The matrix form of the DG scheme applied to the two-dimensional Euler equations reads

$$\left. \frac{d\mathbf{u}^D}{dt} \right|_m = \mathbf{M}^{-1} \left\{ \mathbf{S}_{x_1} \mathbf{f}_{i,1}^D + \mathbf{S}_{x_2} \mathbf{f}_{i,2}^D - \mathbf{b}^{\text{DG}} \right\}_m \quad \text{for } m = 1, \dots, 4, \quad (3.4)$$

where $\mathbf{M}$ is the elemental mass matrix, $\mathbf{S}_{x_1}$ and $\mathbf{S}_{x_2}$ are the elemental stiffness matrices defined in Eq. (2.63), $\mathbf{u}^D$ is the vector of the m -th variable defined on the solution points of a given element (note that we changed notation here; $\mathbf{u}$ is a vector containing the values of a conserved variable on the solution points of a given element, while $\mathbf{u}$

is the vector of the conserved variables), $\mathbf{f}_{i,1}$ and $\mathbf{f}_{i,2}$ are the elemental nonlinear fluxes defined on the solution points of a given element (note the change of notation) and $\mathbf{b}^{\text{DG}}$ is the boundary term (also defined on the solution points of a given element)

$$\mathbf{b}^{\text{DG}} \Big|_m = \int_{\partial\Omega_n} \ell_{ij} \left(\mathcal{H}_i^{\delta I} \cdot \mathbf{n} \right)_m ds \quad \text{for } m = 1, \dots, 4, \quad (3.5)$$

with ℓ_{ij} being a two-dimensional polynomial expansion basis, $\mathbf{n}$ being the outward pointing normal with respect to a given element Ω_n , and $\mathcal{H}_i^{\delta I}$ being the tensor of the numerical inviscid fluxes $\mathcal{H}_i^{\delta I} = [\mathbf{f}_{i,1}^{\delta I}, \mathbf{f}_{i,2}^{\delta I}]$, at the interface between two adjacent elements.

FR approach:

The matrix form of the FR approach applied to Eq. (3.1) reads

$$\mathbf{B}\mathbf{A}(J_n) \frac{d\mathbf{u}^D}{dt} \Big|_m = \left\{ \mathbf{D}_{x_1} \mathbf{f}_{i,1}^D + \mathbf{D}_{x_2} \mathbf{f}_{i,2}^D - \mathbf{b}^{\text{FR}} \right\}_m \quad \text{for } m = 1, \dots, 4, \quad (3.6)$$

where $\mathbf{D}_{x_1}$, $\mathbf{D}_{x_2}$ are the differentiation matrices with respect to x_1 and x_2 , respectively and defined in Eq. (2.84), $\mathbf{f}_{i,1}$, $\mathbf{f}_{i,2}$ are the elemental nonlinear fluxes of the Euler equations and $\mathbf{b}^{\text{FR}}$ is the boundary term

$$\mathbf{b}^{\text{FR}} \Big|_m = \left(\mathcal{H}_i^{\delta C} \cdot \mathbf{n} \right)_m \Psi' = \left[\left(\mathcal{H}_i^{\delta I} \cdot \mathbf{n} \right)_m - \left(\mathcal{H}_i^{\delta DI} \cdot \mathbf{n} \right)_m \right] \Psi' \quad \text{for } m = 1, \dots, 4, \quad (3.7)$$

with Ψ' being the derivative of the correction functions defined in section 2.3.2, $\mathcal{H}_i^{\delta I}$ being the tensor of the numerical inviscid fluxes $\mathcal{H}_i^{\delta I} = [\mathbf{f}_{i,1}^{\delta I}, \mathbf{f}_{i,2}^{\delta I}]$ at the interface between two adjacent elements and $\mathcal{H}_i^{\delta DI}$ being again the tensor of the inviscid fluxes evaluated at the boundary of a given element Ω_n .

As anticipated in chapter 2, one of the crucial features of both the methods and more in general of this class of schemes is the way the boundary term $\mathcal{H}_i^{\delta I}$ is calculated. For the Euler equations and for the advection term of the compressible Navier-Stokes equations it is common to use either exact or approximated Riemann solvers. They both calculate an intermediate numerical flux at the interface between two adjacent elements by using some characteristic information coming from an eigenvalue analysis of the equations and may

contain values coming from the left (+) or right (-) state of a given interface (see figure 2.3)

$$\mathcal{H}_i^{\delta I} \Big|_{\ell} = \mathcal{H}_i^{\delta I}(\mathbf{u}_+^{\delta}, \mathbf{u}_-^{\delta}) \Big|_{\ell} \quad \text{for } m = 1, \dots, 4. \quad (3.8)$$

When adopting a weak strategy for implementing the boundary conditions, the boundary term $\mathcal{H}^{\delta I}$ is also responsible for the boundary conditions to be correctly transferred into the domain via a ghost state. In this thesis we used this approach and the implementations of the boundary conditions is shown in section 3.4.

In addition, the implementation of the compressible solver in Nektar++ has been carried out considering various Riemann solvers, including an exact and various approximated Riemann solvers. In section 3.3, we describe four of the Riemann solvers implemented in Nektar++: exact, HLL and HLLC.

3.2 Compressible Navier-Stokes equations

In contrast to the Euler equations, the compressible Navier-Stokes equations include the effects of fluid viscosity and heat conduction and are consequently composed by an inviscid and a viscous tensor. They are a second-order system of equations, where the viscous tensor depends on the conserved variables and on their first order derivatives. The second-order PDE on a two-dimensional physical domain Ω and in a Cartesian frame of reference (x_1, x_2) reads:

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}_1}{\partial x_1} + \frac{\partial \mathbf{f}_2}{\partial x_2} = 0, \quad (3.9)$$

where $\mathbf{u}$ is the vector of the conserved variables, $\mathbf{f}_1 = \mathbf{f}_1(\mathbf{u}, \nabla_x \mathbf{u})$, and $\mathbf{f}_2 = \mathbf{f}_2(\mathbf{u}, \nabla_x \mathbf{u})$ are the vectors of the fluxes which can also be written as:

$$\begin{aligned} \mathbf{f}_1 &= \mathbf{f}_{i,1} - \mathbf{f}_{v,1}, \\ \mathbf{f}_2 &= \mathbf{f}_{i,2} - \mathbf{f}_{v,2}, \end{aligned} \quad (3.10)$$

where $\mathbf{f}_{i,1}$ and $\mathbf{f}_{i,2}$ are the inviscid fluxes defined in Eq. (3.1) and $\mathbf{f}_{v,1}$, $\mathbf{f}_{v,2}$ are the viscous fluxes which assume the following form:

$$\mathbf{f}_{v,1} = \begin{pmatrix} 0 \\ \tau_{x_1 x_1} \\ \tau_{x_2 x_1} \\ u\tau_{x_1 x_1} + v\tau_{x_2 x_1} + kT_{x_1} \end{pmatrix}, \quad \mathbf{f}_{v,2} = \begin{pmatrix} 0 \\ \tau_{x_1 x_2} \\ \tau_{x_2 x_2} \\ u\tau_{x_1 x_2} + v\tau_{x_2 x_2} + kT_{x_2} \end{pmatrix}, \quad (3.11)$$

where $\tau_{x_1 x_1}$, $\tau_{x_1 x_2}$, $\tau_{x_2 x_1}$, $\tau_{x_2 x_2}$, are the components of the stress tensor¹

$$\begin{aligned} \tau_{x_1 x_1} &= 2\mu \left[\frac{\partial u}{\partial x_1} - \frac{1}{3} \left(\frac{\partial u}{\partial x_1} + \frac{\partial v}{\partial x_2} \right) \right], \\ \tau_{x_2 x_2} &= 2\mu \left[\frac{\partial v}{\partial x_2} - \frac{1}{3} \left(\frac{\partial u}{\partial x_1} + \frac{\partial v}{\partial x_2} \right) \right], \\ \tau_{x_1 x_2} &= \tau_{x_2 x_1} = \mu \left(\frac{\partial v}{\partial x_1} + \frac{\partial u}{\partial x_2} \right), \end{aligned} \quad (3.12)$$

where μ is the dynamic viscosity and it is obtained using the Sutherland's law:

$$\mu = \mu_\infty \left(\frac{T}{T_\infty} \right)^{3/2} \frac{T_\infty + 110.4}{T + 110.4} \quad (3.13)$$

and k is the thermal conductivity.

3.2.1 Numerical discretisation

The numerical discretisation of the compressible Navier-Stokes equations for the inviscid fluxes is identical to that of the Euler equations (the inviscid fluxes of the Euler equations and of the compressible Navier-Stokes are identical) while the additional viscous fluxes are treated in a different manner. Specifically, we use the approach presented in section 2.4 for second-order problems. We split Eq. (3.9) into two first order equations: an auxiliary equation, through which we calculate the derivatives of the conserved variables $\mathbf{u}$, and the principal equation, which is constituted by the original problem (see also Eq. (2.87)). The procedure is therefore divided into two steps:

- the calculation of the spatial derivatives of the auxiliary variables $\mathbf{u}_{aux} = [u, v, T]^T$,

¹Note that, for all the simulations carried out, we used the Stokes hypothesis $\lambda = -2/3$.

necessary to construct the viscous fluxes;

- the computation of the viscous fluxes and the solution of the principal system following either the DG method or the FR approach presented in section 2.4.

For both, DG and FR, the numerical fluxes at the interface between two adjacent elements are calculated using the local discontinuous Galerkin (LDG) method also presented in section 2.4 and here briefly reported for the reader's convenience:

$$\begin{aligned}\mathbf{u}_{aux}^{\delta I} &= \{\{\mathbf{u}_{aux}^{\delta}\}\} \pm \boldsymbol{\beta} \cdot [[\mathbf{u}_{aux}^{\delta}]] \\ \mathcal{H}_v^{\delta I} &= \{\{\mathcal{H}_v^{\delta}\}\} \mp \boldsymbol{\beta} \cdot [[\tilde{\mathcal{H}}_v^{\delta}]] - \gamma [[\mathbf{u}_{aux}^{\delta}]]\end{aligned}\tag{3.14}$$

where

$$\{\{\mathbf{g}\}\} = \frac{\mathbf{g}_+ + \mathbf{g}_-}{2}, \quad [[\mathbf{g}]] = \frac{\mathbf{g}_+ \mathbf{n}_+ + \mathbf{g}_- \mathbf{n}_-}{2}, \quad \boldsymbol{\beta} = \frac{1}{2} \mathbf{n}_+ \tag{3.15}$$

with the quantities $\mathbf{g}_-$ and $\mathbf{g}_+$ being the variable $\mathbf{g}$ on the right and on the left side of the interface between two elements and with $\mathbf{n}_+$ and $\mathbf{n}_-$ being the respective normals (for additional details the interested reader can also refer to Cockburn & Shu (1998a)).

As in the case of the inviscid flux, where the boundary term (i.e. the numerical flux) transfers the boundary conditions into the domain, for the viscous flux the interface fluxes of the auxiliary variables $\mathbf{u}_{aux}^{\delta I}$ and of the primitive system $\mathcal{H}^{\delta I}$ are responsible for the boundary conditions to be correctly transferred into the domain, when a weak approach is adopted.

3.3 Numerical flux: One-dimensional Riemann solvers

As already mentioned before, the one-dimensional Riemann solvers allow the various elements of the DG/FR approach (applied to the Euler equations and to the advection term of the compressible Navier-Stokes equations) to propagate the information and are responsible for the boundary conditions to be correctly transferred into the domain via a ghost state.

In the following we describe some relevant Riemann solvers implemented in Nektar++ and used in the course of this thesis. We will follow a similar formulation as in the book by Toro (2009). In particular, we consider the two-dimensional Euler equations and its associated Riemann problem (figure 3.1), which is composed by two nonlinear waves (dotted lines) associated with the maximum and minimum eigenvalue of the system (shock waves or rarefaction waves), and by two contact waves (dashed line) associated with the two coincident eigenvalues of the problem. The region between the two nonlinear waves is denoted as the

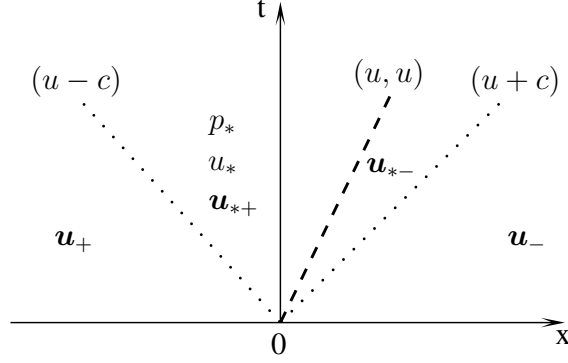


Figure 3.1: Riemann problem for the two-dimensional Euler equations using an ideal gas law.

star region in the following and the left and right states are denoted with the subscripts $+$ and $-$, respectively. The nature of the solution strictly depends on the equation of state. In all the cases considered in this work, we used an ideal gas law.

When using a Riemann solver at a given interface it is necessary to rotate the two-/three-dimensional problem into a one-dimensional problem in the normal direction with respect to the given interface. By exploiting the rotational invariance of the Euler equations it is possible to write

$$[\mathbf{f}_{i,1} \quad \mathbf{f}_{i,2}] \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} = \mathbf{R}^{-1} \mathbf{f}(\mathbf{R}\mathbf{u}), \quad (3.16)$$

where θ is the angles which link the normal frame of reference (with respect to a given interface) to the Cartesian frame of reference, $\mathbf{R}$ is the following rotation matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(\theta) & \sin(\theta) & 0 \\ 0 & -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (3.17)$$

and $\mathbf{R}^{-1}$ is its inverse. From an implementation point of view, it is common to first apply the rotation matrix $\mathbf{R}$ to the variables $\mathbf{u}$ in order to calculate the augmented one-dimensional flux

$$\mathbf{f}(\mathbf{u}_n) = \mathbf{f}(\mathbf{R}\mathbf{u}). \quad (3.18)$$

Successively one rotates the augmented one-dimensional flux back to the cartesian frame of reference by multiplying by the inverse of the rotation matrix using Eq. (3.16). It is important to note here that the final flux obtained is projected into the normal direction with respect

to the interface.

The intermediate step between the two rotations is to calculate the augmented one-dimensional flux. At this stage, different Riemann solvers can be used depending on the performance (dissipation, dispersion properties, stability, computational costs) one desires to obtain.

In the following we briefly describe the three Riemann solvers which have been implemented into Nektar++: an exact Riemann solver and three approximated Riemann solvers, HLL and HLLC.

Exact

The Exact Riemann solver uses the exact solution of the Riemann problem for the Euler equations. It is well-known that there are ten possible wave patterns allowed at an interface and if we use a local frame of reference with respect to the interface, this reduces in solving the Riemann problem in the space-time plane for $x/t = 0$. The first step to perform is therefore to identify which pattern has arisen at a given interface and successively calculate the boundary state $\mathbf{u}_n$ (and consequently the augmented one-dimensional flux $\mathbf{f}(\mathbf{u}_n)$) for $x/t = 0$. In particular we have two macro cases each

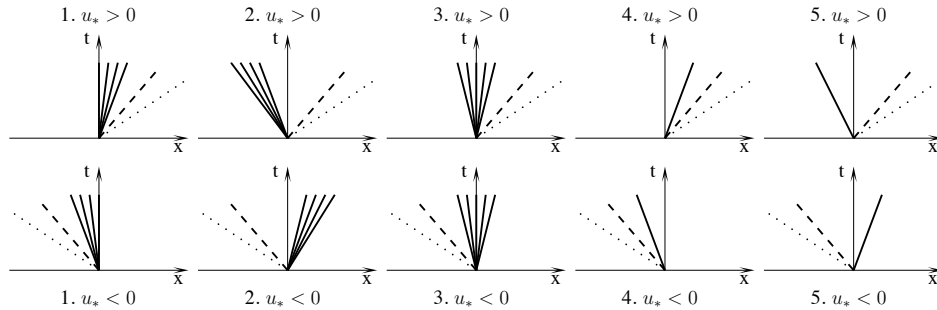


Figure 3.2: Possible wave patterns of the exact Riemann solver for the Euler equation.

of those in turn divided into five sub-cases:

- Positive particle speed in the region between the two nonlinear waves (star region), $u_* \geq 0$
 1. Left rarefaction - $\mathbf{u}_+$: if $p_* \leq p_+$ and $u_+ \geq c_+$ we have a left rarefaction wave with a left state which is supersonic and the state for $x/t = 0$ is the left state. Consequently the boundary fluxes in the normal frame of reference

with respect to a given interface are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ u_+ \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ u_+^2 + p_+ \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ u_+ v_+ \\ \mathcal{H}_E^{\delta I} = u_+ \left[\frac{p_+}{\gamma-1} + \frac{1}{2} \rho_L (u_+^2 + v_+^2) + p_+ \right] \end{cases} \quad (3.19)$$

2. Left rarefaction - $\mathbf{u}_{*+}$: if $p_* \leq p_+$, $u_+ < c_+$, and $u_* < c_+ \left(\frac{p_*}{p_+} \right)^{\frac{\gamma-1}{2\gamma}}$ we have a left rarefaction wave with a left state which is subsonic and the state for $x/t = 0$ is the star left state. The boundary fluxes in this case are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ \left(\frac{p_*}{p_+} \right)^{\frac{1}{\gamma}} u_* \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ \left(\frac{p_*}{p_+} \right)^{\frac{1}{\gamma}} u_*^2 + p_* \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ \left(\frac{p_*}{p_+} \right)^{\frac{1}{\gamma}} u_* v_+ \\ \mathcal{H}_E^{\delta I} = u_* \left[\frac{p_*}{\gamma-1} + \frac{1}{2} \rho_+ \left(\frac{p_*}{p_+} \right)^{\frac{1}{\gamma}} (u_*^2 + v_+^2) + p_* \right] \end{cases} \quad (3.20)$$

3. Left rarefaction - $\mathbf{u}_{+Fan}$: if $p_* \leq p_+$, $u_+ < c_+$ and $u_* \geq c_+ \left(\frac{p_*}{p_+} \right)^{\frac{\gamma-1}{2\gamma}}$ we have a left rarefaction wave with a left state which is subsonic and the state for $x/t = 0$ is the rarefaction fan. The boundary fluxes in this case are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2}{\gamma-1}} \left(\frac{2c_+}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_+ \right) \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2}{\gamma-1}} \left(\frac{2c_+}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_+ \right)^2 + p_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2\gamma}{\gamma-1}} \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2}{\gamma-1}} \left(\frac{2c_+}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_+ \right) v_+ \\ \mathcal{H}_E^{\delta I} = \left(\frac{2c_+}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_+ \right) \left\{ \frac{p_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2\gamma}{\gamma-1}}}{\gamma-1} + \right. \\ \left. + \frac{1}{2} \rho_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2}{\gamma-1}} \left[\left(\frac{2c_+}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_+ \right)^2 + v_+^2 \right] + \right. \\ \left. + p_+ \left[\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} \left(\frac{u_+}{c_+} \right) \right]^{\frac{2\gamma}{\gamma-1}} \right\} \end{cases} \quad (3.21)$$

4. Left shock - $\mathbf{u}_+$: if $p_* > p_+$ and $u_+ \geq c_+ \left(\frac{\gamma+1}{2\gamma} \frac{p_*}{p_+} + \frac{\gamma-1}{2\gamma} \right)^{\frac{1}{2}}$, we have a left shock with the state for $x/t = 0$ being the left state. The associated boundary fluxes

are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ u_+ \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ u_+^2 + p_+ \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ u_+ v_+ \\ \mathcal{H}_E^{\delta I} = u_+ \left[\frac{p_+}{\gamma-1} + \frac{1}{2} \rho_L (u_+^2 + v_+^2) + p_+ \right] \end{cases} \quad (3.22)$$

5. Left shock - $\mathbf{u}_{*+}$: if $p_* > p_+$ and $u_+ < c_+ \left(\frac{\gamma+1}{2\gamma} \frac{p_*}{p_+} + \frac{\gamma-1}{2\gamma} \right)^{\frac{1}{2}}$, we have a left shock with the state for $x/t = 0$ being the left star state. The associated boundary fluxes are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ \left(\frac{\frac{p_*}{p_+} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_+} \frac{\gamma-1}{\gamma+1} + 1} \right) u_* \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ \left(\frac{\frac{p_*}{p_+} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_+} \frac{\gamma-1}{\gamma+1} + 1} \right) u_*^2 + p_* \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ \left(\frac{\frac{p_*}{p_+} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_+} \frac{\gamma-1}{\gamma+1} + 1} \right) u_* v_+ \\ \mathcal{H}_E^{\delta I} = u_* \left[\frac{p_*}{\gamma-1} + \frac{1}{2} \rho_+ \left(\frac{\frac{p_*}{p_+} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_+} \frac{\gamma-1}{\gamma+1} + 1} \right) (u_*^2 + v_+^2) + p_* \right] \end{cases} \quad (3.23)$$

- Negative particle speed in the region between the two nonlinear waves, $u_* < 0$

1. Right rarefaction - $\mathbf{u}_-$: if $p_* \leq p_-$ and $u_- + c_- \leq 0$ then we have a right rarefaction wave and the state in $x/t = 0$ is the right state. The boundary fluxes are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_- u_- \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- u_-^2 + p_- \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- u_- v_- \\ \mathcal{H}_E^{\delta I} = u_- \left[\frac{p_-}{\gamma-1} + \frac{1}{2} \rho_L (u_-^2 + v_-^2) + p_- \right] \end{cases} \quad (3.24)$$

2. Right rarefaction - $\mathbf{u}_{*-}$: if $p_* \leq p_-$, $u_- + c_- > 0$ and $u_* + c_- \left(\frac{p_*}{p_-} \right)^{\frac{\gamma-1}{2\gamma}} \geq 0$, then we have a right rarefaction wave and the state for $x/t = 0$ is the star right state. The boundary fluxes in this case are

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_- \left(\frac{p_*}{p_-} \right)^{\frac{1}{\gamma}} u_* \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- \left(\frac{p_*}{p_-} \right)^{\frac{1}{\gamma}} u_*^2 + p_* \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- \left(\frac{p_*}{p_-} \right)^{\frac{1}{\gamma}} u_* v_- \\ \mathcal{H}_E^{\delta I} = u_* \left[\frac{p_*}{\gamma-1} + \frac{1}{2} \rho_- \left(\frac{p_*}{p_-} \right)^{\frac{1}{\gamma}} (u_*^2 + v_-^2) + p_* \right] \end{cases} \quad (3.25)$$

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3. Right rarefaction - $\mathbf{u}_{-Fan}$: if $p_* \leq p_-$, $u_- + c_- > 0$ and $u_* + c_- \left(\frac{p_*}{p_-}\right)^{\frac{\gamma-1}{2\gamma}} < 0$ we have a right rarefaction wave with a right state which is supersonic and the state for $x/t = 0$ is the rarefaction fan. The boundary fluxes in this case are

$$\left\{ \begin{array}{l} \mathcal{H}_\rho^{\delta I} = \rho_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2}{\gamma-1}} \left(-\frac{2c_-}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_- \right) \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2}{\gamma-1}} \left(-\frac{2c_-}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_- \right)^2 + p_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2\gamma}{\gamma-1}} \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2}{\gamma-1}} \left(\frac{2c_-}{\gamma+1} - \frac{\gamma-1}{\gamma+1} u_- \right) v_- \\ \mathcal{H}_E^{\delta I} = \left(-\frac{2c_-}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_- \right) \left\{ \frac{p_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2\gamma}{\gamma-1}}}{\gamma-1} + \right. \\ \left. + \frac{1}{2} \rho_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2}{\gamma-1}} \left[\left(-\frac{2c_-}{\gamma+1} + \frac{\gamma-1}{\gamma+1} u_- \right)^2 + v_-^2 \right] + \right. \\ \left. + p_- \left[\frac{2}{\gamma+1} - \frac{\gamma-1}{\gamma+1} \left(\frac{u_-}{c_-} \right) \right]^{\frac{2\gamma}{\gamma-1}} \right\} \end{array} \right. \quad (3.26)$$

4. Right shock - $\mathbf{u}_-$: if $p_* > p_-$ and $u_- + c_- \left[\frac{\gamma+1}{2\gamma} \left(\frac{p_*}{p_-} \right) + \frac{\gamma-1}{2\gamma} \right] \leq 0$ we have a right shock with the state for $x/t = 0$ being the right state. The associated boundary fluxes are

$$\left\{ \begin{array}{l} \mathcal{H}_\rho^{\delta I} = \rho_- u_- \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- u_-^2 + p_- \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- u_- v_- \\ \mathcal{H}_E^{\delta I} = u_- \left[\frac{p_-}{\gamma-1} + \frac{1}{2} \rho_- (u_-^2 + v_-^2 + w_-^2) + p_- \right] \end{array} \right. \quad (3.27)$$

5. Right shock - $\mathbf{u}_{*-}$: if $p_* > p_-$ and $u_- + c_- \left[\frac{\gamma+1}{2\gamma} \left(\frac{p_*}{p_-} \right) + \frac{\gamma-1}{2\gamma} \right] > 0$ we have a right shock with the state for $x/t = 0$ being the star right state. The associated boundary fluxes are

$$\left\{ \begin{array}{l} \mathcal{H}_\rho^{\delta I} = \rho_- \left(\frac{\frac{p_*}{p_-} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_-} \frac{\gamma-1}{\gamma+1} + 1} \right) u_* \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- \left(\frac{\frac{p_*}{p_-} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_-} \frac{\gamma-1}{\gamma+1} + 1} \right) u_*^2 + p_* \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- \left(\frac{\frac{p_*}{p_-} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_-} \frac{\gamma-1}{\gamma+1} + 1} \right) u_* v_- \\ \mathcal{H}_E^{\delta I} = u_* \left[\frac{p_*}{\gamma-1} + \frac{1}{2} \rho_- \left(\frac{\frac{p_*}{p_-} + \frac{\gamma-1}{\gamma+1}}{\frac{p_*}{p_-} \frac{\gamma-1}{\gamma+1} + 1} \right) (u_*^2 + v_-^2) + p_* \right] \end{array} \right. \quad (3.28)$$

Note that the pressure in the star region, p_* , is pre-calculated from the following nonlinear equation

$$f(p_*, \mathbf{u}_+, \mathbf{u}_-) = f_+(p_*, \mathbf{u}_+) + f_-(p_*, \mathbf{u}_-) + u_- - u_+ = 0 \quad (3.29)$$

where

$$\left\{ \begin{array}{ll} f_+(p_*, \mathbf{u}_+) = (p_* - p_+) \left[\frac{\frac{2}{(\gamma+1)\rho_+}}{p_* + \frac{(\gamma-1)}{(\gamma+1)}p_+} \right] & \text{if } p_* > p_+ \text{ (shock)} \\ f_+(p_*, \mathbf{u}_+) = \frac{2c_+}{(\gamma-1)} \left[\left(\frac{p_*}{p_+} \right)^{\frac{\gamma-1}{2\gamma}} - 1 \right] & \text{if } p_* \leq p_+ \text{ (rarefaction)} \\ f_-(p_*, \mathbf{u}_-) = (p_* - p_-) \left[\frac{\frac{2}{(\gamma+1)\rho_-}}{p_* + \frac{(\gamma-1)}{(\gamma+1)}p_-} \right] & \text{if } p_* > p_- \text{ (shock)} \\ f_-(p_*, \mathbf{u}_-) = \frac{2c_-}{(\gamma-1)} \left[\left(\frac{p_*}{p_-} \right)^{\frac{\gamma-1}{2\gamma}} - 1 \right] & \text{if } p_* \leq p_- \text{ (rarefaction).} \end{array} \right. \quad (3.30)$$

After having pre-calculated the pressure p_* , the velocity in the star region, u_* , is calculated as follows:

$$u_* = \frac{1}{2}(u_+ + u_-) + \frac{1}{2}[f_-(p_*) + f_+(p_*)] \quad (3.31)$$

HLL

The HLL (Harten, Lax, van Leer) Riemann solver is an approximated solver which assumes a two-wave configuration by neglecting the middle contact wave (i.e. neglecting the shear waves). The possible patterns in this case are three and they are selected through the evaluation of the fastest signal velocities which perturb the initial data, namely, S_+ and S_-

$$\left\{ \begin{array}{l} S_L = \min(u_+ - c_+, u_{Roe} - c_{Roe}) \\ S_R = \max(u_- + c_-, u_{Roe} + c_{Roe}), \end{array} \right. \quad (3.32)$$

where u_{Roe} and c_{Roe} are the Roe averages of the normal speed and of the speed of sound respectively.

1. $S_L \geq 0$

$$\left\{ \begin{array}{l} \mathcal{H}_\rho^{\delta I} = \rho_+ u_+ \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ u_+^2 + p_+ \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ u_+ v_+ \\ \mathcal{H}_E^{\delta I} = u_+ \left[\frac{p_+}{\gamma-1} + \frac{1}{2} \rho_+ (u_+^2 + v_+^2) + p_+ \right] \end{array} \right. \quad (3.33)$$

2. $S_- \leq 0$

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_- u_- \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- u_-^2 + p_- \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- u_- v_- \\ \mathcal{H}_E^{\delta I} = u_- \left[\frac{p_-}{\gamma-1} + \frac{1}{2} \rho_R (u_-^2 + v_-^2) + p_- \right] \end{cases} \quad (3.34)$$

3. $S_+ < 0$ or $S_- > 0$

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \frac{S_- \rho_+ u_+ - S_+ \rho_- u_- + S_- S_+ (\rho_- - \rho_+)}{S_- - S_+} \\ \mathcal{H}_{\rho u}^{\delta I} = \frac{S_- (\rho_+ u_+^2 + p_+) - S_+ (\rho_- u_-^2 + p_-) + S_- S_+ (\rho_- u_- - \rho_+ u_+)}{S_- - S_+} \\ \mathcal{H}_{\rho v}^{\delta I} = \frac{S_- (\rho_+ u_+ v_+) - S_+ (\rho_- u_- v_-) + S_- S_+ (\rho_- v_- - \rho_+ v_+)}{S_- - S_+} \\ \mathcal{H}_E^{\delta I} = \frac{S_- u_+ (E_+ + p_+) - S_+ u_- (E_- + p_-) + S_- S_+ (E_- - E_+)}{S_- - S_+} \end{cases} \quad (3.35)$$

It should be noted that the HLL Riemann solver does not allow for jumps in the tangential velocities as well as in the density. This feature can penalise some problems but can be beneficial for others.

HLLC

The HLLC (Harten, Lax, van Leer + Contact) Riemann solver is an approximated solver which restores a three-wave configuration by contrast to the HLL solver. The possible patterns in this case are 4 and they are selected through the evaluation of the fastest signal velocities which perturb the initial data, namely, S_+ and S_- (computed as in Eq. (3.32)) as well as through the calculation of the middle (contact) wave speed S_M :

$$S_M = \frac{p_- - p_+ + \rho_+ u_+ (S_+ - u_+) - \rho_- u_- (S_- - u_-)}{\rho_+ (S_+ - u_+) - \rho_- (S_- - u_-)} \quad (3.36)$$

1. $S_L \geq 0$

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ u_+ \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ u_+^2 + p_+ \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ u_+ v_+ \\ \mathcal{H}_E^{\delta I} = u_+ \left[\frac{p_+}{\gamma-1} + \frac{1}{2} \rho_L (u_+^2 + v_+^2) + p_+ \right] \end{cases} \quad (3.37)$$

2. $S_- \leq 0$

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_- u_- \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- u_-^2 + p_- \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- u_- v_- \\ \mathcal{H}_E^{\delta I} = u_- \left[\frac{p_-}{\gamma-1} + \frac{1}{2} \rho_R (u_-^2 + v_-^2) + p_- \right] \end{cases} \quad (3.38)$$

3. $S_+ < 0$ and $S_M \geq 0$

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_+ u_+ + S_+ \left[\rho_+ \frac{(S_+ - u_+)}{(S_+ - S_M)} - \rho_+ \right] \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_+ u_+^2 + p_+ + S_+ \left[\rho_+ \frac{(S_+ - u_+)}{(S_+ - S_M)} - \rho_+ u_+ \right] \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_+ u_+ v_+ + S_+ \left[\rho_+ \frac{(S_+ - v_+)}{(S_+ - S_M)} - \rho_+ v_+ \right] \\ \mathcal{H}_E^{\delta I} = u_+ (E_+ + p_+) + S_+ \left\{ \rho_+ \frac{(S_+ - u_+)}{(S_+ - S_M)} \left[\frac{E_+}{\rho_+} + (S_M - u_+) \left(S_M + \frac{p_+}{\rho_+ (S_+ - u_+)} \right) \right] - E_+ \right\} \end{cases} \quad (3.39)$$

4. $S_M < 0$ and $S_- > 0$

$$\begin{cases} \mathcal{H}_\rho^{\delta I} = \rho_- u_- + S_- \left[\rho_- \frac{(S_- - u_-)}{(S_- - S_M)} - \rho_- \right] \\ \mathcal{H}_{\rho u}^{\delta I} = \rho_- u_-^2 + p_- + S_- \left[\rho_- \frac{(S_- - u_-)}{(S_- - S_M)} - \rho_- u_- \right] \\ \mathcal{H}_{\rho v}^{\delta I} = \rho_- u_- v_- + S_- \left[\rho_- \frac{(S_- - v_-)}{(S_- - S_M)} - \rho_- v_- \right] \\ \mathcal{H}_E^{\delta I} = u_- (E_- + p_-) + S_- \left\{ \rho_- \frac{(S_- - u_-)}{(S_- - S_M)} \left[\frac{E_-}{\rho_-} + (S_M - u_-) \left(S_M + \frac{p_-}{\rho_- (S_- - u_-)} \right) \right] - E_- \right\} \end{cases} \quad (3.40)$$

For a more detailed discussion and some additional implementative insights on this topic the interested reader can refer to Toro (2009).

3.4 Boundary conditions

An important aspect when implementing a compressible flow solver is the treatment of the boundary conditions. In this thesis work, we considered (as already mentioned in chapter 2) a weak approach - i.e. we applied the boundary conditions to the fluxes rather than to the conserved variables.

For the inviscid fluxes we explored two approaches. The first approach consists in applying the boundary conditions in an indirect manner through a Riemann solver (Approach A)

whereas the second applies the boundary conditions by calculating the fluxes directly from the known value of the variables at the boundary (Approach B):

$$\text{Approach A :} \quad \mathcal{H}_i^{\delta I} = \mathcal{H}_i^{\delta I}(\mathbf{u}_+^{\delta}, \mathbf{u}_-^{\delta}). \quad (3.41a)$$

$$\text{Approach B :} \quad \mathcal{H}_i^{\delta I} = \mathcal{H}_i^{\delta I}(\mathbf{u}_{BC}^{\delta}). \quad (3.41b)$$

These two approaches are common in the family of methods considered in this thesis and one implementation or the other depends on a series of factors which range from solver architecture, computational costs as well as robustness and effectiveness of the boundary condition implementation for a given problem.

Approach A is less intrusive with respect to the underlying numerics used at the inner interfaces of the numerical discretisation. On the other hand, the use of a Riemann solver for applying the boundary conditions (abbreviated as BCs in the the rest of this chapter) implies the usage of a ghost point where it is necessary to apply a consistent ghost state (indicated with the subscript ‘ $-$ ’ in the rest of this section), which is not always trivial. Additionally, different Riemann solvers have different performance in transferring the BCs into the domain and one Riemann solver can perform better than another depending on the BCs being applied.

While for the advection term (i.e. the inviscid fluxes) we can make a distinction between approach A and B, the only way for applying the boundary conditions to the viscous fluxes in a weak manner is by means of Approach B. A detailed comparison between Approach A and B for the advection term and some considerations on the treatment of the diffusion term have been presented by the author of this thesis in (Mengaldo *et al.*, 2014) and they are not reported here. However, we describe the most robust BCs implementation strategies adopted for both the Euler and compressible Navier-Stokes equations.

As a final remark, we note that the other common way of applying the BCs is by setting directly the boundary solution, *strong* reinforcement, which is not taken into account in the present thesis.

3.4.1 BCs for the Compressible Euler equations

Since the applications which are of interest in this thesis are mainly concerned with external aerodynamics, the BCs considered for the compressible Euler equations are essentially two: farfield BCs and slip wall BCs. The first is concerned with boundaries located in the free-stream, and their main objective is to damp spurious reflections which may arise when an

outgoing wave is passing through the truncated computational domain. The second, slip wall BCs, are applied when the flow hits a solid non-transpiring wall.

3.4.1.1 Farfield BCs

Farfield BCs for the Euler equations have been widely explored in the past decades and the interested reader can refer to Thompson (1987, 1990) and Giles (1990) for a detailed analysis. In this section, we describe a characteristic approach based on the Riemann invariants, which has been implemented in Nektar++. This approach is particularly effective in damping spurious reflections from the boundaries and is also referred to as non-reflective farfield BCs.

According to the hyperbolic nature of the Euler equations the flux evaluated at the boundary is a combination of the information coming from inside and outside the domain. In order to explain how to effectively apply characteristic boundary conditions, we start by defining the Riemann invariants which are derived through the characteristic analysis of the Euler equations.

Specifically, if we consider a normal frame of reference with respect to a given interface between two elements, we can define the normal velocity $u_n = \mathbf{v} \cdot \mathbf{n}$ and the tangential velocity $u_t = \mathbf{v} - u_n \mathbf{n}$ associated to the interface. If we additionally write the Euler equations in terms of primitive (or physical) variables $\mathbf{u}_{\text{pr}} = [\rho, u, v, p]$, we can derive the vector of the eigenvalues as well as the left eigenvector matrix:

$$\boldsymbol{\lambda} = \begin{bmatrix} u_n + c \\ u_n \\ u_n \\ u_n - c \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 0 & 1 & 0 & 1/(\rho c) \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & -1/c^2 \\ 0 & -1 & 0 & 1/(\rho c) \end{bmatrix}. \quad (3.42)$$

Note that u_n and u_t have not to be confused with u , which is the component of the velocity vector in the x_1 -direction in a Cartesian frame of reference.

The eigenvalues $\boldsymbol{\lambda}$ defined in Eq. (3.42) are the velocities at which the information travels along the characteristic lines $d\mathbf{x}/dt$. In particular, the characteristic variables being propag-

ated along the characteristic lines are defined as

$$\mathbf{r} = \mathbf{L} \mathbf{u}_{\text{pr}} = \begin{bmatrix} u_n + \frac{p}{c\rho} \\ -u_t \\ \rho - \frac{p}{c^2} \\ -u_n + \frac{p}{c\rho} \end{bmatrix}, \quad (3.43)$$

and they satisfy the following advection equation

$$\frac{\partial \mathbf{r}}{\partial t} + \text{diag}(\boldsymbol{\lambda}) \cdot \nabla \mathbf{r} = 0, \quad (3.44)$$

where $\text{diag}(\boldsymbol{\lambda})$ represents the diagonal matrix of the eigenvalues $\boldsymbol{\lambda}$. The first and the last equations, associated to the largest and to the smallest eigenvalues, are two genuinely nonlinear fields and they are either shock or rarefaction waves. The second equation represent the constancy of the entropy while the third equation corresponds to the propagation of vorticity waves.

The first and the last equations can be integrated using the isentropic assumption $p/\rho^\gamma = \text{const.}$ producing the following two expressions:

$$r_{\lambda_{\min}} = u_n + \frac{2c}{\gamma - 1}, \quad r_{\lambda_{\max}} = u_n - \frac{2c}{\gamma - 1}, \quad (3.45)$$

which are known as Riemann invariants associated to the incoming $r_{\lambda_{\min}}$ and outgoing $r_{\lambda_{\max}}$ characteristic waves. Having defined the Riemann invariants, we can now define the farfield non-reflective BCs which include two cases: inflow $u_n < 0$ and outflow $u_n > 0$.

Inflow ($u_n < 0$)

In the case of a subsonic inflow region, where the normal Mach number at the inflow boundary is defined as $\text{Ma}_n = |u_n|/c < 1$, the incoming Riemann invariant $r_{\lambda_{\max}}$ is associated to the free-stream values of the conserved variables - i.e. $\mathbf{u}_\infty = [\rho_\infty, (\rho u)_\infty, (\rho v)_\infty, E_\infty]^T$, while the outgoing Riemann invariant $r_{\lambda_{\min}}$ is associated to the values of the conserved variables inside the computational domain $\mathbf{u}_{\text{in}} = [\rho_{\text{in}}, (\rho u)_{\text{in}}, (\rho v)_{\text{in}}, E_{\text{in}}]^T$. Note that the subscript ‘in’ here variables on the interior of the computational domain. Therefore, the

Riemann invariants in Eq. (3.45) become:

$$r_{\lambda_{min}} = u_{n,in} - \frac{2c_{in}}{\gamma - 1}, \quad r_{\lambda_{max}} = u_{n,\infty} + \frac{2c_{\infty}}{\gamma - 1} \quad (3.46)$$

where, $u_{n,\infty} = \mathbf{v}_{\infty} \cdot \mathbf{n}$, $u_{n,in} = \mathbf{v}_{in} \cdot \mathbf{n}$, are the free-stream and interior normal velocities, respectively and $c_{\infty} = \sqrt{\gamma p_{\infty} / \rho_{\infty}}$, $c_{in} = \sqrt{\gamma p_{in} / \rho_{in}}$ are the free-stream and interior values of the speed of sound, respectively.

At the boundary ‘b’, we know that the two Riemann invariants assume a given value that depends on the boundary quantities can write the following relations:

$$r_{\lambda_{min},b} = u_{n,b} + \frac{2c_b}{\gamma - 1} = r_{\lambda_{min}}, \quad r_{\lambda_{max},b} = u_{n,b} - \frac{2c_b}{\gamma - 1} = r_{\lambda_{max}}. \quad (3.47)$$

From Eq. (3.47), we can then calculate the velocity and the speed of sound at the boundary:

$$u_{n,b} = \frac{r_{\lambda_{min}} + r_{\lambda_{max}}}{2}, \quad c_b = \frac{(\gamma - 1)(r_{\lambda_{min}} - r_{\lambda_{max}})}{4}. \quad (3.48)$$

The normal velocity $u_{n,b}$ just calculated can be used to compute the two Cartesian components of the velocity $\mathbf{v}_b = [u_b, v_b]^T$ at the boundary:

$$\mathbf{v}_b = \mathbf{v}_{\infty} + (u_{n,b} - \mathbf{v}_{\infty} \cdot \mathbf{n})\mathbf{n}. \quad (3.49)$$

Since the flow is entering the domain and under the assumption that there are no discontinuities in the solution (e.g. shock waves), the entropy at the boundary is equal to the free-stream entropy and, therefore, we can write the following relation:

$$s_b = \frac{c_{\infty}^2}{\gamma \rho_{\infty}^{\gamma-1}}. \quad (3.50)$$

Now, using Eq. (3.48) for the speed of sound and Eq. (3.50) for the entropy, the density and the pressure at the boundary can be derived as follows:

$$\rho_b = \left(\frac{c_b^2}{\gamma s_b} \right), \quad p_b = \frac{\rho_b c_b^2}{\gamma}. \quad (3.51)$$

Having fully defined the thermodynamic state (ρ_b, p_b) and the velocities $\mathbf{v}_b = [u_b, v_b]^T$ at the boundary (based on the characteristic analysis of the Euler equations) it is possible to

calculate the characteristic boundary state

$$\mathbf{u}_{BC} = \begin{Bmatrix} \rho_{BC} \\ (\rho u)_{BC} \\ (\rho v)_{BC} \\ E_{BC} \end{Bmatrix} = \begin{Bmatrix} \rho_b \\ \rho_b u_b \\ \rho_b v_b \\ E_b \end{Bmatrix}, \quad (3.52)$$

where the total energy E_b is calculated via Eq. (3.3). The subsonic inflow BCs are finally obtained by using Eq. (3.52) to calculate the boundary numerical flux through Eq. (3.41b).

If the flow at the inflow boundary is supersonic, $\text{Ma}_n = |u_n|/c \geq 1$, the above derivation holds, with the only exception that there is no outgoing characteristic wave, thus the outgoing Riemann invariant $r_{\lambda_{max}}$ assumes the following expression:

$$r_{\lambda_{max}} = u_{n,\infty} + \frac{2c_\infty}{\gamma - 1}. \quad (3.53)$$

Outflow ($u_n > 0$)

In the case of a subsonic outflow region, where the normal Mach number at the outflow boundary is defined as $\text{Ma}_n = |u_n|/c < 1$, the incoming Riemann invariant $r_{\lambda_{min}}$ is associated to the free-stream values of the conserved variables while the outgoing Riemann invariant $r_{\lambda_{max}}$ is associated to the values of the conserved variables inside the computational domain. Therefore, the Riemann invariants in Eq. (3.45) become:

$$r_{\lambda_{max}} = u_{n,in} - \frac{2c_{in}}{\gamma - 1}, \quad r_{\lambda_{min}} = u_{n,\infty} + \frac{2c_\infty}{\gamma - 1} \quad (3.54)$$

At the boundary, we can write the same relations as in Eq. (3.47) and calculate the velocity $u_{n,b}$ and the speed of sound c_b at the boundary through Eq. (3.48). The normal velocity $u_{n,b}$ can then be used to compute the two Cartesian components of the velocity $\mathbf{v}_b = [u_b, v_b]^T$ at the boundary using Eq. (3.49).

Since the flow is exiting the domain and under the same assumption made for the inflow case (i.e. no discontinuities), the entropy at the boundary is extrapolated from the interior and is equal to

$$s_b = \frac{c_{in}^2}{\gamma \rho_{in}^{\gamma-1}}. \quad (3.55)$$

The density and the pressure at the boundary can be calculated with Eq. (3.51). Therefore, we can evaluate all the conserved variables as in Eq. (3.52). The flux is then evaluated as in

Eq. (3.41b).

Note that, if the flow at the outflow boundary is supersonic, $\text{Ma}_n = |u_n|/c \geq 1$, there is no incoming characteristic wave and $r_{\lambda_{min}}$ is set as:

$$r_{\lambda_{min}} = u_{n,in} - \frac{2c_{in}}{\gamma - 1}. \quad (3.56)$$

Implementation tip

The implementation carried out in Nektar++ takes automatically into account the sign of the velocity at the boundary (therefore, whether the given boundary is an inflow or an outflow boundary is automatically evaluated) and the local Mach number.

Also to be noted that this implementation is identical to using the following ghost state

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_\infty \\ \rho_\infty u_\infty \\ \rho_\infty v_\infty \\ E_\infty \end{Bmatrix}, \quad (3.57)$$

associated to a boundary numerical flux calculated by a Riemann solver, through Eq. (3.41a). In this case, the Riemann solver takes automatically into account the sign of the eigenvalues and evaluates the flux by combining the inner and outer states. In Nektar++, we have implemented both strategies, which, in terms of computational costs, are very similar.

3.4.1.2 Slip wall BCs

In the case of an inviscid flow hitting a solid surface or wall (with no transpiration), the BCs must prevent the fluid from penetrating it. The slip wall BCs are therefore defined in such a way that the ghost state $\mathbf{v}_- = [u_-, v_-]^T$ is equal to

$$\mathbf{v}_- = \mathbf{v}_{in} - 2(\mathbf{v}_{in} \cdot \mathbf{n})\mathbf{n}, \quad (3.58)$$

where $\mathbf{v}_{in} = [u_{in}, v_{in}]^T$ is the interior velocity being extrapolated for applying the slip wall BCs and $\mathbf{n}$ is the normal with respect to the boundary. This is equivalent to imposing the same tangential component in the ghost point as in the interior one, $u_{t,-} = u_{t,in} = u_{t,+}$, and an opposite normal component of the velocity, $u_{n,-} = -u_{n,in} = -u_{n,+}$, where the subscript ‘t’ denotes the tangential direction and the subscript ‘n’ denotes the normal direction.

The density and internal energy are extrapolated from the interior such that the complete

ghost state is

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_{in} \\ \rho_{in} u_{in} - 2(\mathbf{v}_{in} \cdot \mathbf{n})n_x \\ \rho_{in} v_{in} - 2(\mathbf{v}_{in} \cdot \mathbf{n})n_y \\ E_{in} \end{Bmatrix} \quad (3.59)$$

and the boundary flux is calculated through a Riemann solver using Eq. (3.41a).

The normal component of the velocity evaluated by the Riemann solver is zero since the only non-zero contributions to the flux function are those coming from the pressure in the momentum equations.

3.4.2 BCs for Compressible Navier-Stokes equations

For the compressible Navier-Stokes equations we consider three typologies of BCs: farfield BCs with and without presence of vortical structures passing through the boundaries, inflow/outflow BCs with laminar viscous effects (e.g. the inflow and outflow regions of a flat plate) and no-slip wall BCs both isothermal and adiabatic.

3.4.2.1 Farfield BCs

Farfield BCs for the compressible Navier-Stokes equations are still a very active topic of research. The application of a characteristic approach presented for the Euler equations for farfield boundaries, still works properly for the Navier-Stokes equations in regions where viscosity effects can be neglected. However, viscosity effects become important, such as in the case when vortical structures (e.g. shedding behind an object, turbo machinery wakes, etc.) pass through the farfield boundaries, the characteristic treatment of the BCs presented for the Euler equations fails, generating spurious reflections which pollute the overall solution and may lead to numerical instabilities.

Various techniques have been presented in the literature to minimise and possibly eliminate these spurious reflections. For a review on the topic, the interested reader can refer to Colonius (2004). In Nektar++, we use the characteristic approach presented for the Euler equations for boundaries where the viscosity effects can be neglected, while we have implemented a method based on the so-called sponge terms, for boundaries where viscosity effects may generate the spurious reflections just mentioned. The latter implementation is based on (Israeli & Orszag, 1981) and (Bodony, 2006) and is formulated by modifying the right-hand

side of the compressible Navier-Stokes equations as follows

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}_1}{\partial x_1} + \frac{\partial \mathbf{f}_2}{\partial x_2} = \sigma(\bar{\mathbf{x}})(\mathbf{u}_{ref} - \mathbf{u}), \quad (3.60)$$

where $\sigma(\bar{\mathbf{x}})$ is a damping coefficient defined in a region $\bar{\mathbf{x}}$ in proximity to the boundaries and $\mathbf{u}_{ref}$ is a known reference solution. The length and the shape of the damping coefficient depends on the problem being solved and has been thoroughly investigated by (Mani, 2012).

3.4.2.2 Inflow/Outflow BCs in presence of laminar viscous effects

In the case of an inflow/outflow BCs where laminar viscous effects are important, such as the inflow and outflow regions of a flat plate with a developing boundary layer, to appropriately recover the physical solution we need to apply a different set of BCs than the farfield BCs we have shown in the previous sections. In the following we show the treatment of the advection and of the diffusion terms separately.

Advection term

For the advection term we distinguish the following four different cases.

Subsonic inflow ($\text{Ma}_n = |u_n|/c < 1$)

We impose the density and the velocity on the ghost state and extrapolate the pressure from the interior of the domain:

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_e \\ \rho_e u_e \\ \rho_e v_e \\ \frac{p_{in}}{\gamma-1} + \frac{1}{2}\rho_e(u_e^2 + v_e^2) \end{Bmatrix}. \quad (3.61)$$

We can use this strategy when dealing with a flat plate simulation for instance, where the inflow state is known from the solution of the compressible boundary layer equations.

Supersonic inflow ($\text{Ma}_n = |u_n|/c > 1$)

We impose all the variables on the ghost state:

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_e \\ \rho_e u_e \\ \rho_e v_e \\ \frac{p_e}{\gamma-1} + \frac{1}{2}\rho_e(u_e^2 + v_e^2) \end{Bmatrix}. \quad (3.62)$$

Subsonic outflow ($\text{Ma}_n = |u_n|/c < 1$)

We impose an estimated or exact pressure p_e on the ghost state and extrapolate density and velocity from the interior of the domain:

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_{in} \\ \rho_{in} u_{in} \\ \rho_{in} v_{in} \\ \frac{p_e}{\gamma-1} + \frac{1}{2}\rho_{in}(u_{in}^2 + v_{in}^2) \end{Bmatrix}. \quad (3.63)$$

This approach is particularly accurate when a good pressure estimate is known a priori.

Supersonic outflow ($\text{Ma}_n = |u_n|/c > 1$)

We extrapolate the entire ghost state from the interior values:

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_{in} \\ \rho_{in} u_{in} \\ \rho_{in} v_{in} \\ \frac{p_{in}}{\gamma-1} + \frac{1}{2}\rho_{in}(u_{in}^2 + v_{in}^2) \end{Bmatrix}. \quad (3.64)$$

This approach is also known as do-nothing approach.

The boundary flux is then calculated by Eq. (3.41a) using one of the ghost states above, depending on the boundary. In Nektar++, the identification of a subsonic or a supersonic region is automatic, therefore the user, in this case, needs just to specify if the region is an inflow or an outflow region.

Note also that, all the above implementations are reflective - if an acoustic or a vortical wave hits the boundary this set of inflow/outflow BCs generate spurious reflections. However, when used in conjunction with the previously defined sponge region, the inflow/outflow regions become fully non-reflective.

Diffusion term

Concerning the diffusion term, we need to impose the auxiliary variables, $\mathbf{u}_{aux}$, which are used to calculate the first order derivatives, at the boundary. Since we are using an LDG approach, with a $\beta = 1/2$, we directly impose these values at the boundary as follows (note that we do not make any distinction between the four cases shown for the advection term):

$$\mathbf{u}_{aux_{BC}} = \begin{Bmatrix} u_{BC} \\ v_{BC} \\ T_{BC} \end{Bmatrix} = \begin{Bmatrix} u_{in} \\ v_{in} \\ \frac{p_{in}}{\rho_{in}R} \end{Bmatrix}. \quad (3.65)$$

The velocity at the boundary is left to be free. The boundary viscous flux is then calculated as $\mathcal{H}_v^{\delta I} = \mathcal{H}_v^{\delta I}(\mathbf{u}_{BC}^\delta, (\nabla_{\mathbf{x}} \mathbf{u}^\delta)_{in})$.

Equation (3.65) is used as boundary condition for the auxiliary variables in the LDG method, where the derivatives of u, v, T are calculated, and successively used to evaluate the boundary viscous flux. It is important to note that the boundary viscous flux is computed using the gradients of the inner state since they are unknown at the boundary.

3.4.2.3 Non-slip isothermal and adiabatic wall BCs

For the compressible Navier-Stokes equations a non-slip condition must be applied to the velocity field at a solid wall (with no transpiration). We first consider the treatment of the BCs on the advection term and successively we describe how to apply the BCs to the diffusion term.

Advection term

The most robust and reliable implementation strategy for the no-slip wall BCs on the advection term is achieved by setting the ghost state to

$$\mathbf{u}_- = \begin{Bmatrix} \rho_- \\ (\rho u)_- \\ (\rho v)_- \\ E_- \end{Bmatrix} = \begin{Bmatrix} \rho_{in} \\ -\rho_{in} u_{in} \\ -\rho_{in} v_{in} \\ E_{in} \end{Bmatrix}, \quad (3.66)$$

where the density and the internal energy are extrapolated from the interior while the velocity components are negated. The boundary flux is then calculated through a Riemann solver.

Note that there is no distinction at this level for what concern the isothermal and adiabatic wall BCs. This arises in the implementation of the diffusion term.

Diffusion term

Concerning the diffusion term we impose the boundary conditions on the auxiliary variables $\mathbf{u}_{aux}$, which are used to calculate the first order derivatives. Here, we distinguish between an isothermal and an adiabatic wall.

Isothermal wall

In the case of LDG method in conjunction with an *isothermal wall* with an imposed temperature T_w , the auxiliary variables at the boundary assume the following values:

$$\mathbf{u}_{aux_{BC}} = \begin{Bmatrix} u_{BC} \\ v_{BC} \\ w_{BC} \\ T_{BC} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ T_w \end{Bmatrix}. \quad (3.67)$$

It is possible to appreciate that the velocity at the boundary is set to zero. The final boundary viscous flux is then calculated as $\mathcal{H}_v^{\delta I} = \mathcal{H}_v^{\delta I}(\mathbf{u}_{BC}^\delta, (\nabla_{\mathbf{x}} \mathbf{u}^\delta)_{in})$.

Equation (3.67) is used as boundary condition for the auxiliary variables in the LDG method, where the derivatives of u, v, w, T are calculated, and are successively used to evaluate the boundary viscous flux. Also in this case the boundary viscous flux is computed using the gradients of the inner state since they are unknown at the boundary. An exception is the shear stress part of the boundary viscous flux associated to the energy equation which is set to zero, since it is pre-multiplied by the velocity which is known to be zero at a no-slip wall.

Adiabatic wall

The diffusion term for an *adiabatic wall* differs from the isothermal since we apply the thermal flux instead of the wall temperature. Specifically the temperature at the boundary is extrapolated from the interior and the dependent variables at the wall are

$$\mathbf{u}_{aux_{BC}} = \begin{Bmatrix} u_{BC} \\ v_{BC} \\ w_{BC} \\ T_{BC} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ T_{in} \end{Bmatrix}. \quad (3.68)$$

Equation (3.68) is used as the boundary condition for the auxiliary system in the LDG method.

In order to evaluate the normal flux at the boundary, we need to split the flux in Eq. (3.11) into two parts: one which arises from the shear stresses (superscript ‘ s ’ in Eq. (3.69)), and one which depends on the thermal diffusivity (superscript ‘ td ’ in Eq. (3.69)):

$$\begin{aligned} \mathbf{f}_{v,1}^s &= \begin{Bmatrix} 0 \\ \tau_{xx} \\ \tau_{yx} \\ u\tau_{xx} + v\tau_{yx} \end{Bmatrix}, & \mathbf{f}_{v,1}^{td} &= \begin{Bmatrix} 0 \\ 0 \\ 0 \\ kT_x \end{Bmatrix}, \\ \mathbf{f}_{v,2}^s &= \begin{Bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} \end{Bmatrix}, & \mathbf{f}_{v,2}^{td} &= \begin{Bmatrix} 0 \\ 0 \\ 0 \\ kT_y \end{Bmatrix}, \end{aligned} \quad (3.69)$$

At the boundary, the normal flux arising from the shear stresses is extrapolated from the values obtained in the interior of the domain while the normal flux arising from the thermal diffusivity is set to zero, i.e.

$$k\nabla_{\mathbf{x}}T \cdot \mathbf{n} = 0. \quad (3.70)$$

As for the isothermal BCs, the final boundary viscous flux is calculated by using the following relation: $\mathcal{H}_v^{\delta I} = \mathcal{H}_v^{\delta I}(\mathbf{u}_{BC}^{\delta}, (\nabla_{\mathbf{x}}\mathbf{u}^{\delta})_{in})$.

3.5 Verification of the implementation

The compressible flow solver implemented in Nektar++ and the underlying numerical discretisations by means of the DG and the FR approaches have been thoroughly tested for both the compressible Euler and Navier-Stokes equations. In the next two sections we show a few relevant test cases in order to highlight the main features of discontinuous spectral/ hp element methods applied to nonlinear systems of conservation laws.

3.5.1 Euler equations

We start showing the verification results for the compressible Euler equations. In particular, we consider two test cases, the isentropic vortex test case and a subsonic flow past a cylinder.

The aim of the first test case is to show that the convergence order achieved for the various FR and DG schemes shown in chapter 2 for the unsteady advection diffusion equation are recovered also for a nonlinear system of equations. In addition, the first test case shows the performance of the three different Riemann solvers introduced in section 3.3. The second test case illustrates the generation of numerical entropy when a low-order approximation of the surface of the cylinder is employed and highlights the issues arising from collocated schemes when dealing with nonlinear problems.

3.5.1.1 Isentropic vortex

The exact solution for the isentropic vortex test case is known (Shu, 1998). Specifically, if we assume that the vortex is moving with velocity u_0 and v_0 in the x - and y -direction, respectively, the exact solution reads:

$$\rho = \left(1 - \frac{\beta^2(\gamma - 1)}{16\gamma\pi^2} e^{2(1-r^2)}\right)^{\frac{1}{\gamma-1}} \quad (3.71a)$$

$$u = u_0 - \frac{\beta(y - y_0)}{2\pi} e^{(1-r^2)} \quad v = v_0 + \frac{\beta(x - x_0)}{2\pi} e^{(1-r^2)} \quad (3.71b)$$

$$E = \frac{\rho^\gamma}{\gamma - 1} + \frac{\rho}{2} (u^2 + v^2). \quad (3.71c)$$

where $r = \sqrt{(x - u_0 t - x_0)^2 + (y - u_0 t - y_0)^2}$ and (x_0, y_0) are the coordinates which corresponds to the initial position of the vortex. The aim of the simulations carried out is to recover the exponential order of convergence expected for the FR approach and the DG method in order to show the correctness of the implementation.

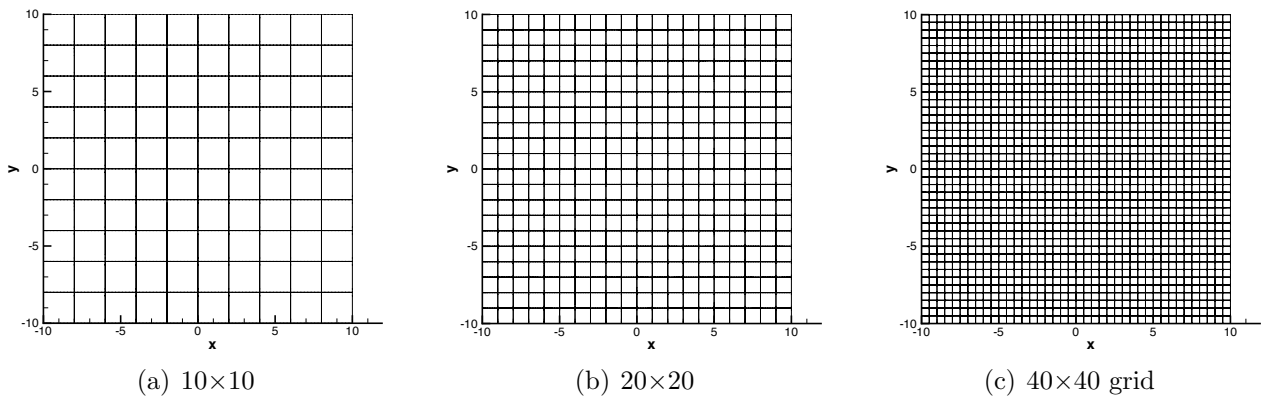


Figure 3.3: Regular meshes adopted for the isentropic vortex test case.

As for the linear advection diffusion equation in chapter 2, we used three different FR schemes, namely the FR_{DG} , FR_{g2} and FR_{SD} schemes (in their collocated version), which have been introduced in section 2.3.2 and both the $\text{DG}_{\text{SEM-LMM}}$ and the $\text{DG}_{\text{SEM-EMM}}$ methods introduced in section 2.3.1. The points chosen were Gauss-Lobatto-Legendre. The simulations have been carried out on the three regular meshes depicted in figure 3.3. The meshes are increasingly finer where the number of elements is equal to $N_x \times N_x$, with $N_x = 10, 20, 40$, respectively. The length of each side is equal to 20. The velocities were set to $(u_0, v_0) = (0, 5)$ and the isentropic vortex, at $t = 0$, was centred in $(x_0, y_0) = (0, 0)$. At the final time $T = 4s$, the vortex returned to its initial position (note that periodic boundary conditions for all the boundaries of the domain). The initial and the final solutions are depicted in figure 3.4.

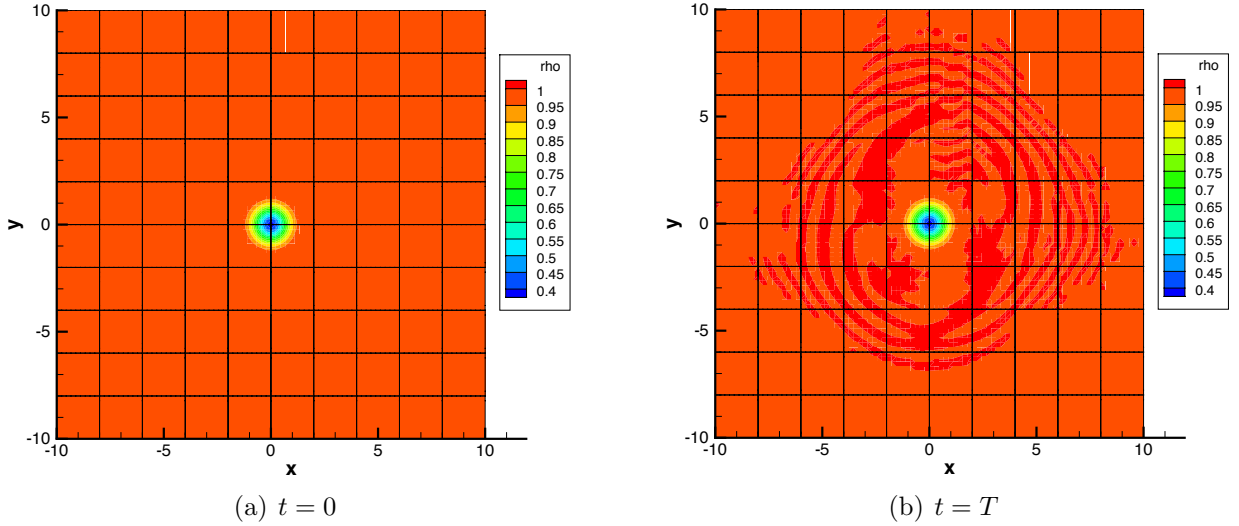


Figure 3.4: Density (ρ) contour for the initial ($t = 0$) and the final time frame ($t = T$) on the coarsest mesh depicted in figure 3.3(a), for $P=8$. Isentropic vortex test case.

The ratio of the specific heats, γ , was set to 1.4 and β was equal to 5.

In figure 3.5(a), we show the L_2 error associated to the density ρ (defined as in Eq. (2.101) in chapter 2) as a function of the mesh size ‘h’ in a logarithmic scale along with the ideal order of convergence (also referred to as order of accuracy) denoted with ‘OOA’. It is possible to appreciate how the optimal order of convergence is achieved for all the polynomial orders considered and for all the DG and FR schemes taken into account.

In figure 3.5(b), we show the L_2 error associated to the density ρ as a function of the polynomial order in a semi-logarithmic scale for all the three meshes employed. The results,

for all the three meshes, shows a good P-convergence, as expected.

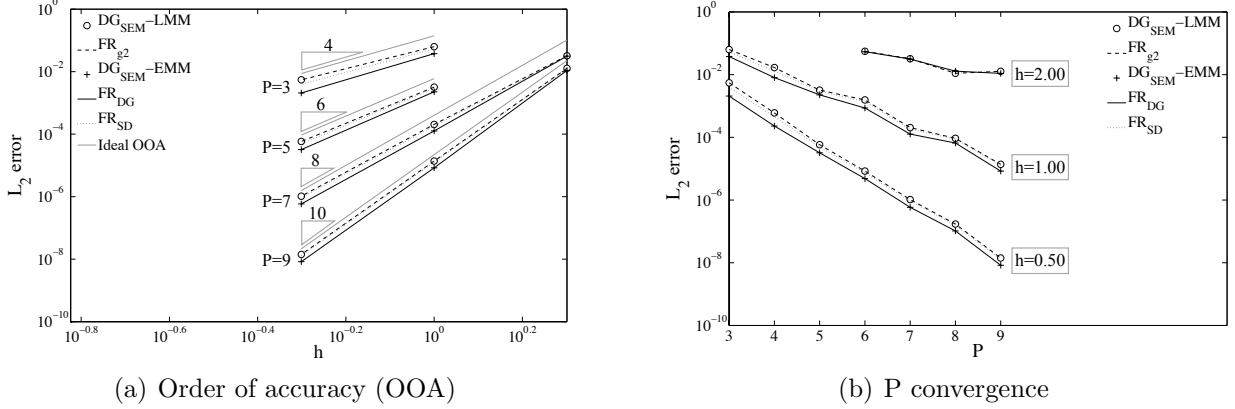


Figure 3.5: Order of accuracy (OOA) and P convergence for the isentropic vortex test case related to the density ρ .

Finally, figure 3.6 shows the behaviour of the three different Riemann solvers detailed in section 3.3. While there are no differences between the exact and HLLC Riemann solvers, it is possible to observe a noticeably different behaviour of the HLL Riemann solver, which provides a higher L_2 error than the other three. This result is expected because the HLL Riemann solver does not take into account the full Riemann problem and therefore is the least accurate (for this problem) among those considered.

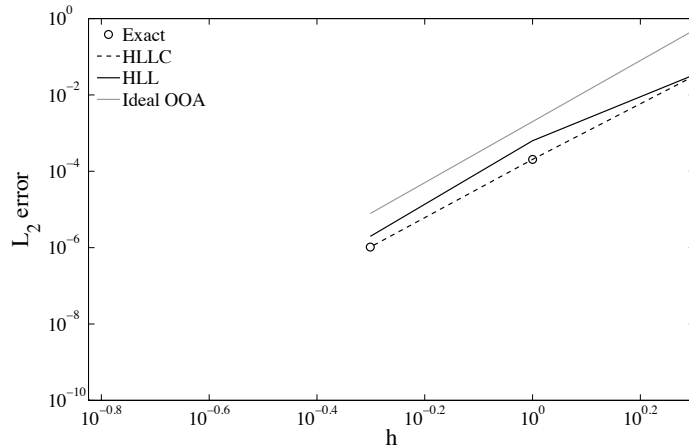


Figure 3.6: Order of accuracy (OOA) related to the density ρ for the three different Riemann solvers presented in section 3.3. Isentropic vortex test case.

3.5.1.2 Subsonic inviscid flow past a cylinder

The subsonic flow past a cylinder is an important test case because shows how the numerical methods considered deal with curved geometries. The solution can in fact be affected by numerically-generated entropy when straight edges are used to describe the cylinder and in general when a low-order approximation of the equations is employed (i.e. in case of spectral element methods, when low polynomial orders are employed).

In addition, the posterior stagnation point is very sensitive to numerical errors (such as integration errors) due to the strong gradients of the solution in that region. In this point, it is possible to see a numerically-generated entropy that can lead to the failure of the simulation. This behaviour can be avoided by using a very fine mesh in proximity to the posterior stagnation point or by using a higher quadrature which lead to reducing the errors arising from the non-exact integration performed when using collocated schemes, such as the case of the nodal DG method or the FR approaches.

The mesh used for the simulations is represented in figure 3.7 and is composed by $N_t \times N_r = 96 \times 44$ elements, where N_t is the number of elements in the tangential direction and N_r is the number of elements in the radial direction (normal to the cylinder surface). The cylinder surface is approximated by a fifth-order spline functions.

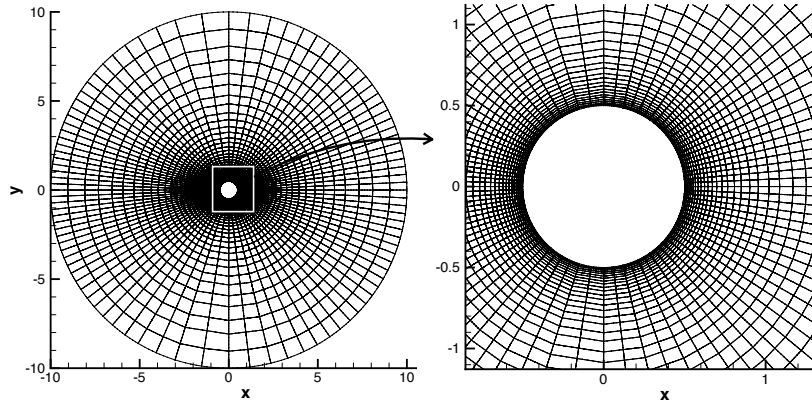


Figure 3.7: Mesh employed for the subsonic inviscid flow past a cylinder test case.

For all the simulations we used a Mach number equal to $Ma_\infty = 0.2$, with the pressure set to $p_\infty = 22600 \text{ Pa}$ and the density equal to $\rho_\infty = 0.364 \text{ kg/m}^3$. For the time-integration we used a fourth-order Runge-Kutta scheme and the time-step was chosen such that the time-integration errors were negligible. The numerical fluxes were obtained through an exact Riemann solver (see section 3.3) and we employed both the DG_{SEM} method with an exact mass matrix (denoted by $DG_{SEM-EMM}$) and the FR_{DG} scheme for the spatial discretisation.

		L_∞ entropy error [$J/(Kg \cdot K)$]				
		P=1, Q=2	P=2, Q=3	P=3, Q=4	P=4, Q=5	P=5, Q=6
Collocated	DG _{SEM} -EMM	X	X	X	X	X
	FR _{DG}	X	X	X	X	X
		P=1, Q=3	P=2, Q=4	P=3, Q=6	P=4, Q=8	P=5, Q=10
Non-collocated	DG _{SEM} -EMM	3.9×10^{-1}	3.0×10^{-2}	4.5×10^{-4}	1.1×10^{-5}	3.4×10^{-6}
	FR _{DG}	3.9×10^{-1}	3.0×10^{-2}	4.5×10^{-4}	1.1×10^{-5}	3.4×10^{-6}

Table 3.1: L_∞ errors associated to the entropy for all the subsonic flow past a cylinder test cases performed at steady state.

Both schemes were run using a collocation projection and a higher order projection. The quadrature points were Gauss-Lobatto-Legendre type.

In table 3.1, we show the L_∞ errors associated to the entropy at steady state. The symbol ‘**X**’ denotes that the simulation failed to converge. It is possible to see how all the tests performed with collocated schemes (i.e. $Q=P+1$, where P is the polynomial order and Q is the number of quadrature points) failed, while using a higher quadrature produced stable simulations. Note that for the higher quadrature tests we used the double of the quadrature points with respect to the polynomial order. This allowed keeping the same order of accuracy for the integration of the nonlinear terms in all the test cases. From the latter perspective, the use of $P=4$ and $Q=6$ for instance, produced an unstable simulation, because the integration errors introduced are higher than the case $P=3$, $Q=6$.

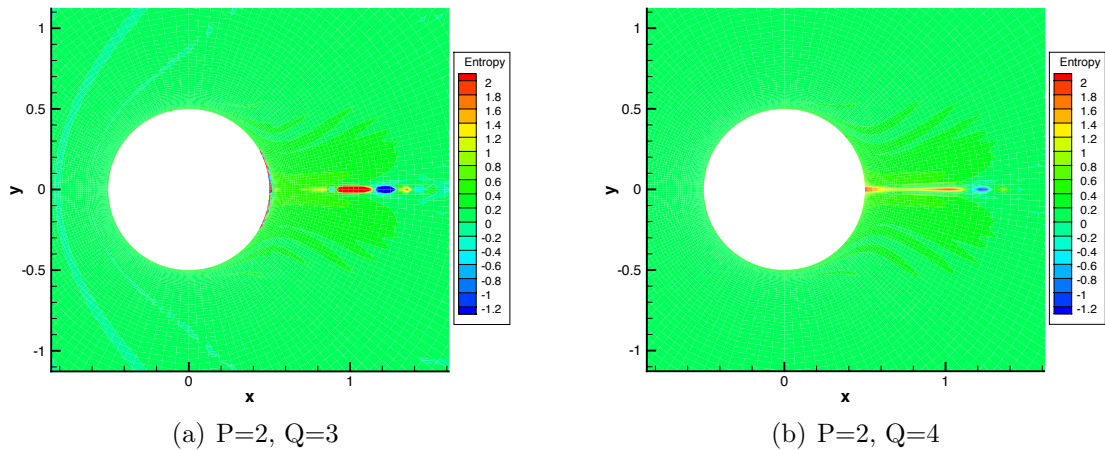


Figure 3.8: Numerical entropy generation at the posterior stagnation point: comparison between collocated figure 3.8(a) and non-collocated figure 3.8(b) DG_{SEM}-EMM methods for $P=2$.

To better illustrate this aspect, in figure 3.8 we show the numerical entropy generated at the posterior stagnation point of the cylinder for $P=2$ at the same physical time. The image on the left (figure 3.8(a)) shows the collocated case, while the image on right (figure 3.8(b)) shows the non-collocated one. For the collocated case it is possible to see a buildup of numerical entropy which will lead to the failure of the simulation after a few time-steps; the non-collocated case instead shows a more contained distribution of the numerical entropy which will be convected downstream and will lead to a stable numerical simulation. Note that, in the two figures, we kept the same colour gradient to highlight the differences in terms of entropy distribution. However, the maximum value of entropy for the collocated case is approximately three orders of magnitude bigger than the non-collocated case.

In figure 3.9 we show the same comparison as in figure 3.8 for $P=5$. In this case, we also show the numerical entropy generated at the wall (figure 3.9(c)) when using a low-order approximation of the geometry equal to $P=1$.

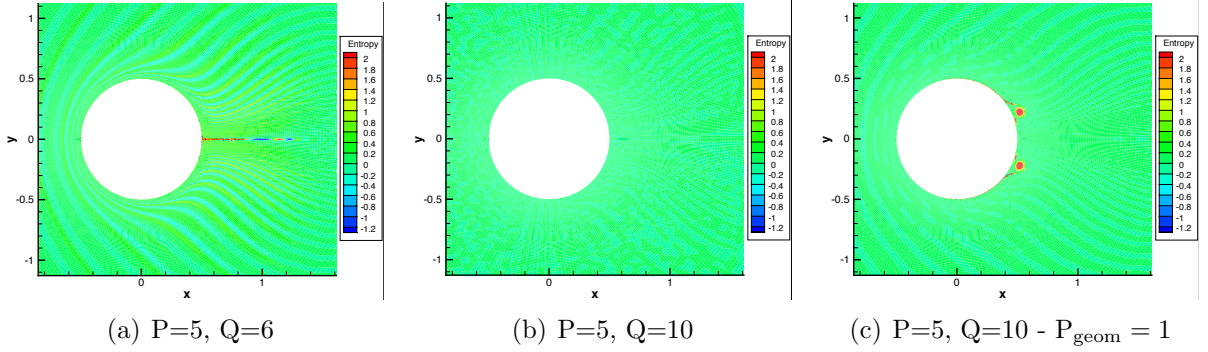


Figure 3.9: Numerical entropy generation at the posterior stagnation point: comparison between collocated figure 3.9(a), non-collocated figure 3.9(b) DG_{SEM} -EMM methods and low-order description of the cylinder surface figure 3.9(c) for $P=5$.

In figure 3.10(a) we show the L_∞ norm of the entropy error calculated on the entire domain as a function of the polynomial order. It is possible to note a consistent decay of the error as the polynomial order increases, as expected. Finally, in figure 3.10(b) we show the numerical entropy at the cylinder surface as a function of the streamwise coordinate x , normalised by the cylinder diameter d_{cylinder} , for three different polynomial orders. These results were obtained using the DG_{SEM} -EMM scheme when the simulation reached the steady state (i.e. the relative variation of the solution between two successive time-step was less than 10^{-15}).

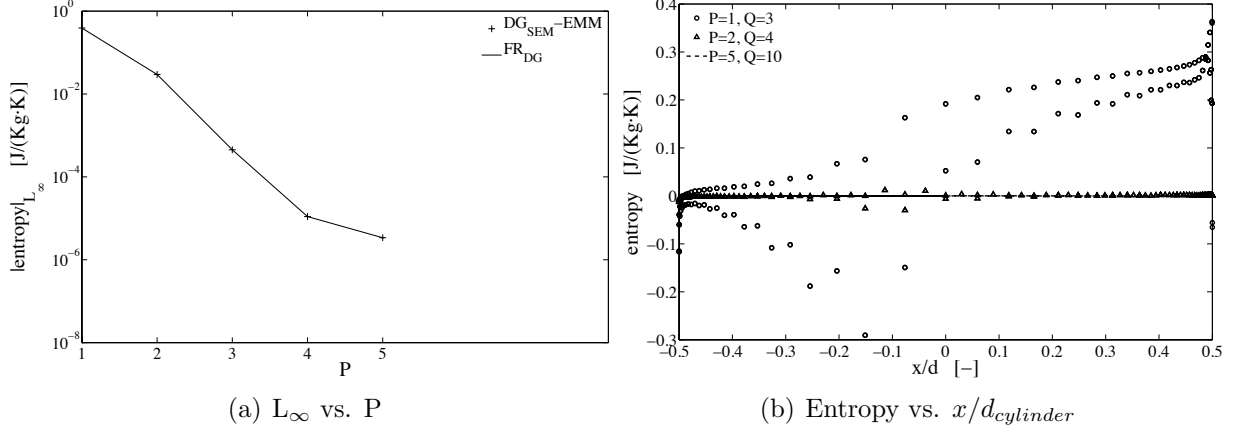


Figure 3.10: Entropy L_∞ error vs. polynomial order and entropy $[\text{J}/(\text{Kg}\cdot\text{K})]$ vs. x/d_{cylinder} for different polynomial orders.

From the results in this section, it should be clear that, when treating nonlinear problems, the under-integration of the nonlinear terms, which arises in the collocated methods, plays a crucial role in the numerical stability of a simulation. This under-integration leads to aliasing errors, which in turn can lead to the failure of the simulation. We have seen that, using a higher quadrature, i.e. using more quadrature points than a collocated method, improves the numerical stability of the simulations carried out (see for example table 3.1). This technique is referred to as consistent integration of the nonlinearities, and it is based on the concept of using a number of quadrature points sufficient to exactly integrate the highest-degree nonlinearity present in the problem (when feasible).

A possible alternative for improving the numerical stability of the simulations (other than using the consistent integration) is to use a better set of quadrature points (such as the Gauss-Legendre points) which permits the exact integration of higher-order functions. Yet, when the nonlinearity degree is too high that even an optimal set of quadrature points does not allow its exact integration, it becomes necessary to use additional quadrature points. This aspect becomes even more critical when deformed or curvilinear meshes are used, because the nonlinearity of the equation(s) may couple with the nonlinear functions describing the deformed elements. In chapter 5, we will thoroughly investigate dealiasing strategies based on the concept of consistent integration. This section provides the general idea behind the motivations for exploring this feature and the next section will further consolidate this aspect.

3.5.2 Compressible Navier-Stokes equations

To verify the implementation of the compressible Navier-Stokes equations, we consider two test cases, the subsonic viscous flow past a flat plate and the subsonic viscous flow past a cylinder. The aim of the first test case is to show that we recover the Blasius solution for a low-Mach number flow, while the aim of the second is to show that we predict the correct Strouhal number.

3.5.2.1 Subsonic viscous flow past a flat plate

The third-order ordinary differential equation governing an incompressible laminar flow over a flat-plate and the associated boundary conditions have been derived by Blasius in 1908 and read

$$2g'''(\eta) + g''(\eta) = 0, \quad g(\eta = 0) = 0; \quad g'(\eta = 0) = 0; \quad g'(\eta = \infty) = 1, \quad (3.72)$$

where η is known as similarity variable and is defined as $\eta = y\sqrt{U_\infty/(\nu x)}$, with x and y being the stream wise and wall normal coordinates, respectively, ν being the kinematic viscosity and U_∞ being the free-stream velocity. Equation (3.72) can be solved numerically and, from its solution, we can derive, for instance, the analytical expression for the skin friction coefficient:

$$C_f \approx \frac{0.664}{\sqrt{Re_x}}, \quad (3.73)$$

where Re_x is the local Reynolds number based on the stream wise location and defined as $Re_x = U_\infty x/\nu$. Note that a compressible solution can be recovered by using the transformation described by Schlichting (1979).

In the following, we show the results obtained on a nearly incompressible flow ($Ma_\infty = 0.1$) past a flat plate, for which we assume the Blasius solution to be valid. In particular, we compare the skin friction coefficient calculated by the Navier-Stokes solver with the one calculated by means of Eq. (3.73).

The computational domain used is depicted in figure 3.11. At the inlet region, located at $x=-0.15$ m, we imposed the inflow characteristic boundary conditions shown in section 3.4.1.1, while at the outlet, located at $x=0.2$ m, we applied the outflow boundary conditions described in section 3.4.2.2. In the region $x \in [-0.15; 0.0]$ m, $y=0$ m, we applied symmetry boundary conditions, while in the region $x \in [0.0; 0.2]$ m, $y=0$ m, we applied isothermal no-slip boundary conditions (described in section 3.4.2.3). Finally, at the top boundary, located at

$y \approx 100\delta_{99}$, where δ_{99} is the thickness of the boundary-layer at the outlet, we used the farfield boundary conditions described in section 3.4.1.1.

The coarser mesh (referred to as mesh 1) is composed by $N_x \times N_y = 69 \times 20$ elements and is the one depicted in figure 3.11. We additionally used a finer mesh (referred to as mesh 2) with approximately the double of the resolution used in mesh 1.

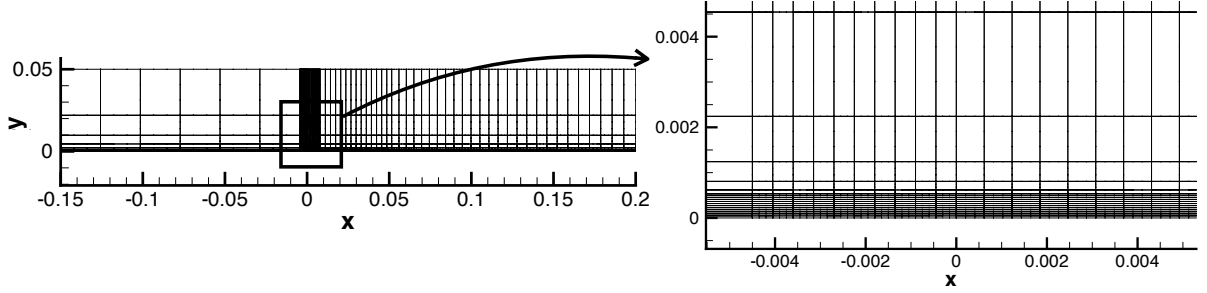


Figure 3.11: Mesh employed for the flat plate simulations.

All the simulations have been carried out using the FR_{g2} method for the spatial discretisation of both the convective and viscous terms. For the numerical fluxes associated to the convective term we used the exact Riemann solver described in section 3.3 and we applied the LDG numerical flux for the diffusion term. The time-integration was performed through a fourth-order Runge-Kutta scheme and the time-step was approximately ten times lower than the time-step limit imposed by the Courant-Friedrichs-Lewy (CFL) condition (Courant *et al.*, 1967).

The Reynolds number chosen was equal to $Re = 1.6 \times 10^6$, based on the length of the flat plate which was equal to $L=0.2$ m, while the Prandtl number Pr was 0.72. The thermodynamics parameters, pressure and density, were $p_\infty = 22600$ Pa, $\rho_\infty = 0.364$ kg/m³.

In figure 3.14, we show the contour of the Mach number and of the y -component of the momentum at the leading edge of the flat plate for mesh 1. From figure 3.12(b) it is possible to see that the resolution at the leading edge is not sufficient to describe the solution and therefore we expect to encounter under-resolution issues, especially if we use the discontinuous spectral/ hp element methods in a collocated fashion.

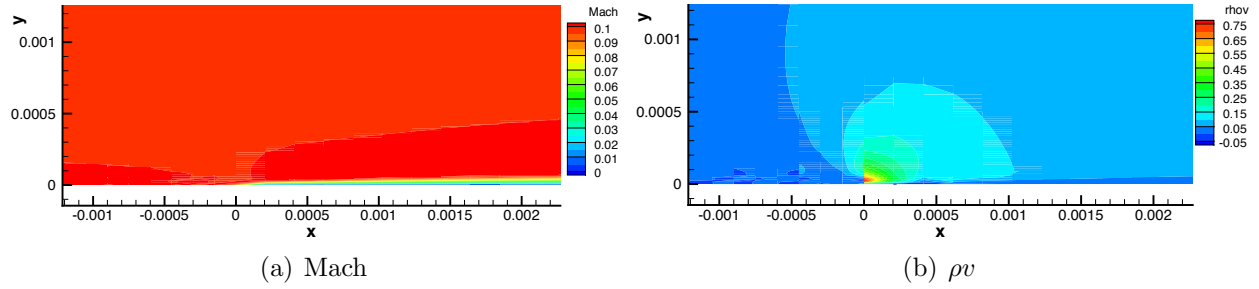


Figure 3.12: Mach and y -component of the momentum at the leading edge of the flat plate.

In table 3.2, we report the differences between the skin friction coefficient derived from the Blasius solution (defined in Eq. (3.73)) and the numerical solution calculated by means of the compressible Navier-Stokes solver implemented in Nektar++, for different combinations of polynomial order and number of quadrature points for the two meshes. This error, also referred to as global L_2 error, represents the difference between the two curves along the entire flat plate (excluding the point $x=0$ m which is singular).

It is possible to see that the simulations carried out using the collocated schemes failed to converge and, when converging (i.e. for $P=1$ and $Q=2$), provided the highest L_2 error. This confirms what expected - the under-resolution at the leading edge couples with the integration errors associated to the collocated schemes leading to unstable simulations.

On the other hand, the non-collocated simulations provided lower L_2 errors and they all converge.

L ₂ errors for C_f				
		P=1, Q=2	P=2, Q=3	P=3, Q=4
Collocated - mesh 1	FR _{g2}	0.64	X	X
Collocated - mesh 2	FR _{g2}	0.31	X	X
		P=1, Q=3	P=2, Q=4	P=3, Q=6
Non-collocated - mesh 1	FR _{g2}	0.50	0.14	0.15
Non-collocated - mesh 2	FR _{g2}	0.22	0.047	0.085

Table 3.2: L_2 errors associated skin friction coefficient C_f for the flat plate simulations.

Note that the errors reported in the table are driven to higher values than those obtained if excluding a few points in proximity to the leading edge. Interestingly, the global L_2 error increases from $P=2$ to $P=3$. This is due to the fact that the under-resolved leading edge

introduces more marked spikes in the solution for higher polynomial order. However, if we measure a local L_2 error downstream of the leading edge, we recover a decreasing trend for the local error when increasing the polynomial order. From this perspective, in table 3.3, we report the L_2 errors at $x=0.02\text{ m}$. It is possible to note that the errors decrease approximately by one order of magnitude than the global L_2 errors presented in table 3.2 and that the errors decrease as the polynomial order increases.

L ₂ errors for C_f at $x = 0.02m$				
		P=1, Q=2	P=2, Q=3	P=3, Q=4
Collocated - mesh 1	FR _{g2}	0.064	X	X
Collocated - mesh 2	FR _{g2}	0.016	X	X
		P=1, Q=3	P=2, Q=4	P=3, Q=6
Non-collocated - mesh 1	FR _{g2}	0.083	0.024	0.0077
Non-collocated - mesh 2	FR _{g2}	0.014	0.0021	0.00029

Table 3.3: L_2 errors associated skin friction coefficient C_f at $x = 0.02m$ ($x/L = 0.1$) for the flat plate simulations.

To illustrate the leading edge under-resolution errors, in figure 3.13, we show the skin friction coefficient calculated for two non-collocated simulations with $P=1$ and $P=3$ against the Blasius skin friction coefficient. It is possible to see that the leading edge is particularly bad resolved for $P=1$ and it becomes well resolved for $P=3$, except for two points that do not match the Blasius curve.

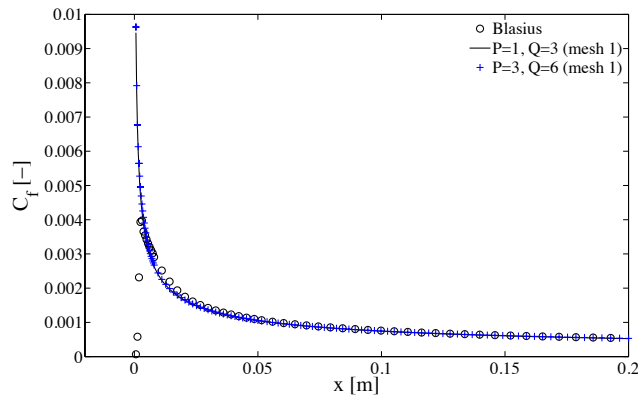


Figure 3.13: Skin-friction coefficient C_f along the flat plate.

Finally, in figure 3.14, we show the x - and the y -component of the velocity, u and v ,

(normalised with respect to the free-stream velocity) as a function of the wall-normal coordinate y . In particular, we compare the Blasius solution with the solutions obtained using the Nektar++ compressible Navier-Stokes solver for $P=1$ and $P=3$. The velocity profiles were extracted at the same x location as that used for calculating the local L_2 errors, $x=0.02$ m. It is possible to note that, while the case $P=1$ provides a poor solution, the case obtained with $P=3$ recovers the Blasius solution well, for both the x - and the y -component of the velocity.

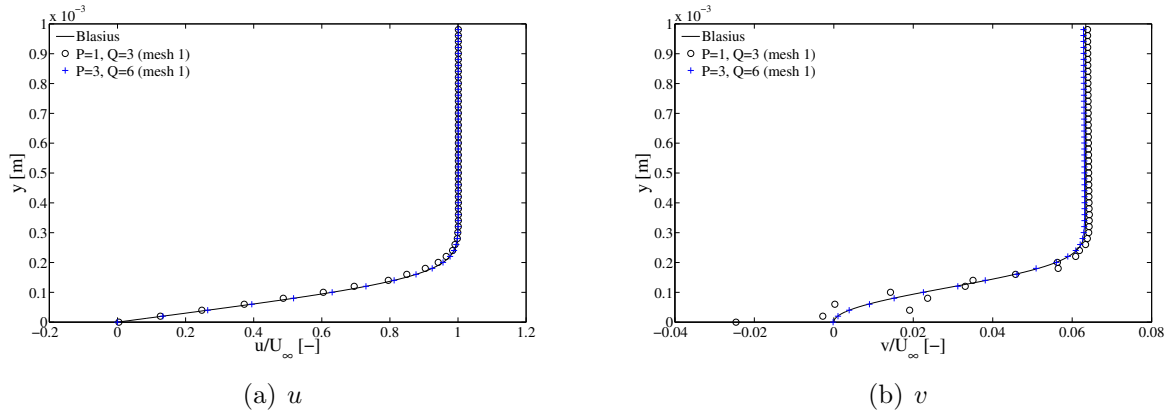


Figure 3.14: X - and y -component of the velocity at $x=0.02$ m.

What seen in this section confirms the preliminary conclusions we have drawn for the inviscid cylinder test case. When we perform an under-resolved simulation, the errors related to the under-integration of the nonlinear terms may lead to unstable simulations. A possible strategy, (as we have seen from the above simulations) is to consistently integrate these nonlinearities in order to enhance the numerical stability.

3.5.2.2 Subsonic viscous flow past a cylinder

The subsonic viscous flow past a cylinder at a Reynolds number equal 100 is a classical test case to verify the implementation of a CFD solver. The mesh used is depicted in figure 3.15 and it is relatively coarse, being composed by 2700 elements. The cylinder surface is approximated by fifth-order polynomials.

The parameters of the simulations are the same as the inviscid flow past a cylinder presented in the previous section. Specifically, we used a Mach number equal to $Ma_\infty = 0.2$, a fourth-order Runge-Kutta time-integration scheme and a time-step approximately ten times lower than the one imposed by the CFL limit. The numerical fluxes were obtained through

an exact Riemann solver (see section 3.3) and we employed the FR_{g2} scheme without using a higher quadrature for integrating the nonlinear terms.

The additional parameters related to the viscous term are the Reynolds number, which was set to $\text{Re}=100$ (based on the diameter of the cylinder $d=1\text{ m}$) and the Prandtl number, which was set to $\text{Pr}=0.72$. Finally, the thermodynamic parameters, pressure and density, were $p_\infty = 22600\text{ Pa}$, $\rho_\infty=0.364\text{ kg/m}^3$.

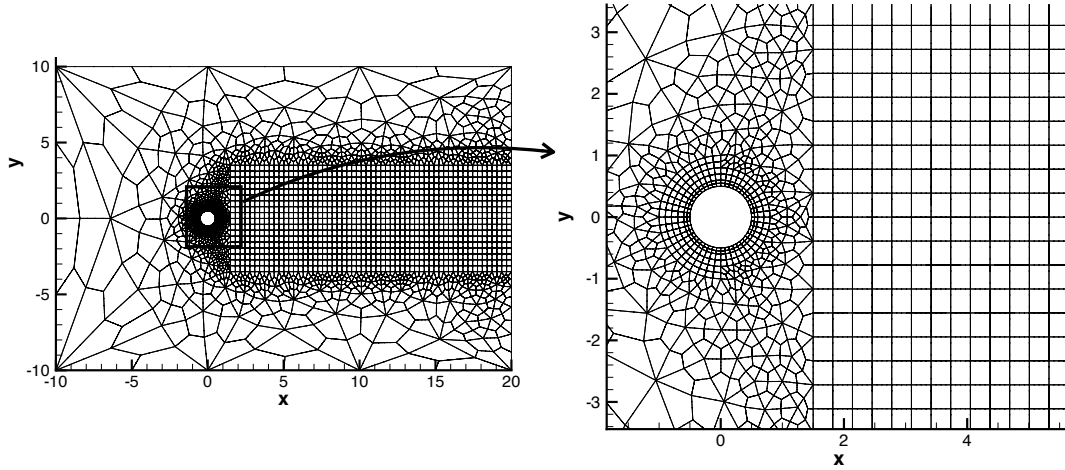


Figure 3.15: Mesh employed for the subsonic viscous flow past a cylinder.

In figure 3.16, we show the z -vorticity and the streamlines. It is possible to appreciate the typical alternate motion of the vortices, whose frequency f_s can be calculated and compared against available experimental and numerical data. Specifically, we measured the so-called Strouhal number, defined as follows:

$$St = \frac{f_s}{U_\infty}, \quad (3.74)$$

where U_∞ is the free stream velocity.

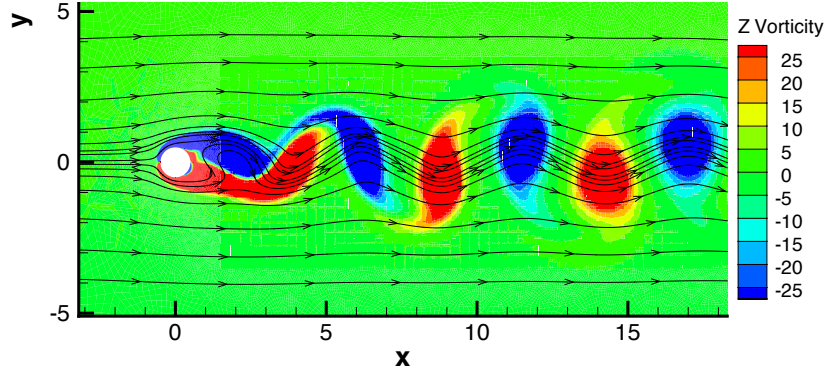


Figure 3.16: Vorticity and streamlines for the subsonic viscous flow past a cylinder.

The four test cases performed are reported in table 3.4 with the associated St number calculated from the numerical experiments. In the table, we also report the numerical results by Lomtev *et al.* (1998) (indicated as ‘Lomtev’) and Burbeau & Sagaut (2002) (indicated as ‘Burbeau’). The agreement with the data produced by other references is satisfactory and it is possible to note a converging trend as the polynomial order increases.

St number [–]						
	P=1, Q=2	P=2, Q=3	P=3, Q=4	P=4, Q=5	‘Lomtev’	‘Burbeau’
FR _{g2}	0.1424	0.169	0.166	0.165	0.165	0.167

Table 3.4: Strouhal number associated to the all the test cases performed.

Also, in contrast to the other numerical experiments performed in section 3.5.2.1 for the flat plate (where the Reynolds number was relatively high), the current set of simulations did not require a higher quadrature of the nonlinear terms (i.e. a consistent integration of the nonlinearities). This is due to the nature of the test case chosen which has a very low Reynolds number.

3.5.3 Summary and discussion

The numerical experiments presented in the previous sections show the need for exploring strategies to improve the numerical stability of the numerical framework developed. Also, they draw the necessity to better understand the connections between the FR approach and the DG scheme in order to be able to deploy similar (if not identical) strategies to both approaches.

Specifically, we have seen how, for under-resolved simulations (which is often the case when considering high-Reynolds number flows), the numerical errors introduced by the under-integration of the nonlinear terms, also referred to as aliasing errors, can easily lead to the failure of the simulation. A possible way to alleviate this issue is the consistent integration of the nonlinearities by higher quadratures. This type of approach naturally fits within the DG numerical framework, given its variational formulation. However, we have seen that it can also be applied to the FR approach - see for example section 3.5.1.2 and 3.5.2.1, where this strategy has been applied to both numerical framework with identical results.

Therefore, a better understanding of the connections between the DG and FR approaches for multidimensional nonlinear conservation laws will allow a clearer view on the implications of applying dealiasing techniques based on the concept of consistent integration of the nonlinearities to the FR approach. Also, a better understanding of the consistent integration in the context of discontinuous spectral/ hp element methods is essential to develop effective strategies to enhance the numerical stability of these methods.

As already mentioned, we will explore the connections between the DG method and the FR approach in the next chapter, while we will detail dealiasing strategies based on the concept of consistent integration of the nonlinearities in chapter 5. The results in this section provide a further insight into the motivations for exploring these two aspects.

Chapter 4

Connections between the DG approach and FR schemes

In this chapter we show the connections between the DG method and high-order flux reconstruction schemes for multidimensional linear and nonlinear conservation laws on irregular and curvilinear grids. In particular, we show the mathematical properties that connects these schemes and we further support the arguments with the results obtained on a set of numerical experiments.

4.1 Introduction

As already mentioned, finding the connections between the DG and FR schemes is crucial for understanding the advantages we can attain by using one or the other method and for exploiting the existing DG machinery for tackling, for instance, the aliasing issues arising in collocated schemes. Also, it allows a better contextualisation of the FR approach within the framework of discontinuous spectral/*hp* element methods.

In this chapter, we analyse three nodal versions of tensor product DG spectral element approximations and three types of FR schemes for solving systems of conservation laws on deformed/curved quadrilateral meshes. The different types of DG approximations result from the choice of the solution nodes of the basis representing the solution and from the quadrature approximation used to integrate the mass matrix and the other terms of the discretisation. The different choices of FR schemes arise from the choice of the correction functions as well as from the number of solution points used for describing the solution polynomial. By considering a general scalar conservation law on an irregular/curvilinear mesh, where

the Jacobian is spatially varying across a given element, we explore mathematically the connections between these discretisations. The findings are further confirmed by the results of some numerical experiments undertaken on two different mesh configurations where, for each configuration, we used three levels of deformation. The numerical study carried out shows also the underlying connections between the DG method and the FR approach when these schemes are subjected to polynomial aliasing arising from the geometry. The essential relationships that are derived in this chapter are summarized in table 4.1.

Flux type	DG _{SEM} -GLL		DG _{SEM}		DG _(Q>P)	
	Lumped mass matrix		Exact mass matrix		Exact mass matrix	
	Linear	Nonlinear	Linear	Nonlinear	Linear	Nonlinear
FR _{g2}	✓*	✓*	✗	✗	✗	✗
FR _{DG}	✗	✗	✓	✓	✗	✗
FR _{DG(Q>P)}	✗	✗	✗	✗	✓	✓

Table 4.1: Connections between different types of DG and FR schemes derived in this work for spatially varying Jacobian elements. ✓ indicates that the schemes are equivalent, whereas ✗ indicates differences between the schemes. *The equivalence holds true for Gauss-Lobatto-Legendre points only.

The notation used in the table for the DG and the FR approaches is the same as in chapter 2 and 3 and is reported in the following for the sake of clarity. Concerning the DG scheme, we consider two DG_{SEM} schemes where the subscript ‘SEM’ denotes the use of a collocation quadrature rule for the inner product of the advection term of the numerical discretisation. The DG_{SEM} scheme in the first column of the table uses a lumped mass matrix (LMM) while the DG_{SEM} scheme in the second column of the table makes use of an exact mass matrix (EMM). The terms ‘lumped’ and ‘exact’ refer to the mass matrix when solving a fully linear problem (i.e. linear partial differential equation and regular grid). Regarding the FR approach, we consider the FR_{g2} and the FR_{DG} schemes introduced in chapter 2. We also propose that, using Q>P, where Q is the number of quadrature points and P is the polynomial order inside each element, further extends the equivalence between FR and DG schemes. In particular, we show that DG and FR_{DG} are identical when using Q>P (we denote these two schemes by DG_(Q>P) and FR_{DG(Q>P)}) for linear and nonlinear flux functions as well as for irregular and curvilinear tensor-product meshes.

The work presented in this chapter is the extension of the work presented by De Grazia *et al.* (2014), where only regular grids were taken into account, and is being considered for

publication in ‘Journal of Scientific Computing’.

This chapter is organised as follows. In section 4.2 we prove that the DG_{SEM} -EMM and the FR_{DG} schemes as well as the DG_{SEM} -LMM and the FR_{g2} schemes are equivalent on irregular and curvilinear grids. In section 4.3 we show some numerical results which support the demonstrations. Finally, in section 4.4 we briefly summarise the chapter and provide some concluding remarks.

4.2 Theory

In this section, we prove that the FR_{DG} scheme and the DG_{SEM} method with exact mass matrix are equivalent on irregular and curvilinear tensor-product meshes when the same approximation of the geometry is employed. We successively demonstrate that also the FR_{g2} scheme and the DG_{SEM} method with lumped mass matrix are equivalent on irregular and curvilinear quadrilateral grids. The demonstrations are carried out on the two-dimensional conservation law defined in Eq. (2.53) (reported here for the readers convenience), that is

$$\frac{\partial u}{\partial t} + \nabla_{\mathbf{x}} \cdot \mathbf{f} = 0, \quad (4.1)$$

within a domain $\Omega \in \mathbb{R}^2$, with $\mathbf{f} = [f_1, f_2]$, where $f = f_1(u)$ and $f_2 = f_2(u)$ are the advection fluxes in the x_1 and x_2 Cartesian directions, respectively. Following what shown in chapter 2, the domain Ω is partitioned into N non-overlapping, quadrilateral elements Ω_n which can be deformed or curved

$$\Omega = \bigcup_{n=1}^N \Omega_n \quad (4.2)$$

and each Ω_n is mapped into a reference quadrilateral element $\Omega_s = [-1, 1] \times [-1, 1]$ in the transformed space $\boldsymbol{\xi} = (\xi_1, \xi_2)$ so that

$$x_i = \Theta_i(\xi_1, \xi_2), \quad i = 1, 2. \quad (4.3)$$

For an arbitrary-shaped straight-sided quadrilateral with vertices A, B, C and D the mapping onto the standard element is defined in Eq. (2.13), while for curvilinear elements the mapping is defined in Eq. (2.14).

4.2.1 FR_{DG} scheme as DG_{SEM} method with EMM

In order to demonstrate the equivalence between the FR_{DG} scheme and the DG_{SEM} method with an exact mass matrix (second column and second row of table 4.1), we start multiplying Eq. (4.1) by a two-dimensional Lagrange polynomial $\ell(\mathbf{x})$ and we integrate the equation over a local element Ω_n

$$\iint_{\Omega_n} \left[\frac{\partial u_n^\delta}{\partial t} + \nabla_{\mathbf{x}} \cdot \mathbf{f}_n^{\delta D} \right] \ell(\mathbf{x}) d\mathbf{x} = 0. \quad (4.4)$$

We successively perform an integration by parts, substitute the boundary terms by the numerical interface fluxes and integrate by parts once more in order to derive the strong form of the DG method

$$\iint_{\Omega_n} \left[\frac{\partial u_n^\delta}{\partial t} + (\nabla_{\mathbf{x}} \cdot \mathbf{f}_n^{\delta D}) \right] \ell(\mathbf{x}) d\mathbf{x} + \int_{\partial\Omega_n} \left[(\mathbf{f}_n^{\delta I} - \mathbf{f}_n^{\delta D}) \cdot \mathbf{n} \right] \ell(\mathbf{x}) ds = 0. \quad (4.5)$$

Note that, in chapter 2, we derived the weak form of the DG method by integrating by parts once and not twice as in this case. However, for the purpose of demonstrating the equivalence between DG_{SEM} and FR_{DG}, the strong form is better suited. In addition, it can be demonstrated that the strong and the weak form of the DG method are identical (Kopriva & Gassner, 2010).

We then expand Eq. (4.5) as follows

$$\iint_{\Omega_n} \left[\frac{\partial u_n^\delta}{\partial t} + \left(\frac{\partial f_{1n}^{\delta D}}{\partial x_1} + \frac{\partial f_{2n}^{\delta D}}{\partial x_2} \right) \right] \ell(\mathbf{x}) d\mathbf{x} + \int_{\partial\Omega_n} \mathbf{n} \cdot \left[\mathbf{f}_n^{\delta I} - \mathbf{f}_n^{\delta D} \right] \ell(\mathbf{x}) ds = 0, \quad (4.6)$$

and map the operations from the local element Ω_n to the reference element Ω_s , in order to highlight the metric terms

$$\begin{aligned} \iint_{\Omega_s} \frac{\partial u^\delta}{\partial t} \ell(\boldsymbol{\xi}) J d\boldsymbol{\xi} + \iint_{\Omega_s} \left[\left(\frac{\partial f_1^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial \xi_1}{\partial x_1} \right) + \left(\frac{\partial f_1^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial \xi_2}{\partial x_1} \right) + \right. \\ \left. + \left(\frac{\partial f_2^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial \xi_1}{\partial x_2} \right) + \left(\frac{\partial f_2^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial \xi_2}{\partial x_2} \right) \right] \ell(\boldsymbol{\xi}) J d\boldsymbol{\xi} + \widehat{\mathbf{b}}^{\text{DG}} = 0, \end{aligned} \quad (4.7)$$

where $\widehat{\mathbf{b}}^{\text{DG}}$ is the boundary term of the DG method. Note that, for the sake of clarity, we have dropped the subscript ‘ n ’ which indicates elemental operations.

We can now simplify Eq. (4.7) by using Eq. (2.15), such that it reads

$$\begin{aligned} \iint_{\Omega_s} \frac{\partial u^\delta}{\partial t} \ell(\boldsymbol{\xi}) J d\boldsymbol{\xi} + \iint_{\Omega_s} \left[\left(\frac{\partial f_1^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial x_2}{\partial \xi_2} \right) - \left(\frac{\partial f_1^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial x_2}{\partial \xi_1} \right) + \right. \\ \left. - \left(\frac{\partial f_2^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial x_1}{\partial \xi_2} \right) + \left(\frac{\partial f_2^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \ell(\boldsymbol{\xi}) d\boldsymbol{\xi} + \widehat{\mathbf{b}}^{\text{DG}} = 0. \end{aligned} \quad (4.8)$$

Considering now the boundary term, $\widehat{\mathbf{b}}^{\text{DG}}$, we can write the extended expression for each edge of the standard element Ω_s . The notation adopted refers to figure 2.2.

Edge ‘B’ (Bottom)

For the bottom edge of the standard element, we can define the normals, $n_{x_{1B}}$, $n_{x_{2B}}$, in the two Cartesian directions x_1 and x_2 and the one-dimensional Jacobian, $J_{1\text{DB}}$, as follows:

$$n_{x_{1B}} = \frac{\partial x_2}{\partial \xi_1} / J_{1\text{DB}}, \quad n_{x_{2B}} = -\frac{\partial x_1}{\partial \xi_1} / J_{1\text{DB}}, \quad J_{1\text{DB}} = \sqrt{\left(\frac{\partial x_2}{\partial \xi_1} \right)^2 + \left(\frac{\partial x_1}{\partial \xi_1} \right)^2}. \quad (4.9)$$

By substituting the quantities in Eq. (4.9) into the boundary term, we then obtain the following expression:

$$\begin{aligned} \int_{\partial\Omega_{s,B}} \left[\left(\mathbf{f}_B^{\delta I} - \mathbf{f}_B^{\delta D} \right) \cdot \mathbf{n}_B \right] \ell(\boldsymbol{\xi}) J_{1\text{DB}} d\widehat{s} = \\ = \int_{\partial\Omega_{s,0}} \left[\Delta f_{1B} \frac{\partial x_2}{\partial \xi_1} - \Delta f_{2B} \frac{\partial x_1}{\partial \xi_1} \right] \ell_i(\xi_1) \ell_j(-1) d\xi_1, \end{aligned} \quad (4.10)$$

where $\Delta f_{1B}^\delta = (f_{1B}^{\delta I} - f_{1B}^{\delta D})$ and $\Delta f_{2B}^\delta = (f_{2B}^{\delta I} - f_{2B}^{\delta D})$.

Edge ‘R’ (Right)

For the right edge of the standard element, the normals in the two Cartesian directions x_1 and x_2 and the one-dimensional Jacobian are defined as follow:

$$n_{x_{1R}} = \frac{\partial x_2}{\partial \xi_2} / J_{1\text{DR}}, \quad n_{x_{2R}} = -\frac{\partial x_1}{\partial \xi_2} / J_{1\text{DR}}, \quad J_{1\text{DR}} = \sqrt{\left(\frac{\partial x_2}{\partial \xi_2} \right)^2 + \left(\frac{\partial x_1}{\partial \xi_2} \right)^2}. \quad (4.11)$$

By substituting the quantities in Eq. (4.11) into the boundary term, we then obtain

the following expression:

$$\begin{aligned} & \int_{\partial\Omega_{s,R}} \left[\left(\mathbf{f}_R^{\delta I} - \mathbf{f}_R^{\delta D} \right) \cdot \mathbf{n}_R \right] \ell(\boldsymbol{\xi}) J_{1D_R} d\widehat{s} = \\ & = \int_{\partial\Omega_{s,R}} \left[\Delta f_{1R}^{\delta} \frac{\partial x_2}{\partial \xi_2} - \Delta f_{2R}^{\delta} \frac{\partial x_1}{\partial \xi_2} \right] \ell_i(1) \ell_j(\xi_2) d\xi_2, \end{aligned} \quad (4.12)$$

where $\Delta f_{1R}^{\delta} = (f_{1R}^{\delta I} - f_{1R}^{\delta D})$ and $\Delta f_{2R}^{\delta} = (f_{2R}^{\delta I} - f_{2R}^{\delta D})$.

Edge ‘T’ (Top)

For the top edge of the standard element, the normals and the one-dimensional Jacobian are:

$$n_{x_{1T}} = -\frac{\partial x_2}{\partial \xi_1} / J_{1DT}, \quad n_{x_{2T}} = \frac{\partial x_1}{\partial \xi_1} / J_{1DT}, \quad J_{1DT} = \sqrt{\left(\frac{\partial x_2}{\partial \xi_1} \right)^2 + \left(\frac{\partial x_1}{\partial \xi_1} \right)^2}. \quad (4.13)$$

By substituting the quantities in Eq. (4.13) into the boundary term, we then obtain the following expression:

$$\begin{aligned} & \int_{\partial\Omega_{s,T}} \left[\left(\mathbf{f}_T^{\delta I} - \mathbf{f}_T^{\delta D} \right) \cdot \mathbf{n}_T \right] \ell(\boldsymbol{\xi}) J_{1DT} d\widehat{s} = \\ & = \int_{\partial\Omega_{s,T}} \left[-\Delta f_{1T}^{\delta} \frac{\partial x_2}{\partial \xi_1} + \Delta f_{2T}^{\delta} \frac{\partial x_1}{\partial \xi_1} \right] \ell_i(\xi_1) \ell_j(1) d\xi_1, \end{aligned} \quad (4.14)$$

where $\Delta f_{1T}^{\delta} = (f_{1T}^{\delta I} - f_{1T}^{\delta D})$ and $\Delta f_{2T}^{\delta} = (f_{2T}^{\delta I} - f_{2T}^{\delta D})$.

Edge ‘L’ (Left)

Finally, for the left edge of the standard element, the normals and the one-dimensional Jacobian can be written as follows:

$$n_{x_{1L}} = -\frac{\partial x_2}{\partial \xi_2} / J_{1DL}, \quad n_{x_{2L}} = \frac{\partial x_1}{\partial \xi_2} / J_{1DL}, \quad J_{1DL} = \sqrt{\left(\frac{\partial x_2}{\partial \xi_2} \right)^2 + \left(\frac{\partial x_1}{\partial \xi_2} \right)^2}. \quad (4.15)$$

By substituting the quantities in Eq. (4.15) into the boundary term, we then obtain

the following expression:

$$\begin{aligned} & \int_{\partial\Omega_{s,L}} \left[\left(\mathbf{f}_L^{\delta I} - \mathbf{f}_L^{\delta D} \right) \cdot \mathbf{n}_L \right] \ell(\boldsymbol{\xi}) J_{1D_L} d\widehat{\mathbf{s}} = \\ & = \int_{\partial\Omega_{s,L}} \left[-\Delta f_{1L} \frac{\partial x_2}{\partial \xi_2} + \Delta f_{2L} \frac{\partial x_1}{\partial \xi_2} \right] \ell_i(-1) \ell_j(\xi_2) d\xi_2, \end{aligned} \quad (4.16)$$

where $\Delta f_{1L}^\delta = (f_{1L}^{\delta I} - f_{1L}^{\delta D})$ and $\Delta f_{2L}^\delta = (f_{2L}^{\delta I} - f_{2L}^{\delta D})$.

If we now assemble the boundary term by summing all the edge contributions, then Eq. (4.8) becomes:

$$\begin{aligned} & \iint_{\Omega_s} \frac{\partial u^\delta}{\partial t} \ell(\boldsymbol{\xi}) J d\boldsymbol{\xi} + \iint_{\Omega_s} \left[\left(\frac{\partial f_1^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial x_2}{\partial \xi_2} \right) - \left(\frac{\partial f_1^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial x_2}{\partial \xi_1} \right) + \right. \\ & \quad \left. - \left(\frac{\partial f_2^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial x_1}{\partial \xi_2} \right) + \left(\frac{\partial f_2^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \ell(\boldsymbol{\xi}) d\boldsymbol{\xi} + \\ & \quad + \int_{\partial\Omega_{s,B}} \left[\Delta f_{1B} \frac{\partial x_2}{\partial \xi_1} - \Delta f_{2B} \frac{\partial x_1}{\partial \xi_1} \right] \ell_i(\xi_1) \ell_j(-1) d\xi_1 + \\ & \quad + \int_{\partial\Omega_{s,R}} \left[\Delta f_{1R} \frac{\partial x_2}{\partial \xi_2} - \Delta f_{2R} \frac{\partial x_1}{\partial \xi_2} \right] \ell_i(1) \ell_j(\xi_2) d\xi_2 + \\ & \quad + \int_{\partial\Omega_{s,T}} \left[-\Delta f_{1T} \frac{\partial x_2}{\partial \xi_1} + \Delta f_{2T} \frac{\partial x_1}{\partial \xi_1} \right] \ell_i(\xi_1) \ell_j(1) d\xi_1 + \\ & \quad + \int_{\partial\Omega_{s,L}} \left[-\Delta f_{1L} \frac{\partial x_2}{\partial \xi_2} + \Delta f_{2L} \frac{\partial x_1}{\partial \xi_2} \right] \ell_i(-1) \ell_j(\xi_2) d\xi_2 = 0, \end{aligned} \quad (4.17)$$

which represents the strong form of the DG_{SEM} method in a standard element Ω_s with the metric terms highlighted.

We remark that we assume the mass matrix to be exact if considering a linear problem, which, for a Gauss-Lobatto-Legendre (GLL) point distribution means calculating Eq. (2.64) on $Q=P+2$ quadrature points, where P is the polynomial order used, while for Gauss-Legendre (GL) points means using $Q=P+1$ quadrature points.

We now consider the FR_{DG} approach, and we show how, starting from its formulation on a standard element, we can recover Eq. (4.17) and thus prove that the DG_{SEM} with an exact mass matrix and the FR_{DG} scheme are identical on deformed/curved elements.

Consider the transformations between the local element Ω_n and the standard element Ω_s ,

introduced in chapter 2 and here reported for the sake of clarity:

$$\widehat{u}^\delta = \widehat{u}^\delta(\boldsymbol{\xi}, t) = Ju_n^\delta(\Theta^{-1}(\boldsymbol{\xi}, t)),$$

$$\widehat{\mathbf{f}}^\delta = \widehat{\mathbf{f}}^\delta(\boldsymbol{\xi}, t) = (\widehat{f}_1^\delta, \widehat{f}_2^\delta) = \left(\frac{\partial x_2}{\partial \xi_2} f_{1,n}^\delta - \frac{\partial x_1}{\partial \xi_2} f_{2,n}^\delta, -\frac{\partial x_2}{\partial \xi_1} f_{1,n}^\delta + \frac{\partial x_1}{\partial \xi_1} f_{2,n}^\delta \right),$$

where J is the Jacobian of the coordinate transformation between the local and standard space and has been defined in Eq. (2.15), while the metric terms $\frac{\partial x_1}{\partial \xi_1}, \frac{\partial x_1}{\partial \xi_2}, \frac{\partial x_2}{\partial \xi_1}, \frac{\partial x_2}{\partial \xi_2}$ can be evaluated either from Eq. (2.13) for arbitrary shaped straight-sided elements or from Eq. (2.14) for curvilinear elements. Note that, as for the DG approach, we have dropped the subscript ‘ n ’.

The FR approach in a reference element Ω_s can be written as

$$\frac{\partial \widehat{u}^\delta}{\partial t} + \frac{\partial \widehat{f}_1^{\delta D}}{\partial \xi_1} + \frac{\partial \widehat{f}_2^{\delta D}}{\partial \xi_2} + \widehat{\mathbf{b}}^{\text{FR}} = 0. \quad (4.18)$$

By using the transformations between the local and the standard element defined above, we can expand Eq. (4.18) in order to highlight the metric terms

$$J \left(\frac{\partial u^\delta}{\partial t} \right) + \frac{f_1^{\delta D}}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial f_1^{\delta D}}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} - \frac{\partial f_2^{\delta D}}{\partial \xi_1} \frac{\partial x_1}{\partial \xi_2} + \frac{\partial f_2^{\delta D}}{\partial \xi_2} \frac{\partial x_1}{\partial \xi_1} + \widehat{\mathbf{b}}^{\text{FR}} = 0, \quad (4.19)$$

where $\widehat{\mathbf{b}}^{\text{FR}}$ is defined as follows

$$\widehat{\mathbf{b}}^{\text{FR}} = \widehat{\Delta f}_{2_B} \Psi'_B(\xi_2) + \widehat{\Delta f}_{1_R} \Psi'_R(\xi_1) + \widehat{\Delta f}_{2_T} \Psi'_T(\xi_2) + \widehat{\Delta f}_{1_L} \Psi'_L(\xi_1), \quad (4.20)$$

with $\Psi'_{B/R/T/L}$ being the derivatives of the correction functions recovering the FR_{DG} scheme and showed in chapter 2 (Eq. (2.73)) and $\widehat{\Delta f}_{2_B}, \widehat{\Delta f}_{1_R}, \widehat{\Delta f}_{2_T}$ and $\widehat{\Delta f}_{1_L}$ being the standard flux jumps on the bottom, right, top and left edge of the standard element, respectively (for the conventions adopted, see also figure 2.2). As done for the DG_{SEM} scheme, we can write explicitly the expression of the boundary term for each edge of the standard element Ω_s .

Edge ‘B’ (Bottom)

The standard flux jump for the bottom edge reads

$$\widehat{\Delta f}_{2_B} = (\mathbf{f}_B^{\delta I} - \mathbf{f}_B^{\delta D}) \cdot (J\mathbf{G}^{-T}\widehat{\mathbf{n}}_B) = \left[-\Delta f_{1_B}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2_B}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right], \quad (4.21)$$

where $\Delta f_{1_B}^\delta = (f_{1_B}^{\delta I} - f_{1_B}^{\delta D})$, $\Delta f_{2_B}^\delta = (f_{2_B}^{\delta I} - f_{2_B}^{\delta D})$, $\mathbf{G}^{-1}$ and J have been defined in

Eq. (2.15) and where we used $\hat{\mathbf{n}}_B = (0, 1)$. In Eq. (4.21), we used the Nanson formula defined in Eq. (2.61) for transforming the flux jump from the physical space to the standard space (for additional details, the interested reader can also refer to Persson (2013)).

Edge ‘R’ (Right)

The standard flux jump for the right edge reads

$$\widehat{\Delta f}_{1_R} = (\mathbf{f}_R^{\delta I} - \mathbf{f}_R^{\delta D}) \cdot (J\mathbf{G}^{-T}\hat{\mathbf{n}}_R) = \left[\Delta f_{1_R}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2_R}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right], \quad (4.22)$$

where $\Delta f_{1_R}^\delta = (f_{1_R}^{\delta I} - f_{1_R}^{\delta D})$, $\Delta f_{2_R}^\delta = (f_{2_R}^{\delta I} - f_{2_R}^{\delta D})$ and where we used $\hat{\mathbf{n}}_1 = (1, 0)$.

Edge ‘T’ (Top)

$$\widehat{\Delta f}_{2_T} = (\mathbf{f}_T^{\delta I} - \mathbf{f}_T^{\delta D}) \cdot (J\mathbf{G}^{-T}\hat{\mathbf{n}}_T) = \left[-\Delta f_{1_T}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2_T}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right], \quad (4.23)$$

where $\Delta f_{1_T}^\delta = (f_{1_T}^{\delta I} - f_{1_T}^{\delta D})$, $\Delta f_{2_T}^\delta = (f_{2_T}^{\delta I} - f_{2_T}^{\delta D})$ and where we used $\hat{\mathbf{n}}_T = (0, 1)$.

Edge ‘L’ (Left)

$$\widehat{\Delta f}_{1_L} = (\mathbf{f}_L^{\delta I} - \mathbf{f}_L^{\delta D}) \cdot (J\mathbf{G}^{-T}\hat{\mathbf{n}}_L) = \left[\Delta f_{1_L}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2_L}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right], \quad (4.24)$$

where $\Delta f_{1_L}^\delta = (f_{1_L}^{\delta I} - f_{1_L}^{\delta D})$, $\Delta f_{2_L}^\delta = (f_{2_L}^{\delta I} - f_{2_L}^{\delta D})$ and where $\hat{\mathbf{n}}_L = (1, 0)$.

If we now substitute the four edge contributions just defined into Eq. (4.19), we can rewrite the FR approach as follows

$$\begin{aligned} J \left(\frac{\partial u^\delta}{\partial t} \right) &+ \frac{f_1^{\delta D}}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial f_1^{\delta D}}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} - \frac{\partial f_2^{\delta D}}{\partial \xi_1} \frac{\partial x_1}{\partial \xi_2} + \frac{\partial f_2^{\delta D}}{\partial \xi_2} \frac{\partial x_1}{\partial \xi_1} + \\ &+ \left[-\Delta f_{1_B}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2_B}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \Psi'_B(\xi_2) + \\ &+ \left[\Delta f_{1_R}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2_R}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \Psi'_R(\xi_1) + \\ &+ \left[-\Delta f_{1_T}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2_T}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \Psi'_T(\xi_2) + \\ &+ \left[\Delta f_{1_L}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2_L}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \Psi'_L(\xi_1) = 0. \end{aligned} \quad (4.25)$$

We now multiply Eq. (4.25) by a two-dimensional Lagrange polynomial $\ell(\boldsymbol{\xi})$ and integrate over the standard domain Ω_s

$$\begin{aligned}
& \iint_{\Omega_s} \frac{\partial u^\delta}{\partial t} \ell(\boldsymbol{\xi}) J \, d\boldsymbol{\xi} + \iint_{\Omega_s} \left[\frac{f_1^{\delta D}}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial f_1^{\delta D}}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} + \right. \\
& \quad \left. - \frac{\partial f_2^{\delta D}}{\partial \xi_1} \frac{\partial x_1}{\partial \xi_2} + \frac{\partial f_2^{\delta D}}{\partial \xi_2} \frac{\partial x_1}{\partial \xi_1} \right] \ell(\boldsymbol{\xi}) \, d\boldsymbol{\xi} + \\
& + \iint_{\Omega_s} \left[-\Delta f_{1B}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2B}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \Psi'_B(\xi_2) \ell_i(\xi_1) \ell_j(\xi_2) \, d\boldsymbol{\xi} + \\
& + \iint_{\Omega_s} \left[\Delta f_{1R}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2R}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \Psi'_R(\xi_1) \ell_i(\xi_1) \ell_j(\xi_2) \, d\boldsymbol{\xi} + \\
& + \iint_{\Omega_s} \left[-\Delta f_{1T}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2T}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \Psi'_T(\xi_2) \ell_i(\xi_1) \ell_j(\xi_2) \, d\boldsymbol{\xi} + \\
& + \iint_{\Omega_s} \left[\Delta f_{1L}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2L}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \Psi'_L(\xi_1) \ell_i(\xi_1) \ell_j(\xi_2) \, d\boldsymbol{\xi} = 0,
\end{aligned} \tag{4.26}$$

where we used the tensor product properties of the Lagrangian basis on a standard quadrilateral element $\ell(\boldsymbol{\xi}) = \ell_i(\xi_1) \ell_j(\xi_2)$.

We can now simplify Eq. (4.26) by using the following relations by Huynh (2007):

$$\int_{-1}^1 \Psi'_L(\xi_1) \ell(\xi_1) \, d\xi_1 = \int_{-1}^1 \Psi'_B(\xi_2) \ell(\xi_2) \, d\xi_2 = -\ell(-1), \tag{4.27}$$

where Ψ_L and Ψ_B are the left and bottom Radau polynomials of order $P+1$ on $\xi_1 \in [-1, 1]$ and $\xi_2 \in [-1, 1]$, respectively, which vanish at $\xi_1 = \xi_2 = 1$. This is due to the orthogonality of the Radau polynomial of order $P+1$ to all the polynomials up to order $P-1$. Analogously, for the top and right correction functions, we can write

$$\int_{-1}^1 \Psi'_1(\xi_1) \ell(\xi_1) \, d\xi_1 = \int_{-1}^1 \Psi'_2(\xi_2) \ell(\xi_2) \, d\xi_2 = \ell(1), \tag{4.28}$$

where Ψ_R and Ψ_T are the right and top Radau polynomials of order $P+1$ on $\xi_1 \in [-1, 1]$ and $\xi_2 \in [-1, 1]$, respectively, which vanish at $\xi_1 = \xi_2 = -1$. Substituting Eq. (4.27) and

Eq. (4.28) into Eq. (4.26), we finally obtain:

$$\begin{aligned}
& \iint_{\Omega_s} \frac{\partial u^\delta}{\partial t} \ell(\boldsymbol{\xi}) J d\boldsymbol{\xi} + \iint_{\Omega_s} \left[\left(\frac{\partial f_1^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial x_2}{\partial \xi_2} \right) - \left(\frac{\partial f_1^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial x_2}{\partial \xi_1} \right) + \right. \\
& \quad \left. - \left(\frac{\partial f_2^{\delta D}}{\partial \xi_1} \right) \left(\frac{\partial x_1}{\partial \xi_2} \right) + \left(\frac{\partial f_2^{\delta D}}{\partial \xi_2} \right) \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \ell(\boldsymbol{\xi}) d\boldsymbol{\xi} + \\
& \quad + \int_{\partial\Omega_{s,B}} \left[\Delta f_{1B} \frac{\partial x_2}{\partial \xi_1} - \Delta f_{2B} \frac{\partial x_1}{\partial \xi_1} \right] \ell_i(\xi_1) \ell_j(-1) d\xi_1 + \\
& \quad + \int_{\partial\Omega_{s,R}} \left[\Delta f_{1R} \frac{\partial x_2}{\partial \xi_2} - \Delta f_{2R} \frac{\partial x_1}{\partial \xi_2} \right] \ell_i(1) \ell_j(\xi_2) d\xi_2 + \\
& \quad + \int_{\partial\Omega_{s,T}} \left[-\Delta f_{1T} \frac{\partial x_2}{\partial \xi_1} + \Delta f_{2T} \frac{\partial x_1}{\partial \xi_1} \right] \ell_i(\xi_1) \ell_j(1) d\xi_1 + \\
& \quad + \int_{\partial\Omega_{s,L}} \left[-\Delta f_{1L} \frac{\partial x_2}{\partial \xi_2} + \Delta f_{2L} \frac{\partial x_1}{\partial \xi_2} \right] \ell_i(-1) \ell_j(\xi_2) d\xi_2 = 0,
\end{aligned} \tag{4.29}$$

which is equivalent to the DG_{SEM} method derived in Eq. (4.17) and concludes the demonstration. We remark that the equivalence presented in this section holds for any point distribution and any quadrature rule as long as the mass matrix is exact.

4.2.2 FR_{g2} scheme as DG_{SEM} method with LMM

In this section, we present the equivalence between the FR_{g2} scheme and the DG_{SEM} scheme with a lumped mass matrix (LMM) on a set of GLL points.

Consider the DG_{SEM} method derived in Eq. (4.17), with solution coefficients represented by Lagrange polynomials at $(P+1) \times (P+1)$ GLL points. If we additionally choose a GLL quadrature for the DG_{SEM} scheme, it is well known that only polynomials up to order $2P - 1$ are exactly integrated (to machine precision). Therefore, the coefficients of the mass matrix, as defined in Eq. (2.64), will contain a numerical quadrature error since each integrand is a polynomial of order $2P$. However, because the Lagrangian basis possesses the property that $\ell_p(\xi_{1q}) = \delta_{pq}$, where δ_{pq} represents the Kronecker delta function, the mass matrix $\mathbf{M}$ obtained using this quadrature rule is diagonal

$$\mathbf{M} = \begin{bmatrix} \ddots & & \\ & w_{ij} J_{ij} & \\ & & \ddots \end{bmatrix}, \tag{4.30}$$

where w_{ij} are the weights of the GLL quadrature rule and J_{ij} are the point wise Jacobians of the transformation between the standard and the local space. This renders the scheme extremely efficient as the mass matrix is now trivially invertible. With this quadrature rule the boundary integrals simplify and each flux jump modifies only its own boundary solution point.

Consider now the FR approach as presented in Eq. (4.26); we can rewrite this expression by splitting the integrals constituting the boundary term into their contributions along ξ_1 and ξ_2 :

$$\begin{aligned}
& \iint_{\Omega_s} \frac{\partial u^\delta}{\partial t} \ell(\boldsymbol{\xi}) J \, d\boldsymbol{\xi} + \iint_{\Omega_s} \left[\frac{f_1^{\delta D}}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial f_1^{\delta D}}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} + \right. \\
& \quad \left. - \frac{\partial f_2^{\delta D}}{\partial \xi_1} \frac{\partial x_1}{\partial \xi_2} + \frac{\partial f_2^{\delta D}}{\partial \xi_2} \frac{\partial x_1}{\partial \xi_1} \right] \ell(\boldsymbol{\xi}) \, d\boldsymbol{\xi} + \\
& + \int_{-1}^{+1} \left[-\Delta f_{1B}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2B}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \ell_i(\xi_1) \, d\xi_1 \int_{-1}^{+1} \Psi'_B(\xi_2) \ell_j(\xi_2) \, d\xi_2 + \\
& + \int_{-1}^{+1} \left[\Delta f_{1R}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2R}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \ell_j(\xi_2) \, d\xi_2 \int_{-1}^{+1} \Psi'_R(\xi_1) \ell_i(\xi_1) \, d\xi_1 + \\
& + \int_{-1}^{+1} \left[-\Delta f_{1T}^\delta \left(\frac{\partial x_2}{\partial \xi_1} \right) + \Delta f_{2T}^\delta \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \ell_i(\xi_1) \, d\xi_1 \int_{-1}^{+1} \Psi'_T(\xi_2) \ell_j(\xi_2) \, d\xi_2 + \\
& + \int_{-1}^{+1} \left[\Delta f_{1L}^\delta \left(\frac{\partial x_2}{\partial \xi_2} \right) - \Delta f_{2L}^\delta \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \ell_j(\xi_2) \, d\xi_2 \int_{-1}^{+1} \Psi'_L(\xi_1) \ell_i(\xi_1) \, d\xi_1 = 0
\end{aligned} \tag{4.31}$$

If we consider a collocation quadrature rule with solution values located at GLL points, the first term of Eq. (4.31) will be constituted by the same diagonal mass matrix as in Eq. (4.30). The volumetric part associated to Eq. (4.31) is also evaluated in the same way as the DG method; therefore, in order for the two schemes to be equivalent, we need to find a correction function which transforms the FR boundary term into the DG one. Specifically, if we use as correction functions those recovering the FR_{g2} scheme, presented in Eq. (2.78), and here reported for demonstration purposes

$$\begin{aligned}
\Psi_B = \Psi_L &= \frac{(-1)^P}{2} \left[L_P - \left(\frac{(P+1)L_{P-1}L_P + PL_{P+1}}{2P+1} \right) \right], \\
\Psi_T = \Psi_R &= \frac{1}{2} \left[L_P + \left(\frac{(P+1)L_{P-1}L_P + PL_{P+1}}{2P+1} \right) \right],
\end{aligned} \tag{4.32}$$

we have that their derivatives vanish at all solution points except at the left/bottom boundar-

ies, if evaluating $\Psi'_L(\xi_1) / \Psi'_B(\xi_2)$, or at the right/top boundaries, if evaluating $\Psi'_R(\xi_1) / \Psi'_T(\xi_2)$ (note that the zeros of Ψ' are located at GLL points). Thus, in the FR_{g2} scheme, the correction flux modifies only the boundary points and we can therefore adopt the following transformations for the integrals involving the correction functions in Eq. (4.31):

$$\begin{aligned}
\int_{-1}^{+1} \Psi'_0(\xi_2) \ell_j(\xi_2) d\xi_2 &= \sum_{k=0}^P [w_k \Psi'_0(\xi_{2_k}) \ell_j(\xi_{2_k})] = -\ell_j(-1) \\
\int_{-1}^{+1} \Psi'_1(\xi_1) \ell_i(\xi_1) d\xi_1 &= \sum_{k=0}^P [w_k \Psi'_1(\xi_{1_k}) \ell_i(\xi_{1_k})] = \ell_i(1) \\
\int_{-1}^{+1} \Psi'_2(\xi_2) \ell_j(\xi_2) d\xi_2 &= \sum_{k=0}^P [w_k \Psi'_2(\xi_{2_k}) \ell_j(\xi_{2_k})] = \ell_j(1) \\
\int_{-1}^{+1} \Psi'_3(\xi_1) \ell_i(\xi_1) d\xi_1 &= \sum_{k=0}^P [w_k \Psi'_3(\xi_{1_k}) \ell_i(\xi_{1_k})] = -\ell_i(-1)
\end{aligned} \tag{4.33}$$

because the derivatives of the correction function in vector form evaluated at the $(P+1) \times (P+1)$ GLL points are the following:

$$\begin{aligned}
\Psi'_0(\xi_2) &= \left[-\frac{1}{w_0}, 0, 0, \dots, 0, 0 \right] \\
\Psi'_1(\xi_1) &= \left[0, 0, \dots, 0, 0, \frac{1}{w_P} \right] \\
\Psi'_2(\xi_2) &= \left[0, 0, \dots, 0, 0, \frac{1}{w_P} \right] \\
\Psi'_3(\xi_1) &= \left[-\frac{1}{w_0}, 0, 0, \dots, 0, 0 \right]
\end{aligned} \tag{4.34}$$

The boundary term of FR_{g2} scheme is equal to the boundary term of the DG_{SEM} scheme with lumped mass matrix and therefore the two schemes are equivalent. Note that this equivalence holds only for GLL points.

4.3 Numerical experiments

In this section we present the numerical results obtained for both linear and nonlinear problems. We used two different mesh configurations: a single-element mesh whose Jacobian

distribution is shown in figure 4.1 and a multi-element mesh whose Jacobian distribution is depicted in figure 4.2. To describe the curved edges, the first mesh configuration (also referred to as *Mesh A*) uses a parabolic function r_1 defined within a one-dimensional reference element

$$r_1 = A_1(1 - \xi^2), \quad \xi \in [-1, 1], \quad (4.35)$$

where A_1 is a constant which determines the amplitude of the deformation. The physical coordinates of each edge were successively calculated by applying the map between the reference ξ and the physical x coordinates

$$x = (1 - \xi)\frac{x_i}{2} + (1 + \xi)\frac{x_{i+1}}{2}, \quad (4.36)$$

where x_i represent the vertices of each edge in the physical space. The constant A_1 was set to 0.2 for the mesh depicted in figure 4.1(a) and to 0.4 for the mesh in figure 4.1(b).

The second mesh configuration (also referred to as *Mesh B*) was generated using a sinusoidal function r_2 also defined within a one-dimensional reference element

$$r_2 = A_2 \sin(\pi\xi), \quad \xi \in [-1, 1], \quad (4.37)$$

where A_2 is a constant which set the deformation amplitude. The physical coordinates of each edge were successively obtained using the map in Eq. (4.36). The constant A_2 was set to 0.1 and to 0.2 for the meshes in figure 4.2(a) and figure 4.2(b), respectively.

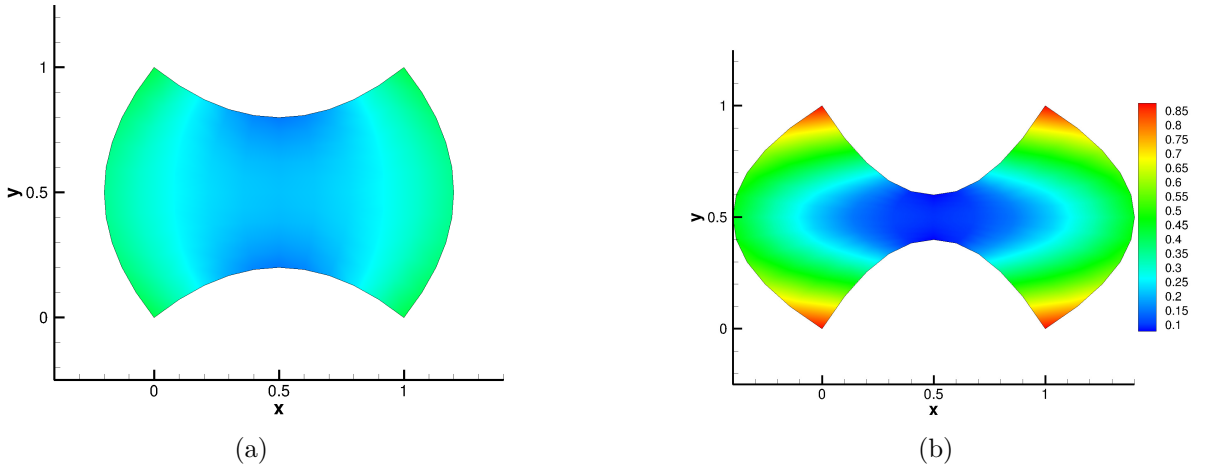


Figure 4.1: Mesh A. Single-element mesh configuration.

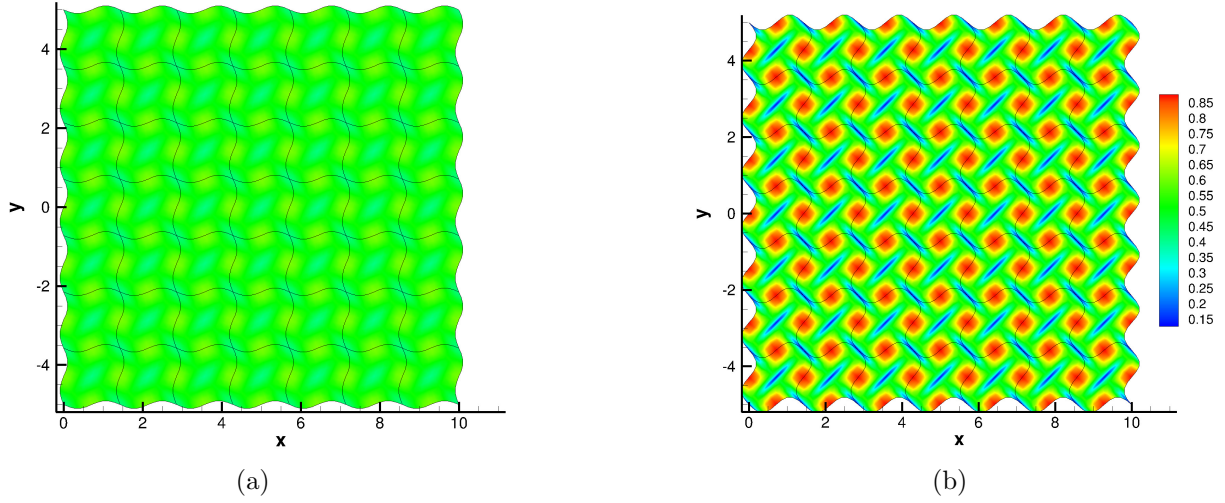


Figure 4.2: Mesh B. Multi-element mesh configuration.

Note that similar meshes were used in Yu & Wang (2013) for studying the accuracy and efficiency of several discontinuous high-order formulations on curved quadrilateral elements.

For both mesh configurations we tested the linear advection equation and the nonlinear compressible Euler equations. The test cases considered span all the possible combinations in table 4.1. In particular, concerning the FR schemes we used:

1. FR_{g2} scheme in its original form (i.e. using a collocation projection);
2. FR_{DG} scheme in its original form (i.e. using a collocation projection);
3. $\text{FR}_{\text{DG}(\text{Q}>\text{P})}$ scheme by using an additional quadrature point to represent the solution (i.e. using a Galerkin projection of the solution).

Regarding the DG scheme we took into account three different versions, namely:

1. DG_{SEM} with a lumped mass matrix (LMM) and a collocation projection of the solution and of the inner product (GLL points only);
 2. DG_{SEM} with an exact mass matrix (EMM) and a collocation projection of the solution and of the inner product;
 3. $\text{DG}_{(\text{Q}>\text{P})}$ with exact mass matrix (EMM) and an additional quadrature point for representing the solution and performing the inner product (i.e. using a Galerkin projection of the solution and of the inner product).
-

We tested two different set of points, Gauss-Lobatto-Legendre (GLL) and Gauss-Legendre (GL) where the interpolations required at the boundaries for GL points were performed in a consistent manner for the various DG and FR schemes considered.

In the following subsections, we will first show the results for the linear problem and successively the results for the nonlinear problem. In both cases, we will make extensive use of the L^2 error defined in Eq. (2.101) and reported here for the reader convenience:

$$E_{L2} = \sqrt{\sum_{i=0}^{Q-1} (u_i - u_{i,exact})^2 w_i}. \quad (4.38)$$

We remind the reader that, u_i is the numerical solution calculated at the i -th quadrature point, $u_{i,exact}$ is the associated exact solution, Q is the total number of quadrature points and w_i are the weights.

4.3.1 Linear problem

The first series of numerical experiments was carried out on the two-dimensional linear advection equation

$$\frac{\partial u}{\partial t} + \mathbf{a} \cdot \nabla_{\mathbf{x}} u = 0, \quad \mathbf{a} = [1, 1]. \quad (4.39)$$

As initial condition, we used

$$u(\Omega, t = 0) = \sin(\tau_x x) \cos(\tau_y y), \quad (4.40)$$

where $\tau_x = \tau_y = 2$ for the first mesh configuration in figure 4.1 and $\tau_x = \tau_y = 0.5$ for the second mesh configuration in figure 4.2. Note that the initial conditions were applied using a collocation projection for FR_{g2} , DG_{SEM} , FR_{DG} while we used a higher order projection (employing an additional quadrature point) for $FR_{DG(Q>P)}$ and $DG_{(Q>P)}$. For both mesh configurations, we applied exact boundary conditions and an explicit fourth-order Runge-Kutta scheme for the time-integration with a final time $T = 2s$ and a time-step sufficiently small to consider the temporal error negligible. We employed four different polynomial orders to discretise the solution. Specifically for the meshes in figure 4.1 we used $P = 10, 11, 12$ and 13 , whilst, for the meshes in figure 4.2, we applied $P = 5, 6, 7$ and 8 . The main parameters used are summarised in table 4.2.

	α	τ_x, τ_y	T	P	T-I
Mesh A	[1, 1]	2, 2	2s	10, 11, 12, 13	RK4
Mesh B	[1, 1]	0.5, 0.5	2s	5, 6, 7, 8	RK4

Table 4.2: Parameter settings for the linear problem and for both the mesh configurations used: Mesh A refers to the meshes represented in figure 4.1; Mesh B refers to the meshes represented in figure 4.2. Note that T-I stands for time-integration scheme.

4.3.1.1 First mesh configuration: Mesh A

Figure 4.3 represents the L^2 error vs. the polynomial order obtained using GLL points for the various schemes tested while figure 4.4 shows the same result when using GL points. Each subfigure in figure 4.3 and figure 4.4 is associated with the two levels of deformation used for Mesh A: figures 4.3(a) and 4.4(a) refer to the mesh depicted in figure 4.1(a) and figures 4.3(b) and 4.4(b) refer to the mesh in figure 4.1(b).

Table 4.3 quantifies up to sixteen digits the results presented in the figures for $P=10$. The results for the other polynomial orders are not tabulated for the sake of compactness and because they provide information similar to table 4.3.

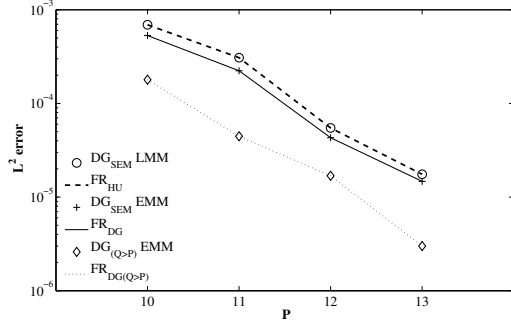
The following equivalences hold true for all the polynomial orders considered as well as for both the point distributions tested:

- FR_{g2} and DG_{SEM} with lumped mass matrix on GLL points;
- FR_{DG} and DG_{SEM} with exact mass matrix;
- $FR_{DG(Q>P)}$ and $DG_{(Q>P)}$.

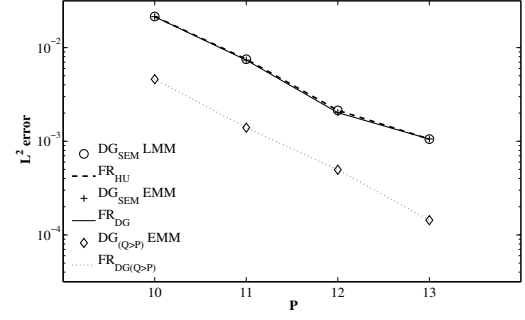
The first two results confirm the demonstrations presented in the previous section. In addition, the last result indicates that these connections are valid also when using additional quadrature points than a standard collocated spectral/ hp element method. Also, the equivalences are up to machine precision for all the deformation levels considered as shown in table 4.3.

From a polynomial aliasing perspective, where the polynomial aliasing is introduced by the curved geometry, it is interesting to note how the magnitude of the error increases as the deformation level of the mesh increases. Specifically, we can see that using either $FR_{DG(Q>P)}$ or $DG_{(Q>P)}$ provides a different result (generally a better L^2 error) for the deformed meshes in the case of GLL points, while, for GL points, the difference is less marked. This is because GL

points possess a higher quadrature precision than GLL points, therefore polynomial aliasing errors introduced by the geometry are already partially adsorbed by the the more powerful quadrature.

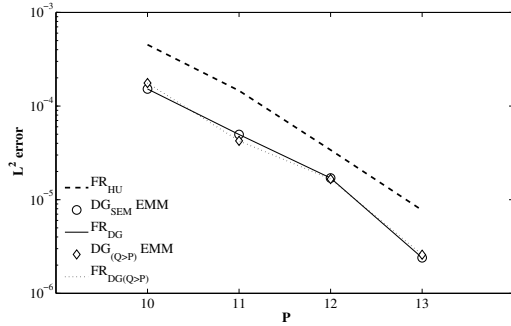


(a) $A_1 = 0.2$

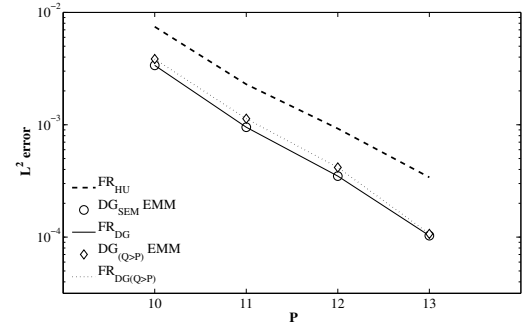


(b) $A_1 = 0.4$

Figure 4.3: L^2 errors vs. polynomial order for the first mesh configuration obtained using the linear advection equation on GLL points.



(a) $A_1 = 0.2$



(b) $A_1 = 0.4$

Figure 4.4: L^2 errors vs. polynomial order for the first mesh configuration obtained using the linear advection equation on GL points.

Method		$\mathbf{FR}_{\mathbf{g2}}$	$\mathbf{DG}_{\mathbf{SEM}}$ LMM
$A_1 = 0.2$	GLL	$6.918095184795 \times 10^{-4}$	$6.918095184794 \times 10^{-4}$
	GL	$4.519754117429 \times 10^{-4}$	—
$A_1 = 0.4$	GLL	$2.15056911125026 \times 10^{-2}$	$2.15056911125026 \times 10^{-2}$
	GL	$7.4522302877056 \times 10^{-3}$	—

Method		$\mathbf{FR}_{\mathbf{DG}}$	$\mathbf{DG}_{\mathbf{SEM}}$ EMM
$A_1 = 0.2$	GLL	$5.311584242439 \times 10^{-4}$	$5.311584242437 \times 10^{-4}$
	GL	$1.520554625620 \times 10^{-4}$	$1.520554625620 \times 10^{-4}$
$A_1 = 0.2$	GLL	$2.13727123365078 \times 10^{-2}$	$2.13727123365078 \times 10^{-2}$
	GL	$3.3786874622380 \times 10^{-3}$	$3.3786874622380 \times 10^{-3}$

Method		$\mathbf{FR}_{\mathbf{DG(Q>P)}}$	$\mathbf{DG}_{(\mathbf{Q>P})}$ EMM
$A_1 = 0.2$	GLL	$1.792971831748 \times 10^{-4}$	$1.792971831749 \times 10^{-4}$
	GL	$1.761116530405 \times 10^{-4}$	$1.761116530402 \times 10^{-4}$
$A_1 = 0.4$	GLL	$4.6056099790686 \times 10^{-3}$	$4.6056099790685 \times 10^{-3}$
	GL	$3.8478160878041 \times 10^{-3}$	$3.8478160878040 \times 10^{-3}$

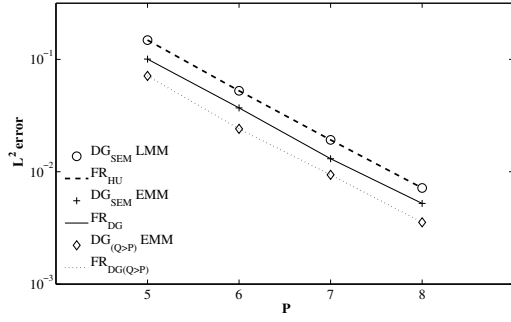
Table 4.3: L^2 errors for the different types of DG and FR schemes tested, in the case of linear advection equation on the first mesh configuration (Mesh A), for polynomial order $P=10$.

4.3.1.2 Second mesh configuration: Mesh B

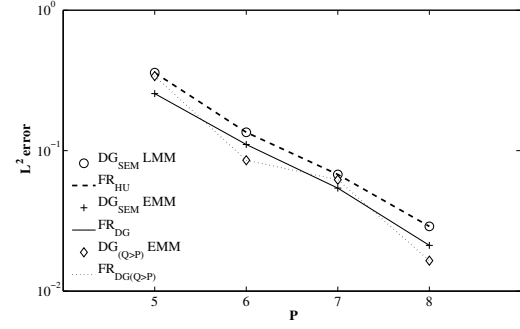
Figure 4.5 represents the L^2 error vs. the polynomial order obtained using GLL points for the various schemes tested, while figure 4.6 shows the same result when using GL points. Each subfigure in figure 4.5 and figure 4.6 is associated with the two levels of deformation used for Mesh A: figures 4.5(a) and 4.6(b) refer to the mesh depicted in figure 4.2(a) and figures 4.5(b) and 4.6(b) refer to the mesh in figure 4.2(b).

Table 4.4 quantifies up to sixteen digits the results presented in the figures for $P=5$. As we can see, the connections presented for the single-element mesh configuration in the previous subsection hold also for the multi-element mesh configuration, Mesh B.

From a polynomial aliasing point of view it is possible to note how the differences between the pairs $\mathbf{FR}_{\mathbf{DG}}$, $\mathbf{DG}_{\mathbf{SEM}}$ with exact mass matrix and $\mathbf{FR}_{\mathbf{DG(Q>P)}}$, $\mathbf{DG}_{(\mathbf{Q>P})}$ are still present although they are less marked for both GLL and GL point distributions.

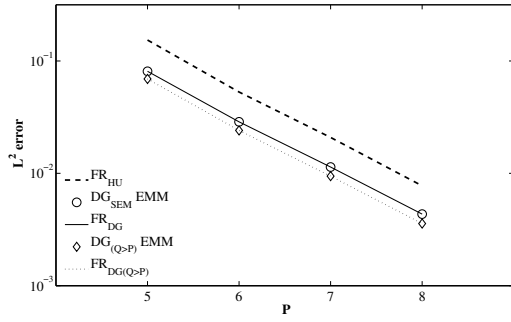


(a) $A_2 = 0.1$

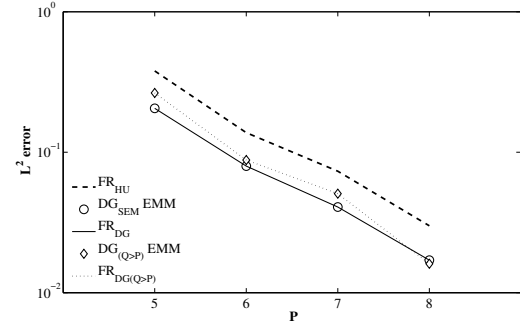


(b) $A_2 = 0.2$

Figure 4.5: L^2 errors vs. polynomial order for the second mesh configuration obtained using the linear advection equation on GLL points.



(a) $A_2 = 0.1$



(b) $A_2 = 0.2$

Figure 4.6: L^2 errors vs. polynomial order for the second mesh configuration obtained using the linear advection equation on GL points.

Method		FR _{g2}	DG _{SEM} LMM
$A_2 = 0.1$	GLL	$1.483031420156138 \times 10^{-1}$	$1.483031420156140 \times 10^{-1}$
	GL	$1.536089072558792 \times 10^{-1}$	–
$A_2 = 0.2$	GLL	$3.589518032507796 \times 10^{-1}$	$3.589518032507806 \times 10^{-1}$
	GL	$3.792730977073956 \times 10^{-1}$	–

Method		FR _{DG}	DG _{SEM} EMM
$A_2 = 0.1$	GLL	$1.006001685998391 \times 10^{-1}$	$1.006001685998393 \times 10^{-1}$
	GL	$8.08508579085947 \times 10^{-2}$	$8.08508579085011 \times 10^{-2}$
$A_2 = 0.2$	GLL	$2.548433855536824 \times 10^{-1}$	$2.548433855536831 \times 10^{-1}$
	GL	$2.052765006831850 \times 10^{-1}$	$2.052765006831897 \times 10^{-1}$

Method		FR _{DG(Q>P)}	DG _(Q>P) EMM
$A_2 = 0.1$	GLL	$7.12574345452700 \times 10^{-2}$	$7.12574345452696 \times 10^{-2}$
	GL	$6.91904507205752 \times 10^{-2}$	$6.91904507205731 \times 10^{-2}$
$A_2 = 0.2$	GLL	$3.395825175351986 \times 10^{-1}$	$3.395825175351990 \times 10^{-1}$
	GL	$2.642066815079172 \times 10^{-1}$	$2.642066815079198 \times 10^{-1}$

Table 4.4: L^2 vs. polynomial order for the different types of DG and FR schemes tested, in the case of linear advection equation on the second mesh configuration (Mesh B), for polynomial order $P=5$.

4.3.2 Nonlinear problem

The second series of numerical experiments were undertaken using the two-dimensional compressible Euler equations introduced in chapter 3 and reported below

$$\frac{\partial \mathbf{u}}{\partial t} + \frac{\partial \mathbf{f}_1^i}{\partial x} + \frac{\partial \mathbf{f}_2^i}{\partial y} = 0,$$

where $\mathbf{u} = (\rho, \rho u, \rho v, E)^\top$ is the vector of the conserved variables and $\mathbf{f}_1^i = \mathbf{f}_1^i(\mathbf{u})$ and $\mathbf{f}_2^i = \mathbf{f}_2^i(\mathbf{u})$ are the vectors of the inviscid fluxes,

$$\mathbf{f}_1^i = \begin{pmatrix} \rho u \\ p + \rho u^2 \\ \rho uv \\ u(E + p) \end{pmatrix}, \quad \mathbf{f}_2^i = \begin{pmatrix} \rho v \\ \rho uv \\ p + \rho v^2 \\ v(E + p) \end{pmatrix}.$$

In the above, ρ is the density, u and v are the velocity components in x and y directions respectively, p is the pressure and E is the total energy

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho(u^2 + v^2),$$

for which we assumed the perfect gas law. Note that γ denotes the constant ratio of specific heats of the gas.

As test case we considered the isentropic vortex problem introduced in chapter 3, whose initial conditions on a two-dimensional grid were

$$\begin{aligned}\rho &= \left(1 - \frac{\beta^2(\gamma - 1)}{16\gamma\pi^2} e^{2(1-r^2)}\right)^{\frac{1}{\gamma-1}} \\ u &= u_0 - \frac{\beta(y - y_0)}{2\pi} e^{(1-r^2)} & v &= v_0 + \frac{\beta(x - x_0)}{2\pi} e^{(1-r^2)} \\ E &= \frac{\rho^\gamma}{\gamma - 1} + \frac{\rho}{2} (u^2 + v^2).\end{aligned}$$

where

$$r = \sqrt{(x - x_0)^2 + (y - y_0)^2},$$

with $\beta = 5$, $R = 1$ and $\gamma = 1.4$. For both mesh configurations, we used an explicit fourth-order Runge-Kutta scheme for the time-integration with a final time $T = 2s$ and a time-step sufficiently small to consider the temporal error negligible. We employed four different polynomial orders to discretise the solution. Specifically for the meshes in figure 4.1 we used $P = 10, 11, 12$ and 13 , while, for the meshes in figure 4.2, we applied $P = 5, 6, 7$ and 8 . The main parameters used are summarised in table 4.5.

	β	R	γ	(x_0, y_0)	(u_0, v_0)	T	P	T-I
Mesh A	5	1	1.4	(0, 0)	(0.5, 0.5)	2s	10, 11, 12, 13	RK4
Mesh B	5	1	1.4	(5, 0)	(0, 5)	2s	5, 6, 7, 8	RK4

Table 4.5: Parameter settings for the nonlinear problem and for both the mesh configurations used: Mesh A refers to the meshes represented in figure 4.1; Mesh B refers to the meshes represented in figure 4.2. Note that T-I stands for time-integration scheme.

4.3.2.1 First mesh configuration: Mesh A

Figure 4.7 represents the L^2 error associated to the density vs. the polynomial order obtained using GLL points for the various schemes tested while figure 4.8 shows the same result when using GL points. Each subfigure in figure 4.7 and figure 4.8 is associated with the two levels of deformation used for Mesh A: figures 4.7(a) and 4.8(a) refer to the mesh depicted in figure 4.1(a) and figures 4.7(b) and 4.8(b) refer to the mesh in figure 4.1(b).

Table 4.6 quantifies up to sixteen digits the results presented in the figures for $P=10$. Also in the case of a nonlinear problem the same equivalences presented before maintain validity up to machine precision as shown in table 4.6.

However, considerations on polynomial aliasing are different because in this case the aliasing sources arise both from the equations themselves, which are nonlinear, and from the geometry. Significant, in particular, is the gap between the pair FR_{DG} , DG_{SEM} and the pair $FR_{DG(Q>P)}$, $DG_{(Q>P)}$ for GLL points. For GL points the differences are less marked as already seen in the linear problem. This is because GL points possess a more powerful quadrature (i.e. the capability of GL points of integrating a higher polynomial order than GLL points).

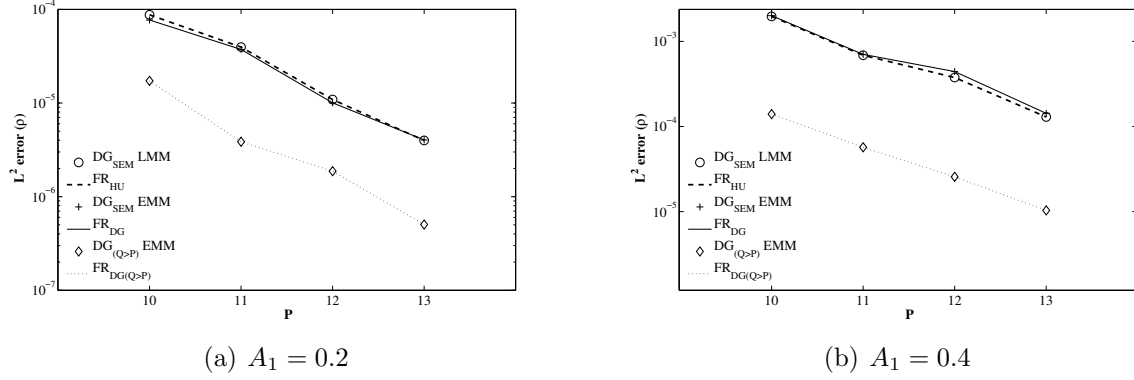
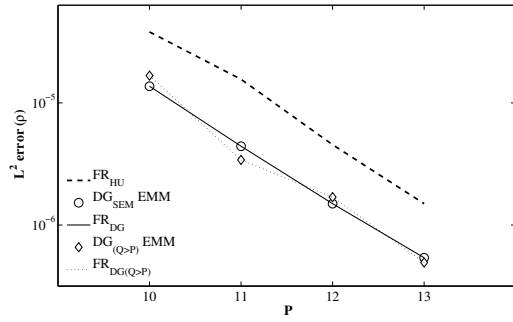
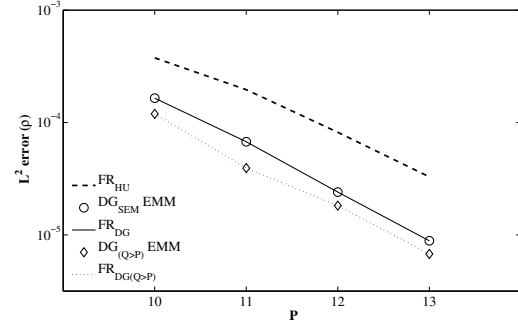


Figure 4.7: L^2 errors vs. polynomial order for the first mesh configuration obtained using the compressible Euler equations using GLL points.



(a) $A_1 = 0.2$



(b) $A_1 = 0.4$

Figure 4.8: L^2 errors vs. polynomial order for the first mesh configuration obtained using the compressible Euler equations using GL points.

Method		$\mathbf{FR_{g2}}$	$\mathbf{DG_{SEM} LMM}$
$A_2 = 0.2$	GLL	$8.74370672196 \times 10^{-5}$	$8.74370672196 \times 10^{-5}$
	GL	$3.81314820750 \times 10^{-5}$	—
$A_2 = 0.4$	GLL	$1.9653166200567 \times 10^{-3}$	$1.9653166200571 \times 10^{-3}$
	GL	$3.766461992222 \times 10^{-4}$	—
Method		$\mathbf{FR_{DG}}$	$\mathbf{DG_{SEM} EMM}$
$A_2 = 0.2$	GLL	$7.70236101694 \times 10^{-5}$	$7.70236101695 \times 10^{-5}$
	GL	$1.36946792400 \times 10^{-5}$	$1.36946792404 \times 10^{-5}$
$A_2 = 0.4$	GLL	$2.0023869182553 \times 10^{-3}$	$2.0023869182552 \times 10^{-3}$
	GL	$1.647596712532 \times 10^{-4}$	$1.647596712532 \times 10^{-4}$
Method		$\mathbf{FR_{DG(Q>P)}}$	$\mathbf{DG_{(Q>P)} EMM}$
$A_2 = 0.2$	GLL	$1.72354832431 \times 10^{-5}$	$1.72354832438 \times 10^{-5}$
	GL	$1.67366736718 \times 10^{-5}$	$1.67366736706 \times 10^{-5}$
$A_2 = 0.4$	GLL	$1.398810392165 \times 10^{-4}$	$1.398810392168 \times 10^{-4}$
	GL	$1.193651546806 \times 10^{-4}$	$1.193651546797 \times 10^{-4}$

Table 4.6: L^2 vs. polynomial order for the different types of DG and FR schemes tested, in the case of compressible Euler equations on the first mesh configuration (Mesh A), for polynomial order $P=10$.

4.3.2.2 Second mesh configuration: Mesh B

Figure 4.9 represents the L^2 error associated to the density vs. the polynomial order obtained using GLL points for the various schemes tested while figure 4.10 shows the same result when using GL points. Each subfigure in figure 4.9 and figure 4.10 is associated with the two levels of deformation used for Mesh B: figures 4.9(a) and 4.10(a) refer to the mesh depicted in figure 4.2(a) and figures 4.9(b) and 4.10(b) refer to the mesh in figure 4.2(b).

Table 4.7 quantifies up to sixteen digits the results presented in the figures for $P=5$. Equal considerations can be made for both connections and polynomial aliasing issues as in the previous subsection.

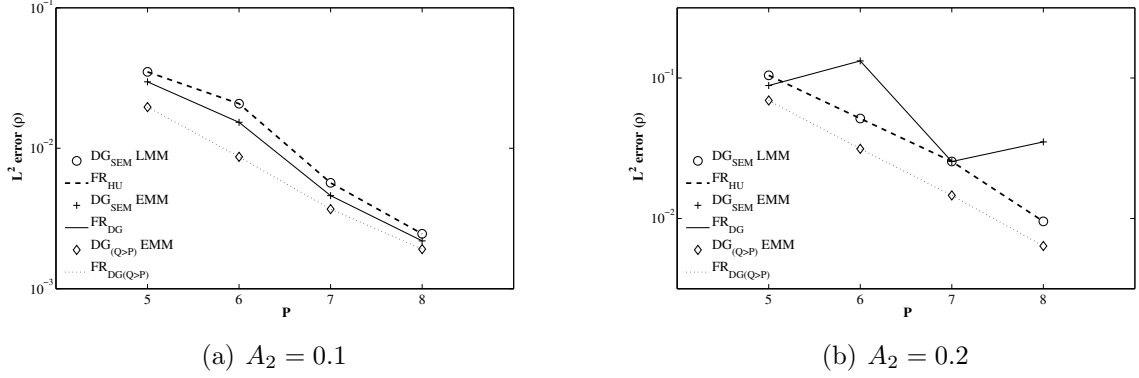


Figure 4.9: L^2 errors vs. polynomial order for the second mesh configuration obtained using the compressible Euler equations using GLL points.

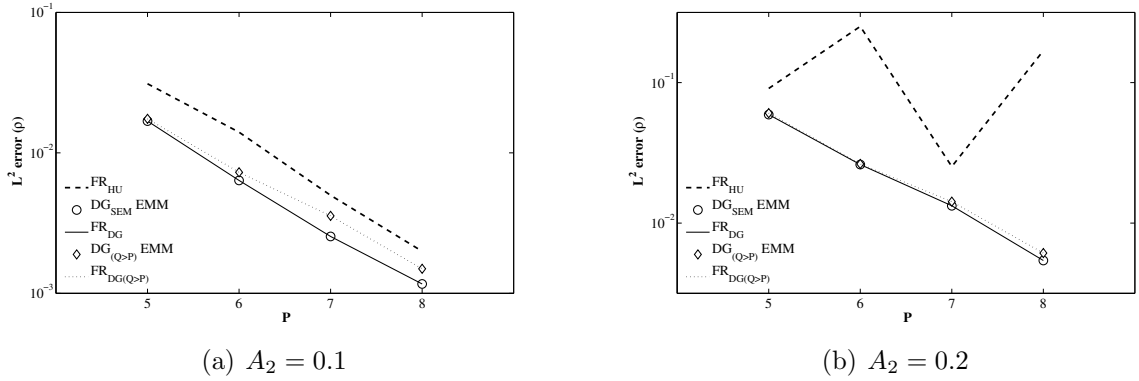


Figure 4.10: L^2 errors vs. polynomial order for the second mesh configuration obtained using the compressible Euler equations using GL points.

Method		FR _{g2}	DG _{SEM} LMM
$A_2 = 0.1$	GLL	$3.49643043963458 \times 10^{-2}$	$3.49643043963231 \times 10^{-2}$
	GL	$3.10369877676844 \times 10^{-2}$	–
$A_2 = 0.2$	GLL	$1.044563879207680 \times 10^{-1}$	$1.044563879207564 \times 10^{-1}$
	GL	$9.08570929838627 \times 10^{-2}$	–

Method		FR _{DG}	DG _{SEM} EMM
$A_2 = 0.1$	GLL	$2.97203508063457 \times 10^{-2}$	$2.97203508063811 \times 10^{-2}$
	GL	$1.68400030314725 \times 10^{-2}$	$1.68400030314833 \times 10^{-2}$
$A_2 = 0.2$	GLL	$8.84471526950831 \times 10^{-2}$	$8.84471526951047 \times 10^{-2}$
	GL	$5.91844810513575 \times 10^{-2}$	$5.91844810513937 \times 10^{-2}$

Method		FR _{DG(Q>P)}	DG _(Q>P) EMM
$A_2 = 0.1$	GLL	$1.96359442119992 \times 10^{-2}$	$1.96359442120232 \times 10^{-2}$
	GL	$1.74520117500541 \times 10^{-2}$	$1.74520117500955 \times 10^{-2}$
$A_2 = 0.2$	GLL	$6.92573585818201 \times 10^{-2}$	$6.92573585818388 \times 10^{-2}$
	GL	$6.06611832896596 \times 10^{-2}$	$6.06611832896908 \times 10^{-2}$

Table 4.7: L^2 vs. polynomial order for the different types of DG and FR schemes tested, in the case of compressible Euler equations on the second mesh configuration (Mesh B), for polynomial order $P=5$.

4.4 Conclusions

In this chapter, we explored the connections between various discontinuous Galerkin methods and high-order flux reconstruction schemes on quadrilateral irregular/curvilinear meshes and the natural underlying implications of polynomial aliasing on these connections. Regarding the DG methods, we considered the DG_{SEM} scheme with a lumped mass matrix (LMM) and a collocation projection of the solution and of the inner product, the DG_{SEM} scheme with an exact mass matrix (EMM) and a collocation projection of the solution and of the inner product and the DG_(Q>P) scheme with exact mass matrix (EMM) and an additional quadrature point for representing the solution and performing the inner product. Concerning the FR schemes, we took into account the FR_{g2} scheme in its original form (i.e. using a collocation projection of the solution), the FR_{DG} scheme in its original form (i.e. using a collocation projection of the solution) and FR_{DG(Q>P)} scheme by using one additional quadrature point to represent the solution.

We found that the connections between discontinuous Galerkin methods and high-order flux reconstruction schemes explored in De Grazia *et al.* (2014) for regular grids generally hold also for irregular/curvilinear meshes. In particular, we mathematically proved the equivalences between FR_{g2} and DG_{SEM} with lumped mass matrix and the connections between

FR_{DG} and DG_{SEM} with exact mass matrix. Both demonstrations were further assessed by numerical experiments on two different mesh configurations at different grid deformation levels and for both linear and nonlinear problems. In addition, we showed numerically how these connections are also valid when applying a higher quadrature $Q > P$ or *consistent integration* of the nonlinearities arising in the problem (i.e. using more quadrature points than a collocation projection: $\text{FR}_{\text{DG}(Q>P)}$ and $\text{DG}_{(Q>P)}$) whose use can help stabilising problems where polynomial aliasing issues can badly affect the solution - this is for example the case of under- or marginally-resolved problems (e.g. high Reynolds number simulations). This result indicates that the aliasing sources for FR_{DG} and DG_{SEM} are identical.

Note that the use of a better point distribution with a more powerful quadrature, such as the GL points, can also alleviate aliasing issues. However, when the nonlinearity cannot be fully described by the GL quadrature or by a more powerful quadrature (Witherden & Vincent, 2014), then the use of additional quadrature points for consistently integrating the nonlinearity sources becomes the only possibility.

In the next chapter, we will better characterise the sources of aliasing in discontinuous spectral/*hp* element methods and we introduce the dealiasing strategies based on the concept of consistent integration of the nonlinearities.

Chapter 5

Dealiasing techniques for spectral/ hp element methods

In this chapter, we investigate the aliasing issues arising in discontinuous spectral/ hp element methods - specifically in the DG method and in the FR approach, identifying the common sources and the role of the various terms appearing in both discretisations. We then present two dealiasing strategies based on the concept of consistent integration of the nonlinearities. The first is a localised approach which targets the nonlinear flux functions in the problem, the second is a global approach which targets both the nonlinear flux functions and the deformed/curved geometry.

5.1 Introduction

Numerical stability is a fundamental requirement for a numerical tool (e.g. a CFD solver) especially when used in an engineering design process where reliability and robustness are key factors in its efficacy in the industrial pipeline. While low order finite element methods demonstrate a usually satisfactory level of numerical stability due to their dissipation properties, high order spectral element methods typically exhibit low dissipation errors, and are therefore affected by a lack of stability which in turn affects their robustness and reliability.

Over the last decade, various attempts have been made to address this problem. Different formulations of the nonlinear terms, for instance, have different numerical stability properties. Blaisdell *et al.* (1996) show that the skew-symmetric form of the convective term results in a reduced amplitude of the numerical errors in comparison to the conservative and nonconservative forms. Examples of the skew-symmetric methodology for discontinuous

Galerkin spectral element methods are presented by Gassner (2013, 2014). Approaches such as spectral vanishing viscosity (Karamanos & Karniadakis, 2000) modify the underlying discretised operators in order to add artificial dissipation at high polynomial orders, thereby reducing high-frequency oscillations in the approximate solution, stabilising the method and making it suitable for high Reynolds number large-eddy simulations (Kirby & Karniadakis, 2002; Pasquetti, 2006; Kirby & Sherwin, 2006*b*; Xu, 2006). Polynomial filtering (Gottlieb & Hesthaven, 2001; Fischer & Mullen, 2001; Fischer & Kruse, 2002; Hesthaven & Kirby, 2008) adopts a similar strategy, in which an interpolation-based filter is applied locally to each element at each timestep to prevent the buildup of unwanted oscillations in the solution field. An alternative route for stabilisation is represented by the variational multiscale schemes (Hughes *et al.*, 1998) which are based on the model used to approximate the coupling between unresolved and partially resolved scales (Wasberg *et al.*, 2009; Farhat *et al.*, 2006; Marras *et al.*, 2012). Finally, Gassner & Beck (2013) show the impact of exponential based modal filtering and over-integration on underresolved high-order turbulent flow computations and Warburton (2013) presents a DG method with guaranteed stability, obtained using a non-polynomial modified basis.

In this chapter, we focus on the concept of over-integration (Maday & Rønquist, 1990; Deville *et al.*, 2002; Kirby & Karniadakis, 2003; Kopriva, 2006; Kirby & Sherwin, 2006*b*), where over-integration refers to the use of more quadrature points than are necessary for a linear operator, but which might equally well be considered as consistent integration of the nonlinear operators.

In the formulation of spectral element methods, on any given element one may choose the number of expansion modes used to represent the approximation of a function, and the number of quadrature points used to calculate numerical estimates of integrals which may appear in the formulation.

A popular choice, as we have seen in previous chapters, is to couple a set of quadrature points with an equal number of polynomials (such as Lagrange polynomials) defined at the same points, leading to a collocation method. There are many examples of this throughout the literature, both in terms of the more traditionally utilised continuous Galerkin (CG) and discontinuous Galerkin (DG) formulations, as well as newer extensions such as the flux reconstruction (FR) technique as presented by Huynh (2007). In collocation methods, while most linear operators can be exactly integrated in this setting depending on the choice of quadrature, integrals of nonlinear terms typically incur numerical error. However, the computational efficiencies that can be attained through the use of a collocation formulation, in

particular the presence of a diagonal mass matrix, often outweigh the numerical error that is incurred.

To illustrate this point, it is well-known that polynomials of order P can be exactly integrated up to machine precision given some minimum number of quadrature points $Q_{\min}$ (Karniadakis & Sherwin, 2005); for example the Gauss-Lobatto-Legendre quadrature, which is widely adopted in the context of spectral element methods, allows one to obtain exact values of integrals for order $2P-3$ polynomials given P quadrature points (Orszag, 1972). However, when nonlinear quantities are calculated at the quadrature points, a collocation projection leads to an error known as *polynomial aliasing*, also referred to as *aliasing error* or simply *aliasing*, being introduced into the obtained integrals due to an insufficient number of quadrature points. This numerical error may be negligible for well-resolved simulations, for example flow simulations at low Reynolds numbers, but can become a critical issue for under- and marginally-resolved simulations such as typically arise in large eddy simulations. In addition, when we have non-polynomial functions (such as the compressible Euler and Navier-Stokes equations fluxes written in conservative form), consistent integration cannot be applied with a specific rule as in the case of polynomial functions. In this case, it is still possible to reduce consistently the aliasing errors but the number of quadrature points required might become computationally too costly.

The aim of this chapter is to illustrate effective and computationally efficient approaches based on consistent integration to resolve the aliasing issues arising in the DG and FR approaches. Specifically, we can write the DG and the FR approach by separating the linear terms from the nonlinear one, as follows:

$$\begin{aligned} \text{DG : } \quad \frac{du}{dt} &= \mathbf{M}^{-1} (\mathcal{L}_V + \mathcal{L}_I + \mathcal{N}_V + \mathcal{N}_I), \\ \text{FR : } \quad \frac{du}{dt} &= \mathcal{L}_V + \mathcal{L}_I + \mathcal{N}_V + \mathcal{N}_I, \end{aligned} \tag{5.1}$$

where u is the approximated solution, $\mathbf{M}$ is the mass matrix, $\mathcal{L}$ are the linear terms and $\mathcal{N}$ are the nonlinear terms that are responsible for the aliasing errors. Both linear and nonlinear terms are composed by a volumetric (V) and an interface (I) contribution. The first, (V), corresponds to the volumetric terms arising in the formulation while the second is related to the interface fluxes defined on the trace space connecting the elements. Note also that, in the term $\mathcal{N}$, we are including both nonlinear and “variable-coefficient¹” fluxes as well as spa-

¹The advection equation with spatially varying coefficients used in section 5.4 is still a linear equation (the advection velocity does not depend on the solution). In this case the aliasing depends on the non-constant

tially varying geometric terms due to a possibly curved geometry. Both, nonlinear/variable-coefficient fluxes and spatially varying geometric terms can result in aliasing errors if not exactly integrated and may generate aliasing-driven numerical instabilities.

To address these aliasing errors, we consider two different dealiasing strategies, both based on the concept of consistent integration. With the first approach we *locally* consistently integrate only the nonlinear terms produced by nonlinear/variable-coefficient fluxes, without addressing aliasing sources arising from the mapping used in deformed/curved meshes. Equation (5.1) can therefore be re-written as:

$$\begin{aligned} \text{DG: } \frac{du}{dt} &= \mathbf{M}^{-1} \Big|_Q \left(\mathcal{L}_V \Big|_Q + \mathcal{L}_I \Big|_Q + \mathcal{N}_V \Big|_{Q_V} + \mathcal{N}_I \Big|_{Q_I} \right), \\ \text{FR: } \frac{du}{dt} &= \mathcal{L}_V \Big|_Q + \mathcal{L}_I \Big|_Q + \mathcal{N}_V \Big|_{Q_V} + \mathcal{N}_I \Big|_{Q_I}, \end{aligned} \tag{5.2}$$

where Q is the number of quadrature points used for the linear term and the mass matrix, and Q_V and Q_I indicate the number of quadrature points used for the consistent integration of the volumetric and the interface part of the nonlinear terms, respectively. We remark the local behaviour of this strategy, which allows us to selectively adjust the quadrature used for the nonlinear terms.

In the second approach, we consistently integrate every term of the numerical discretisation which typically implies the over-integration of the linear terms. In this strategy we can address both PDE-aliasing driven instabilities as well as geometrical-aliasing sources arising from deformed/curved elements. Equation (5.1) becomes:

$$\begin{aligned} \text{DG: } \frac{du}{dt} &= \mathbf{M}^{-1} \Big|_Q \left(\mathcal{L}_V \Big|_Q + \mathcal{L}_I \Big|_Q + \mathcal{N}_V \Big|_Q + \mathcal{N}_I \Big|_Q \right), \\ \text{FR: } \frac{du}{dt} &= \mathcal{L}_V \Big|_Q + \mathcal{L}_I \Big|_Q + \mathcal{N}_V \Big|_Q + \mathcal{N}_I \Big|_Q, \end{aligned} \tag{5.3}$$

where the same number of quadrature points is used *globally* for all the terms. To summarise, the main novel contributions of this chapter are:

- Generalised analysis which encompasses DG and FR discretisations, highlighting the contribution of interface and volumetric terms.
- Local dealiasing strategy which exploits sum-factorisation type approach to improve computational efficiency.

velocity coefficients.

-
- Comparison of geometrical- and PDE-aliasing.

Part of the work presented in this chapter has been published in ‘Journal of Computational Physics’, (Mengaldo *et al.*, 2015a).

This chapter is structured as follows. In section 5.2, we characterise the main aliasing sources arising in DG and FR discretisations, in the context of a scalar conservations law. In section 5.3, we outline the *Local* and *Global* dealiasing strategies we apply to reduce the aliasing errors and improve the stability of the two discretisations considered. In this section, we also describe the implementation details which are required to apply such methodologies within a spectral/*hp* element framework. In section 5.4, we present some applications for each discretisation, by highlighting key examples in compressible and incompressible flows. Finally, in section 5.5, we draw a brief summary and the conclusions.

5.2 Aliasing sources in spectral/*hp* element methods

Aliasing errors in spectral/*hp* element methods typically arise because the terms appearing in the weak form of the equations being discretised are under-integrated. When using Gauss-Lobatto-Legendre (GLL) quadrature points, for instance, the minimum number of quadrature points $Q_{\min}$ necessary to exactly integrate any given order P polynomial $u(\xi) \in \mathbb{P}_P$ up to machine precision is

$$Q_{\min} = \frac{P + 3}{2}. \quad (5.4)$$

In Galerkin methods, we are usually interested in integrating the L^2 inner product of two polynomials (ϕ_p, ϕ_q) (defined in Eq. (2.4)), in order to compute the mass matrix of our discretised problem, where $\phi_p(\xi), \phi_q(\xi) \in \mathbb{P}_P$ are the test functions introduced in chapter 2, which, throughout this work, are the same as the expansion functions employed to represent the solution. By using Eq. (5.4) it is possible to calculate the minimum number of GLL quadrature points necessary for the quadrature to be exact as a function of the polynomial order P , as shown in table 5.1.

Polynomial order P	$Q_{\min}$
$[\phi(\xi)]^2 \in \mathbb{P}_{2P}$	$Q \geq P + 3/2$
$[\phi(\xi)]^3 \in \mathbb{P}_{3P}$	$Q \geq 3P/2 + 3/2$
$[\phi(\xi)]^4 \in \mathbb{P}_{4P}$	$Q \geq 2P + 3/2$

Table 5.1: Number of GLL quadrature points for the GLL quadrature to be exact up to machine precision as a function of the polynomial order of the integrand.

We note that to exactly integrate a linear problem, it is necessary to use $Q_{\min} = P + 2$ GLL quadrature points. For nonlinear problems the number of quadrature points needed increases, and, for quadratic nonlinearities it becomes $Q_{\min} = \frac{3}{2}(P + 1)$ whereas for cubic nonlinearities is $Q_{\min} = 2(P + 1)$. These values are valid as long as the nonlinearities are polynomials. If instead the nonlinearities are non-polynomial functions (for example when we have rational functions such as in the case of the compressible Navier-Stokes equations), the rules presented in table 5.1 no longer hold. In this case, it is necessary to apply a higher number of quadrature points to achieve an exact (or quasi-exact) integration. As mentioned in the introduction to this chapter, the nonlinearities we take into account are of two types:

- related to quasi-linear and nonlinear fluxes, also referred to as *PDE nonlinearities* or *PDE-aliasing sources*.
- related to deformed/curved meshes, also referred to as *geometrical nonlinearities* or *geometrical-aliasing sources*;

In this chapter, we present the results for the DG and FR approaches only. However, the dealiasing techniques presented can be applied to any spectral element method, including the continuous Galerkin approach. In the following, for both, the DG and the FR approach, we highlight the geometrical and the PDE nonlinearities on the two-dimensional conservation law reported below:

$$\frac{\partial u}{\partial t} + \nabla_{\mathbf{x}} \cdot \mathbf{f}(u) = 0, \quad (5.5)$$

where $u = u(\mathbf{x}, t)$ is the conserved variable and $\mathbf{f}(u) = [f_1(u), f_2(u)]^T$ is the flux vector which governs the transport of u , which can be linear, quasi-linear or nonlinear. Note that the form of the advection term in Eq. (5.5) can also affect the numerical stability. There is a rich literature, especially for the incompressible Navier-Stokes equations, where the convective term is represented in a skew-symmetric form in order to enhance stability despite losing conservation. In this regard, Malm *et al.* (2013) detailed the relationship between consistent integration and the skew-symmetric form of the convection operator, pointing out that for any Galerkin-based method the use of consistent integration recovers imaginary eigenvalues (i.e. skew-symmetry of the convective operator) and therefore stability. In this work we do not consider other possible choices for improving stability than consistent integration. Nevertheless, the connections between consistent integration and the skew-symmetric form of the convective term remains an interesting aspect which needs to be further explored (see also Gassner (2013, 2014)).

5.2.1 DG formulation

The matrix form of the weak DG method applied to Eq. (5.5) was derived in chapter 2 and it is reported here for the sake of clarity

$$\begin{aligned} \mathbf{M} \frac{d\mathbf{u}}{dt} + \left[\mathbf{S}_{\xi_2} \Lambda \left(\frac{\partial x_2}{\partial \xi_1} \right) - \mathbf{S}_{\xi_1} \Lambda \left(\frac{\partial x_2}{\partial \xi_2} \right) \right] \mathbf{f}_1(u^\delta) + \\ + \left[\mathbf{S}_{\xi_1} \Lambda \left(\frac{\partial x_1}{\partial \xi_2} \right) - \mathbf{S}_{\xi_2} \Lambda \left(\frac{\partial x_1}{\partial \xi_1} \right) \right] \mathbf{f}_2(u^\delta) + \mathbf{b}^{\text{DG}} = 0. \end{aligned} \quad (5.6)$$

From this equation it is possible to identify the aliasing sources arising in the context of the DG method. Specifically, PDE-aliasing sources are related to the flux functions $\mathbf{f}_1$ and $\mathbf{f}_2$. If they are nonlinear or quasi-linear (i.e. spatially varying coefficient), the overall degree of the integrand will increase leading to PDE-aliasing errors if the number of quadrature points is not incremented to tackle the degree of the nonlinearity.

On the other hand, the mass matrix $\mathbf{M}$ through the Jacobian and the geometric factors $\partial x_i / \partial \xi_j$ appearing in the volumetric and boundary terms are responsible for geometrical aliasing when the mesh is deformed or curved. In these cases in fact, the Jacobian and the geometric factors are spatially varying inside the elements and they are eventually represented through a polynomial of a certain degree depending on the curvature of the edges. The polynomial representing the metric terms couples with the other terms in the discretised equations producing integrand functions of a higher degree. Therefore, the number of quadrature points necessary to integrate exactly these higher-degree functions will increase and if not taken into account will lead to geometrical-aliasing errors.

As already mentioned, finding the exact number of quadrature points necessary for the integration to be exact might not be trivial. In particular, if the nonlinearities are non-polynomial functions, it is not possible to find a fixed number of quadrature points for the exact integration and this may well affect the performance of the solver.

Note also that one of the peculiarities of a discontinuous discretisation like the DG method, is the boundary term. This term introduces aliasing issues and it may well have a bigger or smaller impact than the volumetric term on the overall aliasing errors arising in the simulation being carried out. In this work, we have investigated this issue, by evaluating separately the contribution of the boundary term and of the volumetric term to the overall aliasing errors.

5.2.2 FR formulation

As for the DG method, we show again here the matrix form of the FR approach applied to Eq. (5.5) obtained in chapter 2:

$$\begin{aligned} \mathbf{B}\mathbf{\Lambda}(J_n) \frac{d\mathbf{u}^D}{dt} + \left[(\mathbf{D}_{\xi_1}\mathbf{B})^T \mathbf{\Lambda} \left(\frac{\partial x_2}{\partial \xi_1} \right) - (\mathbf{D}_{\xi_2}\mathbf{B})^T \mathbf{\Lambda} \left(\frac{\partial x_2}{\partial \xi_1} \right) \right] \mathbf{f}_1^D + \\ + \left[(\mathbf{D}_{\xi_2}\mathbf{B})^T \mathbf{\Lambda} \left(\frac{\partial x_1}{\partial \xi_1} \right) - (\mathbf{D}_{\xi_1}\mathbf{B})^T \mathbf{\Lambda} \left(\frac{\partial x_1}{\partial \xi_2} \right) \right] \mathbf{f}_2^D + \mathbf{b}^{\text{FR}} = 0. \end{aligned} \quad (5.7)$$

in order to highlight the nonlinearity sources. As in the case of the DG method, the flux functions $\mathbf{f}_1$ and $\mathbf{f}_2$ are responsible for PDE-aliasing sources if they are either nonlinear or they have spatially varying coefficients (such as the case of the quasi-linear advection equation, where the advection velocities are spatially varying across the domain).

On the other hand, the metric terms $\partial x_i / \partial \xi_j$ and the Jacobians are responsible for the geometrical nonlinearities.

As we have seen in the previous chapter, the DG and the FR approaches are identical for particular choices of the corrections functions. This, in one hand, indicates that the aliasing sources are the same and, when the nonlinearities are polynomials, the aliasing errors can be completely eliminated using the appropriate number of quadrature points. On the other hand, it also indicates that the dealiasing techniques presented in this chapter can be applied to the overall class of FR schemes recovered by the different forms of the corrections functions presented in chapter 2.

Finally, a similar consideration in terms of boundary term holds for the FR approach. Specifically, we have evaluated separately the contribution of the boundary and volumetric terms to the overall aliasing errors for one of the cases considered.

5.3 Dealiasing techniques

In this section we describe the two dealiasing approaches considered in this work. Specifically we present the following two strategies, both based on the concept of consistent integration:

Local dealiasing

It targets PDE-aliasing sources - i.e. those related to the nonlinear (and quasi-linear) fluxes - and it applies a consistent integration of them locally, without considering geometrical-aliasing sources;

Global dealiasing

It targets both PDE- and geometrical-aliasing sources by consistently integrating the nonlinear (and quasi-linear) fluxes, the geometric factors, the mass matrix and the boundary term (i.e. the consistent integration is applied to the overall right-hand side of the equations being discretised).

We remark here that the word *consistent integration* refers to polynomial nonlinearities, where it is effectively possible to know a priori the number of quadrature points for the integration to be exact, thus consistently integrate the nonlinearities. When non-polynomial functions are present, the concept of consistent integration is out of focus because it is not possible to fully control the quadrature error, although it might still be possible for a higher number of quadrature points.

5.3.1 *Local* dealiasing

The first dealiasing technique presented, namely *Local* dealiasing, involves the consistent integration of the nonlinear (quasi-linear) flux terms of Eq. (5.5)). This approach can be applied to both the discontinuous approaches considered in this work and, more generally, it can be applied to any spectral/*hp* element method. As mentioned before, this dealiasing technique addresses the PDE-aliasing errors directly arising from the degree of nonlinearity of the equation(s). It does not tackle geometrical nonlinearities.

In the following, we first outline the one-dimensional case and we successively describe the extension to two-/three-dimensional tensor-product meshes. The implementation presented here has been carried out in the spectral/*hp* element library Nektar++. This implementation, and in particular the extension to multi-dimensional tensor-product meshes for both regular and irregular/curvilinear elements, is computationally efficient for high polynomial orders due to the use of the sum-factorisation technique introduced in chapter 2.

***Local* dealiasing in one dimension**

Consider a one-dimensional approximate solution polynomial $u(\xi)$ of order P represented by a nodal expansion $\ell_i(\xi)$, such as Lagrange polynomials, on a set of $Q_P = P+1$ points:

$$u^{\delta Q_P}(\xi) = \sum_{i=0}^P u_i \ell_i(\xi), \quad Q_P = P + 1. \quad (5.8)$$

The *Local* dealiasing strategy consists of the three steps reported in the following.

A.) *One-dimensional interpolation*

The solution $u^{\delta Q_P}(\xi)$ is interpolated onto a larger set of points $Q = \tilde{P} + 1$ where $\tilde{P} > P$ (note that this operation does not change the order of the polynomial which remains unaltered and equal to P):

$$u^{\delta Q}(\xi_j) = \sum_{i=0}^P u_i \ell_i(\xi_j), \quad 0 \leq j \leq \tilde{P}, \quad Q = \tilde{P} + 1. \quad (5.9)$$

The larger set of points Q required depends on the degree of the nonlinearity and for polynomial functions can be determined a priori (as shown in table 5.1 for GLL points).

Equation (5.9) can be rewritten in compact form by introducing the interpolation matrix $\mathbf{Q}_{Q_P \rightarrow Q}$ of dimensions $Q \times Q_P$ and by using the following matrix-vector multiplication:

$$\mathbf{u}^{\delta Q} = \mathbf{Q}_{Q_P \rightarrow Q} \mathbf{u}^{Q_P}, \quad \mathbf{Q}_{Q_P \rightarrow Q}[i, j] = \ell_i^{Q_P}(\xi_j). \quad (5.10)$$

After having performed this operation, the solution is represented in a richer set of points. The larger set of points not necessarily needs to have the same distribution of the original set of points. Effectively, Gauss-Legendre points does produce better results in terms of evaluation of the nonlinear terms than Gauss-Lobatto-Legendre. Therefore, if the original points are of Gauss-Lobatto-Legendre type for example, it is always possible to interpolate the solution from the original Gauss-Lobatto-Legendre points onto the different (and possibly larger) set of Gauss-Legendre points.

B.) *Collocation product*

The nonlinear term(s) is evaluated on the larger set of points Q using the interpolated solution values $u^Q(\xi_j)$ obtained at step A. If, for simplicity, the nonlinearity is quadratic then the product is calculated as follows:

$$f^{\delta Q}(\xi_j) = u^{\delta Q}(\xi_j) \cdot u^{\delta Q}(\xi_j), \quad 0 \leq j \leq \tilde{P}. \quad (5.11)$$

It is important to remark here that the nonlinearity $f^{\delta Q}(\xi_j)$ can well be a non-polynomial function (such as in the case of the compressible Euler and Navier-Stokes equations). In this case, the aforementioned rules for the exact integration (see table 5.1) no longer hold.

C.) *Galerkin projection*

Finally, we perform a projection onto the original set of points. In the following we use

a Galerkin projection, which is equivalent to a least squares projection of the nonlinear term evaluated on the larger set of points onto the original set of points, so that

$$\left[f^{\delta Q}(\xi) = u^{\delta Q}(\xi_j) \cdot u^{\delta Q}(\xi_j) \right] \xrightarrow{Q \rightarrow Q_P} f^{\delta Q_P}(\xi) \quad (5.12)$$

In a similar manner to step A, this can be interpreted as a matrix-vector multiplication

$$\mathbf{f}^{\delta Q_P} = \mathbf{P}_{Q \rightarrow Q_P} \mathbf{f}^Q, \quad (5.13)$$

where $\mathbf{P}_{Q \rightarrow Q_P}$ is the Galerkin projection matrix that can be written as

$$\mathbf{P} = \mathbf{M}^{-1} \mathbf{Q}_{Q_P \rightarrow Q}^T \mathbf{W}, \quad (5.14)$$

with $\mathbf{M}[i, j] = \int \ell_i^{Q_P} \ell_j^{Q_P} d\xi$ being the mass matrix and $\mathbf{W}[i, i] = \int \ell_i^Q d\xi$ being the diagonal Gauss quadrature weight matrix associated to Q points. Note that the dimensions of $\mathbf{P}_{Q \rightarrow Q_P}$ are $Q_P \times Q$.

To summarise the steps just presented, figure 5.1 depicts a schematic representation of the *Local* dealiasing technique for one-dimensional problems.

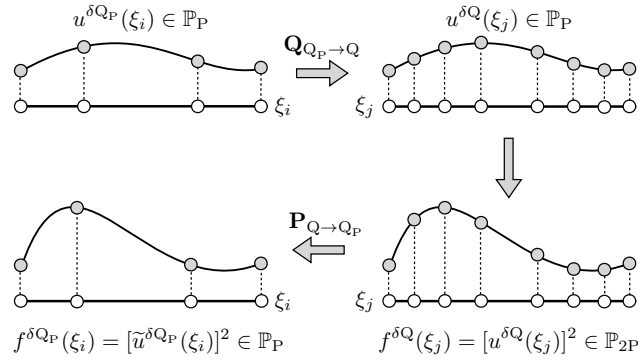


Figure 5.1: Conceptual flow chart of the *Local* dealiasing approach through consistent integration of the nonlinear terms for one-dimensional problems.

This procedure is exact when the iso-parametric mapping $x_i = \Theta_i(\xi_1, \xi_2)$ which describes the coordinates of a physical element Ω_n is affine and therefore the Jacobian is constant. In addition, if $\mathbf{M}$ is diagonal then $\mathbf{M}$ has diagonal components of the weights at Q_P set of points. In this case, the projection is an interpolation matrix with the rows scaled by the inverse of the Q_P quadrature point weights whereas the columns are rescaled by the quadrature point integration weights.

Local dealiasing for two-/three-dimensional tensor-product meshes

For the sake of simplicity, in this section we consider the two-dimensional case and we remark where necessary the differences in terms of computational costs between the two- and the three-dimensional case.

Consider a two-dimensional approximate polynomial solution $u(\xi_1, \xi_2)$ of order P represented through a nodal expansion basis, such as Lagrange polynomials, on a set of $Q_P \times Q_P = (P+1) \times (P+1)$ points

$$u^{\delta Q_P}(\xi_1, \xi_2) = \sum_{i,j=0}^P u_{ij} \ell_i(\xi_1) \ell_j(\xi_2). \quad (5.15)$$

In the two-dimensional (equivalently three-dimensional) case, the dealiasing technique consists of the same conceptual steps as the one-dimensional case. The only difference is that the operations to perform are more numerous and consequently the overall procedure is more costly from a computational point of view. However, the exploitation of the tensor-product properties allows an efficient implementation through the use of the sum-factorisation technique.

A.) Two-/three-dimensional interpolation

The first step involves interpolating the solution $u(\xi_1, \xi_2)$ onto a larger set of $Q \times Q = (\tilde{P}+1) \times (\tilde{P}+1)$ points, where $\tilde{P} > P$:

$$u^{\delta Q}(\xi_{1r}, \xi_{2s}) = \sum_{i,j=0}^{\tilde{P}} u_{ij} \ell_i(\xi_{1r}) \ell_j(\xi_{2s}), \quad 0 \leq r, s \leq \tilde{P}. \quad (5.16)$$

As noted in the one-dimensional case, the number of points Q required depends on the degree of the nonlinearity and on the point distribution chosen. The interpolation in Eq. (5.16) is divided into two (three in the three-dimensional case) sub-steps as shown in figure 5.2. The first sub-step, figure 5.2(a)), is the interpolation of the solution along the ξ_1 -direction whereas the second, figure 5.2(b), is the interpolation along the ξ_2 -direction. This implementation allows a reduction of the floating point operations required.

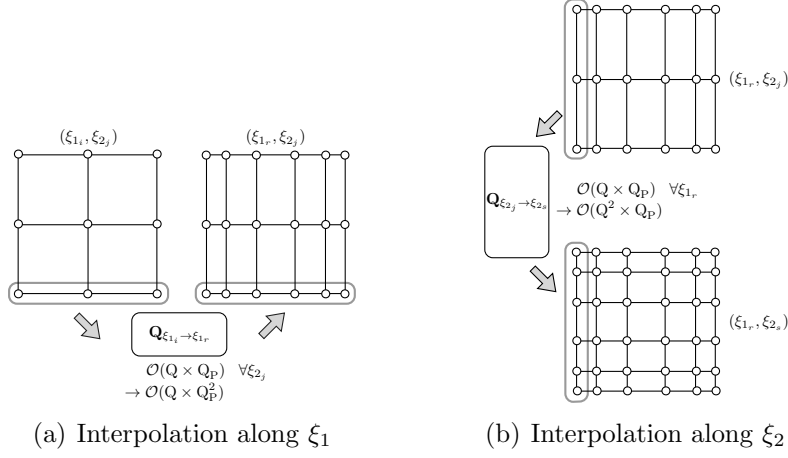


Figure 5.2: Interpolation for two-dimensional tensor-product elements.

Specifically, the overall number of floating point operations required is proportional to $\mathcal{O}(Q \times Q_P^2 + Q^2 \times Q_P)$ while by doing a two-dimensional interpolation for a non-tensor product basis the floating point operations needed become proportional to $\mathcal{O}(Q^2 \times Q_P^2)$. Following a similar procedure for the three-dimensional case gives a number of floating point operations proportional to $\mathcal{O}(Q \times Q_P^2 + Q^2 \times Q_P^2 + Q^3 \times Q_P)$ versus $\mathcal{O}(Q_P^3 \times Q^3)$ operations otherwise required for a full three-dimensional interpolation. Further details of this type of sum factorisation can be found in Karniadakis & Sherwin (2005).

B.) Collocation product

The second step involves evaluating the nonlinear term on the larger set of points $Q \times Q$ using the interpolated values of the solution $u^{\delta Q}(\xi_{1r}, \xi_{2s})$ obtained at step A. Similar to the one-dimensional case, the evaluation of the nonlinear term(s) is achieved by the following collocation product:

$$f^{\delta Q}(\xi_{1r}, \xi_{2s}) = u^{\delta Q}(\xi_{1r}, \xi_{2s}) \cdot u^{\delta Q}(\xi_{1r}, \xi_{2s}), \quad 0 \leq r, s \leq \tilde{P}, \quad (5.17)$$

where we considered a simple quadratic nonlinearity for the sake of simplicity.

C.) Galerkin projection

The third step involves performing a projection of the nonlinear term onto the original set of points $Q_P \times Q_P$. The type of projection used is the Galerkin projection as for the one-dimensional case. However, for the two-/three-dimensional case this operation is divided into two sequential sub-operations as shown in figure 5.3. First we perform a

Galerkin projection in the ξ_2 -direction, figure 5.3(a), and then we perform a Galerkin projection in the ξ_1 -direction figure 5.3(b).

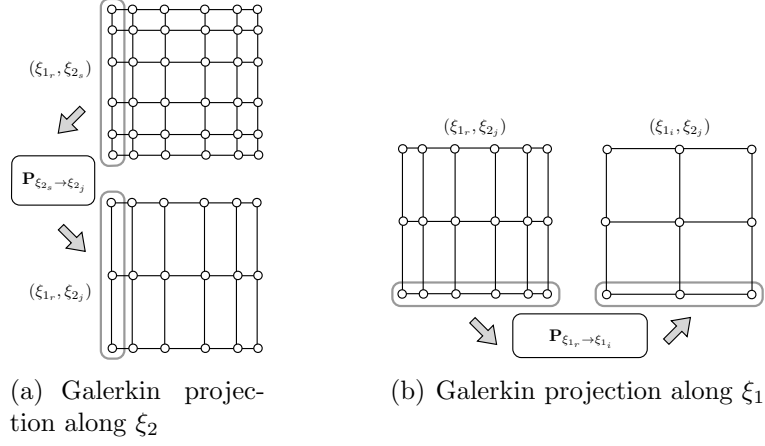


Figure 5.3: Galerkin projection for two-dimensional tensor-product elements.

In figure 5.4 we show an overview of the steps above for the two-/three-dimensional tensor-product case.

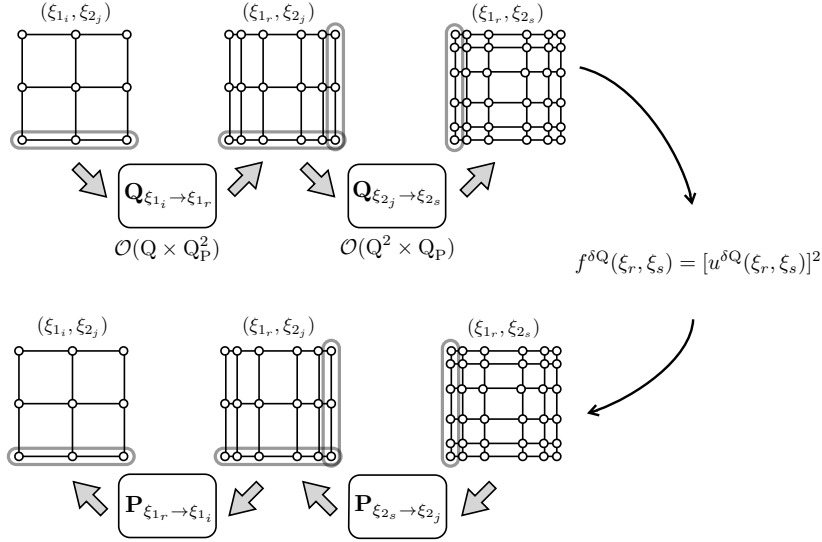


Figure 5.4: Conceptual flow chart of the *Local* dealiasing approach for the two-/three-dimensional tensor-product case.

A similar tensor-product based approach can also be used for triangles in two dimensions as well as prismatic and tetrahedral elements in three dimensions as reported in appendix C.

5.3.2 *Global* dealiasing

The second dealiasing technique proposed, namely *Global* dealiasing, involves the consistent integration of both PDE and geometrical nonlinearities. From this perspective, it is necessary to perform the following steps:

- A.) Interpolate the solution onto a larger set of points, $u^P \rightarrow u^Q$ with $Q > Q_P$;
- B.) Evaluate the nonlinear term(s) on the larger set of points $f(u^{Q_P}) \rightarrow f(u^Q)$;
- C.) Interpolate the geometric factors $\frac{\partial x_i}{\partial \xi_j}$ onto the larger set of points Q ;
- D.) Construct the derivation matrices $\mathbf{D}_{\xi_i}$ using the larger set of points;
- E.) Interpolate the boundary terms $\mathbf{b}^{\text{DG}}/\mathbf{b}^{\text{FR}}$ onto the larger set of points Q ;
- F.) Assemble the overall right-hand side on the larger set of points Q ;
- G.) Project the right-hand side to the original set of points Q_P through a Galerkin projection.

Note that we do not make any distinctions between the one- or two-/three-dimensional cases, nor between different element shapes, since here we do not explicitly exploit the tensor product advantages (although some operators may still have a tensor product form) used in the previous section for the *Local* dealiasing technique. This approach is similar in nature to the one presented by Kirby & Karniadakis (2003). However, here we additionally interpolate the geometric terms, the Jacobians and the geometric factors defined in Eq. (2.15), onto a larger set of points (and therefore the overall integrals are evaluated onto the larger set of points). The *Global* dealiasing technique is more floating point operation demanding than the *Local* dealiasing but it can address both PDE- and geometrical-aliasing errors.

5.4 Numerical results

In this section, we apply the dealiasing techniques outlined above to a simple transport advection equation and to compressible and incompressible flow simulations.

The results obtained on the transport advection equation (for which a periodic solution is known) have the purpose of evaluating quantitatively the effects of the dealiasing techniques in terms of both PDE- and geometrical-aliasing errors.

Specifically, a first set of experiments was performed on regular meshes and the advection equation considered had an advection velocity that was a nonlinear polynomial in space. The

results achieved on this first set of simulation permits, on one hand, a better understanding of the role of the boundary terms in discontinuous numerical discretisations and, on the other hand, the evaluation of the inequalities which lead to exact integration of discretised nonlinear functionals (note that the word ‘nonlinear’ refers to the polynomial representing the advection velocity). It also allows the quantification of the volumetric and boundary contributions to the aliasing errors observed.

A second set of experiments was instead carried out on various deformed and curvilinear meshes in order to show the role of the *Global* dealiasing technique in tackling geometrical-aliasing errors (note that in this case the advection velocity used was constant in order to have only geometrical nonlinearities) and to highlight the link between geometrical- and PDE-aliasing errors.

After these quantitative analyses, we consider some qualitative effects that dealiasing plays in compressible and incompressible flow simulations. Specifically, we carried out a set of simulations of a compressible inviscid flow past a cylinder for different quadrature orders and a simulation of an incompressible flow past a wing tip with and without dealiasing. The results obtained in both cases allows us to show how consistent integration has an effective role in stabilising these simulations. However, even if in the examples presented the dealiasing techniques show a stabilising effect, dealiasing does not guarantee stability and in other flow studies some numerical instabilities were experienced even applying a consistent integration of the flux functions and of the geometric terms. In these cases, for stabilising the simulations, we needed to add some artificial viscosity of the type reported in (Kirby & Sherwin, 2006*b*).

In all the experiments performed we chose Gauss-Lobatto-Legendre points for the consistent integration. The use of a Gauss-Legendre quadrature would have been more efficient for reducing the aliasing issues but, for the scope of this chapter, the results on Gauss-Lobatto-Legendre points are sufficient.

5.4.1 Quantification of PDE-aliasing errors

To understand the effect of the dealiasing techniques presented in the previous section, we considered the following two-dimensional transport equation

$$\begin{cases} \frac{\partial u}{\partial t} + a_x \frac{\partial u}{\partial x} + a_y \frac{\partial u}{\partial y} = 0 \\ u(x, y, t = 0) = \exp\{-41[(x + 0.3)^2 + y^2]\} \\ u(x_b, y_b, t) = 0, \\ a_x = \pi y^{P_{adv}} g(t), \quad a_y = -\pi x^{P_{adv}} g(t), \end{cases} \quad (5.18)$$

where u is the conserved variable, a_x and a_y are the advection velocities along the x - and the y -direction, respectively, x_b and y_b denote the boundaries of the computational domain, P_{adv} is the polynomial order of the advection velocities, thus the polynomial order of the flux and $g(t)$ is the following time dependent periodic function

$$g(t) = \cos(\pi t/T) \quad (5.19)$$

on a time interval $[0, T]$. The solution u evolves in time in such a way that the initial data is recovered at every period T , so that $u(x, y, 0) = u(x, y, T)$. In all the numerical experiments, we selected a period $T = 1$ and a final time equal to $4T$ so that the final solution was identical to the initial one, in order to calculate the error. This strategy is the same as that adopted by LeVeque (1996) and it is widely used to evaluate properties of numerical methods when the exact solution is not available.

For each simulation we used a forward Euler time integration scheme and the time step was chosen to be sufficiently small so as to consider the temporal error negligible². The polynomial order was set to $P = 4$ and the initial condition were given by a collocation projection (i.e. using $Q = 5$ quadrature points).

Throughout the following we will make extensive use of the L_2 error defined in Eq. (2.101) and reported here for the reader's convenience:

$$E_{L2} = \sqrt{\sum_{i=0}^{Q-1} (u_i - u_{i,exact})^2 w_i}. \quad (5.20)$$

²We remark that the results throughout this section hold for any explicit time integration scheme such as 2nd or 4th order Runge-Kutta schemes.

The meshes adopted are depicted in figure 5.5. They are both centred in $(x, y) = (0, 0)$ and the mesh in figure 5.5(a) possesses interfaces between adjacent elements whose coordinates have either positive or negative values, while the mesh in figure 5.5(b) has interfaces whose coordinates assume both positive and negative values.

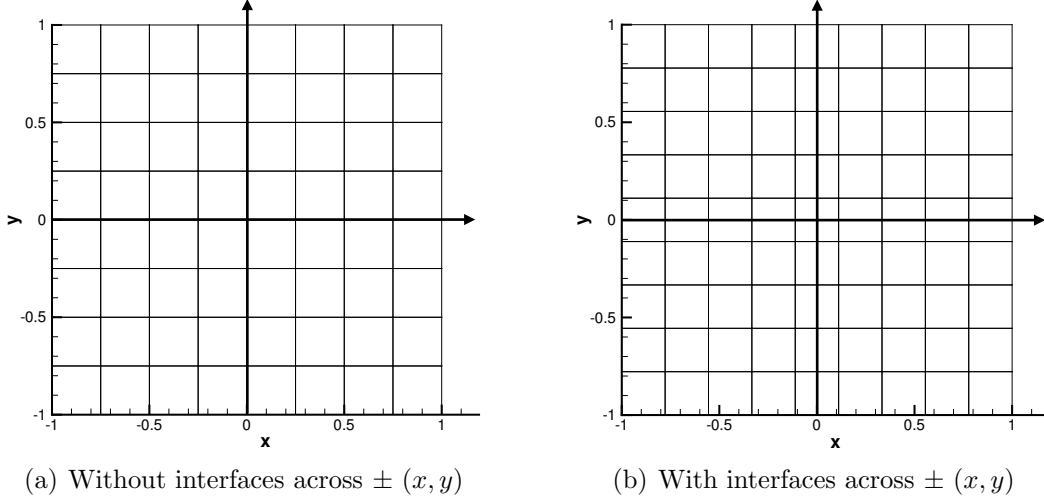


Figure 5.5: Meshes for the two-dimensional linear advection test cases: Mesh A in figure 5.5(a) and Mesh B in figure 5.5(a).

In the following, we first present the comparison between the two different meshes in figure 5.5 for $P_{adv} = 1$. This will provide some insights into the importance of the numerical fluxes (i.e. the boundary term) when dealing with discontinuous discretisations. We successively consider just the mesh in figure 5.5(a) to evaluate the role of advection velocities having an order higher than $P_{adv} = 1$. In addition, we will show the role of the boundary term if compared to the volumetric one when increasing the order of the flux function.

5.4.1.1 $P_{adv} = 1$: Influence of the mesh on the boundary term

In this section we show the behaviour of the *Local* and *Global* dealiasing techniques applied to the FR_{g2} approach and the DG_{SEM} method using an exact mass matrix (EMM) (i.e. a DG_{SEM} method with a mass matrix which was precomputed using $Q=6$ Gauss-Lobatto-Legendre points) for the two meshes in figure 5.5. We remark that ‘EMM’ refers to a problem where no spatially-varying geometric terms are present. If we considered a deformed/curvilinear mesh, the Jacobian of the transformation between the physical and the standard space would have increased the number of points necessary for the mass matrix to be exact. Therefore, using

the *Global* dealiasing strategy, which effectively acts on the mass matrix, would have changed the mass matrix up to the polynomial degree of the geometrical nonlinearity introduced by the mesh. We will further explore this aspect in section 5.4.2.

The first line of table 5.2 refers to the L_2 errors without any dealiasing applied, the successive six lines instead illustrate the L_2 errors for *Local* dealiasing either applied to the volumetric term only or to both the volumetric and the interface term. We remind the reader that Q_V and Q_I refers to additional points added for the consistent integration of the volumetric term and the interface term (boundary term), respectively. Finally, the last three lines show the L_2 errors applying *Global* dealiasing.

From these results it is possible to note that the error does not change (up to machine precision) when using six (or more) quadrature points for the mesh in figure 5.5(a) (Mesh A) and for both the dealiasing techniques presented. We say that the error saturates for $Q=6$. If we instead observe the L_2 errors for the mesh in figure 5.5(b) (Mesh B), a saturation of the error is not achieved even for $Q=8$ when applying *Local* dealiasing which involves the interfaces or when applying *Global* dealiasing (which automatically involves dealiasing the interfaces). This behaviour can be simply explained considering figure 5.6.

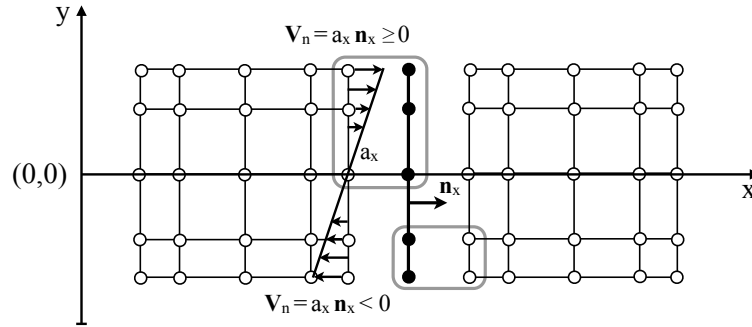


Figure 5.6: Upwind flux at the interface between two elements: Aliasing arising from mixed information coming from the left and the right element.

In fact, for Mesh A, the configuration of the interfaces across the overall domain produces numerical fluxes (therefore boundary terms) which are polynomial functions, since their values come from just one side with respect to a given interface³. In the case of Mesh B instead, the numerical fluxes are formed by values coming from both the adjacent elements sharing a given interface, therefore they are non-polynomial functions and error saturation up to machine precision cannot be expected. This means that, in general, it is not possible to fully

³Note that all the simulations in this and the following section were run using an upwind flux as numerical flux.

control PDE-aliasing errors arising from the boundary term because in this case the function representing the numerical flux does not lie in the polynomial space used for approximating the equation. However, the error does decrease as the number of quadrature points is increased.

In addition, table 5.2 shows that the *Local* (when dealiasing both the boundary and the volumetric terms - lines 5,6 and 7 of the table) and *Global* dealiasing techniques are equivalent for this particular case, because the meshes adopted are regular and therefore the geometric terms are constant (i.e. the geometry does not introduce any nonlinearity).

	Mesh A		Mesh B	
	DG _{SEM} -EMM	FR _{g2}	DG _{SEM} -EMM	FR _{g2}
Q=5	0.00258809184895	0.00517893324999	0.00164993651569	0.00279358970635
Q=5, Q _V = 6, Q _I = 5	0.00245321968401	0.00484792022348	0.00157503364988	0.00265346210971
Q= 5, Q _V = 7, Q _I = 5	0.00245321968401	0.00484792022348	0.00157503364988	0.00265346210971
Q= 5, Q _V = 8, Q _I = 5	0.00245321968401	0.00484792022348	0.00157503364988	0.00265346210971
Q= 5, Q _V = 6, Q _I = 6	0.00238095562954	0.00483526988467	0.00155395894760	0.00266014915428
Q= 5, Q _V = 7, Q _I = 7	0.00238095562954	0.00483526988467	0.00154977027283	0.00265747641942
Q= 5, Q _V = 8, Q _I = 8	0.00238095562954	0.00483526988467	0.00155235541132	0.00265532284115
Q= 6	0.00238095562953	0.00238095562953	0.00155395894760	0.00155395894760
Q= 7	0.00238095562953	0.00238095562953	0.00154977027283	0.00155235541132
Q= 8	0.00238095562953	0.00238095562953	0.00155235541132	0.00155131919159

Table 5.2: L₂ errors for the *Local* and *Global* dealiasing techniques applied to DG and FR formulations for both meshes: Mesh A and Mesh B.

5.4.1.2 $P_{adv} \geq 1$: error saturation

In this subsection we show the results with advection velocities having a degree $P_{adv} \geq 1$, using the same simulation settings as in the previous subsection and employing the mesh in figure 5.5(a) to avoid the aforementioned non-saturation of the error when dealiasing the interfaces.

In figure 5.7, each point represents the difference (in absolute value) between the L₂ error calculated using (Q_V, Q_I) quadrature points and the L₂ error calculated using $(Q_V + 1, Q_I + 1)$ quadrature points for the DG_{SEM}-EMM method and for the FR_{g2} scheme. We note that the machine-precision saturation of the error satisfies the inequality:

$$Q_{\min} \geq P + \frac{P_{adv}}{2} + \frac{3}{2}, \quad (5.21)$$

$$P \geq P_{adv},$$

where $Q_{\min}$ is the minimum number of quadrature points to exactly integrate the PDE nonlinearity associated the polynomial degree of the advection velocity P_{adv} and for a given

expansion order P .

In appendix B, tables B.1, B.2, B.3, and B.4, show the tabulated values associated to figure 5.7. From these results, it is possible to see how, for both the discontinuous formulations, the use of *Local* dealiasing on both the volumetric and interface fluxes was, for all of the cases considered, the best choice in terms of L_2 error among the dealiasing techniques applied. Also note that applying dealiasing does not guarantee a better L_2 error.

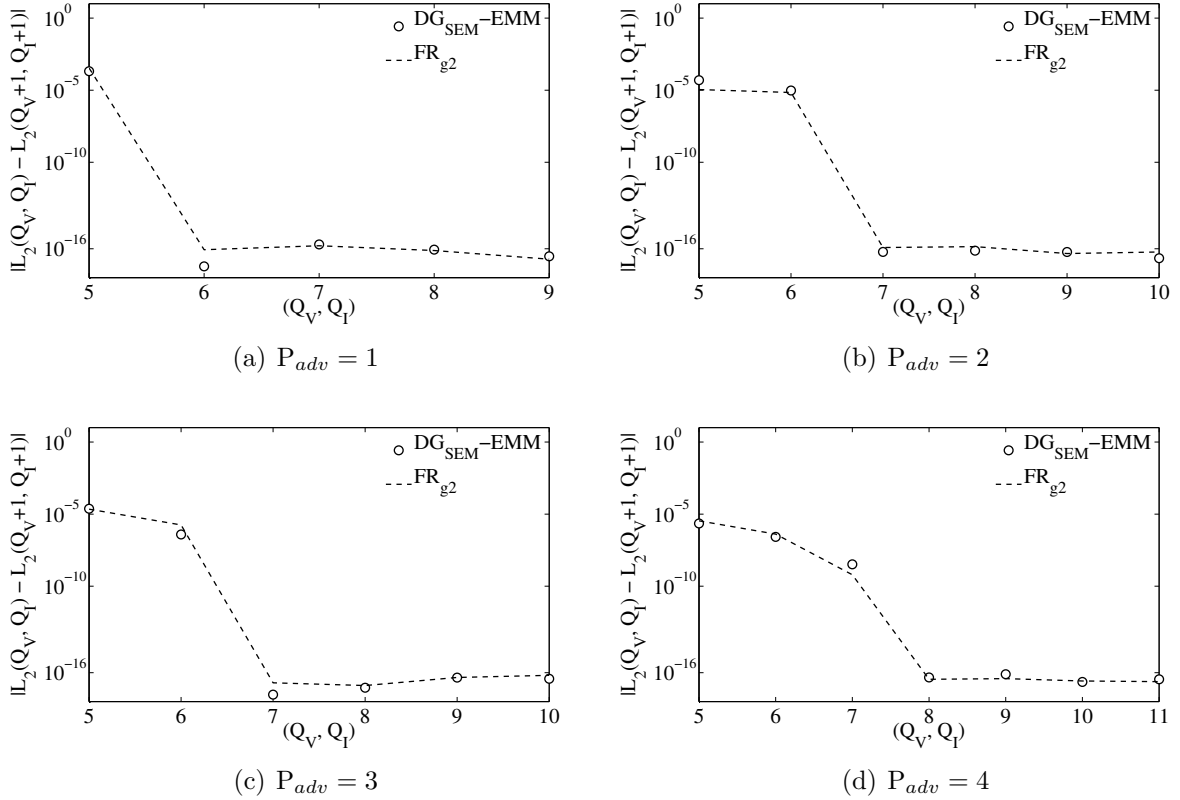


Figure 5.7: Differences between the L_2 errors obtained with (Q_I, Q_V) and $(Q_I + 1, Q_V + 1)$ vs (Q_I, Q_V) using the *Local* dealiasing technique for FR_{g2} and DG_{SEM} with an exact mass matrix.

5.4.1.3 $P_{adv} \geq 1$: boundary term vs. volumetric term

To further quantify the effects of interface vs. volumetric dealiasing introduced in the previous subsection, in table 5.3 we show the comparative contribution of the volumetric and of the interface dealiasing. Specifically, in the first two columns of table 5.3 we show the relative differences between the L_2 errors obtained for *Local* dealiasing of the volumetric flux only

and of both the volumetric and the interface terms:

$$\Delta E_{L2_V} = \frac{(E_{L2}(Q_V)_{sat} - E_{L2}(Q_V, Q_I)_{sat})}{E_{L2}(Q_V, Q_I)_{sat}} \cdot 100, \quad (5.22)$$

for the DG_{SEM} -EMM and FR_{g2} methods. In the above equation, Eq. (5.22), the subscript ‘*sat*’ indicates saturation (i.e. the associated L_2 errors were taken at the point where they saturate, according to figure 5.7).

The second two columns show the relative differences between the L_2 errors obtained for *Local* dealiasing of the interface fluxes only and of both the volumetric and the interface terms:

$$\Delta E_{L2_I} = \frac{(E_{L2}(Q_I)_{sat} - E_{L2}(Q_V, Q_I)_{sat})}{E_{L2}(Q_V, Q_I)_{sat}} \cdot 100, \quad (5.23)$$

again for the DG_{SEM} -EMM and FR_{g2} methods. From this table we can see how the difference between performing only volumetric dealiasing and volumetric plus interface dealiasing increases as P_{adv} increases except for the last case ($P_{adv} = 4$). This is related to the fact that the test case being considered is likely to underestimate the role of the interfaces, because as P_{adv} increases, the magnitude of the advection velocity decreases with the power of the order of the advection velocity P_{adv} (since the mesh is contained within the square $[-1, 1] \times [-1, 1]$). Nevertheless, it is still possible to see that the contribution of the interfaces becomes greater as P_{adv} increases and it is effectively comparable to the contribution of the volumetric dealiasing for $P_{adv} = 2, 3$ and 4.

This result is crucial, because it indicates that, in some cases, the interface dealiasing is as important as the volumetric dealiasing and, therefore the boundary term arising in discontinuous discretisations must be taken into account when implementing a dealiasing strategy.

On the other hand, it is also important to remember that aliasing errors arising in the boundary term cannot be always fully controlled because of their generally non-polynomial nature (as shown in section 5.4.1.1).

	ΔE_{L2_V}		ΔE_{L2_I}	
	DG_{SEM}-EMM	FR_{g2}	DG_{SEM}-EMM	FR_{g2}
$P_{adv} = 1$	3.0351	0.2616	15.4071	7.7022
$P_{adv} = 2$	17.8755	5.5913	15.7231	3.2610
$P_{adv} = 3$	19.2840	10.4100	19.1376	12.4640
$P_{adv} = 4$	7.6039	5.9054	6.4057	3.5643

Table 5.3: Percentage difference between the L_2 errors for *Local* (Q_V), *Local* (Q_I) and *Local* (Q_V, Q_I) dealiasing at error saturation.

5.4.2 Link between geometrical- and PDE-aliasing errors

Having examined the aliasing effects due to the nonlinearities of the flux function (PDE-aliasing errors), in this section we now show how geometrical-aliasing can be linked to PDE-aliasing (despite some differences arising from the fact that geometrical nonlinearities play a slightly different role in the underlying formulation if compared to PDE nonlinearities). We consider a mesh composed of a single standard quadrilateral element $\Omega_{st} = [-1, 1] \times [-1, 1]$, to which we curve either only the top edge, or both the top and left edges. For each configuration, we apply different curvatures to the edges by means of Legendre polynomials of order $P_{geom} = 1, 2, 3$ and 4. In figure 5.8(a) and (5.8(b)), we show the mesh with only the top edge curved (also referred to as Case A) and the mesh with both the top and left edges curved (also referred to as Case B), respectively. Both figures refers to $P_{geom} = 4$.

In each case, the iso-parametric mapping, and therefore the Jacobian, is represented by an expansion of basis functions over the standard element, so that, for example, a $P_{geom} = 4$ expansion contains $5 \times 5 = 25$ different unique modes. Figure 5.9 shows the magnitude of each modal coefficients of the Jacobian for the 4th order case when projected onto an orthonormal basis of order 5 (leading to 36 coefficients), in order to emphasise that the polynomial space is sufficiently large to capture the Jacobian mapping. In particular we note that the Jacobian for both cases is effectively of fourth order, but in the case of figure 5.9(a) we have nonzero modes only in the x -direction (up to the fifth coefficient) whereas in the case of figure 5.9(b) the nonzero modes span both the x - and y -direction (up to the fifth coefficient). The sixth coefficient in both directions is zero, showing adequate support for the Jacobian expansion.

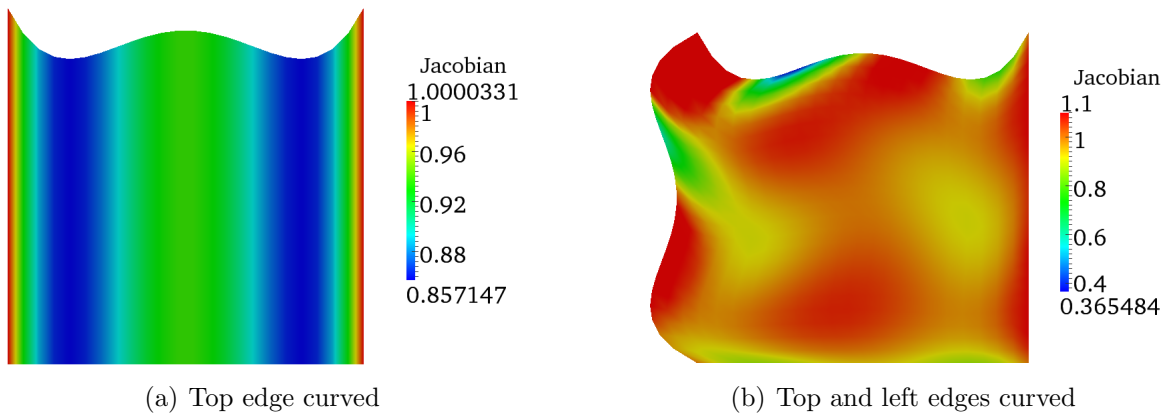


Figure 5.8: Examples of 4th-order meshes employed to investigate geometrical aliasing.

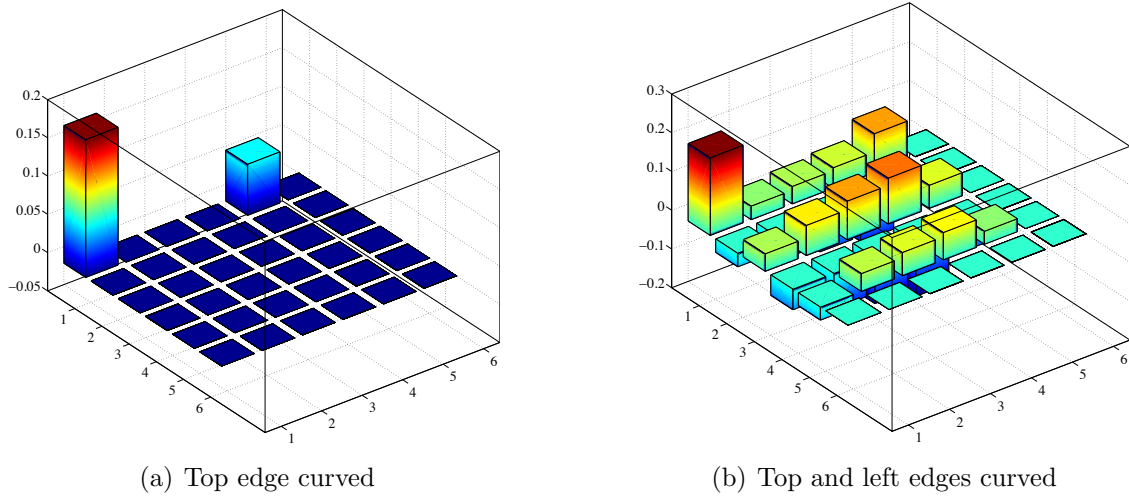


Figure 5.9: Coefficients in an orthonormal expansion basis for the 4th-order meshes showed in figure 5.8.

The test case considered is identical to the one presented in section 5.4.1 (Eq. (5.18)), aside from the use of spatially-constant advection velocities

$$a_x = \frac{\pi}{2} g(t), \quad a_y = -\frac{\pi}{2} g(t), \quad (5.24)$$

where $g(t)$ is defined in Eq. (5.19)). We choose a period $T = 0.5$ and a final time $40T$. For the time-integration we use a 2nd-order Runge-Kutta scheme while the polynomial order was $P = 14$ in order to have a sufficiently resolved problem, and we use a collocation projection of the initial condition. Throughout this section we use the *Global* dealiasing technique in order to target the geometrical aliasing.

We first present estimates of the quadrature necessary for correctly integrating a deformed or curved mesh. For doing so, we take into account the leading order within the problem under investigation, which is the mass matrix in the case of the DG discretisation. In figure 5.10, each point denotes the difference between two mass matrices, the first obtained for Q and the second for $Q+1$ quadrature points:

$$\|\Delta M_{ij}\|_{L2} = \sqrt{\sum_{ij} [M_{ij}(Q) - M_{ij}(Q+1)]^2}. \quad (5.25)$$

Table B.5 in appendix B quantifies the results presented in figure 5.10. We note that the mass matrix does not change (approximately up to machine precision) after we have reached

the minimum number of quadrature points to exactly integrate the polynomial order used for describing the geometry, given by the inequality

$$Q_{\min} \geq P + \frac{P_{geom}}{2} + \frac{3}{2}, \quad (5.26)$$

$$P \geq P_{geom},$$

where $Q_{\min}$ is the minimum number of quadrature points to exactly integrate the mass matrix for a given polynomial of order P_{geom} describing the geometry and for a given expansion order P . Equation (5.26) is identical to Eq. (5.21) with the only exception that P_{geom} now refers to the polynomial order of the geometry deformation. The above result is consistent with the Gauss-Lobatto-Legendre quadrature rules.

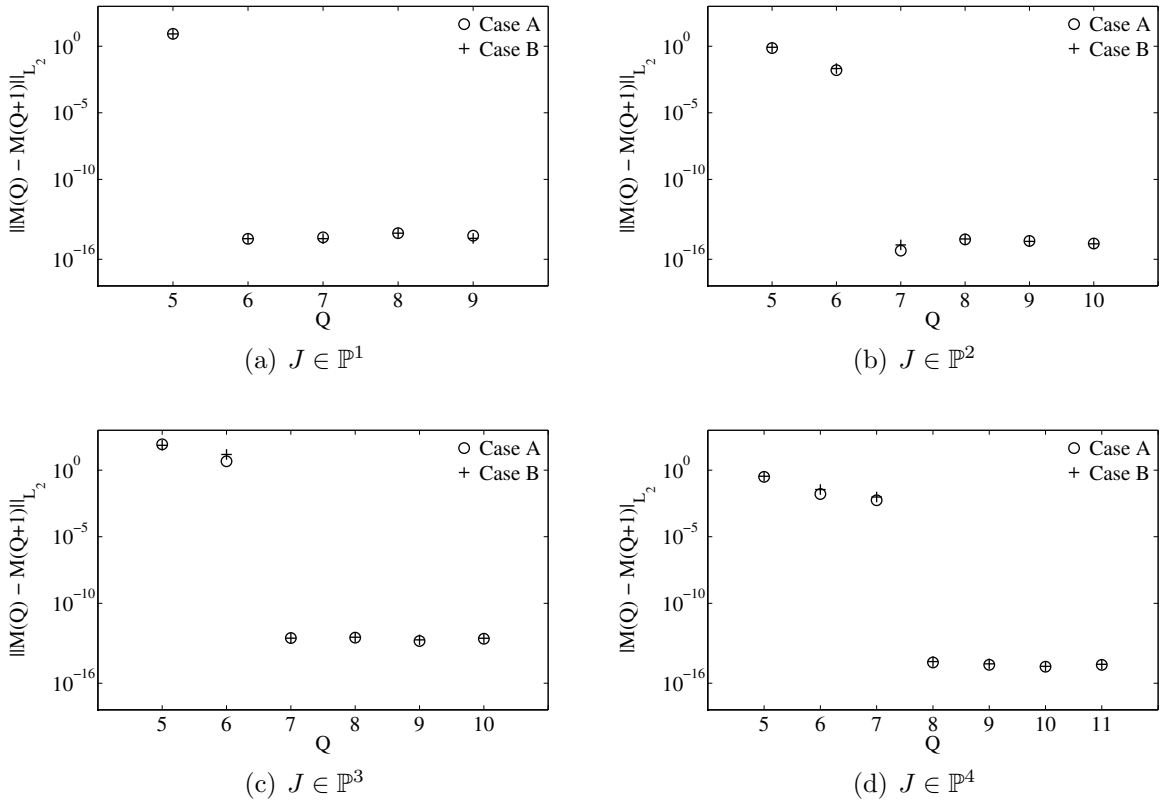


Figure 5.10: L_2 norm of the difference between mass matrices calculated with Q and $Q+1$ GLL points for the four orders considered and using both the mesh configurations shown in figure 5.8. Case A corresponds to figure 5.9(a) and Case B corresponds to figure 5.9(b).

Figure 5.11 shows the differences in the L_2 errors as defined in Eq. (5.20)) instead of

the differences between the mass matrices. We can see how both, the DG and the FR discretisations show machine-precision saturation of the error only for the linear mesh (i.e. $J \in \mathbb{P}^1$). For the other three cases, the saturation level is not at machine precision and it increases as P_{geom} increases. This behaviour is due to the numerical flux (i.e. the boundary term) which might come from both the left (internal domain) and right (boundary condition) sides of a given edge, since the normals are spatially varying across the curved edge(s). Therefore, for complex geometries, it is not possible to fully control the geometrical aliasing arising from the interfaces.

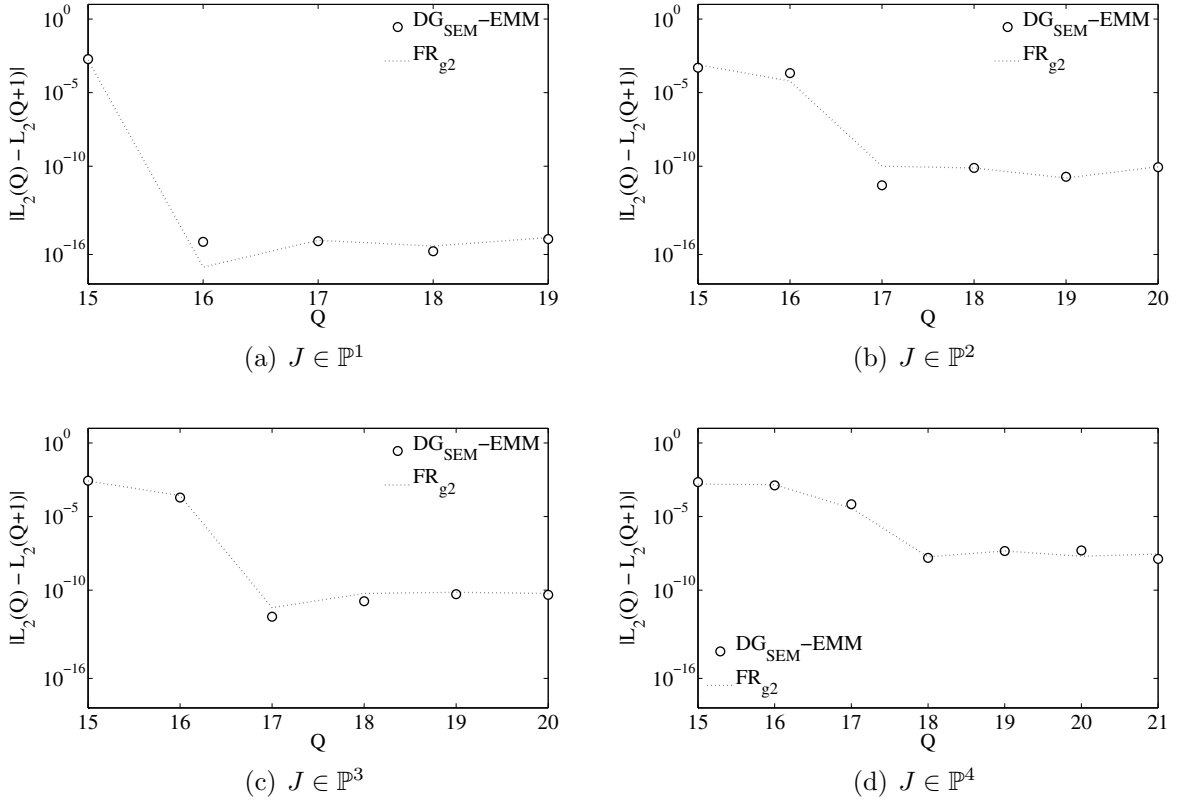


Figure 5.11: Differences between the L_2 errors obtained with Q and $Q+1$ GLL points for the four orders considered using the mesh configuration represented in figure 5.8(a).

5.4.3 Flow applications

Having considered the quantitative effects of the *Local* and *Global* dealiasing techniques, in this section we consider two examples that highlight how stability can be enhanced by using appropriate dealiasing strategies. We begin by considering the compressible Euler test

case described in chapter 3, and show that the use of the *Local* dealiasing technique leads to a control of the numerical entropy generated and stabilises the simulation. Secondly, we consider the Large-Eddy simulation of a NACA 0012 wingtip, and show how dealiasing plays an important role in maintaining the simulation stable.

5.4.3.1 Compressible inviscid subsonic flow past a cylinder

In this section, we present the same flow configuration presented in section 3.5.1.2 - an inviscid compressible flow past a cylinder governed by the compressible Euler equations.

We set the Mach number to 0.2, use an FR_{DG} scheme for the spatial discretisation and a 4th-order Runge-Kutta scheme for the time-integration. The mesh configuration used is the same as the one depicted in figure 3.7, although it is slightly finer having $N_t \times N_r = 178 \times 54$ elements in the tangential- and radial-direction, respectively. The cylinder is described by fifth order splines in order to achieve a high order description of the geometry.

In figure 5.12(a) we show the numerically-generated entropy for the case $P=2$, $Q=3$ without dealiasing techniques applied. As already seen in section 3.5.1.2, we can see a buildup of numerically-generated entropy at the posterior stagnation point, which is sensitive to aliasing-driven instabilities due to the strong gradients of the solution in this region. For this case, the simulation diverged after a few time-steps; however by applying the *Local* dealiasing technique using $Q_V=Q_I=4$, we achieved a stable simulation. In figure 5.12(b) we show a snapshot of the solution field taken at the same value of time as figure 5.12(a). We note that the numerically-generated entropy at the posterior stagnation point is no longer present. The simulation for $P=3$ and $Q=4$ without dealiasing depicted in figure 5.13 was also unstable. As in the previous case (i.e. $P=2$, $Q=3$), the use of the *Local* dealiasing technique stabilised the simulation.

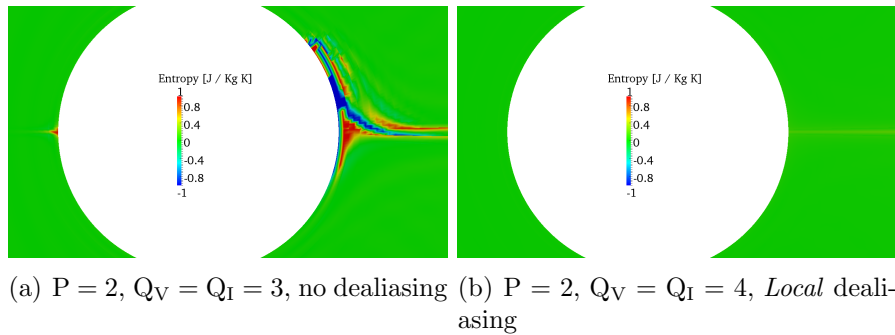


Figure 5.12: Numerically-generated entropy for $P=2$.

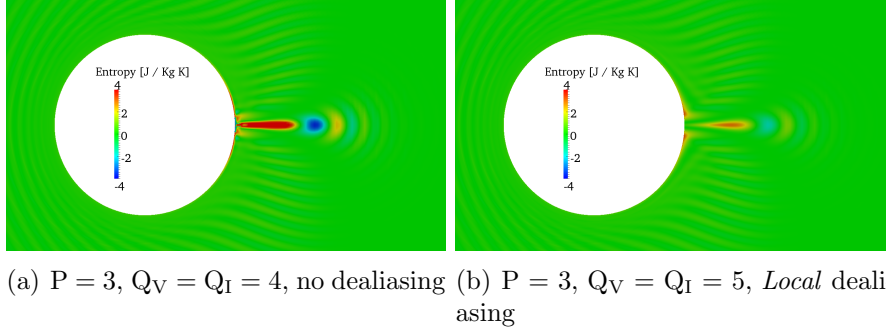


Figure 5.13: Numerically-generated entropy for $P = 3$.

In table 5.4 we report the maximum and minimum values of entropy across the computational domain for all the cases considered (note that the values reported are not steady-state results but refer to the time frame showed in the figures). As we can see, the numerically-generated entropy, which in this case can be seen as a measure of numerical error, roughly saturates at $Q_V=Q_I=6$ for $P = 2$ and at $Q_V=Q_I=8$ for $P=3$. Clearly the saturation is not up to machine precision because of the high-order description of the geometry and the deformed elements throughout the domain which introduce additional ‘nonlinearities’ into the problem. Also, the right-hand side of the compressible Euler equations is composed by rational functions which in turn means that it cannot be represented in a polynomial space, and therefore a machine-precision saturation of the solution cannot be expected. However, a minimisation of the relative error can be seen as Q is increased, and it is associated with the cubic term of the compressible Euler equations which requires $Q \geq 2P + 2$ GLL points to guarantee its exact integration. This also may indicate that the PDE-aliasing is potentially dominant for this problem and does not have a strong coupling with geometrical nonlinearities.

Note that, for both the polynomial orders considered, we also applied the *Global* dealiasing approach not shown here since the results were similar to those obtained using the *Local* dealiasing technique. This further reinforce the role of the PDE-aliasing for this problem.

P=2			
		Entropy-Min [J/Kg K]	Entropy-Max [J/Kg K]
NO		-96.36	94.35
<i>Local</i>	$Q_V=Q_I=4$	-0.86666943708297	1.59384922648788
<i>Local</i>	$Q_V=Q_I=5$	-0.85200075803792	1.54735068797723
<i>Local</i>	$Q_V=Q_I=6$	-0.85130979874330	1.54637692206548
<i>Local</i>	$Q_V=Q_I=7$	-0.85130979868331	1.54637692206548

P=3			
		Entropy-Min	Entropy-Max
NO		-1118.61	4347.60
<i>Local</i>	$Q_V=Q_I=5$	-1.67319315760229	3.49726118185079
<i>Local</i>	$Q_V=Q_I=6$	-1.16291305386484	3.39484597010811
<i>Local</i>	$Q_V=Q_I=7$	-1.13721148529237	3.39967822044820
<i>Local</i>	$Q_V=Q_I=8$	-1.13547005462042	3.39977179691589
<i>Local</i>	$Q_V=Q_I=9$	-1.13517999136988	3.39977189419113

Table 5.4: Maximum and minimum values across the mesh of numerically-generated entropy for both P=2 and P=3.

5.4.3.2 Incompressible viscous flow past a wing tip

In this section we show the results of an incompressible viscous flow past a NACA 0012 wingtip, originally studied experimentally by Chow *et al.* (1997).

The simulation was obtained using the continuous Galerkin (CG) approach, with a velocity correction scheme being used to discretise the incompressible Navier-Stokes equations (Karniadakis *et al.*, 1991). Although the CG approach is not the main target of this thesis, the results presented in this section illustrate how the dealiasing techniques presented can be effectively applied to high-Reynolds flows and they help stabilising the simulations.

The Reynolds number in the experimental studies was set at $Re = 4.6 \times 10^6$. In our current setup, the initial simulations were performed at a lower Reynolds number of $Re = 1.2 \times 10^6$. *Local* dealiasing, when combined with a spectral vanishing viscosity to dampen high-frequency energy buildup, yielded a stable simulation. However, in order to increase the Reynolds number and bring the simulation in line with experimental conditions, we found that without an increase in the global quadrature order (i.e. *Global* dealiasing technique), the simulation quickly became unstable. This highlights the role of geometric as well as PDE aliasing under these conditions for the mesh and polynomial resolution considered.

In figure 5.14 we illustrate the dynamics of the flow by showing the iso-contours of helicity for a P= 3 approximation, where the domain is represented using prismatic elements around the boundary and tetrahedral elements in the rest of the domain. In figures 5.15(a) and 5.15(b), for the same simulation, we represent the aliasing errors. The top image shows up to

30% aliasing error with respect to the magnitude of the non-linear terms while the bottom image shows close up regions of aliasing near the wing surface where we observe up to 600% error, highlighting the crucial nature of dealiasing in this problem.

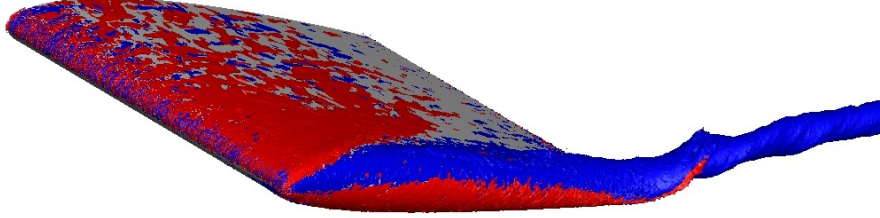


Figure 5.14: Helicity for an incompressible viscous flow over a NACA 0012 wing. [Courtesy of S. Sherwin]

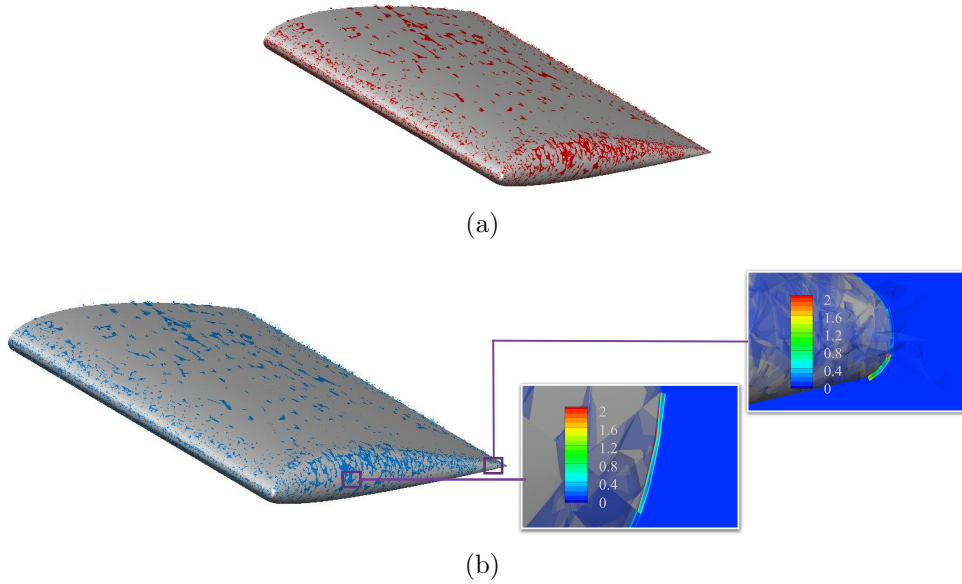


Figure 5.15: Aliasing errors at 30% of the magnitude of the nonlinear terms (figure 5.15(a)) and close up regions near the wing surface of aliasing errors (figure 5.15(b)). [Courtesy of S. Sherwin]

5.5 Conclusions

We have presented two dealiasing approaches based on the concept of consistent integration of the nonlinear terms or over-integration if the integration order is referred to the exact integration of linear operators.

The first technique presented, namely the *Local* approach, targets the PDE-aliasing sources which arise from the nonlinearities contained in the flux function and they are therefore related to the physics of the problem (we remark that also a quasi-linear equation - i.e. an equation with spatially varying coefficient - introduces PDE-aliasing errors). This approach exploits the tensor product structure of the elements to reduce the number of floating point operations, making it computationally efficient at higher polynomial orders. The application of *Local* dealiasing is particularly useful when we have localised nonlinearities (such as in the incompressible Navier-Stokes equations) and geometric aliasing is not significant.

The second technique, namely the *Global* dealiasing approach, involves the use of a richer quadrature order than is typically necessary for linear PDEs for all the terms constituting the right-hand side of the discretised problem and, therefore, it targets both PDE- and geometrical-aliasing sources. The latter arises when deformed or curved elements are used, which is often the case in industrially-relevant problems, where complex geometries are a fairly common feature. In contrast to the first approach, this technique is more computationally intensive (especially from a floating point operation point of view), but can address all the aliasing sources arising in any spectral element method when coupled with a sufficiently large number of quadrature points to tackle the nonlinearities within the problem.

The implementation details of both techniques have been thoroughly explained and we applied them to a linear and a quasi-linear advection equation on both regular and deformed/curvilinear meshes. These first set of numerical experiments had the main aim of quantifying the effects that both dealiasing techniques have on the discontinuous discretisations considered. We then showed how the *Local* and *Global* dealiasing techniques can be applied to compressible and incompressible flow problems improving the numerical stability of the simulations.

Based on the above analysis and supported by the numerical examples showed, the crucial points of this chapter can be summarised as follows:

1. The consistent integration rules presented in Kirby & Karniadakis (2003) are generally true and, for GLL points, can be complemented by the following expression

$$Q_{\min} \geq P_{exp} + \frac{P_{order}}{2} + \frac{3}{2},$$

$$P \geq P_{order},$$

where Q is the minimum number of quadrature points to exactly integrate a given polynomial nonlinearity of order P_{order} and for a given expansion order P . Note that in

the above expression P_{order} can be a combination of geometrical and PDE nonlinearities and therefore the overall degree of the nonlinearity can be higher than that dictated by the flux function.

2. In discontinuous spectral/ hp element methods, it is not generally possible to fully control (i.e. up to machine precision) all aliasing effects, since the boundary terms (through the numerical fluxes at the interfaces between the elements) introduce non-polynomial functions into the discretised problem.
3. The interface dealiasing may play a bigger role as the order of the interface flux function increases, becoming comparable to the volumetric dealiasing contribution. Therefore, when implementing these class of dealiasing techniques, the dealiasing of the boundary term must be taken into account.
4. Geometrical aliasing, in contrast to PDE aliasing, is responsible for modifying the mass matrix and it may therefore change the numerical properties of the discretisation being employed as shown in (Kopriva, 2009).
5. The two dealiasing strategies can be used in a complementary manner and when applied to challenging applications has proven effective and has increased the numerical stability of the simulations. This result is in agreement with the findings in the recent work by Malm *et al.* (2013) where a possible connection between numerical stability and consistent integration has been shown for the incompressible Navier-Stokes equations.

As a future step, one can think to include an efficient dealiasing technique addressing geometrical aliasing sources separately from PDE aliasing, by performing a pre-conditioning of the mesh. Specifically, it is possible to define an appropriate geometrical deformation estimator which carries out an analysis of the mesh a priori and identifies the elements needing additional quadrature points to avoid geometrical aliasing issues. Also to be taken into account in a future analysis is the role of large expansion ratios within a computational grid and how they may affect the numerical stability.

Chapter 6

Compressible boundary layers: Comparison between DNS and triple-deck theory

In this chapter we apply the numerical techniques developed throughout this thesis' work to compressible boundary-layer flows. Specifically, we have made extensive use of the connections found between the DG and the FR schemes in order to exploit the CFL advantages and the reduced computational costs given by the FR_{g2} scheme (or alternatively by the DG_{SEM} scheme with a lumped - i.e. diagonal - mass matrix). This allowed a consistent speed-up of the simulations performed and the coupling of these schemes with the dealiasing techniques (especially the *Local* dealiasing) presented in chapter 5 permitted maintaining numerical stability, thus preventing the simulations to fail due to aliasing-driven instabilities.

In the following we will present a comparison between Direct Numerical Simulation (DNS) and triple-deck theory both applied to high-speed subsonic flows over a two-dimensional Gaussian roughness element (also referred to as hump). The main target of this study is to evaluate the performance of the triple-deck theory when approaching a nonlinear regime, where a separated flow region is present behind the obstacle. The motivations of this work are related to the adoption of the triple-deck theory as an effective reduced model for receptivity studies in aeronautical applications.

Specifically, we investigated three different flow regimes, the first is weakly nonlinear, while the other two are nonlinear and a laminar separation bubble arises downstream of the roughness element. The Reynolds number was sufficiently high to be considered in the regime of aeronautical applications (especially when considering small imperfections at the leading

edge of wings).

The overall DNS process considered here is particularly efficient and allowed running several simulations in order to evaluate the sensitivity of the mesh to the boundaries location and to the mesh resolution. This guarantees a robust reliability and reproducibility of the DNS results. Nevertheless, the simulations performed were relatively expensive and, therefore, limited by the constraints imposed by the computational resources available.

The results presented in the following are published in ‘Journal of Fluid Mechanics’ (Mengaldo *et al.*, 2015b).

This chapter is structured as follows. In section 6.1 we overview the literature concerning boundary-layer flows and the solution strategies adopted in this chapter, namely the triple-deck theory and the DNS. In section 6.2 we detail the model problem considered. In section 6.3 we describe both the triple-deck approximations employed and the DNS process used. In section 6.4 we first present the DNS results in order to gain some insights into the main features of the flow investigated. Successively, we show the comparison between DNS and triple-deck. Finally, in section 6.5, we briefly summarise the main conclusions of the chapter.

6.1 Introduction

Boundary-layer flows have been widely investigated since Prandtl first formulated the boundary layer equations in 1904. Following his work, several contributions have been made during the past century; one of the best-known is (Schlichting, 1979) which summarises the main developments in this field until 1970s. Prandtl’s work provided also the basis of the so-called singular perturbation problem approach, which, in fluid dynamics, is commonly known as the asymptotic triple-deck theory.

The triple-deck theory was specifically designed with the purpose of describing the phenomenon of the boundary-layer separation at large values of the Reynolds number. The basic ideas that lie in the foundation of the triple-deck theory came from experimental studies of the boundary-layer separation, including the separation caused by an impinging shock wave, with the then unexplained phenomenon of ‘upstream influence’. These experiments are summarised by Chapman *et al.* (1958) and early theoretical models to explain the phenomenon are reviewed in a paper by Lighthill (2000). As a formal mathematical theory, the triple-deck theory was put forward by Neiland (1969), Stewartson & Williams (1969), Stewartson (1969), Messiter (1970) and Sychev (1972) and a detailed description of the basic aspects of

the theory can be found in the monograph by Sychev *et al.* (1998). According to the triple-deck theory, the flow near the separation point is described by the Prandtl's boundary-layer equations. The main alteration to the Prandtl's classical theory is that the pressure acting on the boundary layer is not known in advance, but has to be found as a part of the solution, as it is affected by the displacement effect of the boundary layer. The principal advantage of the triple-deck theory is that there are no restrictions on how large the Reynolds number is. In fact, the larger the Reynolds number the more accurate the theory is. However, numerical solution of the triple-deck equations is rather difficult, especially when the separation region is not small. It took a decade before reliable numerical techniques were developed; see Chapter 7 in Sychev *et al.* (1998).

Triple-deck theory assumed a crucial role in predicting flow separation in boundary layers especially when computational resources were not available. Yet, also today, triple-deck theory can be seen as an effective reduced model which allows the fast prediction of the main features of a flow although it can eventually fail when treating complex geometries as commonly demanded in industrial applications. Nevertheless, it can be used as a priori analysis tool within an industrial process for better understanding the underlying physics of a given flow configuration.

On the other hand, the numerical techniques for solving the Navier-Stokes equations also underwent significant improvement over recent decades and the DNS has become a real possibility thanks to the fast expansion of computational resources. A comprehensive summary of the role of the DNS in fluid mechanics until 1998 can be found in Moin & Mahesh (1998) and references therein. More recently, compressible DNS was performed in various works among which we cite (Pirozzoli & Gatski, 2004), (Martin, 2007), (Rizzetta *et al.*, 2010) and (Rizzetta & Visbal, 2014). Recent reviews, mainly devoted to supersonic and hypersonic flows, include (Fedorov, 2011) and (Zhong & Wang, 2012). Despite all of these advances, the task of calculating large Reynolds number separated flows, especially for transonic flow regimes, has been challenging. It is therefore not surprising that there are only few publications where the Navier-Stokes calculations are compared with the triple-deck predictions; see, for example, Hsiao & Pauley (1994).

In this chapter we compare these two analysis tools on a two-dimensional flow at a relatively high Reynolds number. We consider both subsonic and transonic conditions as well as a weakly nonlinear regime and two nonlinear regimes where a laminar separation bubble is present behind the roughness element.

The governing equations for the subsonic triple-deck were formulated by Neiland (1969),

Stewartson & Williams (1969), Stewartson (1969), Messiter (1970) and Sychev (1972), while the transonic version was first formulated by Bodonyi & Kluwick (1977), Bodonyi (1979), Bodonyi & Kluwick (1982) and Bodonyi & Kluwick (1998). In our calculations we employed the special regime described by Timoshin (1990) and Bowles & Smith (1993).

Concerning the DNS, we solved the two-dimensional compressible Navier-Stokes equations presented in chapter 3 by means of the DG_{SEM} method with lumped mass matrix in space (thoroughly investigated in chapter 4 and chapter 5) and an explicit 4th-order Runge-Kutta scheme in time.

6.2 Model problem

The model problem considered to compare the DNS and the triple-deck theory is shown in figure 6.1 and consists of a two-dimensional flat plate equipped with an isolated roughness element.

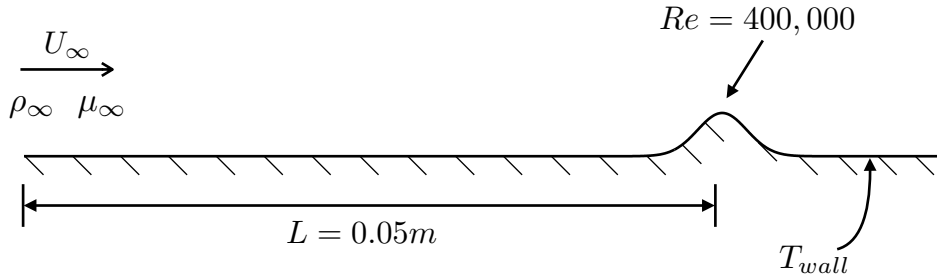


Figure 6.1: Model problem.

The shape of the roughness element is given by the following equation

$$F(x) = h_r e^{-\frac{(L-x)^2}{\beta^2}}, \quad (6.1)$$

where L is the distance from the leading edge of the flat plate to the centre of the roughness element, h_r is the height of the roughness element and β is a parameter which controls the length of the roughness element.

The Reynolds number Re was set equal to 400,000 for all the simulations performed and is defined as follows:

$$Re = \frac{\rho_\infty u_\infty L}{\mu_\infty}, \quad (6.2)$$

where ρ_∞ denotes the free-stream density, U_∞ is the free-stream velocity and μ_∞ refers to the

free-stream dynamic viscosity and L is the previously defined distance between the leading edge of the flat plate and the roughness element.

We took into account two different flow conditions: Case 1, with an associated Mach number¹ $\text{Ma}_\infty=0.5$ (which corresponds to subsonic conditions) and Case 2, with associated Mach number $\text{Ma}_\infty=0.87$ (which corresponds to transonic conditions). The physical parameters adopted for the free-stream conditions in each of the two cases are reported in table 6.1. The free-stream pressure, p_∞ , and density, ρ_∞ , in both cases correspond to an altitude of eleven thousand meters above sea level and they are based on the International Standard Atmosphere (ISA). The free-stream dynamic viscosity, μ_∞ , and thermal conductivity, κ_∞ , for Case 1 (i.e. the subsonic case) do not match exactly the associated values in the ISA model and they were chosen in order to match the fixed Reynolds number adopted in the DNS simulations. The free-stream velocity U_∞ was chosen in such a way that it satisfied the Mach number for the two cases taken into account and we applied isothermal boundary conditions using a temperature T_{wall} at the wall.

The Prandtl number² was fixed and corresponds to the value commonly adopted for air.

	Ma_∞ [—]	ρ_∞ [kg/m^3]	p_∞ [Pa]	μ_∞ [$Pa \cdot m$]	κ_∞ [$W/(m \cdot K)$]	U_∞ [m/s]	T_{wall} [K]	Pr [—]
Case 1	0.50	0.364	22600	$6.710 \cdot 10^{-6}$	0.00936	147.4	216.29	0.72
Case 2	0.87	0.364	22600	$1.167 \cdot 10^{-5}$	0.01629	256.5	258.0	0.72

Table 6.1: Free-stream physical parameters and dimensionless quantities for the subsonic (Case 1) and transonic (Case 2) regimes considered.

For both the flow conditions considered, Case 1 and Case 2, we used three different roughness heights, in order to investigate both weakly nonlinear and nonlinear regimes. The parameters adopted for the shape (i.e. β) and the height (i.e. h_r) of the humps are reported in table 6.2, along with the hump length, ℓ_{999} , that is defined as the length for which the ratio $h(x)/h_r$ becomes smaller than 0.1%. In table 6.2 we also report the ratios between the height of the hump h_r and the corresponding thickness of the compressible boundary layer at the roughness location, $\tilde{\delta}_{999_L}^1$ for Case 1 and $\tilde{\delta}_{999_L}^2$ for Case 2.

¹The Mach number is defined as the ratio between the free-stream speed of sound c_∞ and the free-stream velocity U_∞ : $\text{Ma}_\infty = c_\infty/U_\infty$.

²The Prandtl number is defined as the ratio of kinematic viscosity ν to thermal diffusivity $\alpha = \kappa/(\rho C_p)$, that is: $\text{Pr} = C_p \mu / \kappa$, where C_p is the specific heat at constant pressure.

	h_r [m]	β [m]	ℓ_{999} [m]	h_r/ℓ_{999} [%]	$h_r/\tilde{\delta}_{999_L}^1$ [%]	$h_r/\tilde{\delta}_{999_L}^2$ [%]
Hump A	$0.5 \times 3.88 \cdot 10^{-5}$	$1.7961 \cdot 10^{-4}$	9.60×10^{-4}	2.02	4.20	3.90
Hump B	$1.0 \times 3.88 \cdot 10^{-5}$	$1.7961 \cdot 10^{-4}$	9.60×10^{-4}	4.04	8.40	7.80
Hump C	$1.5 \times 3.88 \cdot 10^{-5}$	$1.7961 \cdot 10^{-4}$	9.60×10^{-4}	6.06	12.6	11.7
					Case 1	Case 2

Table 6.2: Set of humps employed for the simulations.

The three hump heights explored range from a non-separated flow to a well established separation bubble behind the obstacle.

6.3 Numerical and theoretical approaches

In this section we first present the numerical settings and the process adopted for the DNS and successively we briefly explain the two triple-deck formulations applied to the model problem, namely the subsonic and the transonic formulation showing also the transformations needed to compare the DNS and the triple-deck results.

6.3.1 DNS setup

The governing equations considered for the direct numerical simulations are the unsteady two-dimensional compressible Navier-Stokes equations, as presented in chapter 3, Eqs. (3.9, 3.10, and 3.11).

The spatial numerical discretisation of the advection term used is the DG_{SEM} method with a lumped mass matrix, for which, we showed the connections with the FR_{g2} scheme in chapter 4. The use of this scheme allowed a larger time-step, approximately the double than the DG_{SEM} scheme with an exact mass matrix (or equivalently the FR_{DG} scheme) and permitted a consistent speed-up of the simulations because of the diagonal nature of the mass matrix. Note that we used the ‘exact’ Riemann solver to calculate the numerical fluxes at the element interfaces and we exploited the weak strategy presented in chapter 3 for applying the boundary conditions. The diffusion term was discretised with the local discontinuous Galerkin approach (LDG) presented in chapter 2. The polynomial order used was $P = 3$ for all the simulations carried out and we applied a local dealiasing technique for the first

stages of the simulations, in order to avoid numerical instabilities in proximity to the hump location. The temporal discretisation used was an explicit 4th-order Runge-Kutta scheme.

The DNS process adopted is depicted in figure 6.3. We first performed a DNS of a flat plate without roughness element (flat plate domain $\overline{OBCD}$ as sketched in figure 6.3) on a relatively coarse mesh. For this simulation we applied the following set of boundary conditions in a weak manner (the boundary conditions in the following has been presented in chapter 3):

- Region $\overline{OA}$: slip;
- Region $\overline{AB}$: no-slip;
- Region $\overline{BC}$: pressure outflow;
- Region $\overline{CD}$: farfield;
- Region $\overline{DO}$: inflow.

To guarantee that the results were reliable, we compared them with the compressible boundary layer solutions (available from a third party software developed within the LFC-UK group) observing an excellent agreement.

After having obtained the steady state data on the flat plate, we interpolated the solution onto the embedded domain around the roughness element represented in figure 6.3. As initial and boundary conditions we used the values obtained in the flat plate simulation. The embedded mesh had a larger resolution than the flat-plate mesh and to verify that the mesh resolution was sufficient, we performed a mesh resolution study (for Case 1, Hump C, which was the most challenging configuration) obtaining a converged result. Specifically, the relative maximum difference (normalised with respect to the respective absolute maximum values) between the finest mesh and a mesh with approximately half of the resolution was less than 1% for both the wall shear stress τ_{xy} and the pressure gradient dp/dx . We also assured that the influence of the boundary conditions on the embedded domain did not have any effect on the region in proximity to the hump. Specifically, the perturbation of the flow due to the hump was damped much prior to the boundaries themselves. For the cases where the flow was most distorted, Hump C, the maximum difference between the compressible similarity solution and the flow calculated by the compressible Navier-Stokes solver in proximity to the boundaries was less than 0.03% (normalised with respect to the free-stream values).

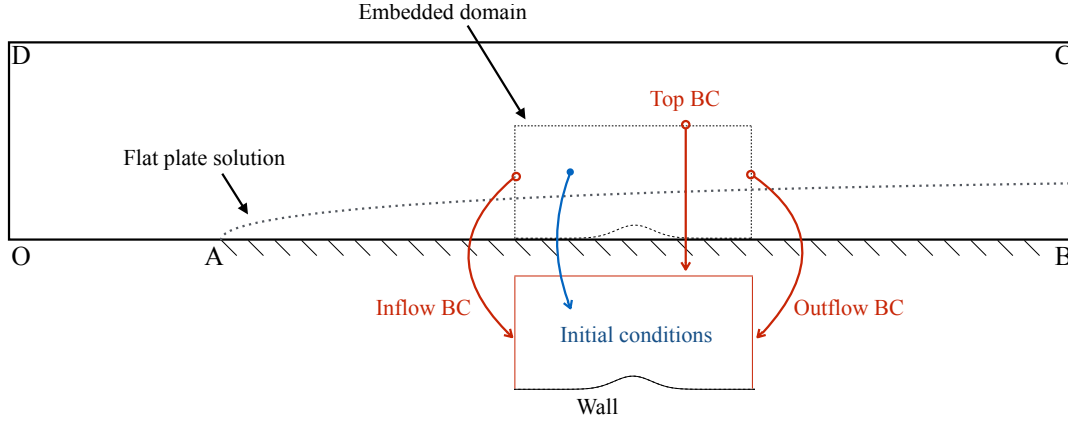


Figure 6.2: Sketch of the process adopted to perform the direct numerical simulations.

The final embedded mesh was constituted by 500×66 elements where we employed a stretching technique in both the wall-normal and the streamwise directions. As already mentioned, we used a solution polynomial of order three within each element and, therefore, the number of solutions points in the wall-normal direction was equal to 264 (220 of which contained within the boundary layer at the roughness location) while in the streamwise direction was equal to 2000. The minimum Δy in proximity to the wall was equal to $2 \times 10^{-6} m$. In the streamwise direction the maximum resolution near the roughness element was equal to $1 \times 10^{-5} m$. The description of the roughness element geometry was obtained using a 5th-order spline to achieve an accurate representation.

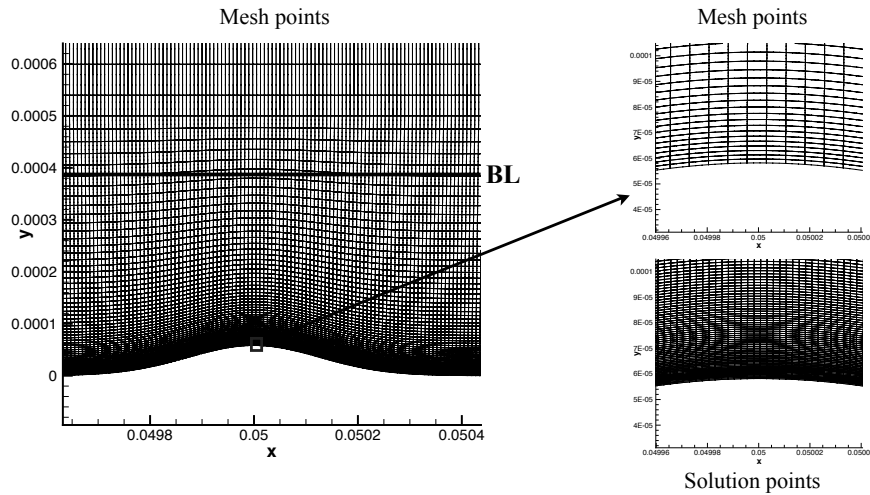


Figure 6.3: Example of mesh adopted for the DNS simulations.

Note that the steady-state criterion adopted is the following:

$$\frac{\|g^t - g^{t+\Delta t}\|_{L_\infty}}{g_\infty \Delta t} < 1 \cdot 10^{-6}, \quad (6.3)$$

where g is a generic variable, g_∞ is the related free-stream value and Δt is the time interval between which the relative error has been calculated. In section 6.4.1, we will show how, in the most challenging case (i.e. Hump C, subsonic), this approach allowed finding that the unsteadiness observed was due to a mesh resolution issue rather than being a physical feature of the problem.

The DNS process described made the simulations feasible in terms of computational costs and represents an efficient way for performing DNSs on the configurations investigated.

6.3.2 Triple-deck formulation

The scales adopted in the triple-deck formulation are depicted in figure 6.4. Specifically, the subsonic case is represented in figure 6.4(a) while the transonic case in figure 6.4(b). In the following we briefly summarise the formulation used and the transformations necessary to compare the DNS and triple-deck data.

We first write the set of equations for the lower deck (see figure 6.4) in the canonical form

$$\begin{aligned} u_0 \frac{\partial u_0}{\partial x} + v_0 \frac{\partial u_0}{\partial y} &= -\frac{\partial p_0}{\partial x} + \frac{\partial^2 u_0}{\partial y^2}, \\ \frac{\partial u_0}{\partial x} + \frac{\partial v_0}{\partial y} &= 0, \end{aligned} \quad (6.4)$$

complemented by the following set of boundary conditions:

$$\begin{aligned} u_0 = v_0 &= 0 && \text{at } y = 0 \\ u_0 &= y + \dots && \text{as } x \longrightarrow -\infty \\ u_0 &= y + A_0(x) + \dots && \text{as } y \longrightarrow \infty, \end{aligned} \quad (6.5)$$

where $A_0(x)$ is the displacement function, x_0, y_0 are the streamwise and wall-normal coordinates and u_0, v_0 are the associated velocities, p_0 is the pressure and the subscript ‘0’ denotes leading order dimensionless quantities which are defined in the asymptotic expansions used and defined in sections 6.3.2.1 and 6.3.2.2. The interaction law between the boundary layer (lower deck) and the potential flow in the upper deck has the form of the Hilbert integral

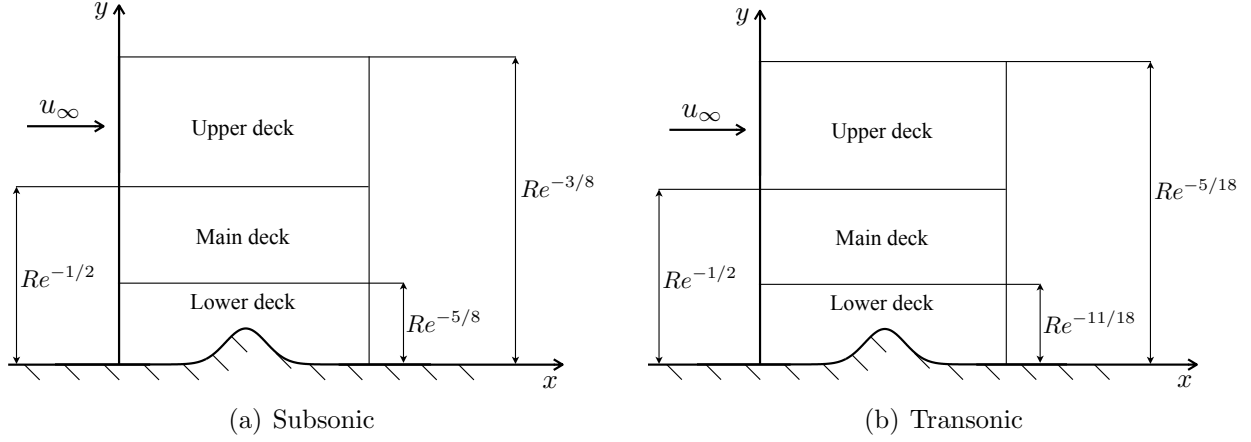


Figure 6.4: Three-tiered structure of the interaction region for the subsonic (figure 6.4(a)) and transonic (figure 6.4(b)) cases.

for subsonic flows. In the transonic case, the interaction law is instead represented by the following integral

$$p_0(x_0) = \frac{1}{\pi\sqrt{m}} \int_{-\infty}^{\infty} \frac{F'(s) - A'_0(s)}{s - x_0} ds, \quad (6.6)$$

where m is known as Kármán-Guderley parameter (m was equal to 1.3 in all the transonic cases taken into account) and is defined as

$$M_\infty^2 = 1 - mRe^{-1/9}, \quad (6.7)$$

where M_∞ is the free-stream Mach number and $F(x_0)$ is the shape of the roughness element defined in section 6.2. Note that in Eq. (6.6), for the transonic case, the upper deck steady problem is described by the Transonic Small Perturbation (TSP) equation disregarding the time derivative (for additional details the interested reader can refer to Cole & Cook (1986)). In the subsonic case the upper deck is instead solved through the Laplace equation for the pressure.

To solve the steady triple-deck problem numerically we discretise our two-dimensional domain with a set of mesh points at generic coordinates $(x_{0,i}, y_{0,j})$, where $(i = 1, \dots, N_{x_0}; j = 1, \dots, N_{y_0})$. Following Kravtsova & Ruban (2005), (where the numerical approach adopted here is detailed), we define the displacement function vector $\mathbf{A}_{0,i}$ and the pressure vector $\mathbf{p}_{0,i}$ at each point $x_{0,i}$. The vectors $\mathbf{A}_{0,i}$ and $\mathbf{p}_{0,i}$ define the values of the displacement function and of the pressure at all the points along the wall-normal direction y_0 . For a given displacement

function $\mathbf{A}_{0,i}$, the interaction law in Eq. (6.6) (equivalently the interaction law for the subsonic case) allows the calculation of the *inviscid pressure gradient* $d\mathbf{p}_0/dx_0|_{inv}$. For the same $\mathbf{A}_{0,i}$, it is also possible to calculate the *viscous pressure gradient* $d\mathbf{p}_0/dx_0|_v$ by means of the lower deck equations (6.4 and 6.5). The final set of implicit equations to be solved to find the displacement function $\mathbf{A}_{0,i}$ is the difference between the *inviscid* and the *viscous pressure gradients*

$$\Phi_0(\mathbf{A}_{0,i}) = \left. \frac{d\mathbf{p}_0}{dx_0} \right|_{inv}(x_{0,i}) - \left. \frac{d\mathbf{p}_0}{dx_0} \right|_v(x_{0,i}) = 0, \quad \text{for } i = 1, \dots, N_{x_0}. \quad (6.8)$$

Equation (6.8) is solved using the Newton-Raphson method by iterating on the displacement function $\mathbf{A}_{0,i}$. For additional details, the interested reader can also refer to Sychev *et al.* (1998).

All the triple-deck calculations were performed using a 2^{nd} -order finite-difference method on a 801×301 non-uniform mesh with minimum step sizes $\Delta x_0 = 0.005$ and $\Delta y_0 = 0.005$ in the streamwise and wall-normal directions respectively. Note that to guarantee independence of the numerical results from the mesh resolution we performed a mesh refinement study by using a finer grid (one order of magnitude greater resolution than the Δx_0 and Δy_0 provided above). The relative maximum difference (normalised with respect to the respective absolute maximum values) between the results on the coarser and finer grids was less than 0.1% for both the wall shear stress τ_{xy} and pressure gradient dp/dx .

The asymptotic dimensionless expansions of the variables (denoted in Eq. (6.4) by the subscript ‘0’) used in the subsonic and transonic regimes are different and they are reported in the following two subsections.

6.3.2.1 Subsonic regime

If we consider the typical triple-deck scales around an isolated roughness element in a subsonic regime (figure 6.4(a)) we can expand the streamwise and wall-normal directions as follows

$$x = L \left(1 + Re^{-\frac{3}{8}} \lambda^{-\frac{5}{4}} \mu_0^{-\frac{1}{4}} \rho_0^{-\frac{1}{2}} \alpha^{-\frac{3}{4}} x_0 \right), \quad y = L Re^{-\frac{5}{8}} \lambda^{-\frac{3}{4}} \mu_0^{\frac{1}{4}} \rho_0^{-\frac{1}{2}} \alpha^{-\frac{1}{4}} (y_0 + F(x_0)), \quad (6.9)$$

where x_0 and y_0 indicate the dimensionless quantities used as asymptotic expansions in the triple-deck approach. Note that we applied the Prandtl transposition theorem for the vertical coordinate and we assumed that the hump was fully contained within the lower deck.

Concerning the dimensional velocities (u, v) we used the following asymptotic expansions

$$u = u_\infty Re^{-\frac{1}{8}} \lambda^{\frac{1}{4}} \mu_0^{\frac{1}{4}} \rho_0^{-\frac{1}{2}} \alpha^{-\frac{1}{4}} u_0 + \dots, \quad v = u_\infty Re^{-\frac{3}{8}} \lambda^{\frac{3}{4}} \mu_0^{\frac{3}{4}} \rho_0^{-\frac{1}{2}} \alpha^{\frac{1}{4}} \left(v_0 + u_0 F'(x_0) \right) + \dots, \quad (6.10)$$

where u_0 and v_0 are the velocities related to the triple-deck approach. Finally, the dimensional thermodynamic quantities were expanded as follows

$$p = p_\infty + \rho_\infty u_\infty^2 Re^{-\frac{1}{4}} \lambda^{\frac{1}{2}} \mu_0^{\frac{1}{2}} \rho_0^{-\frac{1}{2}} \alpha^{-\frac{1}{2}} p_0 + \dots \quad (6.11)$$

$$\rho = \rho_\infty \rho_0(x_0, 0) + \dots, \quad \mu = \mu_\infty \mu_0(x_0, 0) + \dots,$$

where $\alpha = \sqrt{1 - \text{Ma}_\infty^2}$, p is the pressure, ρ is the density and μ is the dynamic viscosity. Note that these variable transformations were used to compare DNS and triple-deck data in the subsonic case.

6.3.2.2 Transonic regime

In the transonic regime (figure 6.4(b)), the scales for the streamwise and wall-normal coordinates become

$$x = L \left(1 + Re^{-\frac{1}{3}} \lambda^{-\frac{4}{3}} \mu_0^{-\frac{1}{3}} \rho_0^{-\frac{1}{3}} x_0 \right), \quad y = L Re^{-\frac{11}{18}} \lambda^{-\frac{7}{9}} \mu_0^{\frac{2}{9}} \rho_0^{-\frac{4}{9}} (y_0 + F(x_0)), \quad (6.12)$$

where, again, we assumed that the hump was fully contained within the lower deck. Concerning the dimensional velocities (u, v) we used the following asymptotic expansions

$$u = u_\infty Re^{-\frac{1}{9}} \lambda^{\frac{2}{9}} \mu_0^{\frac{2}{9}} \rho_0^{-\frac{4}{9}} u_0 + \dots, \quad v = u_\infty Re^{-\frac{7}{18}} \lambda^{\frac{7}{9}} \mu_0^{\frac{7}{9}} \rho_0^{-\frac{5}{9}} \left(v_0 + u_0 F'(x_0) \right) + \dots \quad (6.13)$$

Finally, the dimensional thermodynamic quantities assumed the same expressions as in the subsonic case with the exception of the pressure that was expanded as follows:

$$p = p_\infty + \rho_\infty u_\infty^2 Re^{-\frac{2}{9}} \left(\lambda^{\frac{4}{9}} \mu_0^{\frac{4}{9}} \rho_0^{\frac{1}{9}} \right) p_0 + \dots \quad (6.14)$$

The Kármán-Guderley parameter was instead

$$m = (1 - \text{Ma}_\infty^2) \lambda_0^{-\frac{2}{9}} \mu_0^{-\frac{2}{9}} \rho_0^{\frac{4}{9}} Re^{\frac{1}{9}} \dots \quad (6.15)$$

Note that these variable transformations were used to compare DNS and triple-deck data in the transonic case.

6.4 Numerical results

In this section, we first present the DNS results, in order to show the comparison between the various hump heights considered at the two flow regimes. We also present the issues encountered when dealing with Hump C, Case 1 in terms of steadiness of the separation bubble and we make some considerations concerning this aspect.

We successively focus on the comparison between the DNS and the triple-deck theory results, highlighting the main differences in terms of streamwise pressure gradient dp/dx and shear stress τ_{xy} as well as in terms of detachment point (D), reattachment point (R) and length of the separation bubbles when present.

6.4.1 DNS results

6.4.1.1 Overview

In figure 6.5 and 6.6, we show dp/dx and τ_{xy} at the wall for the three hump heights investigated, namely Hump A, Hump B and Hump C, and for both the subsonic and the transonic regimes. We can see that both the parameters taken into account have a very similar behaviour between the subsonic (figure 6.5(a) and figure 6.6(a)) and the transonic (figure 6.5(b) and figure 6.6(b)) case (note that the scales of the subfigures are different). Specifically, it is possible to see that the distortion of the boundary layer solution increases as the hump height increases and we can observe that for Hump B and C in the subsonic case and for Hump C in the transonic case, the flow develops a small separation bubble behind the roughness element (the wall shear stress τ_{xy} becomes in fact negative in a small region behind the hump).

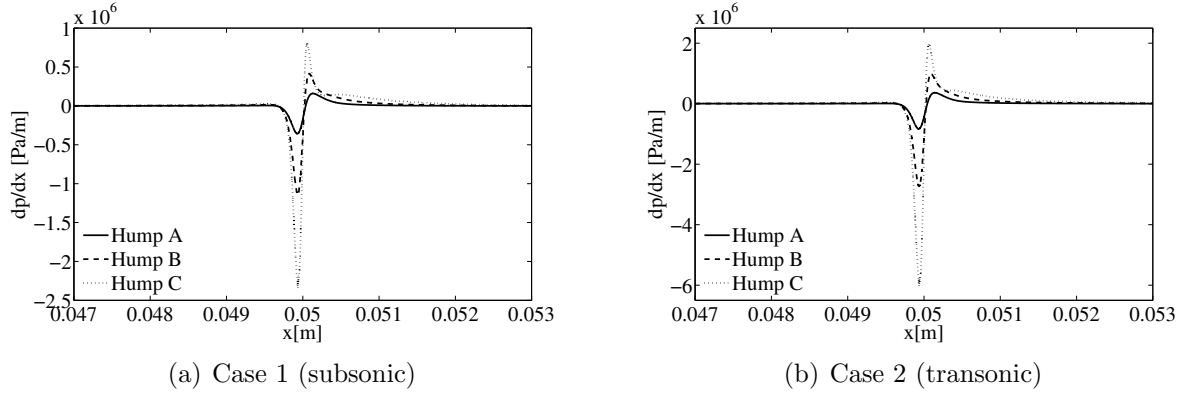


Figure 6.5: Comparison of $\frac{dp}{dx}$ (dimensional values) varying the hump heights for the two cases considered: Case 1 (subsonic), figure 6.5(a) and Case 2 (transonic) figure 6.5(b).

It is also possible to note that, for Hump C, in both the subsonic and the transonic case, the distortion of the dp/dx behind the hump assumes a slightly different shape with a small change in curvature than Hump A and B. This can indicate that, by increasing the hump height further, we may encounter the development of additional nonlinear effects which could potentially lead to the unsteadiness of the separation bubble.

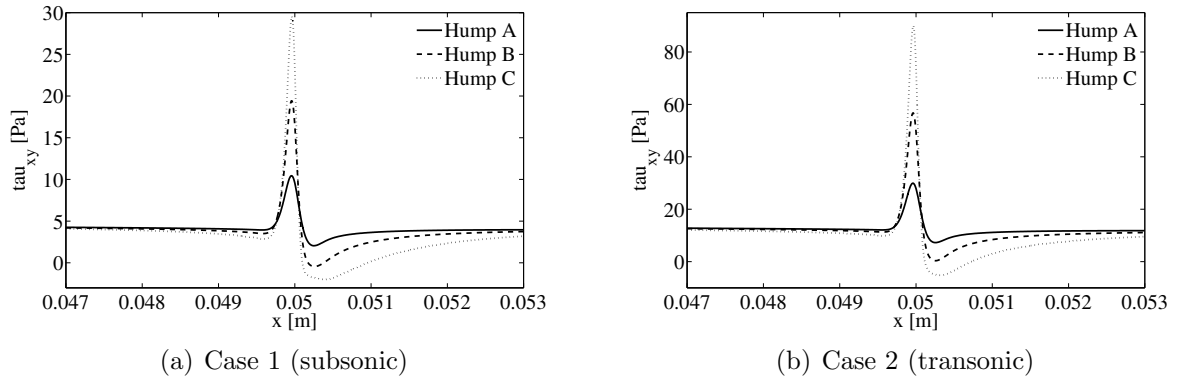


Figure 6.6: Comparison of τ_{xy} (dimensional values) varying the hump heights for the two cases considered: Case 1 (subsonic), figure 6.6(a) and Case 2 (transonic) figure 6.6(b).

In figure 6.7(b) we show the pressure field for Hump C in the subsonic (figure 6.7(a)) and transonic (figure 6.7(b)) case. It is possible to see that the transonic case have a bigger impact on the pressure than the subsonic case (note that the contours have a different range between the two subfigures).

On the other hand, in figure 6.8 we show the separation bubbles for Hump C in the subsonic (figure 6.8(a)) and transonic (figure 6.8(b)) regime. In this case it is possible to note that the larger separation bubble, therefore the bigger impact on the base flow, takes place for the subsonic case. This is related to the iso-thermal boundary condition applied to the wall. A cold wall, like the one adopted in the subsonic case, makes the boundary layer thinner than a hot wall because it changes the viscosity locally through the Sutherland's law, thus the local Reynolds number is also changed.

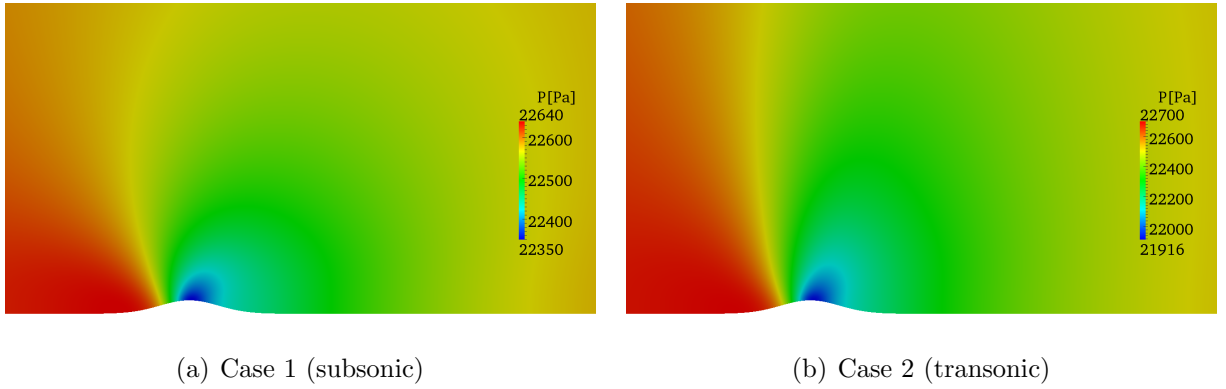


Figure 6.7: Comparison of P (dimensional values) varying the hump heights for the two cases considered: Case 1 (subsonic), figure 6.7(a) and Case 2 (transonic) figure 6.7(b).

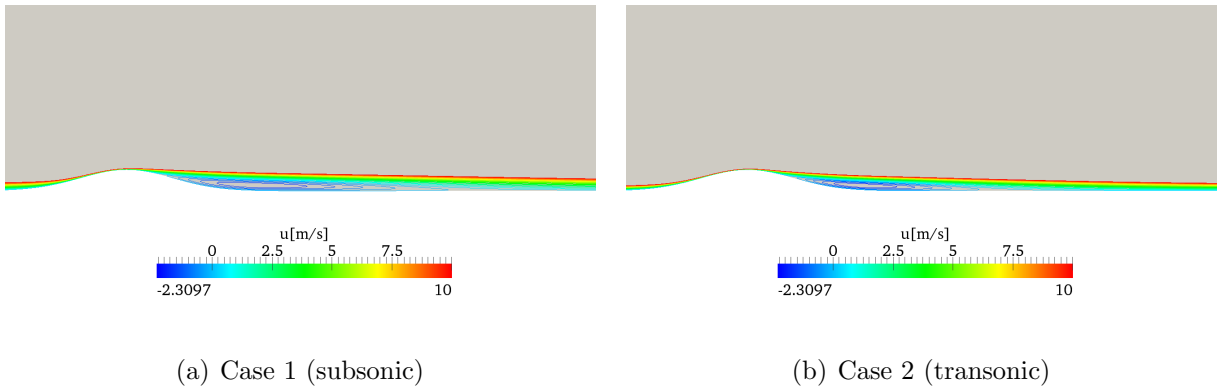


Figure 6.8: Comparison of the separation bubbles for Hump C hump between the two cases considered: Case 1 (subsonic), figure 6.8(a) and Case 2 (transonic) figure 6.8(b).

To summarise, Hump B and C in the subsonic case present a laminar separation bubble (LSB) behind the roughness element (which, in the case of Hump B is very small), while only

Hump C in the transonic case develops an LSB. In addition, the distortion of the pressure (with respect to the flat plate solution) is bigger for the transonic case than the subsonic one.

Based on this considerations, in terms of comparisons, we expect a larger difference between DNS and triple-deck for the transonic case for what concern the two parameters taken into account, namely dp/dx and τ_{xy} , because the pressure distortion enters into the boundary-layer and the higher the transverse pressure gradient (dp/dy) the larger the difference with the triple-deck theory, where dp/dy is neglected³.

On the other hand, a larger difference in terms of separation bubble length is expected for Hump C - subsonic case, because is the largest LSB and the triple-deck theory may face some issues, especially when dealing with large separation regions.

6.4.2 Steadiness of the separation bubbles

All the results presented in section 6.4.1.1 show steady separation bubbles. However, it is important to understand the reliability of the results obtained in order to exclude the possibility that small numerical noise, insufficient resolution of the mesh and other factors, induced by the numerics rather than the physics, can influence the behaviour of the separation bubble. From this perspective, we performed two different simulations for the most challenging case: Hump C - Case 1 (subsonic). The first had a spatial resolution which was approximately four times smaller than the second. The results can be seen in figure 6.9. If we look at the stream wise pressure gradient dp/dx (figure 6.9(a)), it is not possible to appreciate any difference between the two curves (the blue line indicates the coarser mesh while the black dots represent the better-resolved mesh).

³Note that the differences between DNS and triple-deck can also be caused by other factors neglected in the triple-deck theory and not necessarily only by the transverse pressure gradient. However, dp/dy is likely to play an important role.

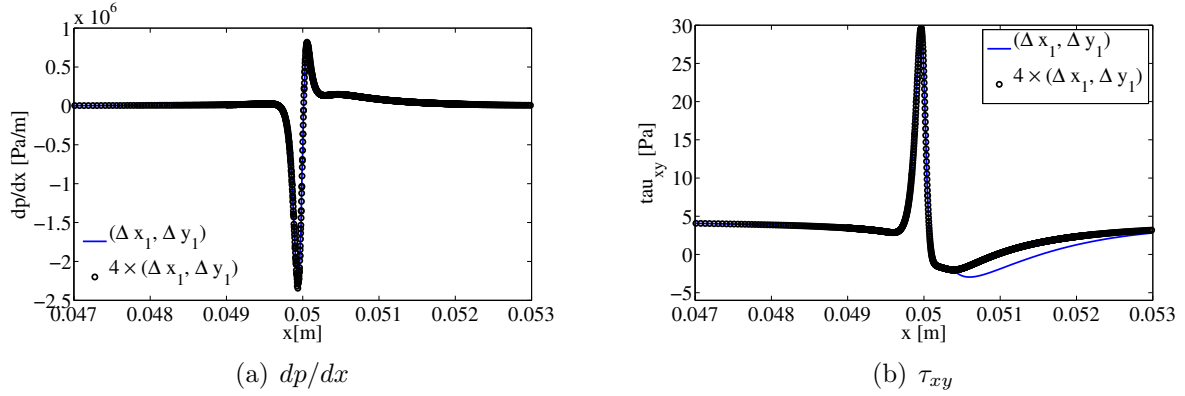


Figure 6.9: Wall parameters dp/dx (figure 6.9(a)) and τ_{xy} (figure 6.9(b)) for Case 1 (subsonic) Hump C using a coarse (blue line) and a fine mesh (black dots).

Regarding the wall shear stress τ_{xy} (figure 6.9(b)) instead, it is possible to see a significant mismatch between the two meshes. Specifically, in the coarser mesh (blue line), the separation bubble started becoming unsteady with the reattachment point moving forward and backward (the blue line plotted in figure 6.9(b) is taken at the same time frame - i.e. after ≈ 38 convective times - as the black dots). This, as anticipated, is due to insufficient mesh resolution. If we in fact plot the difference between two time frames (figure 6.10), we can see that there is a region where the solution is polluted and strongly unsteady. This region corresponds exactly at the point where the wall shear stress changes curvature.

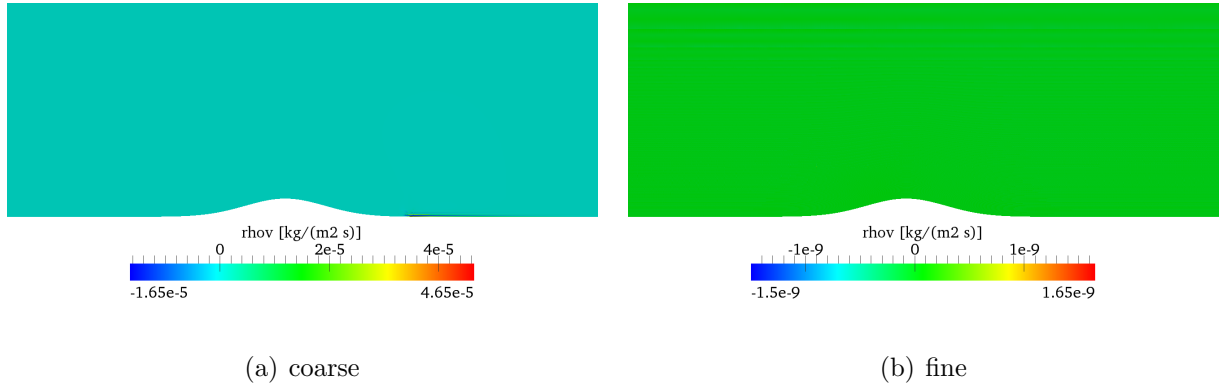


Figure 6.10: Contours of the difference between two time frames of ρv for the coarser mesh (figure 6.10(a)) and for the finer mesh (figure 6.10(b)) for Case 1 (subsonic), Hump C.

Since this test case is the most challenging from the perspective of the LSB, the resolution

chosen for all the other simulations is the same as this test case, although it might well be excessive for some of the test cases.

6.4.3 Comparison between DNS and triple-deck

In this subsection, we present the comparisons between DNS and triple-deck. The results shown in the figures take into account only Hump C. We however report all the data associated to the other humps in table 6.3.

In figure 6.11, we show the comparison of dp/dx and τ_{xy} for Hump C, Case 1: Mach number $M = 0.50$, wall temperature $T_{wall} = 216.29K$.

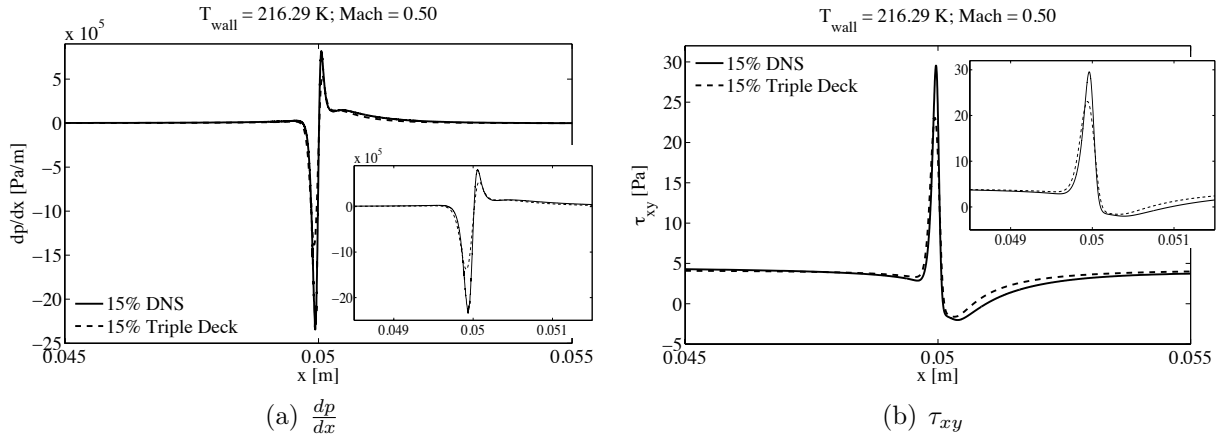


Figure 6.11: Comparison of $\frac{dp}{dx}$ and τ_{xy} (dimensional values) between triple-deck and DNS for Hump C with $Ma_\infty = 0.50$ and $T_{wall} = T_\infty$ (Case 1, subsonic).

In figure 6.12, we show the same quantities for Case 2: Mach number $M = 0.87$, wall temperature $T_{wall} = 258.0K$ (see also table 6.2).

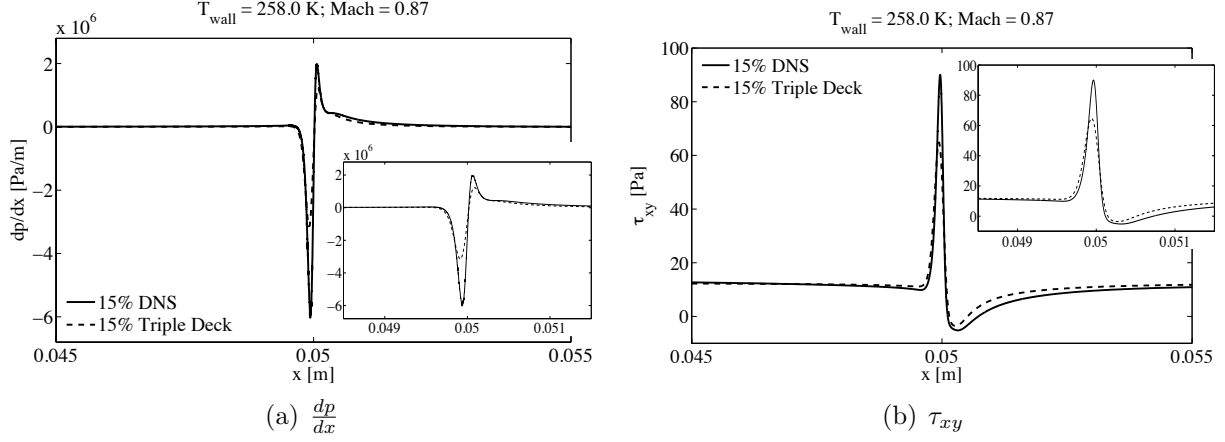


Figure 6.12: Comparison of $\frac{dp}{dx}$ and τ_{xy} (dimensional values) between triple-deck and DNS for Hump C with $\text{Ma}_\infty = 0.87$ and $T_{\text{wall}} = 258.0 \text{ K}$ (Case 2, transonic).

From the figures presented it is possible to see an overall good agreement between triple-deck and DNS data except in proximity to the centre of the roughness element, where large discrepancies are observed (as it can be seen from the inset images, which are a close up view of this region). These discrepancies are expected in the asymptotic analysis because in proximity to the roughness element we observed a larger acceleration of the flow in the DNS results if compared to the triple-deck data. Also, the deceleration of the flow behind the hump is larger in the DNS data if compared to the triple-deck one. These features generate sharper and higher peaks in terms of both wall shear stress τ_{xy} and longitudinal pressure gradient dp/dx . We also observed that the wall-normal pressure gradient in the DNS data has a contribution which is neglected in the triple-deck theory and which might affect the behaviour of the two quantities investigated and contribute to the discrepancies seen, although some other factors can also play a role.

In table 6.3, we quantified the differences between DNS and triple-deck data at four distinct locations along the streamwise direction: P1= 0.046 [m], P2= 0.049 [m], P3= 0.051 [m] and P4= 0.054 [m], for both the wall shear stress τ_{xy} and the longitudinal pressure gradient dp/dx . We also measured the maximum difference between DNS and triple-deck data (reported in the last column of the table). All the differences are normalised with respect to the maximum absolute value obtained in the DNS results for each case considered.

		$ \Delta dp/dx /\max(dp/dx_{DNS})[\%]$				
	h_r	P1	P2	P3	P4	$\max(\Delta)$
Case 1	Hump A	≈ 0.01	≈ 0.16	≈ 0.52	< 0.01	≈ 27.54
	Hump B	≈ 0.02	≈ 0.18	≈ 0.87	≈ 0.01	≈ 38.28
	Hump C	≈ 0.03	≈ 0.19	≈ 1.09	≈ 0.05	≈ 44.35
Case 2	Hump A	≈ 0.03	≈ 0.19	≈ 0.78	≈ 0.06	≈ 29.90
	Hump B	≈ 0.04	≈ 0.20	≈ 1.04	≈ 0.08	≈ 44.34
	Hump C	≈ 0.08	≈ 0.22	≈ 1.08	≈ 0.09	≈ 54.08

		$ \Delta \tau_{xy} /\max(\tau_{xy_{DNS}})[\%]$				
	h_r	P1	P2	P3	P4	$\max(\Delta)$
Case 1	Hump A	≈ 1.69	≈ 0.17	≈ 1.12	≈ 1.56	≈ 11.63
	Hump B	≈ 0.87	≈ 0.26	≈ 2.26	≈ 1.03	≈ 20.38
	Hump C	≈ 0.49	≈ 0.60	≈ 3.67	≈ 1.06	≈ 25.56
Case 2	Hump A	≈ 1.75	≈ 0.17	≈ 1.19	≈ 1.71	≈ 12.47
	Hump B	≈ 0.86	≈ 0.28	≈ 2.21	≈ 1.22	≈ 23.90
	Hump C	≈ 0.28	≈ 0.83	≈ 3.26	≈ 1.20	≈ 30.34

Table 6.3: Difference between DNS and triple-deck in the dp/dx and τ_{xy} variables normalised with respect to their maximum absolute values obtained from the DNS data at 4 different locations along the flat plate: P1= 0.046 [m], P2= 0.049 [m], P3= 0.051 [m], P4= 0.054 [m]. The last value is the maximum difference between DNS and triple deck data, also normalised with respect to the DNS maximum absolute value.

We can see how the difference between DNS and triple-deck data in terms of dp/dx and τ_{xy} in proximity to the hump (points P2, P3 and maximum difference) increases as the height of the roughness element increases in both subsonic and transonic regimes. In particular we estimated a maximum difference in terms of dp/dx in proximity to the centre of the roughness element comprised between 27% (for $h_r = 0.05\delta_{99}$, subsonic case) to more than 54% (for $h_r = 0.15\delta_{99}$, transonic case) while, moving slightly off from the roughness centre these differences were consistently lower with an order of magnitude below 2%. The maximum differences in terms of τ_{xy} were comprised between 11% to 30% (also in this case normalised with respect to the maximum absolute value of τ_{xy} obtained from the DNS data) in proximity to the centre of the roughness element. These differences consistently decreased moving either upstream or downstream, with maximum differences in these regions below 4%. These differences are consistent with the comments made in section 6.4.1, where we observed a much stronger distortion of the base flow in term of dp/dx and τ_{xy} for the transonic case.

In table 6.4, we quantified the differences in terms of separation bubble lengths, and

	h_r	Separation		$ \Delta(D, R, \mathcal{L}) [\%]$		
		DNS	Triple-deck	D	R	$\mathcal{L}$
Case 1	Hump A	X	X	-	-	-
	Hump B	✓	✓	5.94	9.79	15.73
	Hump C	✓	✓	2.50	26.35	28.85
Case 2	Hump A	X	X	-	-	-
	Hump B	X	X	-	-	-
	Hump C	✓	✓	2.81	23.54	26.35

Table 6.4: Difference between DNS and triple-deck in terms of the detachment point (D), reattachment point (R) and length ($\mathcal{L}$) of the separation bubbles - normalised quantities.

the differences in terms of detachment (D) and reattachment (R) points between DNS and triple-deck. These differences are normalised with respect to the length ℓ_{999} of the hump (we considered a tolerance $\mathcal{T} = 0.001h_r$ at which we measured the length of the hump; this gave us $\ell_{999} \approx 0.00096$ [m]). It is possible to note that the underlying qualitative physics of the problem is well captured by both approaches. In particular, separation happens for the same test cases in both DNS and triple-deck. The differences in terms of detachment point is relatively contained (under 6% for $h_r = 0.10\delta_{99}$ - subsonic case and under 3% for $h_r = 0.15\delta_{99}$ subsonic and transonic cases). The major gap between the two approaches is related to the reattachment point which in turn influences the length of the separation bubble. We can note that this gap increases as the height of the hump increases as seen for τ_{xy} and dp/dx . This is likely due to the local nature of the triple-deck theory which may not capture large separated regions properly (from a quantitative point of view). In addition, the local behaviour of the pressure in proximity and prior to the reattachment point can also have an impact. In fact, in the DNS results we observe a larger streamwise pressure gradient in proximity to and behind the hump if compared to the triple-deck theory. Also, the wall-normal pressure gradient is not negligible in this region. They both (the streamwise and the wall-normal pressure gradients) strongly influence the deceleration of the flow and therefore the position of the reattachment point (see for example Stewartson (1970) and Korolev *et al.* (2002)).

Although the gap related to the reattachment point and therefore to the length of the separation bubble, we consider the agreement between the two approaches satisfactory.

6.5 Conclusions

In this chapter we compared two substantially different approaches, namely DNS and asymptotic triple-deck theory, for calculating the flow past an isolated roughness element on a flat plate with separation behind the element in some situations. The calculations were performed in a compressible regime and both subsonic and transonic conditions were taken into account. We considered three different roughness element heights, ranging from a non-separated flow to a laminar separation bubble behind the obstacle. The study has been carried out by comparing two quantities, the pressure gradient dp/dx at the wall and the wall shear stress τ_{xy} , as well as by evaluating the location of the detachment and reattachment points and the length of the separation bubble where present.

The results obtained for dp/dx and τ_{xy} show an overall good agreement between DNS and triple-deck data except for their maximum and minimum values. This behaviour can be expected since we observe a non-zero wall-normal pressure gradient in the DNS data if compared to the triple-deck results (where a zero pressure gradient along the wall-normal direction is assumed) and a different flow acceleration/deceleration in proximity to the hump between triple-deck and DNS data. This aspect can well affect the local behaviour of the wall shear stress and of the longitudinal pressure gradient, even though some other nonlinear effects can also contribute to the differences observed. In addition, triple-deck relies on the assumption that Re is large, therefore we expect a better agreement if we increased Re , although the Reynolds number adopted in this work is relevant for aeronautical applications. It is also possible to note that the discrepancy between DNS and triple-deck data increases as the height of the roughness element increases and therefore when nonlinear effects become more relevant (see table 6.3 for example).

Concerning the separation bubbles, both DNS and triple-deck predict separation for the same test cases. This indicates that the underlying physics of the problem is well captured in both approaches (in the limit of two-dimensional separation bubbles). Specifically, the detachment point compares well while some rather large differences are present for the reattachment point, likely due to the different longitudinal and wall-normal pressure distributions between DNS and triple-deck data.

This study demonstrates that the asymptotic triple-deck theory captures correctly the qualitative physics and the main flow features in practical aeronautical applications and therefore it can be effectively used as a reduced model in this field. This is of particular use in estimates of receptivity and in separation prediction studies, where the quantities we have

investigated, dp/dx and τ_{xy} are very relevant (see for example Ruban *et al.* (2013); Rizzetta *et al.* (1978))⁴. On the other hand, from a quantitative perspective, the triple-deck theory, is not entirely reliable when considering a nonlinear regime and its use in this context has to be considered carefully, especially if accurate quantitative calculations are required.

This work also constitutes the rationale for further studies on the connections between two approaches historically developed in two different research communities and it allows a more theoretical insight into the DNS data as well as a more applied viewpoint for the triple-deck community.

Finally, the results presented here show that the numerical approaches developed in this thesis work, namely the DG and the FR approaches, presented in chapter 2, along with the connections found in chapter 4 and the dealiasing strategies shown in chapter 5, constitute a robust, reliable and, ultimately, efficient tool for simulating the class of problems investigated in this chapter. Also, the strategy for implementing the boundary conditions presented in chapter 3 has proven effective to tackle these problems, enhancing the overall robustness of the DNS process.

⁴Note that, the velocity profiles are also important for stability and receptivity calculations. However, the two quantities analysed provide a satisfactory insight into the separation characteristics of the flow and they can also be used in receptivity analyses.

Chapter 7

Conclusions and future developments

This thesis focusses on the development and analysis of discontinuous spectral/*hp* element methods - specifically the discontinuous Galerkin (DG) method and the flux reconstruction (FR) approach - and on their applications to compressible boundary-layer flows.

After having implemented the FR approach into the spectral/*hp* element library Nektar++, where an existing implementation of the DG method was present, we have explored the links between the DG method and the FR approach, identifying new underlying connections between the two classes of schemes for nonlinear multidimensional systems of conservation laws on both regular and irregular/curvilinear tensor-product meshes (chapter 4). These connections have helped a better understanding of the FR approach and contextualised it in terms of the wider class of discontinuous spectral/*hp* element methods. Specifically, we have demonstrated that the DG_{SEM} method with an exact mass matrix (also referred to as DG_{SEM} -EMM) is equivalent to the FR_{DG} method for nonlinear equations on both regular and irregular/curved tensor-product meshes. We have also proved that the DG_{SEM} method, with a lumped mass matrix (also referred to as DG_{SEM} -LMM) is equivalent to the FR_{g2} scheme.

All these schemes have been considered in their collocated form - i.e. when a collocation projection of the equations is used - which allows a significant reduction of the computational costs associated to a full-integration of the nonlinear terms. This form, despite being particularly efficient, has a major shortcoming: it produces under-integration errors (also referred to as aliasing errors/issues) when applied to a nonlinear problem. Aliasing issues, in turn, can lead to numerical instabilities and in the most dramatic scenario to the failure of the simulation. This aspect has crucial implications regarding the applicability of these methods to industrial-type problems, where challenging geometries and high-Reynolds numbers hamper the adoption of these attractive methods because of their lack of numerical stability

which translates in a shortage of reliability, highly needed in an industrial context. From a more academic perspective, the study of these issues enlighten the issues associated to aliasing errors in discontinuous spectral/ hp element methods, providing crucial insight into the technicalities needed to tackle these problems. Therefore, after having identified these new connections we have gone further, exploring the implications in terms of aliasing that these connections could have. This allowed us to establish that the aliasing sources in the DG method and in the FR approach are identical and can be recast into the wider class of aliasing issues affecting discontinuous spectral/ hp element methods in general, without a real distinction between the DG and the FR approaches. Hence, the techniques adopted in the more established continuous Galerkin (CG) and DG scientific communities can be equally applied to the FR approach and provide identical results in alleviating aliasing issues. From this perspective, we have investigated the role of consistent integration of the nonlinearities (also referred to as over-integration of the linear terms) for both the DG and FR approaches, proposing two techniques (chapter 5).

The first is referred to as *Local* dealiasing and it targets the nonlinear terms present in the equations being considered (also referred to as PDE nonlinearities or PDE-aliasing sources). This technique tackles ‘locally’ the aliasing issues by means of a new efficient implementation based on the sum-factorisation technique.

The second is referred to as *Global* dealiasing and targets both the the nonlinear terms present in the equations and the additional nonlinearities introduced by deformed/curved elements possibly present in a given mesh (also referred to as geometrical nonlinearities or geometrical-aliasing sources).

By means of these two dealiasing techniques, we explored the role of the boundary terms arising in discontinuous spectral/ hp element methods and compared their importance with respect to the volumetric terms (associated to the volume integrals in the DG method and to the derivatives of the discontinuous flux in the FR approach). From this point of view, we have established that the aliasing arising from the boundary terms becomes more critical as the nonlinearity of the flux function increases. We have also found that a full control on the boundary aliasing sources is not possible because of the rational functions underlying this terms when deformed/curvilinear elements are used. Finally, we have linked the geometrical-aliasing sources to the PDE-aliasing sources, showing how their contributions can become comparable in particular circumstances. These results provide crucial foundations for the development of existing and new dealiasing strategies for discontinuous spectral/ hp element methods.

After having developed and analysed the underlying numerics of the DG and the FR approaches, we have applied this numerical framework to a compressible boundary-layer flow, particularly relevant to the Laminar Flow Control (LFC-UK) centre (chapter 6). This problem involved a high-speed subsonic flow past a flat plate equipped with an isolated Gaussian-shaped roughness element. The spatial resolution required was highly demanding because of the high-Reynolds number considered and the numerical approach needed to produce reliable results to be compared against asymptotic triple-deck data generated at the LFC-UK centre. From this perspective, we used the connections found between the DG and the FR approach in order to identify the best compromise in terms of accuracy and CFL restrictions, that is the FR_{g2} scheme, and we used the *Local* dealiasing technique to address the aliasing issues arising at the first stages of the simulations. This combination allowed us to perform the simulation maintaining the collocated nature of the FR_{g2} scheme while enhancing its numerical stability. The results obtained on this flow configuration indicates that this class of schemes is very attractive for boundary-layer flow problems and using the connections and dealiasing strategies developed in the course of this thesis can overcome one of the major shortcoming of this class of schemes that is the numerical stability.

Ultimately, we have developed an efficient and effective numerical framework for compressible flow simulations, with particular focus on boundary-layer flows, and we have provided a more comprehensive view on the FR and DG approaches, recasting them into the broader class of discontinuous spectral/*hp* element methods. In particular, we have shown how, by exploiting the connections between the FR and DG schemes we can attain critical advantages in terms of time-step restrictions (the use of the FR_{g2} or the $\text{DG}_{\text{SEM-LMM}}$ scheme allows the bigger time-step across the methods investigated), aliasing properties (the consistent integration of the nonlinearities can be equally applied to both class of methods providing identical results) and implementation (having an existing implementation of the DG method into a library, automatically allows recovering the FR_{g2} scheme by under-integrating the mass matrix).

We also speculate that the overall family of DG methods can be re-formulated to recover a wide-range schemes (as the FR approach) with different accuracy and time-restriction properties. This aspect might indicate the way forward for future research efforts in the DG scientific community.

Some other points have remained unaddressed in this work. This includes the extension of the new connections to simplex element, the role of the GPUs vs. CPUs. in the performance of the two classes of schemes, the benefits of using other dealiasing strategies such as more

stable forms of the nonlinear terms, more optimal quadrature points, polynomial filtering compared to the consistent integration and the related computational costs.

As a final remark, the entire implementation work - the DG diffusion operator, the FR advection and diffusion terms and the compressible flow solver developed - are now part of the spectral/ hp element library Nektar++ and available to the user-community of the code. Note that also the efficient extension to three-dimensional problems by means of a Fourier basis in one (homogenous) direction, briefly introduced in appendix A, has been part of the implementation work and is now into the Nektar++ library.

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Appendix A

Efficient extension to three-dimensional problems: FR- and DG-Fourier approximations

The FR- and DG-Fourier approximations consist of discretising a three-dimensional problem by combining a Fourier expansion with the spectral/*hp* element method. In particular, the $x_1 - x_2$ plane is discretised through a spectral/*hp* element discretisation - in our case, either FR or DG - while, the x_3 -direction is assumed to be homogenous and is represented through a Fourier series. This methodology was first presented by Karniadakis (1990) for incompressible flows and it has been successively applied to heat transfer in complex geometries by Amon (1995) and extended to cylindrical coordinates by Blackburn & Sherwin (2004). All these references employ a continuous Galerkin spectral/*hp* element approximation in the $x_1 - x_2$ plane. A DG-Fourier approximation was first proposed by Sherwin (1999) and has been recently applied to rotating sliding meshes by Ferrer & Willden (2012). There is no existing literature regarding the FR approach coupled with a Fourier expansion although the machinery behind this approximation is essentially identical to the DG-Fourier approach.

In the following, we detail the main steps needed to implement either an FR- or a DG-Fourier approximation for the following first order three-dimensional conservation law

$$\frac{\partial u(x_1, x_2, x_3; t)}{\partial t} + \nabla \cdot \mathbf{f}(u) = 0, \quad (\text{A.1})$$

where

$$\mathbf{f}(u) = [f_1(u), f_2(u), f_3(u)] \quad (\text{A.2})$$

is the tensor which describes the transport of the conserved variable $u(x_1, x_2, x_3; t)$ and each $i - th$ column is the advection flux associated to the x_i -direction.

The first step is to represent the conserved variable in terms of a Fourier transform

$$u(x_1, x_2, x_3; t) = \sum_r u_r(x_1, x_2; t) e^{i\beta_r x_3}, \quad (\text{A.3})$$

with $i = \sqrt{-1}$ and $\beta_r = 2\pi r / L_{x_3}$.

If we now substitute Eq. (A.3) into Eq. (A.1), we obtain

$$\begin{aligned} \frac{\partial}{\partial t} \left[\sum_r u_r(x_1, x_2; t) e^{i\beta_r x_3} \right] &+ \frac{\partial}{\partial x_1} \left[f_1 \left(\sum_r u_r(x_1, x_2; t) e^{i\beta_r x_3} \right) \right] + \\ &+ \frac{\partial}{\partial x_2} \left[f_2 \left(\sum_r u_r(x_1, x_2; t) e^{i\beta_r x_3} \right) \right] + \\ &+ \frac{\partial}{\partial x_3} \left[f_3 \left(\sum_r u_r(x_1, x_2; t) e^{i\beta_r x_3} \right) \right] = 0. \end{aligned} \quad (\text{A.4})$$

Equation (A.4) can be reduced further by expanding the derivatives and by gathering the term ' $e^{i\beta_r x_3}$ '

$$\begin{aligned} \sum_r e^{i\beta_r x_3} \left\{ \frac{\partial}{\partial t} \left[u_r(x_1, x_2; t) \right] &+ \frac{\partial}{\partial x_1} \left[f_1 \left(\sum_r u_r(x_1, x_2; t) \right) \right] + \right. \\ &+ \frac{\partial}{\partial x_2} \left[f_2 \left(\sum_r u_r(x_1, x_2; t) \right) \right] + \\ &\left. + i\beta_r f_3 \left(\sum_r u_r(x_1, x_2; t) \right) \right\} = 0, \end{aligned} \quad (\text{A.5})$$

which, in compact form, becomes

$$\sum_r e^{i\beta_r x_3} \left\{ \frac{\partial}{\partial t} \left[u_r(x_1, x_2; t) \right] + \nabla_{(x_1, x_2)} \cdot \mathbf{f}_{(x_1, x_2)} + i\beta_r f_3 \left(\sum_r u_r(x_1, x_2; t) \right) \right\} = 0, \quad (\text{A.6})$$

where $\nabla_{(x_1, x_2)} = [\partial/\partial x_1, \partial/\partial x_2]$ and $\mathbf{f}_{(x_1, x_2)} = [f_1(u), f_2(u)]$. At this stage, we need to decompose each two-dimensional plane, which corresponds to a given value of r , into N non-overlapping subdomains, and transform each subdomain into a standard element. Using the weak form of the DG advection operator shown in section 2.3.1 and the final form of the advection FR operator shown in section 2.3.2, we obtain the final expressions for the DG-Fourier and FR-Fourier methods:

DG-Fourier advection operator

$$\begin{aligned} \iint_{\Omega_s} \frac{\partial \widehat{u}_r^\delta}{\partial t} \ell(\boldsymbol{\xi}) \, d\boldsymbol{\xi} - \iint_{\Omega_s} \left[\widehat{f}_{1,r}^\delta \frac{\partial \ell(\boldsymbol{\xi})}{\partial \xi_1} + \widehat{f}_{2,r}^\delta \frac{\partial \ell(\boldsymbol{\xi})}{\partial \xi_2} \right] d\boldsymbol{\xi} + \int_{\partial\Omega_s} \mathbf{n} \cdot \left[\widehat{\mathbf{f}}_r^{\delta I} \ell(\boldsymbol{\xi}) \right] d\boldsymbol{\xi} + \\ + i \iint_{\Omega_s} \beta_r f_{3,r} \ell(\boldsymbol{\xi}) d\boldsymbol{\xi} = 0. \end{aligned} \quad (\text{A.7})$$

FR-Fourier advection operator

$$\begin{aligned} \frac{d\widehat{\mathbf{u}}}{dt} = \sum_{i,j=0}^{i,j=P} \widehat{f}_{1,i,j}^{\delta D} \frac{\partial \ell_i(\xi_1)}{\partial \xi_1} \ell_j(\xi_2) + \sum_{i,j=0}^{i,j=P} \widehat{f}_{2,i,j}^{\delta D} \ell_i(\xi_1) \frac{\partial \ell_j(\xi_2)}{\partial \xi_2} + \\ + \left((\widehat{\mathbf{f}} \cdot \widehat{\mathbf{n}})_L^{\delta I} - \widehat{f}_{1,L}^{\delta DI} \right) \frac{\partial \Psi_L}{\partial \xi_1} + \left((\widehat{\mathbf{f}} \cdot \widehat{\mathbf{n}})_R^{\delta I} - \widehat{f}_{1,R}^{\delta DI} \right) \frac{\partial \Psi_R}{\partial \xi_1} + \\ + \left((\widehat{\mathbf{f}} \cdot \widehat{\mathbf{n}})_B^{\delta I} - \widehat{f}_{2,B}^{\delta DI} \right) \frac{\partial \Psi_B}{\partial \xi_2} + \left((\widehat{\mathbf{f}} \cdot \widehat{\mathbf{n}})_T^{\delta I} - \widehat{f}_{2,T}^{\delta DI} \right) \frac{\partial \Psi_T}{\partial \xi_2} + \\ + i \beta_r \widehat{f}_{3,r}^{\delta D} \ell(\boldsymbol{\xi}), \end{aligned} \quad (\text{A.8})$$

The methodology for second order PDEs is identical to the above with the exception that, in the case of a second order flux, we need to split the system into a principal and an auxiliary equation and follow the steps in section 2.4.1 and 2.4.2. For the sake of brevity, the description of the DG- and FR-Fourier approximations for second order problem is not reported here.

Appendix B

Tabulated L_2 errors

In this appendix we report the tabulated values of the L_2 errors associated to the results shown in chapter 5. Specifically, tables B.1, B.2, B.3, B.4 show the L_2 errors associated to the test case with higher advection velocities. Table B.5 reports the differences in terms of L_2 norm between mass matrices calculated with different number of quadrature points.

$P_{adv} = 1$ (Mesh A)		
	DG _{SEM} -EMM	FR _{g2}
$Q = 5$	0.00258809184895	0.00517893325001
$Q = 5, Q_V = 6, Q_I = 5$	0.00245321968401	0.00484792022350
$Q = 5, Q_V = 7, Q_I = 5$	0.00245321968401	0.00484792022350
$Q = 5, Q_V = 5, Q_I = 6$	0.00274779163543	0.00520769016825
$Q = 5, Q_V = 5, Q_I = 7$	0.00274779163543	0.00520769016825
$Q = 5, Q_V = 6, Q_I = 6$	0.00238095562954	0.00483526988468
$Q = 5, Q_V = 7, Q_I = 7$	0.00238095562954	0.00483526988468

Table B.1: L_2 errors for *Local* strategy applied to the DG and FR formulations. Comparative evaluation of the influence of interface dealiasing for $P_{adv} = 1$.

$P_{adv} = 2$ (Mesh A)		
	DG _{SEM} -EMM	FR _{g2}
$Q = 5$	0.00162175780389	0.00255493783002
$Q = 5, Q_V = 6, Q_I = 5$	0.00200593990694	0.00273091929261
$Q = 5, Q_V = 7, Q_I = 5$	0.00195899758668	0.00270213303084
$Q = 5, Q_V = 8, Q_I = 5$	0.00195899758668	0.00270213303084
$Q = 5, Q_V = 5, Q_I = 6$	0.00192599663912	0.00264115695244
$Q = 5, Q_V = 5, Q_I = 7$	0.00192322522336	0.00264249943410
$Q = 5, Q_V = 5, Q_I = 8$	0.00192322522336	0.00264249943410
$Q = 5, Q_V = 6, Q_I = 6$	0.00167138551873	0.00256594327336
$Q = 5, Q_V = 7, Q_I = 7$	0.00166192034988	0.00255904806344
$Q = 5, Q_V = 8, Q_I = 8$	0.00166192034988	0.00255904806344

Table B.2: L_2 errors for *Local* strategy applied to the DG and FR formulations. Comparative evaluation of the influence of interface dealiasing for $P_{adv} = 2$.

$P_{adv} = 3$ (Mesh A)		
	DG _{SEM} -EMM	FR _{g2}
$Q = 5$	0.00145932994542	0.00170667104239
$Q = 5, Q_V = 6, Q_I = 5$	0.00180851746188	0.00195005809316
$Q = 5, Q_V = 7, Q_I = 5$	0.00171296996980	0.00185789594691
$Q = 5, Q_V = 8, Q_I = 5$	0.00171296996980	0.00185789594691
$Q = 5, Q_V = 5, Q_I = 6$	0.00171031388518	0.00188971296917
$Q = 5, Q_V = 5, Q_I = 7$	0.00171086707088	0.00189247015965
$Q = 5, Q_V = 5, Q_I = 8$	0.00171086707088	0.00189247015965
$Q = 5, Q_V = 6, Q_I = 6$	0.00143565622568	0.00167138551874
$Q = 5, Q_V = 7, Q_I = 7$	0.00143604324583	0.00168273243243
$Q = 5, Q_V = 8, Q_I = 8$	0.00143604324583	0.00168273243243

Table B.3: L_2 errors for *Local* strategy applied to the DG and FR formulations. Comparative evaluation of the influence of interface dealiasing for $P_{adv} = 3$.

$P_{adv} = 4$ (Mesh A)		
	DG _{SEM} -EMM	FR _{g2}
$Q = 5$	0.00126816984360	0.00131289321292
$Q = 5, Q_V = 6, Q_I = 5$	0.00140946714795	0.00141416909853
$Q = 5, Q_V = 7, Q_I = 5$	0.00136256229206	0.00138729451635
$Q = 5, Q_V = 8, Q_I = 5$	0.00136245047508	0.00138724135413
$Q = 5, Q_V = 9, Q_I = 5$	0.00136245047508	0.00138724135413
$Q = 5, Q_V = 5, Q_I = 6$	0.00134288748995	0.00134093637296
$Q = 5, Q_V = 5, Q_I = 7$	0.00134727708510	0.00135650237293
$Q = 5, Q_V = 5, Q_I = 8$	0.00134727957789	0.00135657503799
$Q = 5, Q_V = 5, Q_I = 9$	0.00134727957789	0.00135657503799
$Q = 5, Q_V = 6, Q_I = 6$	0.00126591291017	0.00130947841396
$Q = 5, Q_V = 7, Q_I = 7$	0.00126617520357	0.00130988788959
$Q = 5, Q_V = 8, Q_I = 8$	0.00126617189683	0.00130988726422
$Q = 5, Q_V = 9, Q_I = 9$	0.00126617189683	0.00130988726422

Table B.4: L_2 errors for *Local* strategy applied to the DG and FR formulations. Comparative evaluation of the influence of interface dealiasing for $P_{adv} = 4$.

Case A: $[M(J)_{Q+1} - M(J)_Q]_{L_2}$				
	$J \in \mathcal{P}^1$	$J \in \mathcal{P}^2$	$J \in \mathcal{P}^3$	$J \in \mathcal{P}^4$
$Q_{min} - Q_{min} + 1,$	0.7244	0.3144	0.1773	0.1176
$Q = 4$	0.0160			
$Q = 5$	4.4222e-16			
$Q = 6$		0.0162		
$Q = 7$		0.0055		
$Q = 8$		3.6038e-15	0.0135	
$Q = 9$			0.0079	
$Q = 10$			0.0027	0.0105
$Q = 11$			1.2322e-12	0.0080
$Q = 12$				0.0047
$Q = 13$				0.0016
$Q = 14$				3.4482e-15

Case B: $[M(J)_{Q+1} - M(J)_Q]_{L_2}$				
	$J \in \mathcal{P}^2$	$J \in \mathcal{P}^4$	$J \in \mathcal{P}^6$	$J \in \mathcal{P}^8$
$Q_{min} - Q_{min} + 1,$	0.8368	0.3472	-	-
$Q = 4$	0.0210			
$Q = 5$	1.1943e-15			
$Q = 6$		0.0351		
$Q = 7$		0.0096		
$Q = 8$		4.2100e-15		

Table B.5: L_2 norm of the difference between mass matrices calculated with Q and $Q+1$ GLL points for the four orders considered and using both the mesh in figure 5.8.

Appendix C

Dealiasing within triangular and tetrahedral regions

Throughout this paper, most of the applications demonstrate how the local and global dealiasing techniques can be applied in the setting of quadrilateral and hexahedral elements. In this appendix, we show that the same arguments are also applicable in the case of tensor-product triangular elements. An extension of this argument to prismatic and tetrahedral elements follows easily.

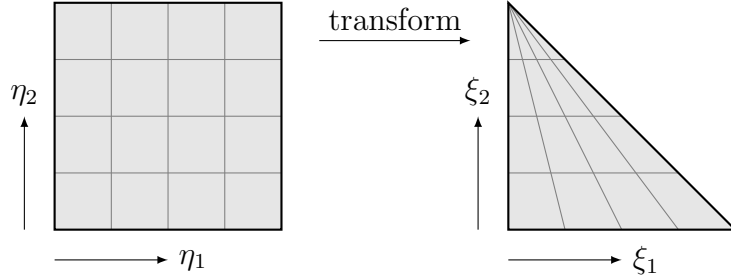


Figure C.1: Illustrative diagram describing Duffy transformation between collapsed coordinates $(\eta_1, \eta_2) \in [-1, 1]^2$ and Cartesian coordinates $(\xi_1, \xi_2) \in \Omega_{\text{st}}$.

We consider a tensor product of two one-dimensional hierarchical hp expansion basis functions ψ^a and ψ^b , so that in the standard triangle $\Omega_{\text{st}} = \{(\xi_1, \xi_2) \mid \xi_1 \in [-1, 1], \xi_1 + \xi_2 \leq 0\}$, an approximate solution u^δ may be written as an expansion

$$u^\delta(\xi_1, \xi_2) = \sum_{p=0}^P \sum_{q=0}^{P-p} \hat{u}_{pq} \psi_p^a(\eta_1) \psi_{pq}^b(\eta_2)$$

where the collapsed coordinates $(\eta_1, \eta_2) \in [-1, 1]^2$ are given by the Duffy transformation

$$\eta_1 = 2 \frac{1 + \xi_1}{1 - \xi_2} - 1, \quad \eta_2 = \xi_2.$$

The use of this collapsed coordinate system within the standard quadrilateral region $[-1, 1]^2$ leads to a tensor product of quadrature points within the standard triangular region, as shown in figure C.1.

Although this transformation is singular at the top vertex of the triangle, the use of Gauss-Radau quadrature in the η_2 direction, in which the top vertex is excluded, leads to a formulation without geometric singularities. Additionally, Gauss-Radau points with a choice of $\alpha = 1$ and $\beta = 0$ naturally incorporate the Jacobian term which appears in integrals over the standard triangular region when weighted by a constant factor of $\frac{1}{2}$ Karniadakis & Sherwin (2005).

In this setting, the sum-factorised dealiasing methods described in section 5.3.1 can be utilised for triangular elements by applying the stated tensor product logic to the grid of (η_1, η_2) quadrature points, without applying the Duffy transformation to obtain the desired dealiasing effect in the simplex region. Mathematically this approach projects the nonlinear terms down to the tensor product space spanned by the (η_1, η_2) space which is typically richer than the space spanned by the triangular expansion in the (ξ_1, ξ_2) space. However since the inner product using the collapsed coordinates spans the (η_1, η_2) space, this is sufficient to ensure the non-linear product is correctly integrated. This logic extends to three dimensions for both prismatic elements and tetrahedra when coupled with suitable extensions of the Duffy transformation to a hexahedral region.